

Toledo — Equation Library Catalogue

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Contents

Part 1 — Genesis-first catalogue

A.1 — A.1 The root of reality is not a domain

tier: untagged; status: current

$\delta_R = (a \# b)$ composes two retained differences; the running retained state is the triple $\mathfrak{S}_n = (G_n, \Lambda_n, \mathcal{T}_n)$ -- graph, labels, tape. [[Reality is retained before it is divided into domains]]

A.10 — A.10 Number of fields $\neq$ number of domains ★

tier: untagged; status: current

One field is not one domain; two fields are not two domains. Several fields can be a single domain if the full coupling between them is jointly necessary for one readout: $\mathcal{D}_\alpha = (\Phi, \Psi, \chi, \dots) / \sim_\alpha$. "Do we need two fields?" is a sufficiency/closure question (A.6's gate), not a field-counting question; must be settled by whether the candidate state is sufficient (A.6) and the partition closes exactly (A.4), never by counting symbols in the state vector.

A.11 — A.11 Domains can change over time

tier: untagged; status: current

$\mathcal{P}_{\{\alpha, n\}} \neq \mathcal{P}_{\{\alpha, n+1\}}$ -- a partition can develop a defect, a cell splitting $C_a \rightarrow C_{\{a, 1\}} \sqcup C_{\{a, 2\}}$, and when the equivalence that justified a coarse cell breaks, refinement is mandatory, not optional. Coarsening back is permitted only when freshly proven to preserve dynamics, readout, and residue.

A.12 — A.12 Cross-domain relationships

tier: untagged; status: current

$K_{\{\alpha \leftarrow \beta\}}: \mathcal{D}_\beta \times \mathcal{T} \rightarrow \mathcal{D}^\wedge_\alpha$ $\epsilon_{\{\alpha \leftarrow \beta\}} = d_\Delta(\mathcal{D}_\alpha, \mathcal{D}^\wedge_\alpha)$. A bridge between domains need not be exact; when it is not, it must carry a stated, non-zero recovery error $\epsilon_{\{\alpha \leftarrow \beta\}}$, never a silent claim of equivalence. [[translatable $\neq$ exactly equivalent]]

A.2 — A.2 What a domain is

tier: untagged; status: current

Let $\ddagger_n$ be a state proposed with translation in mind, F_n its dynamics, Q_α a question, 0_α the readout that question demands, and $q_{\{\alpha, n\}}$ the candidate translation map. Then $\mathcal{D}_{\{\alpha, n\}} = \ddagger_n / \sim_{\{\alpha, n\}}$. [[Domain = minimal sufficient dynamically closed translation]]

A.3 — A.3 A domain is relative to its question — but not arbitrary

tier: untagged; status: current

$Q_\alpha \neq Q_\beta$ can force $\mathcal{D}_\alpha \neq \mathcal{D}_\beta$. There is no "absolute domain" that exists independently of some readout asking for it; every candidate partition still has to pass the exact quotient gate, and a partition that fails it is not a domain, whatever question motivated proposing it.

A.4 — A.4 The Exact Domain Gate

tier: untagged; status: current

$q_{\{\alpha, n+1\}} \circ F_n = F_{\{\alpha, n\}} \circ q_{\{\alpha, n\}} \quad 0_{\{\alpha, n\}} = 0_{\{\alpha, n\}} \circ q_{\{\alpha, n\}} \quad q_{\{\alpha, n\}}(z) = q_{\{\alpha, n\}}(z') \implies 0_{\{\alpha, n:n+L\}} = 0_{\{\alpha, n:n+L\}}(z', u)$

A.5 — A.5 State is not domain

tier: untagged; status: current

$\mathfrak{S}_n \neq \mathfrak{F}_{\{\alpha, n\}} \neq \mathcal{D}_{\{\alpha, n\}}$ -- retained reality, the representation chosen to work with, and the quotient discovered on top of that representation are three distinct objects.

Code	Dom.	Statement	Status
A.5/E.01.v1	E	$A_n \neq r_n \neq x_n$; root retention $\neq$ belief	current
A.5/E.02.v1	E	state $\neq$ meaning $\neq$ experience $\neq$ choice $P^{\wedge}H_t \rightarrow Q_{\{H, t\}} \text{---Decompose_AI} \rightarrow Q^{\wedge}\{AI\}_{\{t+1\}} \text{---T_AI} \rightarrow H_{\sim\{t+1\}} \text{---}/\sim_{\{D, t\}} \rightarrow H^{\wedge}\{disc\}_{\{t+1\}} \text{---Gate} \rightarrow U_{\{t+1\}} \text{---Spawn} \rightarrow F_{\{t+1\}} \rightarrow (Q_{\{t+2\}}, H_{\sim\{t+2\}}) \text{ or } \text{---}u^* \rightarrow \delta_{\{t+1\}} \text{---revise} \rightarrow P_{\{t+1\}}$	current
A.5/E.03.v1	E	$D_{\Phi}(x_0) = \{d_0_{\{x'\}} = (x_0, x') : \Phi(x') \neq \Phi(x_0)\}$; $P(d) = (V_d, E_d, \tau_d)$	current
A.5/E.04.v1	E	Role Separation: generation, truth, evidential support, reliability, understanding, possession, endorsement, accountability, credibility, and institutional authorization are distinct epistemic roles. No pair may be identified merely because one human agent often occupies both.	current
A.5/E.05.v1	E	Representationality $\neq$ Selectivity	current
A.5/E.06.v1	E	§8: Exposure $\neq$ felt intensity $\neq$ retention $\neq$ improvement. §11: external pattern $\neq$ meaning; meaning $\neq$ explicit naming; shared stimulus $\neq$ shared inner state; affective intensity $\neq$ epistemic warrant; accessibility $\neq$ truth; resonance $\neq$ understanding; retention $\neq$ benefit; verbalization $\neq$ lossless translation; same surface form $\neq$ same reader-relative meaning.	current

Code	Dom.	Statement	Status
A.5/E.07.v1	E	EMG-15: order $\neq$ rhythm, repetition $\neq$ rhythm, rhythm $\neq$ meaning, rhythm $\neq$ retention. EMG-20: release $\neq$ transformation, intensity $\neq$ truth, shock $\neq$ insight, barrier-crossing $\neq$ improvement. EMG-27: phenomenon $\neq$ experience; event trace $\neq$ meaning; meaning-giving $\neq$ truth-making; pre-explicit significance $\neq$ fully-formed proposition; naming $\neq$ passive reporting; resonance $\neq$ identity; resonance $\neq$ truth; rhythm $\neq$ repetition count; rhythm $\neq$ meaning; accessibility $\neq$ warrant; semantic attraction $\neq$ evidence; informational work $\neq$ physical energy; shock $\neq$ coherent insight; release $\neq$ durable transformation; retention $\neq$ improvement; Human-AI amplification $\neq$ synergy; coupled performance $\neq$ Human Return. EMG-29: $\mu^{\text{aff}} \neq \mu^{\text{epi}}$, $\mu^{\text{auto}} \neq \mu^{\text{epi}}$, $\mu^{\text{prag}} \neq \text{truth}$.	current
A.5/E.08.v1	E	successful adaptation = mental health [explicitly REJECTED]	current
A.5/E.09.v1	E	usable $\neq$ true; T_U down $\Rightarrow$ W(H) up $\Rightarrow$ truth	current
A.5/H.01.v1	H	H_t -L_H- > Q_t, Q_t $\neq$ H_t (Pre-Prompt Human State Principle); session-indexed re-statement H_{s,0} -L_H- > Q_{s,0}	current
A.5/H.02.v1	H	P^{live}_{H,t} defined; TopicEntry $\in$ {LiveProblem, OpenExploration, RoutineDelegation}; Problem-First $\Rightarrow$ {Agenda Ownership, Bounded Relevance, World-Side Testability, Observable Human Return}; Problem-First $\neq$ Problem-Only	current
A.5/H.03.v1	H	U_{n+1} = \Pi_{U \geq 0} [(I - D_U) U_n + J_{reinforce,n} + J_{release,n}]; sufficient condition J_{release} - J_{reinforce} $\geq \epsilon_{release} > 0$; release requires $z_n \in V_A$, $O_E(z_n) \neq 0$, $\ U_n\ \rightarrow 0$; nonbinding mode $v_{free} \in \ker B_I$ with $O_E(v_{free}) \neq 0$	current
A.5/H.04.v1	H	$\Sigma_{\{H+AI\}} = \frac{D^{\{use\}}_{\{H+AI\}}(B,Q)}{\max\{D^{\{use\}}_H(B,Q), D^{\{use\}}_{\{AI\}}(B,Q), 1\}}$; non-collapse: more AI output $\neq$ more epistemic diversity, more epistemic diversity $\neq$ better warrant	current
A.5/H.05.v1	H	$\pi^{\text{choice}}_{\{A,t\}} \in \Pi^{\text{live}}_{\{A,t\}}(g)$; $\pi^{\text{act}} \neq \pi^{\text{choice}}$ possible, Y_{obs} = O_q(H_{0:T}) $\neq$ H_{0:T}; possible $\neq$ feasible $\neq$ live $\neq$ chosen $\neq$ enacted $\neq$ observed	current
A.5/H.06.v1	H	L_{A,t+1} $\neq$ L_{A,t} (if a residue is retained); at world-system scale: $\Lambda^{\text{live}}_{H,t} = \lambda_1 B_t + \lambda_2 X_t + \lambda_3 P^{\text{plural}}_t + \lambda_4 r^{\text{H}}_{t-1} - \lambda_5 D_t - \lambda_6 C^{\text{I}}_{t-\delta} \Lambda^{\text{live}}_{H,t}$	current

Code	Dom.	Statement	Status
A.5/H.07.v1	H	AI-first fluency $\neq$ human baseline; explanation $\neq$ verification; resistance quality $\neq$ resistance accessibility; uncertainty signal $\neq$ truth; output count $\neq$ epistemic diversity; trust $\neq$ appropriate re- liance; self-confidence $\neq$ competence; amplification $\neq$ warrant; retention $\neq$ improvement; augmentation $\neq$ synergy; joint gain $\neq$ human return; user preference $\neq$ epistemic direction	current
A.5/H.08.v1	H	$\Delta\text{Performance_AI} > 0 \not\Rightarrow \Delta\text{H_return} > 0$ (equivalently: Assisted performance $\neq$ Unaided Human Return)	current
A.5/H.09.v1	H	Exposure $\neq$ Retention $\neq$ Improvement	current
A.5/H.10.v1	H	AI fluency $\neq$ human baseline; explanation $\neq$ verification; out- put count $\neq$ epistemic diversity; exposure $\neq$ retention $\neq$ improvement; trust in AI $\neq$ calibrated confidence in self; augmentation $\neq$ synergy $\neq$ Human Re- turn; retained gain $\neq$ net expansion; provenance $\neq$ warrant; persistence $\neq$ truth	current
A.5/H.11.v1	H	Anchor $\rightarrow$ Expand $\rightarrow$ Oppose $\rightarrow$ Discriminate $\rightarrow$ Verify $\rightarrow$ Integrate $\rightarrow$ Remove $\rightarrow$ Return; $Q_t\text{-AI} \rightarrow \{Q_i, K_i\}$; Rival Generation / Boundary Test; Oppose $\neq$ manufacture	current
A.5/H.12.v1	H	$L(c) = \{\text{Decision, Risk, Money, Health, Legal, Publication, Irreversible Action}\}$; $V = \{\text{primary source, data, experiment, calculation, expert, independent}$ method, world outcome $\}$; Source $\neq$ AI synthesis $\neq$ Human inference $\neq$ Unverified candidate	current
A.5/H.13.v1	H	$E := \text{event}$; $C_e := (S_e, R_e, A_e, \tau_e)$; $T^\wedge := I(E C_acc)$; $T^\wedge \neq E$; general: $T^\wedge(k) = I(E C_acc^\wedge(k))$; epoch 3 (LLM): $T^\wedge(3) = I(E C_acc^\wedge(3)) + \Delta C$ via dialogue; $E \rightarrow C_acc^\wedge(0) \rightarrow \dots \rightarrow C_acc^\wedge(3)$	current
A.5/H.14.v1	H	G (referential grounding); Ge (experiential grounding); Emb (embodiment); $G \neq Emb$	current
A.5/H.15.v1	H	Resources $\neq$ Access, Access $\neq$ Capability, Capability $\neq$ Realized Opportunity; Access(z) $\neq$ Control(z); Equal AI Access $=/= >$ Equal Capability Conversion	current
A.5/H.16.v1	H	$B^\wedge_{\{i,n\}} \subseteq \{\text{Knowledge, Skill, Language, Tool, Resource, Time, Network, Credential, Pe}$ Observed Difficulty $\neq$ Skill Deficit	current
A.5/H.17.v1	H	$C^\wedge_{\text{cand}}\{i,n\} = \text{Gen}(P^\wedge_{\text{live}}, B^\wedge_{\text{bar}}, K^\wedge_{\text{life}}, R^\wedge_{\text{return}})$; $C^\wedge_{\text{live}}\{i,n\} = \{c \in C^\wedge_{\text{cand}} : \text{Endorse}_i(c) = 1\}$; Proactive Suggestion $\neq$ Human Goal Own- ership; Capability Advancement $\neq$ AI Goal Authority; Scaffolding $\neq$ Control	current
A.5/H.18.v1	H	$\Omega^\wedge_{\text{real}}\{i,n\} = G_O(R^\wedge_{\text{return}}\{H,i,n\}, K^\wedge_{\text{life}}\{i,n\}, C^\wedge_{\text{cred}}\{i,n\}, \text{Net}\{i,n\}, \text{Perm}\{i,n\})$; Credential $\neq$ Capability; Market Legibility $\neq$ Human Worth; Skill $\rightarrow$ Evidence of Skill $\rightarrow$ Recognition $\rightarrow$ Opportunity	current

Code	Dom.	Statement	Status
A.5/H.19.v1	H	Outcome_C–Outcome_A $\neq$ Effect_HCA; Schooling+Human-Owned Inquiry+Supervised Adaptive AI+Projects+Human Return+Opportunity Bridges	current
A.5/M.01.v1	M	$S_n \neq Z_{\{D,n\}} \neq D_{\{D,n\}}$ (D=HCA or MEMK)	current
A.5/M.02.v1	M	$P_{\text{local}} \not\subset D_{\text{AI}}$; MultiAIConsensus= $\neq$ >GeographicCompleteness; $DVP^*=DVP+GDA$; $S_G=L \times G \times M \times B \times F$; ManuscriptStrength>IdentityDependence; ConversionPlan_i={Global_i,Thai_i}; $K2, \text{Global}+K2, \text{Thai} \Rightarrow \text{RouteDiversity} \uparrow$; FrictionValue_e;	current
A.5/M.03.v1	M	GlobalAsset $\rightarrow$ ThaiLegibility $\rightarrow$ ThaiFriction neighboring evidence $\neq$ formal-variable validation $\neq$ truth of the integrated theory; reachability $\neq$ accessibility; speed $\neq$ quality/warrant; AI output volume $\neq$ epistemic diversity; Attraction $\neq$ Accessibility $\neq$ Warrant $\neq$ Truth; generation speed $\neq$ hypothesis quality	current
A.5/M.04.v1	M	$G1K \neq G2K \Rightarrow L(\tau_U G1K, Q) \neq L(\tau_U G2K, Q)$	current
A.5/M.05.v1	M	$\text{Ess}(d) = \bigcap \{p \in \Pi(d)\} V(p)$; $\Pi_{\{\Delta v\}}(d) = \{p \in \Pi(d) : v \notin V(p)\}$; Corollary (misrouted defeat under silent lift): if $\text{Ess}(d) \Delta \hat{\text{Ess}}(d) \neq \emptyset$, a node-specific defeater aimed at that symmetric difference produces a represented vulnerability that differs from d's actual dependency vulnerability.	current
A.5/M.06.v1	M	$A \neq x \neq \mu \neq E \neq M \neq \text{Bel} \neq p \neq \sigma_K(p)$; $A_i = A_j \Rightarrow x_i = x_j$; {Belief-Strength, BeliefDistribution, Auth, Pow, Val_E}(p) typed-distinct; StatusOfSpeaker $\neq$ StatusOfClaim, Auth(p) $\neq$ Val_E(p), Pow(p) $\neq$ Val_E(p); $\sigma_K(p) \neq \Pi_{\text{prac}}(p)$; Bel_global(p) $\neq$ K_global(p); $A_A \neq I_A \neq \text{Phen}_A \wedge \text{str} \neq P_A \wedge \text{lived} \neq \text{Own}_A \neq \text{Coh}_A \neq \text{Val}_A$; $R_A(1) \neq \text{Claim}_A \neq \text{Identity}_A$, Identity_A $\neq$ ActionCandidate_A $\neq$ CommittedAction_A; CurrentReadout+SelectionPriority+SelectionStability+SelfReport $\Rightarrow G_A \wedge \text{MR}$ (no-free-governance)	current
A.5/M.07.v1	M	Credit != EpistemicValue	current
A.5/M.08.v1	M	Friction != Fellowship	current
A.5/M.09.v1	M	SelfExperience != GeneralEvidence	current
A.5/M.10.v1	M	PositionalAccess != PopulationAuthority	current
A.5/M.11.v1	M	CommunityTrust != Representativeness	current
A.5/M.12.v1	M	DVP $\Rightarrow$ K2	current
A.5/M.13.v1	M	ManyModels $\Rightarrow$ Independence	current
A.5/M.14.v1	M	MechanicalValidity != SemanticValidity	current
A.5/M.15.v1	M	SourceExistence != ClaimSupport	current
A.5/M.16.v1	M	Friendship != IndependentEvidence	current
A.5/M.17.v1	M	Correspondence != PeerReview	current
A.5/M.18.v1	M	IntellectualAffinity != Truth	current

Code	Dom.	Statement	Status
A.5/M.19.v1	M	ActivationAction != CreditEvent	current
A.5/M.20.v1	M	RawSpeed(down) ==> V_C(down)	current
A.5/M.21.v1	M	L_H != Truth, L_V != Truth	current
A.5/M.22.v1	M	MultiAIConsensus ==> GeographicCompleteness	current
A.5/M.23.v1	M	DoubleBlind = Bonus; DoubleBlind != Requirement	current
A.5/M.24.v1	M	K2,Global != K2,Thai	current
A.5/M.25.v1	M	InterventionCreator != SoleEvaluator	current
A.5/M.26.v1	M	PracticeExperience != PopulationEvidence	current
A.5/M.27.v1	M	A_t != C_t^scholarly	current
A.5/M.28.v1	M	M_attention != M_truth, M_attention != M_K2	current
A.5/M.29.v1	M	NoHumanAvailable ==> ResearchStop	current
A.5/M.30.v1	M	Prestige ==> APCApproval; APCApproval ==> FieldFit + CreditYield + BudgetFit	current
A.5/M.31.v1	M	DisclosurePenalty ==> Concealment; DisclosurePenalty ==> BetterProvenance + BetterHumanDefence + VenueFit	current
A.5/M.32.v1	M	AIContribution != EpistemicResponsibility	current
A.5/S.01.v1	S	$x' = F(x, C)$; $x(t) \in V$; $d/dt[d(x(t), V)] < 0$ (persistence regulation); $C_{t+1} = G(x_t, C_t)$; $\partial C / \partial x \neq 0$ (necessary, not sufficient); Agency-Condition: $\partial Tp / \partial C \cdot \partial C / \partial x > 0$; Proto-Agency: $\partial C / \partial x = 0$; L3: $C_{t+1} = G(x_t, C_t, H_t)$; L4: $C_{t+1} = G(x_t, C_t, x_{t+k})$; L5: $G_{t+1} = M(G_t)$	split
A.5/S.02.v1	S	no $H_{org, n+1} \neq H_{org, n}$ attributable to the processing	current
A.5/S.03.v1	S	$\Pi^{\text{live}}_{A,t}(g) \subseteq \Pi^{\text{feas}}_{A,t}(g) \subseteq \Pi^{\text{phys}}_{A,t}(g)$; $L_{A,t}(g) = \{(\pi, \lambda^{\text{live}}_{A,t}(\pi g)) : \pi \in \Pi^{\text{feas}}_{A,t}(g)\}$; $\Pi^{\text{live}}_{A,t}(g) = \{\pi : \lambda^{\text{live}}_{A,t}(\pi g) \geq \tau_{\text{live}}\}$; $\pi^{\text{choice}}_{A,t} \in \Pi^{\text{live}}_{A,t}(g)$; $\pi^{\text{act}} \neq \pi^{\text{choice}}$ possible, $Y_{\text{obs}} = O_q(H_{0:T}) \neq H_{0:T}$; possible $\neq$ feasible $\neq$ live $\neq$ chosen $\neq$ enacted $\neq$ observed	split
A.5/S.04.v1	S	$x' = F(x, C)$	current
A.5/S.05.v1	S	$x(t) \in V$, where $V \subseteq S$ is the viable region of the state space S , for all times in the interval of observation	current
A.5/S.06.v1	S	$d/dt [d(x(t), V)] < 0$ (in expectation), where $d(x, V)$ is the distance between state x and the viable region V	current
A.5/S.07.v1	S	$C_{t+1} = G(x_t, C_t)$	current
A.5/S.08.v1	S	$\partial C / \partial x \neq 0$	current
A.5/S.09.v1	S	$\partial Tp / \partial C \cdot \partial C / \partial x > 0$, where Tp is the expected persistence time of the system within the viable region V	current
A.5/S.10.v1	S	$\partial C / \partial x = 0$	current
A.5/S.11.v1	S	$C_{t+1} = G(x_t, C_t, H_t)$, where H_t is the system's interaction history	current
A.5/S.12.v1	S	$C_{t+1} = G(x_t, C_t, x_{t+k})$, where x_{t+k} represents predicted future states	current
A.5/S.13.v1	S	$G_{t+1} = M(G_t)$	current

Code	Dom.	Statement	Status
A.5/S.14.v1	S	$\Pi_{\text{live}}\{A,t\}(g) \subseteq \Pi_{\text{feas}}\{A,t\}(g) \subseteq \Pi_{\text{phys}_t}(g), \pi^{\text{choice}} \in \Pi_{\text{live}}\{A,t\}(g)$	current
A.5/S.15.v1	S	$L_{\{A,t\}}(g) = \{(\pi, \kappa_{\{A,t\}}(\pi g)) : \pi \in \Pi_{\text{feas}}\{A,t\}(g)\}$	current
A.5/S.16.v1	S	$\Pi_{\text{live}}\{A,t\}(g) = \{\pi : \kappa_{\{A,t\}}(\pi g) \geq \tau_{\text{live}}\}$	current
A.5/S.17.v1	S	$\pi^{\text{choice}}_{\{A,t\}} \in \Pi_{\text{live}}\{A,t\}(g)$	current
A.5/S.18.v1	S	$\pi^{\text{act}} \neq \pi^{\text{choice}}$ possible; $Y_{\text{obs}} = \text{Oq}(H_0:T)$; $Y_{\text{obs}} \neq H_0:T$	current
A.5/S.19.v1	S	possible $\neq$ feasible $\neq$ live $\neq$ chosen $\neq$ enacted $\neq$ observed	current
A.5/S.20.v1	S	$\Pi_{\text{live}}\{A,t\}(g) \subseteq \Pi_{\text{feas}}_A(g; h, z, T, B)$	current
A.5/W.01.v1	W	$B^{\text{scarce}}_{\{i,t\}} = (R^{\text{house}} + R^{\text{land}} + R^{\text{energy}} + DS)/Y$; $\Gamma^{\text{eff}}_{\{i,t\}} = \max\{0, \Gamma^{\text{net}}_{\{i,t\}} - B^{\text{scarce}}_{\{i,t\}}\}$; machine abundance $\neq$ low rent burden $\neq$ effective material freedom	split
A.5/W.02.v1	W	$P_t = \langle P^{\text{econ}}_t, P^{\text{info}}_t, P^{\text{coerc}}_t \rangle$; $P^B_t \propto \Gamma^{\text{eff}}_t A^{\text{corr}}_{\{H,t\}} \Lambda^{\text{live}}_{\{H,t\}} X^H_{\text{tr}} H_t$; $P^E_t = P_E(P_t, \dot{G}^{\text{conv}}, D^H_t, 1 - \Gamma^{\text{eff}}_t)$; $\dot{I}_t = F_I(P^B_t, P^E_t, \text{state capacity, rules, shocks})$; ownership $\neq$ informational power $\neq$ coercive power	split
A.5/W.03.v1	W	AI capability $\neq$ validated knowledge; assisted performance $\neq$ human learning; augmentation $\neq$ synergy; formal options $\neq$ live possibilities; access $\neq$ credible exit; income transfer $\neq$ future ownership; material security $\neq$ agency; market concentration $\neq$ domination; productive necessity $\neq$ social necessity	split
A.5/W.04.v1	W	$B^{\text{scarce}}(i,t) = (R^{\text{house}}(i,t) + R^{\text{land}}(i,t) + R^{\text{energy}}(i,t) + DS(i,t)) / Y(i,t)$	current
A.5/W.05.v1	W	$\Gamma^{\text{eff}}(i,t) = \max\{0, \Gamma^{\text{net}}(i,t) - B^{\text{scarce}}(i,t)\}$	current
A.5/W.06.v1	W	machine abundance $\neq$ low rent burden $\neq$ effective material freedom	current
A.5/W.07.v1	W	$P_t = \langle P^{\text{econ}}_t, P^{\text{info}}_t, P^{\text{coerc}}_t \rangle$	current
A.5/W.08.v1	W	P^B_t proportional-to Γ^{eff}_t $A^{\text{corr}}(H,t) \Lambda^{\text{live}}(H,t) X^H_{\text{tr}} H_t$	current
A.5/W.09.v1	W	$P^E_t = P_E(P_t, \dot{G}^{\text{conv}}, D^H_t, 1 - \Gamma^{\text{eff}}_t)$	current
A.5/W.10.v1	W	$\dot{I}_t = F_I(P^B_t, P^E_t, \text{state capacity, rules, shocks})$	current
A.5/W.11.v1	W	ownership $\neq$ informational power $\neq$ coercive power	current
A.5/W.12.v1	W	AI capability $\neq$ validated knowledge	current
A.5/W.13.v1	W	assisted performance $\neq$ human learning	current
A.5/W.14.v1	W	augmentation $\neq$ synergy	current
A.5/W.15.v1	W	formal options $\neq$ live possibilities	current
A.5/W.16.v1	W	access $\neq$ credible exit	current
A.5/W.17.v1	W	income transfer $\neq$ future ownership	current
A.5/W.18.v1	W	material security $\neq$ agency	current
A.5/W.19.v1	W	market concentration $\neq$ domination	current

Code	Dom.	Statement	Status
A.5/W.20.v1	W	productive necessity != social necessity	current

A.6 — A.6 Sufficiency before discovery

tier: untagged; status: current

Sufficiency $\text{Suff}_{\{\alpha, L\}}(\ddagger_{\alpha}^{\wedge} \text{cand})$ must be checked before a partition is ever searched for. Mandated order: Retention $\rightarrow$ State Proposal $\rightarrow$ Sufficiency Audit $\rightarrow$ Partition Discovery $\rightarrow$ Exact Quotient. If insufficient: $\ddagger_{\alpha}^{\wedge}(k+1) = \ddagger_{\alpha}^{\wedge}(k) \boxplus \Delta \ddagger_{\text{missing}}$ -- enlarge the retained state; never compensate by redefining the output's meaning or substituting a target-domain law for the missing retained data.

A.7 — A.7 The no-early-collapse law

tier: untagged; status: current

[[Refinement cannot recover distinctions absent from its input]] If an encoding E maps $E(z) = E(z')$ while $z \neq z'$, no downstream solver can ever split them back apart. [[compress only after equivalence is audited]]

A.8 — A.8 Merging structure must not erase history

tier: untagged; status: current

$\gamma \equiv_S \gamma' \not\equiv_R \gamma$ -- sharing a structural node is not the same as being identical at the level of record. Formation and exact recoverability of a signature Σ_{α} : $\Sigma_{\alpha}(\mathfrak{S}) = (O_{\alpha}(\mathfrak{S}), Q_{\alpha}(\mathfrak{S}))$
 $\Lambda_{\alpha} \circ \Sigma_{\alpha} = I_{\mathfrak{S}}$

Code	Dom.	Statement	Status
A.8/E.01.v1	E	No epistemic discrimination without provenance. Operational form: (i) What does the attributed source distinguish? (ii) Which distinction does the claim add? (iii) Which provenance path makes that addition licit?	current
A.8/E.02.v1	E	(origin, procedure, dependence, checks, answerability) $\rightarrow \Delta \text{Cr}(p)$; $\text{RPE} = \text{Cr}(p E, R, A, O1) - \text{Cr}(p E, R, A, O2)$	current
A.8/E.03.v1	E	Provenance Relevance Constraint: provenance may rationally alter epistemic standing insofar as it changes total evidence, reliability, dependence, assurance, accountability, or another specified epistemically relevant condition; it has no additional free-standing force from redescribing the origin of an otherwise fixed epistemic route.	current
A.8/H.01.v1	H	$I_s = \langle \Delta M_s, E^{\wedge} \text{decisive}_s, U^{\wedge} \text{remain}_s, \text{Next}_s \rangle$	current
A.8/H.02.v1	H	Expansion:=maximize candidate diversity and discriminability; Contraction:=prune by evidence, provenance, stakes, and action relevance	current
A.8/H.03.v1	H	$A^{\wedge} \text{HCA}_i = \langle \text{Gain_CTSA}, \text{Loss}, \text{Transfer}, \text{Ownership}, \text{Burden}, \text{BarrierChange}, \text{Opportunity} \rangle$ $\text{ATE_HCA}(x) = E[Y(1) - Y(0) K^{\wedge} \text{life} = x]$	current
A.8/H.04.v1	H	$u^{\wedge} \text{diag} = \arg\max_u [IG_B(u) - \lambda_C \text{Cost}(u) - \rho \text{Risk}(u)]$; $u^{\wedge} \text{adv} = \arg\min_u E[\dots]$; $\pi^* \in \arg\max_{\pi} E[\Delta H_s \pi]$; $\pi^{\wedge} \text{perf} \in \arg\max_{\pi} E[P^{\wedge} \text{assist}_s \pi]$	current
A.8/M.01.v1	M	$\Delta A_{\text{past}} = 0$	current

Code	Dom.	Statement	Status
A.8/M.02.v1	M	$\Lambda(e_i) = \langle \text{Prov_i}, \text{Tier_i}, \text{Def_i}, \text{Reader_i}, \text{Falsifier_i} \rangle$	current
A.8/M.03.v1	M	Status $\in \{\text{Source}, \text{AI-Synthesis}, \text{Human-Inference}, \text{Candidate}, \text{Decision}\}$	current
A.8/M.04.v1	M	PRC(e_i) $\in \{0,1\}$; C_t^valid = $\Sigma_{iw_ie_i} \text{PRC}(e_i)$	current
A.8/M.05.v1	M	Cost_{K2,r} = (Cash_r + $\lambda_{HH_r} + \lambda_{LL_r}$) / (Depth_r + ϵ); ExpectedK2Yield_j = (P(Review_j)Depth_jFit_j) / (Cash_j + Prep_j + Latency_j + ϵ); EffectiveK2 = Depth $\times$ Relevance $\times$ Independence	current
A.8/M.06.v1	M	K0_start = I0 + R0 + N0 + P0; E_t = f(Q_t, C_t, V_t, U_t, X_t, T_t); PosCap = Access $\times$ Language $\times$ SituatedObservation $\times$ TranslationCapacity $\times$ Trust; NetPractice = P_A - (Co + Bi)	current
A.8/M.07.v1	M	$\mu_H, f \approx 0$, $\mu_A \gg \mu_H, f$; EIC $\Rightarrow$ BuildSyntheticFormationInfrastructure; AI speed $\rightarrow$ synthetic criticism $\rightarrow$ K1 $\rightarrow$ human correction	current
A.8/M.08.v1	M	$\Lambda = \min(\mu_L, \mu_H, \mu_E, \mu_P, \mu_C)$; V_c = (H $\cdot$ L $\cdot$ T)^(1/3); D_e = A $\cdot$ (1 - V_c); PublicOutputVelocity $\leq$ VerificationCapacity	current
A.8/M.09.v1	M	A_s = AssociationStrength(Author, Problem); A_s(t+1) > A_s(t); Paper1 $\rightarrow$ Theme; Paper2 $\rightarrow$ SameTheme + NewMechanism; Paper3 $\rightarrow$ EmpiricalTest; Paper4 $\rightarrow$ BoundaryExtension	current
A.8/M.10.v1	M	FlagshipConcept $\rightarrow$ Preprint $\rightarrow$ Conference $\rightarrow$ Journal $\rightarrow$ EmpiricalTest $\rightarrow$ ComparativeExperiment; CreditLeverage_i = (Σ_j CreditEvent_ij) / (CoreIntellectualInvestment_i + ϵ); PC_i = JournalQuality $\times$ ProgrammeFit $\times$ ContributionStrength $\times$ UptakePotential (superseded)	current
A.8/M.11.v1	M	k_t = ν_{tf_coh} , tL_g, tp_int, tm_leg, tu_conv, tPc_u, tPc_m; $\beta_D = (\Sigma w_dE_d) / (\Sigma w_dE_d + \Sigma w_pE_p + \epsilon)$; IncreaseMintRate $\Rightarrow \beta_D \geq \beta_min \wedge IntegrityClean \wedge XenonLow \wedge ConversionLogOn$; $\rho_R = N^{ind}_{\leq 6m} / N^{cit}_{> 6m}$; X_t = 1(O_sub) + 2(E_known) + 3(C_stale); BR_t = $N^{\{ind-reuse\}}_t / N^{\{terminal\}}_t$	current
A.8/M.12.v1	M	r = A ϵ - δ ; V = (1/2)r^TW_r	current
A.8/M.13.v1	M	AI candidate $\rightarrow$ OriginalSource $\rightarrow$ ClaimMatch $\rightarrow$ VerifiedCitation; Reject $\rightarrow$ Objection $\rightarrow$ Revision $\rightarrow \Delta Q$; SCRAM = FreezeNewRelease + Correction + ReAudit	current
A.8/M.14.v1	M	H_g = (questions defended without AI) / 10	current
A.8/M.15.v1	M	B_year = B_conference + B_ethics + B_software + B_data + B_publication + B_travel; MissingResource $\Rightarrow$ ProjectHold; H_planned > H_sustainable $\Rightarrow$ PortfolioShrink; PublicWIP ≤ 3 ; Priority_i = (I_iN_iE_iF_iY_iS_i) / (D_i + C_i + Frag_i + COI_i + ϵ)	current
A.8/M.16.v1	M	PracticeObservation $\rightarrow$ Hypothesis; Claim <- Literature + ExternalEvidence + ComparativeEvidence; ClaimScope $\leq$ SamplingScope	current
A.8/M.17.v1	M	E (provenance existence): P(d) nonempty or the claim is marked provenance-indeterminate. A (provenance attribution): a credited source S must pass the licensing test for d. D (provenance disclosure): every non-source node on d's essential dependency path must remain declared or recoverable.	current

Code	Dom.	Statement	Status
A.8/M.18.v1	M	Epistemic overreach: a claim-level distinction lacks an adequate provenance path, is attributed to a node that does not license it, or depends on an essential augmentation concealed in the representation. Silent lift: with $\lambda(d)$ the actual essential dependency set and $\lambda^\wedge(d)$ the represented one, a silent lift occurs when an essential node v is omitted, or a source S is credited with d despite failing its licensing condition.	current
A.8/M.19.v1	M	$P(D +) = 0.90(0.01) / [0.90(0.01)+0.09(0.99)] \approx 0.0917$; diagram $(y, K^\sim) + \pi + L \rightarrow \text{belief/action}$	current
A.8/S.01.v1	S	P1: $r=d=x=9/10$, channel always open, every reason voiced $\Rightarrow f_P1=1$, $p_P1=729/1000$. P2: channel open w.p. $1/5$, every open-channel reason voiced $\Rightarrow f_P2=1/5$, $p_P2=729/5000$. Both induce the identical record $\Pr(\text{complaint})=1/10$, so the record alone supports only the interval $[729/5000, 729/1000]$.	current

A.9 — A.9 Meaning is not domain

tier: untagged; status: current

$\mathfrak{S} \neq 0_\alpha(\mathfrak{S}) \neq Y_\alpha \neq P_\alpha$ -- a domain is the structure that makes something readable; meaning is what a decoder, context, memory, or criterion does with that readable structure afterward.

A13.ArchitectureBox — A.13 The architecture-around-DRL box

tier: untagged; status: current

$\delta_R \Rightarrow \mathfrak{S}_n \rightarrow [E_\alpha] \ \mathfrak{f}_\alpha^{\wedge \text{cand}} \rightarrow [\text{Suff}_{\{\alpha, L\}}] \ \mathfrak{f}_\alpha \rightarrow [C \in \{1, 0, \perp\}] \ \text{Adm}_n \rightarrow [\text{FixRefine}(F_n, 0_\alpha, \text{Inv}_\alpha, c_n)]$
 $\mathcal{P}_\alpha \rightarrow [q_\alpha] \ \mathcal{D}_\alpha \text{ subject to } q_{\{n+1\}} F_n = F_{\mathfrak{f}_n} q_n, 0_\alpha = 0_{\mathfrak{f}_\alpha} q_n, \text{Inv}_\alpha = \text{Inv}_{\mathfrak{f}_\alpha} q_n \rightarrow [\mathbb{G}_n(G_n, c_n, \mathcal{T}_n)]$
 $(\Phi_n, \Psi_n) \rightarrow [S_{\text{DRL}}] \ r_{\text{RD}} \rightarrow [H_{\text{cal}}, U_\alpha] \ y_\alpha^{\wedge \text{known}} \rightarrow [\text{Checker}] \ \epsilon_\alpha \rightarrow [\Pi_\alpha] \ P_\alpha. \ S_{\text{DRL}}$ (the DRL-Telegraph recurrence of Part II) appears here exactly once, untouched, as the fixed core the other seven gates are built around.

A13.DefectVector — A.13 The Defect Vector

tier: untagged; status: current

$\epsilon_\alpha = (\epsilon_{\text{suff}}, \epsilon_{\text{dyn}}, \epsilon_{\text{read}}, \epsilon_{\text{inv}}, \epsilon_{\text{bridge}}, \epsilon_{\text{cal}})^T$. An exact claim requires $\epsilon_\alpha = 0$ in every component; anything else is approximate and must name which component(s) are non-zero rather than reporting a single blended error term.

A13.SummaryBox — A.13 Standalone summary box

tier: untagged; status: current

$\delta_R \Rightarrow \mathfrak{S}_n \rightarrow [E_\alpha] \ \mathfrak{f}_\alpha^{\wedge \text{cand}} \rightarrow [\text{Suff}] \ \mathfrak{f}_\alpha \rightarrow [\text{FixRefine}] \ \mathcal{P}_\alpha \rightarrow [q_\alpha] \ \mathcal{D}_\alpha = \mathfrak{f}_\alpha / \mathcal{P}_\alpha \text{ subject to } q_{\{\alpha, n+1\}} F_n = F_{\mathfrak{f}_{\{\alpha, n\}}} q_{\{\alpha, n\}}, 0_{\{\alpha, n\}} = 0_{\mathfrak{f}_{\{\alpha, n\}}} q_{\{\alpha, n\}} \rightarrow [\Sigma_\alpha] \ (0_\alpha, Q_\alpha) \rightarrow [\Pi_\alpha] \ P_\alpha. \ [[\text{Reality is retained before it is translated}]] \ [[\text{Domain = the minimal translation that still preserves the differences that question needs}]] \ [[\text{Meaning arises after readout; it does not sit at the root}]] \ [[\text{Never reduce before proving that what is being merged does not differ with respect to that question}]]$

AI_ethics_special_case — V.9 Social — AI ethics as a special case of q_social (readout-selection structure, not a universal moral fact)

tier: finite_diagnostic; status: current

An AI has no morality of its own. Ethics is a readout-selection structure built from five ingredients: the data retained, the set of answers actually accessible to the system, the selection rule applied over that set, the values of whoever defines the system, and the audit/revision process that can later reopen any of the first four. The goal for an AI: (a) disclose which reading frame it is answering from, (b) preserve every affected human party's standing ability to correct or object, (c) adapt its reading to local culture without erasing dignity or voice. Corrigibility is restated as boundary-observance via the same J-eta feedback term (Face 7; VIII.5's Omega_H pipeline).

APP-B.1 — APP-B.1 Conserved pairing charge of the (Φ, Ψ) reader-record system (Paper I)

tier: Th_coqc; status: current

$H = \sum_k (\Phi_k \cdot \pi^* \Psi_k - \Psi_k \cdot \pi^* \Phi_k)$ conserved along the DRL flow (built from the antisymmetric η/ω pairing already posited in the core.) Machine-checked here (formal/InfoRetentionLagrangian_attempt.v, PR #185, axiom-free over $\mathbb{Q}$): discrete Euler-Lagrange equivalence (T1_el_psi_node1/node2, T1_el_phi_node1) and the Legendre D-cancellation identity (T2_D_cancellation, showing H is D-free), for the 3-ring / T=3-4-slice scope only. Exact conservation of H along a full trajectory when $D \neq 0$ is NOT proved (source numerics show $O(\Delta\theta^2)$ charge drift); no generalization off the 3-ring exists in this repo.

APP-B.3 — APP-B.3 Frontier candidates from Paper VI (unreviewed, same-author)

tier: Dr; status: current

Paper VI ("Universal Readout Closure") states three Gates absent from this core: 1. Self-interaction curvature unit $\rightarrow \times 4$ doubling -- a geometric-unit argument (Weitzenböck-type) yielding a factor-4 doubling. [Open] 2. Spin-statistics exchange sign $(-1)^{2s}$ -- recovered from finite framed dynamics rather than from a relativistic field-theory axiom. [Open] 3. $d = 3$ selector -- a local-directional-feed criterion selecting three spatial dimensions. [Open] Status: candidate frontier, NOT canon -- same-author; non-peer-reviewed self-closeout using Th (weaker than Th_coqc), disclaims proof-assistant formalization; recorded as [Dr]-declared / [Open] candidates awaiting independent review. (Footnote: Paper III's [Th_sym] (G, Ω) -uniqueness result matches the core's posited η/ω matrices in form but answers a formal-uniqueness question, not the open empirical falsifier T1; noted as [Dr] context only.)

AdmissibilitySquare — VI.6 — Admissibility square (InfoQuotientCompressionExactness)

tier: Th_coqc; status: current

$q_{\{n+1\}} \cdot F_n = F_{\#_n} \cdot q_n$; "is not a metaphor for this commuting requirement -- it is the commuting requirement"

AllOrderCharacter_v1.0 — All-order character closure (v1.0) — exact SU(3) Weyl integral window

tier: Th_coqc; status: current

$u(\kappa)$, $v(\kappa)$ closed to all orders from exact SU(3) Weyl integrals (deterministic quadrature, no truncation): $u = c_3/(3 c_0)$, $v = c_8/(8 c_0)$, $\chi_8 = |\chi_3|^2 - 1$ from 3 tensor $3_{\text{bar}} = 1 + 8$, recursion $c_0' = 2 c_3$; the all-order window is $0 < \kappa < 0.053583974745\dots$, matching the v0.9 low-order series to $\sim 4e-6$ (not a fit). High-precision numerical closure, not a rigorous interval proof.

Atlas.AlphaQG — IV.2 The Atlas — quantum-gravity diagnostic

tier: untagged; status: current

$\alpha_{\text{QG}} = G \cdot E^2 / (\hbar c^5) = (E/E_{\text{Planck}})^2$ -- a calculator/diagnostic of how far a given energy E sits from the Planck energy, expressed as a dimensionless ratio; it flags when quantum-gravity effects become non-negligible, it does not by itself derive or solve quantum gravity.

Atlas.ScaleTable — IV.2 The Atlas — 85 orders of magnitude scale table

tier: untagged; status: current

Planck scale $\tau_{\text{c}} \sim 5.4 \times 10^{-44}$ s; Electron $\sim 6.4 \times 10^{-22}$ s; Proton $\sim 3.5 \times 10^{-25}$ s; Quark (top) $\sim 2.5 \times 10^{-27}$ s; Muon $\sim 3.5 \times 10^{-24}$ s; Nuclear $\sim 10^{-23}$ s; Molecular $\sim 10^{-14}$ – 10^{-8} s; Cell $\sim 10^{-3}$ – 10^0 s; Human $\tau_{\text{c}}^{\text{H}} \sim 0.1$ – 3 s; Social $\sim 10^0$ – 10^8 s; Civilizational $\sim 10^9$ – 10^{11} s; Geological $\sim 10^{12}$ – 10^{15} s; Cosmological $\sim 4.35 \times 10^{17}$ s (Hubble time). The atlas contains 37 disciplines, 220 entries (src/anse_spine/tau_c/tau_c_master.py).

B.1 — B.1 The Domain-Neutral Problem Object

tier: untagged; status: current

$\mathfrak{P}_{\text{RD}} = (\Gamma_0, \delta_{\text{R}}, \mathfrak{f}^{\text{cand}}, \text{F}, \text{Adm}, \mathcal{B}, \mathcal{T}, \mathcal{Q}, \mathcal{O}, \mathcal{H}, \mathcal{K})$. Γ_0 initial history · δ_{R} the distinctions retained · $\mathfrak{f}^{\text{cand}}$ candidate state · F transition · Adm which transitions are admissible · $\mathcal{B}$ boundary/cut · $\mathcal{T}$ tape/memory · $\mathcal{Q}$ the question · $\mathcal{O}$ readout · $\mathcal{H}$ the time/intervention horizon · $\mathcal{K}$ the budget on access, resolution, and reproduction. At this stage there is still no domain name and no target unit.

B.2 — APP-B.2 Linear-pair honesty caveat (Paper I)

tier: untagged; status: current

In the linear case the record field decouples: $\Psi \equiv 0 \Rightarrow \text{H} \equiv 0$. So the narrative "the record absorbs exactly what the reader loses" is [Open] in the linear regime -- it has content only once nonlinear coupling is retained.

B.2.Step1 — B.2 Step 1 — Phenomenon Capture

tier: untagged; status: current

$\Gamma_{\text{n}} = (\text{e}_0 \dots \text{e}_{\text{n-1}})$; separate what is observed, what changes, what persists, what has left the observation window, what is recorded, and what is simply not yet known. Starting from an assumed mechanism name is forbidden -- that would presuppose the answer.

B.2.Step10 — B.2 Step 10 — Cross-Domain Signature Extraction

tier: untagged; status: current

$\Xi_{\alpha} = (\mathcal{P}_{\alpha}, \mathcal{F}_{\alpha}, \mathcal{C}_{\alpha}, \mathcal{B}_{\alpha}, \mathcal{T}_{\alpha}, \mathcal{O}_{\alpha})$; a bridge $\text{K}_{\{\beta \leftarrow \alpha\}}$: $\mathcal{D}_{\alpha} \rightarrow \mathcal{D}_{\beta}$ is real only where K $\mathcal{F}_{\alpha} = \mathcal{F}_{\beta}$ K and $\mathcal{O}_{\beta}$ $\text{K} = \mathcal{D}_{\{\beta \leftarrow \alpha\}}$ $\mathcal{O}_{\alpha}$, with $\mathcal{D}_{\{\beta \leftarrow \alpha\}}$ an honest decoder translating the result, never a bolted-on mechanism engineered to force agreement.

B.2.Step2 — B.2 Step 2 — Domain Vocabulary Stripping

tier: untagged; status: current

Replace domain-specific nouns with their information-theoretic role: particle/molecule/cell/person → retained unit; force/reaction/signal/thought → transition/coupling; energy/concentration/belief → state/readout; environment/solvent/society → boundary/context; genetic history/memory → lineage/tape; measurement/report/behavior → readout/report. This does not assert that stripped phenomena are "the same"; it lets structurally comparable phenomena be compared without being fooled by vocabulary.

B.2.Step3 — B.2 Step 3 — Question & Claim Declaration

tier: untagged; status: current

$\mathcal{Q}_{\alpha} = (\text{target}, \text{horizon}, \text{resolution}, \text{interventions}, \text{tolerance})$, declared before a domain is constructed, not fitted afterward to match a domain already assumed.

B.2.Step4 — B.2 Step 4 — Retained-State Proposal

tier: untagged; status: current

$\sharp_{\alpha}^{\text{cand}} = E_{\alpha}(\mathbb{G})$. [[candidate representation $\neq$ discovered domain]]

B.2.Step5 — B.2 Step 5 — State-Sufficiency Audit

tier: untagged; status: current

$\text{Suff}_{\{\alpha, L\}}(\sharp_{\alpha}^{\text{cand}})$ must check at least: readout sufficiency, dynamical sufficiency, intervention sufficiency, delayed-return/memory sufficiency, and coupling sufficiency. If insufficient: $\sharp^{(k+1)} = \sharp^{(k)} \boxplus \Delta \sharp_{\text{missing}}$ -- never compensated by adding a formula that manufactures the missing answer.

B.2.Step6 — B.2 Step 6 — Field & Coupling Discovery

tier: untagged; status: current

The number of fields must be discovered, never fixed in advance: $X_n = (X^{(1)} \dots X^{(m)})^T$, $m_{\alpha} = \min\{m : 0_{\alpha} \text{ and } F \text{ close on } X^{(1:m)}\}$. Forcing several fields down to one merely to match a preferred reduced-form equation is forbidden -- this is the extraction-time twin of A.10.

B.2.Step7 — B.2 Step 7 — Automatic Partition Discovery

tier: untagged; status: current

$\mathcal{P}_{\{\alpha, n\}} = \text{FixRefine}(\sharp_{\{\alpha, n\}}, F_n, 0_{\alpha}, Q_{\alpha})$; states share a cell exactly when their local labels, interaction profiles, transition signatures, and future-readout-under-intervention all agree.

B.2.Step8 — B.2 Step 8 — Exact Quotient Audit

tier: untagged; status: current

$q_{\{\alpha, n+1\}} F_n = F_{\sharp_{\{\alpha, n\}}} q_{\{\alpha, n\}}$ and $0_{\{\alpha, n\}} = 0_{\sharp_{\{\alpha, n\}}} q_{\{\alpha, n\}}$; when only approximate, the defects ϵ_{dyn} , ϵ_{read} must be declared. Calling an approximate quotient "exact" is a category error, not a rounding choice.

B.2.Step9 — B.2 Step 9 — Admissible Synthesis

tier: untagged; status: current

$\text{Adm} = T \wedge I \wedge B \wedge [0_{\rho} = 0]$; this does not enumerate every possible world, and computational nodes may still be shared, but lineage may never be deleted in the process.

B.2a.ConservationLedger — B.2a A generic conservation ledger

tier: Dr; status: current

Independent of any domain-specific conserved quantity claimed elsewhere in this book (e.g. the bR ledger's $I_Q = I_B + 0_C + 0_P + 0_B$, V.13/Face 10), the synthesis-graph state itself obeys a bookkeeping identity between successive steps: $C(\mathbb{G}_{n+1}) - C(\mathbb{G}_n) = J_n^{\text{in}} - J_n^{\text{out}} + J_n^{\text{created}} - J_n^{\text{obstructed}}$ [Dr]. This ledger is architecture, not a physics claim: it reports graph, lineage, tape, currents, and operator powers for a semantic card to audit against its own declared conserved quantity -- it does not, by itself, assert that any particular physical quantity is conserved.

B.2a.NormalFormLineage — B.2a Normal-form sharing and the lineage graph

tier: finite_diagnostic; status: current

A candidate $u \star v$ that passes Adm is reduced to a normal form and folded into the running synthesis graph and lineage ledger: $g_c = \text{NF}(u \star v)$; $G_{n+1} = G_n \cup \{g_c\}$, $\Lambda_{n+1} = \Lambda_n \boxplus \ell_c$ [finite_diagnostic]. If that normal form already exists in G_n , the solver shares the structural node -- it does not duplicate it -- but still records the new lineage that reached it: $\gamma \equiv_S \gamma' \not\Rightarrow \gamma =_R \gamma'$ [Dr] (structural sharing $\neq$ retained-lineage identity). A candidate that fails Adm is never constructed at all, and an obstruction certificate is appended to the tape instead.

B.2a.SynthesisComplexity — B.2a Synthesis complexity

tier: finite_diagnostic; status: current

$D_{\text{syn}}(n) = |NF_{\text{adm}}(G_n)|$. For occupation normal forms over a partition: $D_{\text{syn}} = \Pi_c (|C_c| + 1)$ [finite_diagnostic]. The discovery process reports this count but never enumerates the full lineage space to compute it -- a possible-lineage space of size 2^N is exactly the kind of unrestricted branching the copy licence (I.1a) forbids generating without admissibility. Any polynomial-regime statement about D_{syn} 's growth remains a theorem candidate, not a proven bound.

B.3 — B.3 Cross-Domain Equivalence Levels

tier: untagged; status: current

L0 surface analogy · L1 shared variables / relations · L2 shared transition signature · L3 readout-preserving map · L4 dynamically commuting quotient · L5 recoverable bidirectional equivalence. Two equations that merely look alike are L0/L1 at best -- resemblance of form is not shared mechanism.

B.4 — B.4 Native-unit and final-unit translation

tier: untagged; status: current

$r_{\text{native}} \in \mathbb{U}_{\text{RD}}$; $y_{\alpha}^{\text{known}} = U_{\alpha}(r_{\text{native}}; C_{\alpha}^{\text{cal}})$. U_{α} converts units, names an observable, and applies a declared calibration; it must never introduce a new predictive mechanism in the process, and it must state the range over which it is valid. [[native computation $\neq$ unit conversion $\neq$ semantic report]]

B.5 — B.5 The Maker–Checker Epistemic Firewall

tier: untagged; status: current

[[A solver cannot certify its own unrestricted interpretation]] -- a condition on the reliability of inference itself, true regardless of domain. The Maker-Checker firewall governs whether the party making a claim is even structurally capable of being unbiased about it (distinct from VI.7's fail-able gate law, which governs whether any single gate is honest evidence).

B.5.Auditor — B.5.6 Auditor/Reporter

tier: untagged; status: current

Result $\rightarrow$ Scope Audit $\rightarrow$ Bounded Claim. [[computed $\neq$ checked $\neq$ interpreted $\neq$ claimed]]

B.5.Checker — B.5.5 Checker

tier: untagged; status: current

$(H_{\text{freeze}}, \hat{Y}, Y_{\text{heldout}}, \mathcal{R}_0) \rightarrow \mathcal{V}$. A Checker verifies the hash, coverage, invariants, any claimed exact-quotient property, baselines and controls, error and uncertainty, and the claim boundary, and rules strictly against the criteria frozen in advance. A Checker must never: swap models, retune parameters, flip a score's sign, cherry-pick favorable cases, or move a threshold -- any of these, once outcomes are visible, after the freeze.

B.5.Maker — B.5.3 Maker

tier: untagged; status: current

$(\mathcal{R}_0, \mathcal{X}_{\text{allowed}}) \rightarrow (\mathcal{P}^{\wedge}, F^{\wedge}, \hat{Y}, \Pi_{\text{proof}})$. A Maker must never see: held-out outcomes, checker-only labels, a threshold derived from the answer, a mechanism flag that is itself the answer, or any report produced before freeze. The condition is exact: $\text{Info}(\text{Maker}; Y_{\text{checker}}) = 0$.

B.5.PredictionFreeze — B.5.4 Prediction Freeze

tier: untagged; status: current

$H_{\text{freeze}} = \text{Hash}(\mathcal{P}^\wedge, F^\wedge, \hat{Y}, \theta, \text{source}, \mathcal{R}_0)$, covering code, config, parameters, the prediction itself, score orientation, normal form, and dependency versions. A hash does not by itself prove no leakage occurred, but it does let anyone verify that nothing changed after the answer was opened.

B.5.ProtocolDeviation — B.5.7 Protocol Deviation

tier: untagged; status: current

$\mathcal{T}_{\{n+1\}} = \mathcal{T}_n \boxplus \text{DeviationRecord}$ (append-only): when a deviation was found, what changed, why, whether forbidden data was touched, and whether the claim boundary was affected. A deviation record is never overwritten, only appended.

B.5.Registrar — B.5.2 Registrar/Curator

tier: untagged; status: current

$\mathcal{R}_0 = (Q, \text{allowed inputs}, \text{forbidden inputs}, \text{metrics}, \text{thresholds}, \text{failure rules})$. The Registrar fixes the claim boundary, separates inputs a Maker is allowed to see from inputs reserved to the Checker only, states tolerances, records deviations, and freezes a version and hash of everything above -- and never selects a model on the basis of an outcome it has already seen.

B.6 — B.6 The Refinement Loop

tier: untagged; status: current

$\{ \in \{F_{\text{state}}, F_{\text{partition}}, F_{\text{dynamics}}, F_{\text{readout}}, F_{\text{bridge}}, F_{\text{statistics}}, F_{\text{leakage}}\} \{ \rightarrow \text{record} \rightarrow \text{refine hypothesis} \rightarrow \text{new freeze} \rightarrow \text{new untouched check} \}$. Re-scoring a refined hypothesis against the same previously-seen data and reporting that as an independent confirmation is never permitted -- a refinement earns a fresh freeze and a fresh, untouched check, every time.

B.7 — B.7 The full assembled process

tier: untagged; status: current

$\delta_R \rightarrow \text{Phenomenon Capture} \rightarrow \text{Vocabulary Stripping} \rightarrow Q_\alpha + \text{Claim Boundary} \rightarrow \mathbb{I}_\alpha^{\text{cand}} \rightarrow \text{Suff}_{\{\alpha, L\}} \rightarrow \text{Field/Coupling Discovery} \rightarrow \mathcal{P}_\alpha = \text{FixRefine} \rightarrow \text{Exact Quotient Audit} \rightarrow \text{Adm} + \text{NF} \rightarrow \Xi_\alpha \rightarrow K_{\{\beta \leftarrow \alpha\}} + \epsilon_{\text{bridge}} \rightarrow \text{Registrar} \rightarrow \text{Maker} \rightarrow H_{\text{freeze}} \rightarrow \text{Checker} \rightarrow U_\alpha \rightarrow \Pi_\alpha \rightarrow \text{Bounded Report} \rightarrow \mathcal{T}_{\{n+1\}}$

B.8 — B.8 Cross-reference — Gate 7 of A.13

tier: untagged; status: current

A.13's Gate 7 (Calibration Firewall) states, at full generality and before any domain is named, the same condition B.4's U_α and this Part's H_{freeze} /Checker machinery enforce in practice: a native retained-domain readout r_{RD} is not a physical observable until an independently checked, frozen calibration (H_{cal} , verified by a Checker who never saw the held-out set beforehand) has been applied to it. B.4/B.5 are Gate 7 operationalized as a procedure; Gate 7 is B.4/B.5 stated once as a principle that holds regardless of which domain's units U_α happens to be converting into.

B1 — VI.4 — B1 Landauer bound

tier: untagged; status: current

CPU nucleus runs near thermodynamic floor; LLM is $\sim 10^6 \times$ above it.

B2 — VI.4 — B2 Readout bound

tier: untagged; status: current

abstain/decide cheaply; only contested steps pay.

B3 — VI.4 — B3 Regime bound

tier: untagged; status: current

stay over-damped ($k < k_c$); no wasted oscillation/retry.

B4 — VI.4 — B4 Repair bound

tier: untagged; status: current

preserve repairability so re-planning is cheap, never a full restart.

BiologyDomain_T2_closed — V.19 Registered Domain: Biology — endogenous operator (T2) closure by operator-memory augmentation

tier: Open; status: current

T2 endogenous $L_R[\Theta]$: I-only fails, augmented (I, Θ) closes. [CLOSED]

BiologyDomain_heredit — V.19 Registered Domain: Biology — heredity quotient / lineage-count frequency change (BIO-G4)

tier: untagged; status: current

BIO-G4 heredity $q_H \rightarrow$ lineage counts $N_{\{n+1\}} = B_n N_n \rightarrow$ frequency change [CLOSED count]; worked fixture $B = \text{diag}(2,1)$, $N_0 = (1,1) \rightarrow p_1 = (2/3, 1/3)$, no fitness variable.

BiologyDomain_living_unit — V.19 Registered Domain: Biology — living-unit fixed point (BIO-G3)

tier: untagged; status: current

BIO-G3 living-unit $V_A = \Gamma(V_A)$ under a declared boundary contract; no-free-repair ledger; mortality = irreversible exit from V_A .

Code	Dom.	Statement	Status
BiologyDomain_living_Bunit/B.0001	Compound microscope	total magnification $M = M_{\text{objective}} * M_{\text{eyepiece}}$	current
BiologyDomain_living_Bunit/B.0002	Surface-area-to-volume ratio of a cube	$SA/V = 6/\text{side}$	current
BiologyDomain_living_Bunit/B.0003	Population density	$\text{density} = \text{count}/\text{area}$	current
BiologyDomain_living_Bunit/B.0004	Body mass index BMI	$\text{BMI} = \text{mass_kg}/\text{height_m}^2$	current
BiologyDomain_living_Bunit/B.0005	Mendelian monohybrid cross	recessive-phenotype probability $P(aa) = p_{\text{maternal}} * p_{\text{paternal}}$	current
BiologyDomain_living_Bunit/B.0006	Cardiac output	$CO = HR * SV$	current
BiologyDomain_living_Bunit/B.0007	Hardy-Weinberg equilibrium,	homozygous-recessive genotype frequency q^2	current
BiologyDomain_living_Bunit/B.0008	Allele frequency from genotype counts	$q = (2n_{aa} + n_{Aa}) / (2(n_{AA} + n_{Aa} + n_{aa}))$	current
BiologyDomain_living_Bunit/B.0009	Chi-square goodness-of-fit statistic	$\chi^2 = \sum((O_i - E_i)^2 / E_i)$	current
BiologyDomain_living_Bunit/B.0010	Exponential (Malthusian) population growth	$N = N_0 \exp(r t)$	current
BiologyDomain_living_Bunit/B.0011	Logistic population growth with carrying capacity	$N = K / (1 + ((K - N_0) / N_0) \exp(-r t))$	current
BiologyDomain_living_Bunit/B.0012	Lincoln-Petersen mark-recapture population estimator	$N = M C / R$	current
BiologyDomain_living_Bunit/B.0013	Plant/cell water potential	$\Psi = \Psi_s + \Psi_p$	current

Code	Dom.	Statement	Status
BiologyDomain_living_Bunit/B.1	Fick's first law of diffusion rate = D A (dC/dx)	current	
BiologyDomain_living_Bunit/B.1	Respiratory quotient RQ = CO2_produced/O2_consumed	current	
BiologyDomain_living_Bunit/B.1	Goldman-Hodgkin-Katz multi-ion membrane potential Vm = (RT/F) ln((P_K K_out+P_Na Na_out+P_Cl Cl_in)/(P_K K_in+P_Na Na_in+P_Cl Cl_out))	current	
BiologyDomain_living_Bunit/B.1	Hill equation cooperative ligand binding Y = S^n/(S0^n+S^n) (Vmax-scaled variant also serves the PhD allosteric-enzyme row)	current	
BiologyDomain_living_Bunit/B.1	Lotka-Volterra predator-prey interior equilibrium (x*, y*) = (gamma/delta, alpha/beta)	current	
BiologyDomain_living_Bunit/B.1	One-compartment pharmacokinetic clearance CL = k Vd	current	
BiologyDomain_living_Bunit/B.2	One-compartment IV-bolus area under the curve AUC = C0/k	current	
BiologyDomain_living_Bunit/B.2	Mutation-selection balance, recessive allele q_hat = sqrt(mu/s)	current	
BiologyDomain_living_Bunit/B.2	Kleiber's law, 3/4-power metabolic scaling BMR = a M^0.75	current	
BiologyDomain_living_Bunit/B.2	Jukes-Cantor phylogenetic distance d = -3/4 ln(1-4p/3)	current	
BiologyDomain_living_Bunit/B.2	Moran/Wright-Fisher genic-selection fixation probability p_fix = (1-exp(-2 Ne s p0))/(1-exp(-2 Ne s))	current	
BiologyDomain_living_Bunit/B.2	Kingman coalescent expected TMRCA E[TMRCA] = 2(1-1/n) Ne	current	
BiologyDomain_living_Bunit/B.2	Breide equation, selection-only term Delta_zbar = Cov(w,z)/wbar	current	
BiologyDomain_living_Bunit/B.2	Lotka-Volterra interior-equilibrium Jacobian eigenvalue magnitude \\\Im(lambda)\\ =sqrt(alpha gamma)	current	
BiologyDomain_living_Bunit/B.2	Wima-Delbrück fluctuation-test expected mutant number m = N mu	current	
BiologyDomain_living_Bunit/B.2	Karlin-Altschul local-alignment E-value E = K m n exp(-lam S)	current	
BiologyDomain_living_Bunit/B.3	Approaching-sound auditory time-dilation effect sizes (+15% approach / -6% recede)	current	

BlindMatterSearch_v1.6 — Blind matter search (v1.6) — forced matter skeleton

tier: *Th_cogc*; status: *current*

Enumerate representation multiplicities over the minimal carrier alphabet $\{1, 3, 3\bar{3}\} \times \{1, 2\}$ under gauge-consistency gates (color $[SU(3)]^3$ anomaly, global- $SU(2)$ parity, closure, no-vectorlike, minimality) The unique minimum up to total conjugation is $(3, 2) + 2(3\bar{3}, 1) + (1, 2) + (1, 1)$, $N=5$, $D_{total}=15$ -- the two colored singlets and the colorless doublet are forced, not listed; names are attached AFTER. Coq proves the unbounded minimality $D_{total} \geq 15$ (via lia, not just a finite scan).

BlockClosure_v0.7 — b=2 block, first correlated shell (v0.7)

tier: *Th_cogc*; status: *current*

Exact block integral $K_{\{B2\}}$, $c_R^{\{(2)\}}$, $\rho_{t^{\text{full}}(2)}$ are defined (finite tensor contraction). First correlated shell (single cube-bumps) is exact: in 4D each plaquette has $2(D-2)=4$ bump directions, minimal surface $A_{min}=4$, giving 16 single-bumps (area $4 \rightarrow 8$), so $\rho_{\{1, \text{geom}\}(2)} = u^4 (1+16u^4)$ and correlation defect $\epsilon_{\text{geom}}^{\{(1)\}} = (1/4) \log(1+16u^4) > 0$ -- correlations HELP triality survive,

but only at $O(u^4)$. First-shell certificate $20e * u^4 (1+16u^4) < 1 \Leftrightarrow u < 0.34915$ (later shown in v0.9 to be an invalid convergence criterion). A plain Metropolis estimator FAILED (signal $\approx$ noise, FAIL_ESTIMATOR_NOISE).

BridgeCommute — VI.6 — Cross-domain bridge commuting condition

tier: untagged; status: current

$T_{\{a \rightarrow b\}} \cdot F\#_a = F\#_b \cdot T_{\{a \rightarrow b\}}$ (and preserves the readout); a bridge between two domains is real only if the translation commutes

Code	Dom.	Statement	Status
BridgeCommute/H.01.vIII		$T_{\{j\} \rightarrow \{i\}} \cdot F\#_H^j = F\#_H^i \cdot T_{\{j\} \rightarrow \{i\}}$ (commutation criterion for 'closure-meaningful' cross-agent communication)	current

BridgeFormula.MPositedFinding — IV.1 companion finding — M is posited, not derived

tier: untagged; status: current

M is POSITED, not derived -- eight independent attempts to force M out of a deeper generator failed; the only place M's presence is actually exercised is the quantum leaf (V.1), where a discretized D/M comparison against a QuTiP reference gave a residual around 7.6×10^{-4} . τ_c is discrete and prior; mass is its readout; M is a positable coefficient that quantum alone currently exercises.

BridgeFormula.MassReadout — IV.1 The Universal Bridge Formula — mass readout direction

tier: untagged; status: current

$m = \hbar / (2 c^2 \tau_c)$ [readout direction -- mass is what τ_c LOOKS LIKE once you insist on asking "how much stuff is this?"]

BridgeFormula.TauC — IV.1 The Universal Bridge Formula — τ_c primitive

tier: untagged; status: current

$\tau_c = \hbar / (2 m c^2)$ [N4 SCALE BUS -- connects every domain by memory time]. τ_c is the primitive; mass is a readout of τ_c -- founder-locked, 2026-07-21.

CenterConfinement_v0.3 — Center-sector confinement (v0.3) — Wilson area law

tier: Th_cogc; status: current

The root curvature action $\kappa \sum |u_p - 1|^2$ gives $\langle u_p \rangle = (1-r)/(1+2r) = q$, hence $\langle W(C) \rangle = q^{|A(C)|}$ and the area law $-\log \langle W \rangle = \sigma A$, $\sigma = -\log(q) > 0$, so $V(R) = \sigma R$ -- separation cost grows linearly, derived not assumed (control $\kappa \rightarrow \infty \Rightarrow \sigma \rightarrow 0$). Scope: Z_3 -only, 2D, no continuum.

ChemDomain_ledger — V.16 Registered Domain: Chemistry — conservation ledger

tier: untagged; status: current

ledger $A(n_1-n_0)=0$ (conservation) -- the free structural carrier N^G under a frozen registry, the exact source-word count homomorphism, the finite permutation-sufficiency gate, conditional decomposition uniqueness, refinement/coarsening commutation, and structural sharing without lineage erasure.

Code	Dom.	Statement	Status
ChemDomain_ledger/CC01.v1		Amount of substance $n = m/M$ (using IUPAC atomic-weight molar masses)	current

Code	Dom.	Statement	Status
ChemDomain_ledger/CC02.v1		Percent composition by mass $\%X = n_X M_X / M_{\text{compound}} * 100$	current
ChemDomain_ledger/CC03.v1		Mass-mass stoichiometry from a balanced equation	current
ChemDomain_ledger/CC04.v1		Molarity $c = n/V$	current
ChemDomain_ledger/CC05.v1		Dilution $C_1 V_1 = C_2 V_2$	current
ChemDomain_ledger/CC06.v1		pH scale $pH = -\log_{10}([H^+])$	current
ChemDomain_ledger/CC07.v1		Limiting-reagent product yield (compare n_i/coeff_i , take the minimum)	current
ChemDomain_ledger/CC08.v1		Empirical-formula mole ratio from percent composition	current
ChemDomain_ledger/CC09.v1		Molecular-formula multiplier $k = M_{\text{molecular}}/M_{\text{empirical}}$	current
ChemDomain_ledger/CC10.v1		Graham's law of effusion $\text{rate1}/\text{rate2} = \sqrt{M_2/M_1}$	current
ChemDomain_ledger/CC11.v1		Hess's law of constant heat summation	current
ChemDomain_ledger/CC12.v1		Reaction enthalpy from average bond dissociation energies $dH = \text{sum}(\text{broken}) - \text{sum}(\text{formed})$	current
ChemDomain_ledger/CC13.v1		Law of mass action / equilibrium constant $K_c = [B]/[A]$, ICE-table solution	current
ChemDomain_ledger/CC14.v1		Le Chatelier's principle (quantitative re-equilibration)	current
ChemDomain_ledger/CC15.v1		Henderson-Hasselbalch equation $pH = pK_a + \log_{10}([A^-]/[HA])$	current
ChemDomain_ledger/CC16.v1		Acid-base titration equivalence-point stoichiometry $C_{\text{acid}} V_{\text{acid}} = C_{\text{base}} V_{\text{base}}$	current
ChemDomain_ledger/CC17.v1		Solubility product K_{sp} , molar solubility of a 1:1 salt $s = \sqrt{K_{sp}}$	current
ChemDomain_ledger/CC18.v1		Standard electrochemical cell potential $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$, standard reduction potentials	current
ChemDomain_ledger/CC19.v1		Faraday's laws of electrolysis $m = I t M / (n F)$	current
ChemDomain_ledger/CC20.v1		Bimolecular (mixed second-order) rate law $\text{rate} = k[A][B]$	current
ChemDomain_ledger/CC21.v1		Raoult's law $P = x_{\text{solvent}} P_0$	current
ChemDomain_ledger/CC22.v1		Colligative property closed form $dT = i K m$ (boiling-point elevation / freezing-point depression, one closed form for both via $K=K_b$ or $K=K_f$)	current
ChemDomain_ledger/CC23.v1		Van der Waals real-gas equation of state $P = nRT/(V-nb) - a n^2/V^2$	current
ChemDomain_ledger/CC24.v1		Clausius-Clapeyron equation $\ln(P_2/P_1) = -(dH_{\text{vap}}/R)(1/T_2 - 1/T_1)$	current
ChemDomain_ledger/CC25.v1		Gibbs free energy $dG = dH - T dS$	current
ChemDomain_ledger/CC26.v1		Van 't Hoff equation $\ln(K_2/K_1) = -(dH/R)(1/T_2 - 1/T_1)$	current
ChemDomain_ledger/CC27.v1		Nernst equation $E = E_0 - (RT/nF) \ln Q$	current
ChemDomain_ledger/CC28.v1		Arrhenius equation, two-temperature activation energy $E_a = -R \ln(k_2/k_1)/(1/T_2 - 1/T_1)$	current
ChemDomain_ledger/CC29.v1		Zero-order integrated rate law $[A] = [A]_0 - k t$	current
ChemDomain_ledger/CC30.v1		Second-order integrated rate law $1/[A] = 1/[A]_0 + k t$	current
ChemDomain_ledger/CC31.v1		Beer-Lambert law $A = \epsilon b c$	current

Code	Dom.	Statement	Status
ChemDomain_ledger/CC32.v1		Bragg's law $n \lambda = 2 d \sin(\theta)$	current
ChemDomain_ledger/CC33.v1		Born-Landé lattice energy $U = -(N_A M z_1 z_2 e^2)/(4 \pi \epsilon_0 r_0)(1-1/n)$	current
ChemDomain_ledger/CC34.v1		Rigid-rotor rotational constant $B = h/(8 \pi^2 \mu r^2)$	current
ChemDomain_ledger/CC35.v1		Debye-Hückel limiting law $\log_{10}(\gamma) = -A z^2 \sqrt{I}$	current
ChemDomain_ledger/CC36.v1		Michaelis-Menten enzyme kinetics $v = V_{\max} [S]/(K_m + [S])$	current
ChemDomain_ledger/CC37.v1		Eyring transition-state theory $k = (k_B T/h) \exp(-dG^\ddagger/(R T))$	current
ChemDomain_ledger/CC38.v1		Marcus electron-transfer theory $k = A \exp(-(dG_0 + \lambda)^2/(4 \lambda k_B T))$	current
ChemDomain_ledger/CC39.v1		Langmuir adsorption isotherm $\theta = K P/(1 + K P)$	current
ChemDomain_ledger/CC40.v1		Brunauer-Emmett-Teller (BET) multilayer adsorption isotherm $V/V_m = C x/((1-x)(1-x+Cx))$	current
ChemDomain_ledger/CC41.v1		Zero-point-energy model of the primary kinetic isotope effect $k_H/k_D = \exp(hc(w_H - w_D)/(2 k_B T))$	current
ChemDomain_ledger/CC42.v1		Stokes-Einstein diffusion coefficient $D = k_B T/(6 \pi \eta r)$	current

ColorSU3_v0.2 — Color=3 and SU(3) from ordered tape (v0.2)

tier: Th_coqc; status: current

Order-sensitivity $\Rightarrow$ antisymmetry, and closed-loop start-independence $\Rightarrow (-1)^{k-1} = 1 \Rightarrow k$ odd (derived, not posited) $\Rightarrow$ minimal $k=3 \Rightarrow \dim V=3$; preserving the load + triple record $\Rightarrow$ SU(3), the common phase $c^3=1 \Rightarrow$ the Z_3 center, $\dim \mathfrak{su}(3) = 9-1 = 8$; single channels carry no invariant readout, pair and triple do (kinematic color-neutrality).

CommutingSquare.Bridge — IV.5 The Commuting-Square Bridge Criterion — bridge commutation

tier: Th_coqc; status: current

$T_{\{a \rightarrow b\}} \cdot F\#_a = F\#_b \cdot T_{\{a \rightarrow b\}}$ (readout preserved across the bridge)

CommutingSquare.Quotient — IV.5 The Commuting-Square Bridge Criterion — admissibility square, one quotient step

tier: Th_coqc; status: current

$q_{\{n+1\}} \cdot F\#_n = F\#_n \cdot q_n$ (the admissibility square, one quotient step). This is exactly the classical Kemeny-Snell lumpability condition, given a machine-checked name: `InfoQuotientCompressionExactness`.

CompatibilityRule — VII.3 — Lane compatibility rule

tier: untagged; status: current

`is_semantic_compatible(a_lane, b_lane)  $\rightarrow$  bool`; dimension-compatible but lane-incompatible $\rightarrow$ gate closes with warning, not error

ConfinementCertificate_v0.5 — Confinement certificate (v0.5) — computable confinement certificate

tier: Th_coqc; status: current

Derive ρ_t (triality retention = character integral) and $\mu_4 \leq 20e$ (4D plaquette entropy) FROM the action $S_p = \kappa * ||U-I||^2$ (with $||U-I||^2 = 6 - 2 \text{Re Tr } U$ on $SU(3)$), making confinement a computable certificate $c_t = \mu_4 * \rho_t < 1$: rigorous for $0 < \kappa < 0.0020252$ (all reps, $\rho_t \leq e^{\{9 \kappa\} - 1}$), candidate $\kappa \sim 0.053$.

ConnectionRootStream — VIII.2 — Connection to the Root Stream ($\tau_c \rightarrow \tau_c^H$ pipeline + memory kernel)

tier: Dr; status: current

τ_c (universal) $\dashv \bullet \tau_c^H$ (human canonical readout timescale) $\dashv$ memory kernel $K_{\tau^H}(s) = (1/\tau_c^H) e^{-s/\tau_c^H}$ $\dashv$ readout $M_H[n] = K_H \theta(E[n]) + \eta_H[n]$ $\dashv$ phenomenalize $P_H[n] = \Phi_H(M_H[n])$ $\dashv$ resonance gain $G_i^H = 1 / (1 + (\omega_i \tau_c^H - 1)^2)$ $\dashv$ policy $\pi^* = \text{argmin } E||\text{Res}||^2 + \lambda C + \rho \text{Risk} - \nu \text{Val} + \mu \text{RepairLoss}$. τ_c^H is the species-level instantiation of the universal causal-memory coordinate τ_c ; channel-specific timescales (τ_{vis} , τ_{aud} , τ_{motor} , τ_{social}) are downstream projections, not separate foundations.

CorrectedCertificate_v0.9 — Corrected certificate (v0.9) — linear convergence criterion, representation-tail closure

tier: Th_cogc; status: current

Self-caught correction: v0.7's $20e * u^4(1+16u^4) < 1 \iff u < 0.34915$ was not a valid convergence criterion -- $u^4(1+16u^4)$ is the minimal-surface amplitude, not the per-plaquette ratio; entropy grows with area, so the sum $\sum_A N_A u^A$ converges iff $\mu_4 \cdot u_{\text{hat}} < 1$ (LINEAR). From $G > 0$ the internal automorphism group is compact, the link intertwiner $P = \int_G \rho(h) dh$ is an orthogonal projector with $||P|| \leq 1$ (a contraction), so $u_{\text{hat}} \leq u/(1-8v)$ for $8v < 1$; corrected window $\kappa \sim 0.05358$.

CrossDomainLineage — VIII.7 — Cross-Domain Lineage of a Single Percept

tier: Dr; status: current

$I_{\text{raw}} = I_A + O_S + O_T + O_E + O_C$, where I_A is the information that survives all the way to ω_A at the policy-facing end of the Ω_H pipeline, and O_S, O_T, O_E, O_C are the named obstruction terms lost at S_H, T_H, E_H, C_H respectively -- exactly parallel in form to $I_Q = I_B + O_C + O_P + O_B$ in the physical (bR) lineage case.

CutCurrent — DRL-Telegraph apparatus: cut current decomposition

tier: finite_diagnostic; status: current

$\mathcal{J}_C = \mathcal{J}_{\text{transport}} + \mathcal{J}_{\text{return}} + \mathcal{J}_{\text{readout}} + \mathcal{J}_{\text{tape}}$

DRLAction — DRL-Telegraph apparatus: the DRL discrete action

tier: Dr; status: current

$S_{\text{DRL}} = \sum_n \mathbb{L}^n$; $\mathbb{L}^n = (1/\Delta t) \Delta \Phi_n^T M_n \Delta \Psi_n + \frac{1}{2}(\Phi_n^T D_n \Delta \Psi_n - \Psi_n^T D_n \Delta \Phi_n) - \Delta t [K_n \Phi_n^T G_n \Psi_n + \Psi_n^T \nabla V_n(\Phi_n) - J_n^T \Psi_n]$

DomainDiscoveryEngine — VI.5 — Domain-Discovery Engine (L3 made automatic)

tier: finite_diagnostic; status: current

Given a raw tape of (N,S) state transitions, discovers: (a) how many variables the domain minimally needs -- via an exact-gate closure test; (b) whether channels interact -- via mixed second-difference detection; (c) the minimal exact update law -- solved over $\mathbb{Q}$ (exact rationals, no float drift); (d) conservation laws -- read off the discovered update law.

EQ-001 — Primordial difference

tier: Ax; status: current

$\exists a, b : a \neq b$

Code	Dom.	Statement	Status
EQ-001/B.01.v1	B	$a \neq b$ is retained as a distinction; the distinction is asymmetric, giving an ordering direction. The same shared root as relativity/ and quantum/, not a new axiom.	current
EQ-001/B.02.v1	B	The asymmetric ordering yields ordered transitions with a positive retention time $\tau_c > 0$.	current
EQ-001/B.03.v1	B	Retention discretizes into ticks; the ordered-tape carrier $G \cdot$ keeps order ($ab \neq ba$), giving the finite carrier the BIO-G1 gate operates on.	current
EQ-001/B.04.v1	B	The count carrier N^G (multiset/count quotient of $G \cdot$) FAILS to keep dynamics well-defined: $\nu(ab) = \nu(ba) = (1,1)$ but $\nu(F(ab)) = (2,0) \neq \nu(F(ba)) = (0,2)$ under the future-response signature F . Exact witness in biology_closure_v0_1.py section BIO-G1.	current
EQ-001/B.05.v1	B	Because the count carrier N^G loses the distinction the future-response signature F needs, the ORDERED quotient q_{seq} (which keeps $ab \neq ba$) closes exact: dynamics are unambiguous under q_{seq} where they are ambiguous under N^G . STRUCTURALLY_CLOSED. 'sequence' is a carrier property, not yet a real amino-acid or nucleotide chain.	current
EQ-001/B.06.v1	B	The intervention-response equivalence class $q_F(w) = O(w)$ (order-of-positions readout) is invariant under the swap intervention for the passing case, while the coarser keep-first-letter quotient q_{first} breaks output factorization (a failing control). q_F CLOSES as function; order and function are demonstrably distinct quotients.	current
EQ-001/B.07.v1	B	A two-component state (integrity x , reserve e) updates under a no-free-repair ledger $x' + e' = x + e + j - d$ for contract (d, j) and repair policy a , setting up the fixed-point search for BIO-G3.	current
EQ-001/B.08.v1	B	A self-maintaining fixed point $V_A = \text{Gamma}(V_A)$ exists under the no-free-repair ledger: exact witness $(1,1)$ with contract $\{(0,0), (1,1)\}$, policy $a=d$, stays $(1,1)$ and remains in V_A (PASSING); a sustained deficit contract $(d, j) = (1,0)$ drives integrity to 0, exiting V_A (FAILING). CONDITIONALLY_CLOSED on the boundary contract.	current

Code	Dom.	Statement	Status
EQ-001/B.09.v1	B	The failing control of BIO-G3 (sustained $(d,j)=(1,0)$) exhibits integrity $x \rightarrow 0$ with no return path in the finite fixture, licensing 'mortality' as a structural READOUT: irreversible exit from V_A . Exact witness $x_2=0$.	current
EQ-001/B.10.v1	B	A heredity quotient q_H (descendant-signature equivalence) closes structurally: a coarse parent-quotient that merges classes with different licensed replication counts breaks the successor-count function (FAILING control, $2 \neq 1$), while descendant-signature q_H keeps successor counts well-defined (PASSING). NO fitness term is introduced as a root variable anywhere in this rule.	current
EQ-001/B.11.v1	B	From licensed replication-count matrix B and initial counts N_0 , $N_1=B \cdot N_0$ gives a frequency readout $p_1=N_1/\sum(N_1)$ purely from lineage counts. Exact witness $B=\text{diag}(2,1)$, $N_0=(1,1) \rightarrow N_1=(2,1) \rightarrow p_1=(2/3,1/3)$. No external evolution law used.	current
EQ-001/B.12.v1	B	The I-only quotient q_I fails to distinguish $z_1=(1,1,0)$ and $z_2=(1,1,1)$ BEFORE the endogenous step (q_I identical), but their images differ AFTER ($z_1'=(1,1,0)$, $z_2'=(2,0,1)$) – q_I loses a distinction the dynamics needs. The augmented quotient $q_aug=(I,\Theta)$ distinguishes z_1, z_2 exactly and CLOSES. CLOSED_BY_AUGMENTATION.	current
EQ-001/B.13.v1	B	Exact witness: $tstep(1,1,0)=(1,1,0)$, $tstep(1,1,1)=(2,0,1)$; $q_I(z_1)=q_I(z_2)=(1,1)$ before, but $(1,1,0)_{\{1:2\}} \neq (2,0,1)_{\{1:2\}}$ after – the I-only quotient loses the distinction that forces the T2 augmentation.	current
EQ-001/B.14.v1	B	Geometric decay, turnover-is-production, and homeostasis-balance theorems compile axiom-clean and root-native, backing the BIO-G3 living-unit substrate with a discrete-algebra homeostasis model.	current
EQ-001/B.15.v1	B	Setpoint relaxation (one-step and n-step error decay toward a fixed setpoint) theorems compile axiom-clean and root-native, backing the BIO-G3 living-unit substrate's dynamic-response quality.	current
EQ-001/B.16.v1	B	A bistable window and critical-slowness theorem compile axiom-clean and root-native, backing the BIO-G3 living-unit substrate's stability-boundary quality.	current
EQ-001/B.17.v1	B	Coupling-energy, locked-iff-energy-zero, and one-way-coupling-breaks-conservation theorems compile axiom-clean and root-native, backing the T2 endogenous-operator substrate's coupling quality.	current

Code	Dom.	Statement	Status
EQ-001/B.18.v1	B	BIO-G2 closes q_F as an abstract functional quotient; whether q_F corresponds to a real protein, enzyme, or catalytic function requires a calibrated encoding from retained state to biochemical observables. No such encoding exists yet.	unverified
EQ-001/B.19.v1	B	BIO-G3 closes V_A as an abstract self-maintaining fixed point; whether V_A corresponds to a real cell or organism requires a calibrated encoding. No such encoding exists yet.	unverified
EQ-001/B.20.v1	B	A reproduction operator R_kappa is architecturally sketched as a branch off the living-unit fixed point V_A (a self-maintaining unit producing a new self-maintaining unit), but its construction and closure are declared, not derived or witnessed in <code>biology_closure_v0_1.py</code> .	unverified
EQ-001/B.21.v1	B	BIO-G4 closes q_H and the frequency read-out $p_{\{j,n+1\}}$ structurally; whether this frequency change corresponds to real biological selection or fitness requires a calibrated encoding. No fitness variable is ever introduced as a root quantity.	unverified
EQ-001/B.22.v1	B	The lineage-frequency trend (B-LIN) can be READ as an 'evolution' trajectory, but this reading is a declared interpretation, not a derived biological-evolution law – 'evolution' here is a structural count trend, not natural selection.	unverified
EQ-001/B.23.v1	B	B-SUB1's discrete-algebra homeostasis theorems are Coq-green as FORMAL claims; whether they describe real physiological homeostasis is a declared clinical reading (Dr, not root-derived), open pending calibration.	unverified
EQ-001/B.24.v1	B	B-SUB2's setpoint-relaxation theorems are Coq-green as FORMAL claims; whether they describe real clinical recovery/relapse dynamics is a declared reading (Dr), open pending calibration.	unverified
EQ-001/B.25.v1	B	B-SUB3's bistable-window theorem is Coq-green as a FORMAL claim; whether it describes a real bistable disease state is a declared reading (Dr), open pending calibration.	unverified
EQ-001/B.26.v1	B	B-SUB4's coupling-energy theorems are Coq-green as FORMAL claims; whether they describe real physiological coupling (hormonal, neural, etc.) is a declared reading (Dr), open pending calibration.	unverified
EQ-001/B.27.v1	B	The domain-discovery battery is real and runnable (PASS), but has been run only on SYNTHETIC tapes – no real event-resolved biological data has been passed through it. The bridge to real data is open.	unverified

Code	Dom.	Statement	Status
EQ-001/B.28.v1	B	The bacteriorhodopsin (bR) lineage ledger and its CLI are narrated and architected in the core, but NO real event-resolved run has ever been executed. This is architecture-only, not evidence.	unverified
EQ-001/B.29.v1	B	TARGET, not established: a calibrated encoding mapping retained root state to real biological observables (sequence identity, functional identity, unit identity, heredity identity). This is the single central open bottleneck licensing all real-domain semantics; none of BIO-G1..G4/T2 supply it.	unverified
EQ-001/B.30.v1	B	TARGET, not established: derivation of any real biological result from the root through real (measured, event-resolved) data. End-to-end root \to real-biology through event-resolved data = 0%.	unverified
EQ-001/B.31.v1	B	DNA/genome as a real molecular carrier is a destination the ordered-sequence carrier (BIO-G1) might eventually be calibrated to reach – never a premise anywhere in this registry.	unverified
EQ-001/B.32.v1	B	A real cell or organelle is a destination the living-unit closure (BIO-G3) might eventually be calibrated to reach – never a premise anywhere in this registry.	unverified
EQ-001/B.33.v1	B	Real enzyme kinetics (e.g. Michaelis-Menten-shaped catalysis) is a destination the functional quotient (BIO-G2) might eventually be calibrated to reach – never a premise. Michaelis-Menten remains a prohibited active token (see DRIFT_CONTRACT.json).	unverified
EQ-001/B.34.v1	B	Fitness / natural selection is explicitly FORBIDDEN as a root variable anywhere in BIO-G4/B-LIN; it remains a destination requiring a calibrated encoding, fully open.	unverified
EQ-001/B.35.v1	B	Real population-level ecological dynamics beyond the abstract lineage-count trend (B-LIN/B-Y5) are fully open – no root-native construction exists. Lotka-Volterra remains a prohibited active token.	unverified
EQ-001/B.36.v1	B	The LINE-2 textbook curriculum solver (45/45 checklist, tier textbook_closure/finite_diagnostic, no Th_coqc) sits outside this DAG entirely as a CHECKER. This node exists ONLY to make the boundary explicit and permanently red – it must never be promoted, folded in, or counted toward the 47.2%/62.5% LINE-1 score.	unverified
EQ-001/C.01.v1	C	If the declared closed boundary preserves ledger $L(n)=A \ n$, then $A(n1-n0)=0$.	current
EQ-001/C.02.v1	C	A change delta preserves the ledger iff delta is in $\ker(A)$.	current

Code	Dom.	Statement	Status
EQ-001/C.03.v1	C	If columns of N form a basis for the admitted kernel subspace, every admitted change has coordinates $\mathbf{x}_i$ with $\delta = N \mathbf{x}_i$.	current
EQ-001/C.04.v1	C	Admitted coordinates satisfy $n_0 + N \mathbf{x}_i \geq 0$ and all declared capacity/boundary inequalities.	current
EQ-001/C.05.v1	C	For a finite state set, repeatedly splitting cells by declared readout and successor-cell signatures terminates in an exact closed refinement.	current
EQ-001/C.06.v1	C	If composition is a free commutative monoid and R is a monoid homomorphism, then $R(c) = \sum_i c_i R(e_i)$.	current
EQ-001/C.07.v1	C	A quotient fixed point can hide source-state cycles or currents; therefore fixed quotient readout alone establishes only observational stationarity.	current
EQ-001/C.08.v1	C	Candidate mathematical languages must be frozen and reported before held-out evidence is opened.	current
EQ-001/C.09.v1	C	Use integer/Fraction arithmetic for the finite proof witnesses in v0.901.	current
EQ-001/C.10.v1	C	Freeze generator identities, aliases, source profiles, contexts, and permitted transformations before quotient admission.	current
EQ-001/C.11.v1	C	For a frozen finite generator registry G , N^G with coordinatewise addition and zero is a free commutative monoid.	current
EQ-001/C.12.v1	C	Counting registered generators maps empty source word to zero and concatenation to coordinatewise addition.	current
EQ-001/C.13.v1	C	If equal occupation vectors imply equal registered readout and successor signatures in every frozen profile, the count quotient exactly factors those profiles.	current
EQ-001/C.14.v1	C	Within N^G and a frozen independent generator basis, occupation coordinates uniquely specify structural decomposition.	current
EQ-001/C.15.v1	C	A declared nonnegative-integer coarsening map P commutes with composition: $P(c+d) = Pc + Pd$.	current
EQ-001/C.16.v1	C	The marked quotient stores structural occupation separately from an append-only lineage sidecar; equal structural projections need not identify marked histories.	current
EQ-001/C.17.v1	C	Enumerate all registered same-count pairs and compare exact frozen signatures.	current
EQ-001/E.01.v1	E	A mathematical theory of communication	current
EQ-001/E.02.v1	E	Szilárd's information-work engine	current
EQ-001/E.03.v1	E	Landauer's erasure principle	current
EQ-001/E.04.v1	E	It from bit	current
EQ-001/E.05.v1	E	Stinespring dilation	current
EQ-001/E.06.v1	E	Completely positive maps	current
EQ-001/E.07.v1	E	States, Effects, and Operations (Kraus)	current

Code	Dom.	Statement	Status
EQ-001/P.01.v1	P	$a \neq b$ is retained as a distinction; the distinction is asymmetric, giving an ordering direction. Identical to relativity's RB-P0 – this is the one shared root, not a new axiom.	current
EQ-001/P.02.v1	P	The asymmetric ordering yields ordered transitions with a positive retention time $\tau_c > 0$.	current
EQ-001/P.03.v1	P	Retention discretizes into ticks $t = n \cdot \Delta_\theta$; a finite causal graph is built on those ticks.	current
EQ-001/P.04.v1	P	The retained-distinction axioms FORCE the finite retention operator L_R 's form – not merely posit it. Machine-checked, axiom-free, re-compiled 2026-07-21 (coqc -q -R . DQG).	current
EQ-001/P.05.v1	P	Coarse-graining L_R over the ticks gives the spine second-order stepper $M \cdot d^2 + D \cdot d + K$, shared verbatim with the relativity domain's RB-P2/RB-P3.	current
EQ-001/P.06.v1	P	The spine stepper's characteristic equation is $M \cdot s^2 + D \cdot s + K \cdot \lambda = 0$, exact-rational, parametrized by a mode label λ .	current
EQ-001/P.07.v1	P	$\lambda_c = D^2 / (4 \cdot M \cdot K)$. The discriminant $\text{disc}(\lambda) = D^2 - 4 \cdot M \cdot K \cdot \lambda$ is negative (complex-conjugate root pair, oscillatory) for $\lambda > \lambda_c$, positive (overdamped) for $\lambda < \lambda_c$, and exactly zero at $\lambda = \lambda_c$ (critical). Exact rational witness: $M=D=K=1 \rightarrow \lambda_c=1/4$; $\lambda=1$ gives $\text{disc}=-3 < 0$ (oscillatory); $\lambda=1/8$ gives $\text{disc}=1/2 > 0$ (overdamped); $\lambda=\lambda_c$ gives $\text{disc}=0$ (critical).	current
EQ-001/P.08.v1	P	Any asymmetric seed matrix R_0 decomposes uniquely as $R_0 = \text{DiagPart} + \text{SymOff} + \text{SkewOff}$. Exact rational witness with a 3x3 R_0 reproduces R_0 exactly from the three parts; SkewOff is genuinely nonzero (torsion/circulation branch).	current
EQ-001/P.09.v1	P	The seed torsion/circulation branch is exactly SkewOff ; the set of admissible SkewOff parts forms a group and the construction extends to rank N , axiom-free.	current
EQ-001/P.10.v1	P	The torsion/circulation branch produces genuine (non-trivial, non-decomposable-further) mixing among retained components – a finite, exact-rational witness with a nonzero SkewOff block.	current

Code	Dom.	Statement	Status
EQ-001/P.11.v1	P	An orientation J on a closed real oriented mode-pair with $J^2 = -I$ and G -adjoint $J^\dagger G = -J$ (G -skew under the positive retention metric G) EARNs the complex coordinate $\psi = u + J \cdot v$ as a READOUT – i is a consequence of a closed real structure, never a premise. Exact rational witness: $J = [[0, -1], [1, 0]]$, $G = I_2$, $J^2 = -I_2$, $J^\dagger = -J$. FAILING control: $J_{\text{bad}} = I_2$ (identity) has $J_{\text{bad}}^2 = +I_2 \neq -I_2$, and is correctly REFUSED the complex coordinate (NO_COMPLEX_QUANTUM_COORDINATE).	unverified
EQ-001/P.12.v1	P	$N_Q(\psi) = \langle \psi, G \psi \rangle$ is ≥ 0 for all ψ and $= 0$ iff $\psi = 0$, on the exact rational witness ($G = I_2$, $\psi = (3, 4)$ gives $N_Q = 25$; $\psi = 0$ gives $N_Q = 0$).	current
EQ-001/P.13.v1	P	An evolution operator U satisfying $U^\dagger G U = G$ preserves N_Q exactly. Exact witness: $U = J$ (the complexification-gate orientation itself, reused as a rotation), $N_Q(U \cdot \psi) = N_Q(\psi) = 25$.	current
EQ-001/P.14.v1	P	Retained memory (discrete algebra on the retention operator) FORCES a mass-like readout, axiom-free, machine-checked. This rule establishes the formal/algebraic claim only; whether this readout IS physical mass is a separate, explicitly YELLOW bridge (see Q-Y5). NOT counted as an independent node in the 32-node Quantum DAG closure audit – it is the supporting formal citation folded into Q-Y5’s yellow accounting (matching the verifier’s own ‘memory-before-mass’ yellow-list item).	current
EQ-001/P.15.v1	P	The discrete connection $U_{\{j \leftarrow i\}} = V_j^{-1} \cdot V_i$, the curvature certificate $K_C = U_C - I$ (two finite transports failing to commute, nonzero exact witness), and the supporting discrete-geometry chain (metric-compatible curvature, 2nd Bianchi, Gauss-Bonnet, Riemann pair symmetry, rational $SO(3)$ curvature) are folded in here from the relativity domain’s OB-03/OB-04 as shared root-native structure, deduplicated (same underlying L_R operator, not re-derived).	current

Code	Dom.	Statement	Status
EQ-001/P.16.v1	P	InfoQuantumRelativityUnification.v (box_quad_is_spine_residual, spine_dispersion_iff_box_quad_vanishes) and InfoSeedFeedsQuantumRelativity_attempt.v compile axiom-free, BUT box_quad INHERITS the imported Minkowski signature from InfoLorentzInvariance.box_quad. This is therefore a bounded ALGEBRAIC IDENTITY between two independently-posed constructions, never a root-native derivation of QM/SR unification.	unverified
EQ-001/P.17.v1	P	InfoTelegraphHorizonUnification_attempt.v / InfoTelegraphCrossover_attempt.v prove disc_is_spine_discr, lam_c_is_spine_lambda_c, quantum_iff_above_horizon – the spine’s OWN classical/quantum crossover discriminant $\lambda_c = D^2/4MK$, axiom-free. This is a real, machine-checked result, but the BRIDGE from ‘above/below this discriminant threshold’ to ‘this regime IS quantum’ is not established – the theorem names a threshold, not a semantic identification.	unverified
EQ-001/P.18.v1	P	Under the DECLARED (not derived) identification of the spine’s own front-speed-squared with K/M, the oscillatory-regime split of Q-OS1 can be read as an oscillator-parameter statement. This identification is conditional and declared, exactly parallel to relativity’s OC-01 (graph front-speed vs measured-speed) – it is NOT derived from the root.	unverified
EQ-001/P.19.v1	P	Q-G1’s discrete metric/curvature chain is root-native and Coq-green as GEOMETRY, but its bridge toward a quantum-regime interpretation (e.g. curvature as any kind of quantum-gravitational coupling) is not established – naming it here only as a formal-closed / bridge-partial connective, matching relativity’s own OB-01 caveat (operator-to-metric direction only, reverse forbidden).	unverified
EQ-001/P.20.v1	P	Q-M1 establishes the FORMAL algebraic claim (retained memory forces a mass-like readout) as Coq-green. Whether that readout IS real physical mass/inertia is a DECLARED, not derived, correspondence – exactly parallel to relativity’s OC-02 (mass-memory declared correspondence).	unverified

Code	Dom.	Statement	Status
EQ-001/P.21.v1	P	A formal CPTP (completely-positive, trace-preserving) channel CARD exists as a target definition (the abstract shape a quantum channel would need to satisfy), but the bridge FROM the root's own reversible G-orthogonal evolution (Q-N2) TO a general CPTP map (which must also cover irreversible/open-system dynamics) is OPEN – not yet constructed or gated.	unverified
EQ-001/P.22.v1	P	Q-C1 establishes complexification for ONE closed real oriented mode-pair, exact witness. General existence and uniqueness of a root-derived complex psi across all admissible retained states is OPEN.	unverified
EQ-001/P.23.v1	P	A candidate architecture for a quantum quotient q_Q (mirroring relativity's q_g / chem's q_D pattern) is sketched, but the commuting-square condition $q_Q \circ F_full = F_Q \circ q_Q$ has NOT been proven or even witnessed on a finite fixture. This is the central open bottleneck for the whole domain.	unverified
EQ-001/P.24.v1	P	Whether a mixed-state description (density-matrix-shaped object) arises from tracing out a hidden lineage of the retained-distinction root is OPEN – no finite witness exists.	unverified
EQ-001/P.25.v1	P	Whether a composition operation $(x)_kappa$ on retained subsystems, derived from the no-free-copy law, reproduces tensor-product-shaped behavior is OPEN. No finite witness exists; this is distinct from and does not assume any imported tensor product.	unverified
EQ-001/P.26.v1	P	TARGET, not established: uniqueness of the Born weight $p= \psi ^2$ across ALL admissible refinements. <code>quantum_closure_v0_1.py</code> section E is <code>FINITE_DIAGNOSTIC</code> evidence FOR the quadratic form only: on one exact refinement ($25=9+16$), the quadratic weight ($p=2$) is refinement-invariant while a linear weight ($p=1$) is not ($5 \neq 7$, correctly failing). This is evidence, not a uniqueness proof across all refinements/bases.	unverified
EQ-001/P.27.v1	P	Whether a root-native conditional-update rule (projection-postulate-shaped) exists is OPEN – no finite witness, no gate constructed.	unverified
EQ-001/P.28.v1	P	An RDL_Heisenberg-shaped formal guard exists and SURVIVED the hollow-card audit (it correctly refuses to fire on non-quantum inputs), but it is a boundary/consistency check, not a root-derivation of an uncertainty principle. Kept OPEN pending a root-native construction.	unverified

Code	Dom.	Statement	Status
EQ-001/P.29.v1	P	A CHSH/Tsirelson-shaped formal guard exists as a boundary check (it can flag classical-vs-quantum-shaped correlation bounds), but it is not a root derivation of quantum correlation, entanglement, or the Tsirelson bound. Kept OPEN, depends on the still-open Q-X4 composition.	unverified
EQ-001/P.30.v1	P	A particle ontology (localized, countable excitations with mass) is OPEN – Q-M1/Q-Y5 establish only a formal mass-like readout from memory algebra, declared not derived as physical mass.	unverified
EQ-001/P.31.v1	P	Spin and the spin-statistics theorem are fully OPEN. Q-S1's SkewOff torsion branch is NOT spin – no rotation-group representation, half-integer quantization, or statistics connection has been constructed.	unverified
EQ-001/P.32.v1	P	A full QFT / effective-field-theory closure requires the JOINT construction of the quantum quotient q_Q , the relativity domain's quotient q_g , and a geometry quotient $q_geometry$ – none of which are individually complete, let alone jointly commuting. Fully OPEN.	unverified
EQ-001/P.33.v1	P	A conditional superposition gate exists ONLY for a fixed, pre-specified pair of orthogonal readout operators on the single closed-pair witness of Q-C1 – linear combinations of the two witness components stay in the closed pair and N_Q is additive on them (Pythagorean, exact rational). This is NOT a general superposition principle: it does not extend to arbitrary bases, arbitrary numbers of components, or non-fixed operators without further construction.	unverified
EQ-001/P.34.v1	P	$a \setminus neqb$ is retained as a distinction; the distinction is asymmetric, giving an ordering direction.	current
EQ-001/P.35.v1	P	Coarse-graining the stepper gives a telegraph process with finite front speed $v = \sqrt{D/\tau_c}$, derived from finite retention, not inserted.	current
EQ-001/P.36.v1	P	The finite front speed bounds a causal cone $ x \leq v \cdot t$.	current
EQ-001/P.37.v1	P	Define null coordinates $n_+ = v \cdot t + x$, $n_- = v \cdot t - x$ from the cone's own two edges.	current
EQ-001/P.38.v1	P	$Q_v = n_+ \cdot n_- = v^2 t^2 - x^2$, composed from the cone's own null edges, not an imported interval.	current
EQ-001/P.39.v1	P	$B(\delta_1 + \delta_2) = B(\delta_1) + B(\delta_2)$ is imposed and forces linearity of B over Q (derived, not separately assumed).	current
EQ-001/P.40.v1	P	The observer map B must send the cone to itself (preserve $ x \leq v \cdot t$ under the map).	current

Code	Dom.	Statement	Status
EQ-001/P.41.v1	P	The observer map B preserves the forward/backward orientation of the cone.	current
EQ-001/P.42.v1	P	A pure observer change preserves the RD causal-cell count $n_+ \cdot n_- (Q_v)$. This is the one new internal gate added by this closure; it is proposed, not derived from the root backbone.	unverified
EQ-001/P.43.v1	P	O4 forces $n_+' = \kappa^{-1} \cdot n_+$, $n_-' = \kappa \cdot n_-$, $\kappa > 0$, so that $Q_v' = Q_v$ automatically.	unverified
EQ-001/P.44.v1	P	Relative velocity $\beta = u/v$; $\kappa^2 = (1+\beta)/(1-\beta)$; $\Gamma_R = (\kappa+1/\kappa)/2 = 1/\sqrt{1-\beta^2}$. Exact witness: $v=5/4, u=3/4 \rightarrow \beta=3/5, \kappa=2, \Gamma_R=5/4$.	unverified
EQ-001/P.45.v1	P	$x' = \Gamma_R \cdot (x - u \cdot t)$, $t' = \Gamma_R \cdot (t - u \cdot x/v^2)$. Exact witness: event $(t=8, x=5) \rightarrow (t'=7, x'=-5/4)$; $Q_v = Q_v' = 75$.	unverified
EQ-001/P.46.v1	P	$t'/t = 1/\Gamma_R$ and $L = L_0/\Gamma_R$ follow from OA-09 with no new law inserted.	unverified
EQ-001/P.47.v1	P	$\Delta t' = -\Gamma_R \cdot u \cdot \Delta x / v^2$ follows from OA-09 with no new law inserted.	unverified
EQ-001/P.48.v1	P	$u_{21} = (u_1 + u_2)/(1 + u_1 \cdot u_2/v^2)$ follows from OA-09; exact witness stays sub-cone ($ u_{21} < v$).	unverified
EQ-001/P.49.v1	P	$\hat{g}^{ij} \sim 1/2 \cdot \text{Hess}(\sigma_{LR})$, the retention operator's principal symbol read as a metric. This direction carries the master core's [Th_coqc] base tier (Face 8). The reverse direction (spectrum-alone $\rightarrow$ metric) is explicitly forbidden.	current
EQ-001/P.50.v1	P	The geometry-only quotient Θ lacks the variables to close on its own (Θ_{n+1} depends on Φ, Ψ , not Θ alone); the minimal closed state is $Z_n = (\Phi_n, \Phi_{n-1}, \Psi_n, \Psi_{n-1}, \Theta_n)$. Basis covariance under this minimal state is only PARTIALLY closed.	current
EQ-001/P.51.v1	P	$U_{j<i} = V_j^{-1} \cdot V_i$ from the basis-change identity; transport defect $o_{i \rightarrow j} = \phi_j - U_{j<i} \cdot \phi_i$.	current
EQ-001/P.52.v1	P	Loop transport $U_C = U_C(\text{loop}) - I$ defines curvature K_C : local transports failing to commute (finite witness: $U_x = [[1,1],[0,1]]$, $U_y = [[1,0],[1,1]] \Rightarrow U_C = [[0,1],[-1,3]]$, $K_C = [[-1,1],[-1,2]] \neq 0$).	current
EQ-001/P.53.v1	P	Free path selects $O(P) = \sum_e o_e^T G_j o_e$ minimal; witness $\phi_i = [1,0]$, $\phi_f = [1,1]$: path $x \rightarrow y$ obstruction 0, path $y \rightarrow x$ obstruction 1, minimizer selects $x \rightarrow y$.	current

Code	Dom.	Statement	Status
EQ-001/P.54.v1	P	The geometry-field feedback loop closes as a time-unrolled DAG (not circular), satisfying $q_LG \circ F_full = F_LG \circ q_LG$, with a shared singular fail-state $perp_singular$.	current
EQ-001/P.55.v1	P	CLOSED in v0.2 by the Null-Transport Factorization Gate: the local causal-cell rate relates to a remote record rate via the internal lapse $N = \sqrt{\det B}$ ($d\theta_i = N d\theta_o$, $\nu_o = N \nu_i$), with no named metric inserted as a premise.	unverified
EQ-001/P.56.v1	P	The diagonal cone map $B = \text{diag}(a,b)$ factors uniquely as $B = N \cdot \text{diag}(\chi^{-1}, \chi)$, with $N = \sqrt{ab} = \sqrt{\det B}$ (the internal LAPSE) and $\chi = \sqrt{b/a}$ (relative-observer factor). Pure observer change $\Rightarrow \det B = 1 \Rightarrow N = 1$; geometry-dependent read-out $\Rightarrow N = \sqrt{\det B}$. Exact witness: $a=2, b=8 \Rightarrow \det B=16, N=4, \chi=2$.	unverified
EQ-001/P.57.v1	P	On a stationary path, $d\theta_i = N_{\{i o\}} d\theta_o$ (θ = retained transition count), so $\nu_o = N \nu_i$ and $d\theta_o = d\theta_i / N$. No external light/gravity/metric law is used; 'frequency shift' here means relative retained-step rate only.	current
EQ-001/P.58.v1	P	Determinants multiply under composition of diagonal observer maps, so $N_{\{j o\}} = N_{\{j i\}} \cdot N_{\{i o\}}$, and $N_P = \text{product of } N_e \text{ along a path } P$. Around a loop C , the normalization holonomy $H_N(C) = \text{product of } N_e \text{ over the loop}$ is the scalar side of curvature. Exact witness: $B_{\{i o\}} = \text{diag}(2,8)$ ($N=4$), $B_{\{j i\}} = \text{diag}(3,27)$ ($N=9$), composed $B_{\{j o\}} = \text{diag}(6,216)$ ($\det=1296, N=36=9 \cdot 4$).	current
EQ-001/P.59.v1	P	$N=0$ iff $\det B=0$, a finite rank-loss boundary of the observer map (not a statement involving infinity). At $N=0$ the translation B has no inverse; the exterior cannot reconstruct the interior. Clone/agency-horizon connects here as a READOUT equivalence only. Exact witness: $B_h = \text{diag}(0,5)$, $\det=0$, $N=0$, singular (no inverse).	unverified

Code	Dom.	Statement	Status
EQ-001/P.60.v1	P	From the DeclaredFormula $\tau_c(U) = \pi \cdot c/a_{\text{local}}$, $a_{\text{local}} = \pi \cdot c/\tau_c(U)$. The ob- server reads $a_o = N_{\{i o\}} \cdot a_{\text{local}}$; at the horizon, $\kappa_R := \lim_{\{i \rightarrow H\}} N_{\{i o\}} a_{\text{local}}(i)$, if finite. CORRECT STATUS: $a_{\text{local}} \neq \kappa_R$ directly, but $\kappa_R = \text{Normalize}_N(a_{\text{local}})$; 'surface gravity' requires EXTERNAL se- mantic calibration – only the INTERNAL bridge (κ_R as a normalized readout of a_{local}) is repaired here, not the physical identification. Exact witness: $a_{\text{local}}=7/2$, $N=4 \Rightarrow \kappa_R=14$, finite, $\kappa_R \neq a_{\text{local}}$.	current
EQ-001/P.61.v1	P	The graph front-speed v is declared (not de- rived) as playing the role a measured propaga- tion speed would play; the identification with any specific physical speed is not established.	current
EQ-001/P.62.v1	P	Retention memory is declared (not derived) as playing a mass-like role; no mass-energy law is derived from the root.	current
EQ-001/P.63.v1	P	$r_s = 2 \cdot G \cdot E/c^4$, admitted only as a declared calculator identity, not a derived theorem of this domain.	current
EQ-001/P.64.v1	P	$\tau_c = \pi \cdot c/a$, admitted only as a de- clared calculator identity, not a derived theo- rem of this domain.	current
EQ-001/P.65.v1	P	CLOSED in v0.2: the horizon half by GC- 04 ($N=0 \leq \det B=0$, finite rank-loss boundary) plus GC-05 (Unruh-fix internal bridge $\kappa_R = \text{Normalize}_N(a_{\text{local}})$); the metric-source half by the Geome- try Stationarity Gate (GD-01..GD-04), a source/record field Ψ driving a Theta- evolution that passes a commuting square AND has a failing control case (exact fixture $\Theta_{\{n+1\}}=3/8$, $R_{\Theta}=0$; failing control $\Theta_{\text{wrong}}=7/8$, $R_{\Theta}=1 \neq 0$).	current
EQ-001/P.66.v1	P	Since $G[\Theta] = G_0 + \sum_a \Theta^a G_a$, $dG/d\Theta^a = G_a$. The geometry source is $S_{\Theta, n}^a = \Phi_n^T G_a \Psi_n$ – reader x record changing the operator geometry reads. This is NOT posited as energy/mass/stress. Exact witness: $G_1 = [[0,1],[1,0]]$, $\Phi_n = [1,0]$, $\Psi_n = [0,1]$ $\Rightarrow S_{\Theta} = 1$.	current

Code	Dom.	Statement	Status
EQ-001/P.67.v1	P	The living-geometry state Θ obeys the same retained-action grammar as the rest of the root: $S_{\text{closed}} = S_{\text{DRL}} + \sum_n [(1/2) dt] d\Theta_n^T M_{\Theta} d\Theta_n - dt U_{\Theta}(\Theta_n)$. Stationarity in Θ gives $M_{\Theta} d^2_t \Theta_n + \text{grad } U_{\Theta}(\Theta_n) + K S_{\Theta,n} = 0$, with explicit stepper $\Theta_{n+1} = 2\Theta_n - \Theta_{n-1} - dt^2 M_{\Theta}^{-1} [\text{grad } U_{\Theta}(\Theta_n) + K S_{\Theta,n}]$. Exact fixture ($\Theta_{n-1}=0$, $\Theta_n=1/2$, $M_{\Theta}=2$, $K=1$, $dt=1$, $U_{\Theta}=\Theta^2/4$) with $S_{\Theta}=1$ gives $\Theta_{n+1}=3/8$ and action residual $R_{\Theta}=0$. Then $G_{n+1}=G[\Theta_{n+1}]$ and $g_{n+1}^{ij}=(1/2)$ Hess $\sigma_{G[\Theta_{n+1}]}$ – operator\to metric only, spectrum-alone still forbidden.	current
EQ-001/P.68.v1	P	The minimal closed state is $Z_n=(\Phi_n, \Phi_{n-1}, \Psi_n, \Psi_{n-1}, \Theta_n, \Theta_{n-1})$; $Z_{n+1}=F_{\text{MSG}}(Z_n)$. The commuting square $q_{\text{MSG}} \circ F_{\text{full}} = F_{\text{MSG}} \circ q_{\text{MSG}}$ holds, but the metric-only quotient q_g LACKS the variables to close alone – two states with equal metric but different Φ , Ψ , Θ_{n-1} give different next metrics. Time-unrolled, this is a DAG, not circular.	current
EQ-001/P.69.v1	P	Four named fail states for the geometry stationarity gate: $\text{FAIL_GEOMETRY_PIVOT}$ ($\det M_{\Theta}=0$ / low pivot), $\text{FAIL_METRIC_ADMISSIBILITY}$ (bad-signature/degenerate metric), FAIL_NULL_MIXING (null edges mixed into a stationary claim), HORIZON_BOUNDARY (lapse=0, ties to GC-04). No silent regularization; no inverse computed across a horizon.	current

EQ-002 — Distinguishability needs a distinguisher

tier: Ax ; status: *current*

$\exists A : A$ discriminates $E_1 \neq E_2$

Code	Dom.	Statement	Status
EQ-002/E.01.v1	E	$r_n = R_H(x_n H_n, c_n)$; extended with an explicit question index in Before Meaning, Before Choice: $\rho^{\wedge} H_n = R_H(x_n H_n, c_n, Q_n)$	current
EQ-002/E.02.v1	E	$R_A = O_A(W; \Pi_A) (\neq W)$; $m(A) \neq \rho(A)$	current
EQ-002/E.03.v1	E	$M_A(E) = (T_A \circ \Pi_A)(S_A(E))$	current

Code	Dom.	Statement	Status
EQ-002/E.04.v1	E	$\mu_n = (\mu_n^{\text{aff}}, \mu_n^{\text{prag}}, \mu_n^{\text{auto}}, \mu_n^{\text{conc}}, \mu_n^{\text{epi}})$	current
EQ-002/E.05.v1	E	$\ell_n = L_H(E_n, \mu_n H_n, c_n);$ $\mu_n \rightarrow E_n \rightarrow \ell_n \rightarrow \mu_{n+1} \rightarrow E_{n+1}$	current
EQ-002/E.06.v1	E	$m_{t+1}(e) = \rho m_t(e) + 1[e_t = e];$ $a_{\{Q,t\}}(e) = [\Phi_{\{Q,t\}}(s) - \Phi_{\{Q,t\}}(s')]_+;$ $\kappa_{\text{sem}_{t+1}}(e Q) \propto \kappa_{\text{sem},(0)_t}(e Q) \cdot \exp(\beta a + \mu m - \nu c + \xi)$	current
EQ-002/E.07.v1	E	$\epsilon_{\text{tot}} > 0$ (Genesis Constraint-First); decomposed independently as $\epsilon_{\text{tot}} = \epsilon_{\text{clock}} + \epsilon_{\text{cross}} + \epsilon_{\text{sel}} + \epsilon_{\text{map}} + \epsilon_{\text{self}}$ (Structural Error theorem, Mind as Information Horizon)	current
EQ-002/E.08.v1	E	$k_{\text{epi}}(t) = \text{New}_{t+1} / \max(1, F_t);$ $k_{\text{epi}} < 1$ contractive, $k_{\text{epi}} \approx 1$ critical, $k_{\text{epi}} > 1$ expanding	current
EQ-002/E.09.v1	E	$x_{\{i,n\}} = \text{Access}(A_n; O_i, L_i, T_i, R_i, C_i)$	current
EQ-002/E.10.v1	E	$O_A[n] = \Pi_A(E[n]); \text{enc}_A(O_A)[n] = T_A(O_A[n])$	current
EQ-002/E.11.v1	E	$D_n = \{D_n^{\text{first}}, D_n^{\text{beh}}, D_n^{\text{neural}}, D_n^{\text{world}}\}$	current
EQ-002/E.12.v1	E	Friction, not magic: institutional certification has epistemic force insofar as institutions produce reliable epistemic friction (criticism, validation, replication, robustness testing, archival continuity, correction, and answerability). Certification does not create truth by status alone.	current
EQ-002/H.01.v1	H	$B[n] \rightarrow H_{\text{body}}[n] \rightarrow N[n] \rightarrow S[n] \leftrightarrow A[n] \rightarrow \pi[n] \rightarrow U[n] \rightarrow \Phi[n+1],$ with each arrow a discrete update Φ_* ; independently paralleled by $x_n \rightarrow \zeta_n \rightarrow v_n \rightarrow \rho_n \rightarrow U_n \rightarrow \beta_n \rightarrow I_{n+1} \rightarrow a_n \rightarrow S_{n+1}$ (contact $\rightarrow$ valence $\rightarrow$ drive $\rightarrow$ identity-binding $\rightarrow$ appropriated experience $\rightarrow$ action $\rightarrow$ renewed state)	current
EQ-002/H.02.v1	H	$A_{\{i,n\}} = q_A(Z_{\{i,n\}}; Q_A, O_A, c_n);$ $\text{Aut}(F_A, O_A) = \{h:$ $O_A \circ h = O_A, \quad h \circ F_A = F_A \circ h\};$ $I_n = q_{\text{comp}}(M_n \oplus \text{Bel}_n \oplus \Theta_n \oplus \text{Roles}_n \oplus \text{BodyTool}_n \oplus \text{SocialLineage}_n)$	current
EQ-002/H.03.v1	H	Retained Difference $\rightarrow$ Human Readout $\rightarrow$ Live Problem $\rightarrow$ Barrier Readout $\rightarrow$ Candidate Routes $\rightarrow$ Human Endorsement $\rightarrow$ Adaptive Scaffold $\rightarrow$ Practice $\rightarrow$ Withdrawal $\rightarrow$ Human Return $\rightarrow$ Novel Transfer $\rightarrow$ World Feedback $\rightarrow$ Opportunity Conversion	current
EQ-002/H.04.v1	H	$\alpha_t(o) \in \{\text{HUMAN}, \text{AI}, \text{JOINT}\}$ for $o \in \Omega = \{q_{\text{sem}}, \Pi_P, \Pi_Q, B, a, m, \kappa, \Gamma_Q, U_D, \Pi_R\}$	current
EQ-002/M.01.v1	M	$\Phi: X \rightarrow Z$ is readout-admissible relative to $R: X \rightarrow Y$ iff constant on every fiber of R : $R(x) = R(x') \implies \Phi(x) = \Phi(x')$; equivalently $\ker R \subseteq \ker \Phi$, or $\Phi = \bar{g} \circ R$ (Factorization Theorem). Stochastic form: $C = G \circ K^*$ (access admissibility via a Markov kernel); data-processing inequality $I(X; Y_{j+1}) \leq I(X; Y_j)$; non-collapse $R^* \neq R^\sim, K^* \neq K^\sim$ (actual source relation vs claimant's model of it).	current

Code	Dom.	Statement	Status
EQ-002/M.02.v1	M	Coh_effective=Coh_latent×L_g; L_g^proxy=(public assets with one-click programme path)/(public assets); K1→RelatedAssetDiscovery→LongerExposure→K2 opportunity	current
EQ-002/M.03.v1	M	Readout_{Q,O,c}(S) = z, z ≠ S	current
EQ-002/W.01.v1	W	C^{10}_{j,t}, C^{50}_{j,t}, C^{90}_{j,t}; I^H_{j,t}=C^{90}_{j,t}-C^{10}_{j,t}	split
EQ-002/W.02.v1	W	C^{10}_{(j,t)}, C^{50}_{(j,t)}, C^{90}_{(j,t)}	current
EQ-002/W.03.v1	W	I^H_{(j,t)} = C^{90}_{(j,t)} - C^{10}_{(j,t)}	current

EQ-003 — Asymmetry is forced

tier: untagged; status: current

$A \rightarrow B \neq B \rightarrow A$ (a directed admissible-transition relation on the retained graph, not material implication)

EQ-004 — Temporal ordering

tier: untagged; status: current

$t(s) := \min \#steps(s_0 \rightarrow s) \in \mathbb{N}$

EQ-005 — Persistence / causal memory

tier: Ax; status: current

$\tau_c > 0$

EQ-006 — Discrete world before continuum

tier: Ax; status: current

$t = n\Delta\theta$, $n \in \mathbb{N}$, $\Delta\theta > 0$ ($\Delta\theta=0$ undefined; continuum is a controlled readout limit)

EQ-007 — Finite causal propagation speed

tier: Ax; status: current

$v = \sqrt{(D/\tau_c)} < \infty$

EQ-008 — Retained graph-Laplacian operator L_R

tier: Ax; status: current

$L_R := D_W - W$ (D_W = weighted degree matrix, W = retained adjacency; full operator on a multimode retained state, not a scalar)

EQ-009 — RD unit definition

tier: Ax; status: current

1 RD := one retained-distinction record

Code	Dom.	Statement	Status
EQ-009/E.01.v1	E	$z^\sim = C z \setminus \text{to } z' = \sqrt{1-\gamma} z^\sim, \setminus \rho$ $= -\sqrt{\gamma} z^\sim \setminus \text{to } \Psi_{n+1} = \Psi_n$ $\setminus \text{oplus } \setminus \rho_n$	current

Code	Dom.	Statement	Status
EQ-009/E.02.v1	E	T1 conservation [Cyrillic text omitted in the print catalogue – see the JSON/site entry for the source text]: $Q(z_n) + \Sigma Q(\rho_j) = Q(z_{\{0\}})$ — rel. error $< 10^{-1} \cdot 10^{-2}$, decay ω [Thai text omitted in the print catalogue – see the JSON/site entry for the source text] $(1 - \gamma)^n < 10^{-9}$	current
EQ-009/E.03.v1	E	T4 = [Thai text omitted in the print catalogue – see the JSON/site entry for the source text]: [Thai text omitted in the print catalogue – see the JSON/site entry for the source text] envelope [Thai text omitted in the print catalogue – see the JSON/site entry for the source text] $\gamma \rightarrow D/M$	current

EQ-010 — Domain encoder

tier: Ax; status: current

$x_domain - [Enc_Q] \rightarrow x_RD$ (encoder: domain quantity $\rightarrow$ RD coordinate)

EQ-011 — Domain decoder

tier: Ax; status: current

$y_RD - [Dec_Q] \rightarrow y_domain$ (decoder: RD coordinate $\rightarrow$ calibrated domain quantity)

EQ-012 — Resource-logic retention judgment

tier: Ax; status: current

$\Gamma \vdash_{\{\alpha, \rho, \kappa\}} \phi$

EQ-013 — No unrestricted contraction

tier: Ax; status: current

$A \not\Rightarrow A \otimes A$

EQ-014 — Copy licence

tier: Ax; status: current

$!_{\kappa} A \vdash A^{\otimes m}, m \leq \kappa$

EQ-015 — Universal spine PDE (mixed-tier term breakdown)

tier: untagged; status: current

$M \partial_t^2 \Phi + D \partial_t \Phi + K \cdot L_R \Phi + \nabla V(\Phi) = J - \eta$ [M $\partial^2 \Phi$: Dr $\cdot$ D $\partial \Phi$: finite_diagnostic $\cdot$ K $\cdot$ L $\cdot$ R Φ : Th_coqc (admissibility skeleton) $\cdot$ η : finite_diagnostic/Open]

Code	Dom.	Statement	Status
EQ-015/B.01.v1	B	setpoint_is_fixed (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/B.02.v1	B	one_step_error (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/B.03.v1	B	n_step_error (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/B.04.v1	B	disc_even (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/B.05.v1	B	cuspid_factor (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/B.06.v1	B	fold_from_double_root (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/B.07.v1	B	bistable_window_dec (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified

Code	Dom.	Statement	Status
EQ-015/B.08.v1	B	critical_slowing_marginal (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/B.09.v1	B	rest_iff_critical (Coq theorem name only; no LaTeX statement extracted in this pass – statement text lives in the .v file body, out of BBL-195/198 source-inventory scope, and reading it would require opening solver arc (private)’s Coq source beyond the inventory JSON this build was scoped to)	unverified
EQ-015/E.01.v1	E	$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n)$; typed as $\mu_n = (\mu_n^{\text{aff}}, \mu_n^{\text{prag}}, \mu_n^{\text{auto}}, \mu_n^{\text{conc}}, \mu_n^{\text{epi}})$	current
EQ-015/E.02.v1	E	$\xi_n = (\xi_n^+ + \xi_n^-, \xi_n^0)$; $\Xi_n = \alpha_n^+ + P_n^+ + \alpha_n^- - P_n^- + \alpha_n^0 P_n^0$; $G_{\sim\{\mu,n\}} = G_{\{\mu,n\}} \circ (I + \Xi_n)$; $\epsilon_{\{\Xi,n\}} = d_O(G_{\{\mu,n\}}(z_n), G_{\sim\{\mu,n\}}(z_n))$	current
EQ-015/E.03.v1	E	$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n)$	current
EQ-015/E.04.v1	E	$E_n = \Phi_E(x_n, \mu_n, \gamma^{\mu_n}, c_n)$; central identity: Experience = phenomenon-as-meaningfully-read	current
EQ-015/E.05.v1	E	$\ell_n = L_H(E_n, \mu_n \mid H_n, c_n)$, with E_n, μ_n permitted before stable ℓ_n ; recursive form: $\mu_n \rightarrow E_n \rightarrow \ell_n \rightarrow \mu_{n+1} \rightarrow E_{n+1}$	current
EQ-015/E.06.v1	E	$I_{\{H,n\}} = \text{Retrieve}(M_{\{H,n\}} \mid c_n, Q_n)$; $\text{Res_H}(n) = C_H(E_n^{\text{cur}}, I_{\{H,n\}} \mid c_n, Q_n) \in [0,1]$; $\text{Res} \neq \text{Identity}$, $\text{Res} \neq \text{Truth}$, $\text{Res} \neq \text{Retention}$, $\text{Res} \neq \text{Improvement}$	current
EQ-015/E.07.v1	E	$\Delta W_{\{j,N\}} = \Sigma_{\{n=1\}}^N \eta_n \text{-eligibleGradient_n}$; $W_n^{\text{eff}} = W_{\{\text{info},n\}} \cdot g(\text{Res_H}(n), \text{eligibility}, \text{context})$; $\Sigma_{\{k \leq n\}} W_k^{\text{eff}} \geq \Delta V_{\{\text{old} \rightarrow \text{new}\}}$	current
EQ-015/E.08.v1	E	$H_{n+1} = U_H(H_n, \text{Retain}(E_n, \mu_n, r_n), \delta^{\text{world}_{n:n+1}})$ — restated with an added other-experience term in later Human-AI chapters as $H_{n+1} = U_H(H_t, E^{\{\text{AI}\}}_{\{H,t\}}, \delta^{\{\text{world}\}}, X^{\{\text{other}\}})$	current
EQ-015/E.09.v1	E	$V_A[n] = \text{Align}(M_A[n], \theta_W \mid D)$; $E[V_A[n+1] \mid \text{Rsn_A}, D] > E[V_A[n] \mid D]$	current
EQ-015/E.10.v1	E	When external generation of coherent structure increases without bound, observable production ceases to serve as a unique indicator of internal organization; regime transition: accumulation $\rightarrow$ discrimination	current
EQ-015/E.11.v1	E	$K_A(D,t) := V_A^D(M_A(t), \theta_D)$; $V_A^D = w1P + w2I + w3S + w4R + w5L$, $\Sigma w_i^D = 1$	current
EQ-015/E.12.v1	E	$M_A[n] = K_A \cdot \theta(E[n]) + \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}}$	current

Code	Dom.	Statement	Status
EQ-015/E.13.v1	E	$S_{\{A,t\}} := q_{\{sem,A,\Omega_t\}}(Z_t)$	current
EQ-015/E.14.v1	E	No Bare Pedigree: a source label is epistemically incomplete reporting. The relevant object is the tuple of production procedure, selection conditions, dependencies, checks, error model, inferential role, and answerability.	current
EQ-015/E.15.v1	E	$M_A[n] \neq \theta(E[n]); \quad S_{\{A,t+1\}} \neq S_{\{A,t\}}$ is admissible and expected; reachable(H)=1 $\Rightarrow$ Acc(H Q) high; wholesome $\neq$ true, unwholesome $\neq$ false; $a_{\{Q,t\}}(H_i)$ high $\Rightarrow$ $W_t(H_i)$ high, $m_t(H_i)$ high $\Rightarrow$ $W_t(H_i)$ high, $Acc(H_i)$ high $\Rightarrow$ $W_t(H_i)$ high; Attraction $\neq$ Warrant, Momentum $\neq$ Truth; $H_i \sim \{U_{\{D,t\}}\} H_j$ $\Rightarrow$ a_G $H_i \sim \{U_{\{D,t+1\}}\} H_j$; up $\Rightarrow$ $W_G(H)$ up, m_G up $\Rightarrow$ truth; Attraction $\neq$ Accessibility $\neq$ Warrant $\neq$ Truth	current
EQ-015/H.01.v1	H	$H_t \rightarrow \{L_H\} Q_t; \quad Q_t \rightarrow AI_t \rightarrow Y_t$ $\rightarrow \{R_H\} \quad E^{\{AI\}}_{\{H,t\}};$ $H_{\{t+1\}} = U_H(H_t, E^{\{AI\}}_{\{H,t\}}, \delta^{\{world\}}, X^{\{other\}})$	current
EQ-015/H.02.v1	H	$A0 = \langle P0, M0, U0, E0, F0, S0 \rangle$ (latest, DCP eq.12); super-sedes $H0^* = (P0, M0, U0, E0, \Phi0, \kappa0)$ and $ENTRY_{\{H \rightarrow AI\}} = H0^* \wedge V0 \wedge C0 \wedge R^*$	current
EQ-015/H.03.v1	H	$H_t \rightarrow L_H \rightarrow Q_t; \quad Q_t \rightarrow AI_t \rightarrow Y_t$ $\rightarrow R_H \rightarrow \quad E^{\{AI\}}_{\{H,t\}};$ $H_{\{t+1\}} = U_H(H_t, E^{\{AI\}}_{\{H,t\}}, \delta^{\{world\}}, X^{\{other\}});$ $L_{\{A,t+1\}} \neq L_{\{A,t\}}$	current
EQ-015/H.04.v1	H	$Q_t \rightarrow AI_t \rightarrow Y_t \rightarrow R_H \rightarrow E^{\{AI\}}_{\{H,t\}};$ $H_{\{t+1\}} = U_H(H_t, E^{\{AI\}}_{\{H,t\}}, \delta^{\{world\}}, X^{\{other\}})$	current
EQ-015/H.05.v1	H	$Z_{dlg[s,n+1]} = F \#_{dlg}(Z_{dlg[s,n]}, u_H[s,n], u_{AI}[s,n], T[s,n]);$ $\chi_{recip[s,n,L]} = D_{recip[s,n,L]} / \Sigma[s,n] $	current
EQ-015/H.06.v1	H	$\Delta O_n^{fast} = B_n A_n; \quad g_n =$ $\Gamma_n(s_n, r_n^{err}, q_n, v_n, c_n, p_n) \in$ $\mathbb{Q}[0,1]; \quad \Delta O_n^{ret} = g_n \cdot \Delta O_n^{fast};$ $O_H[n+1] = O_H[n] + \Delta O_n^{ret} + \epsilon_n;$ finite-bottleneck postulate $0 < m_n < d_n;$ $rank_{\mathbb{Q}}(B_n A_n) \leq m_n$ (proved)	current
EQ-015/H.07.v1	H	$\Pi^{live}_{\{A,t\}}(g) \subseteq \Pi^{feas}_{\{A,t\}}(g) \subseteq$ $\Pi^{phys}_t(g)$	current
EQ-015/H.08.v1	H	$L_{\{A,t\}}(g) = \{(\pi, \lambda^{live}_{\{A,t\}}(\pi g)) : \pi \in \Pi^{feas}_{\{A,t\}}(g)\};$ $\Pi^{live}_{\{A,t\}}(g) = \{\pi : \lambda^{live}_{\{A,t\}}(\pi g) \geq \tau_{live}\}$	current
EQ-015/H.09.v1	H	$p^*_{\{A,g\}}(h,z;T,B,P) = \max_{\pi \in \Pi^{wit}_A(g;h,z;T,B,P)}$ $\Pr \pi_P(Read_g \cap D_g \cap X_g \cap F_g)$	current
EQ-015/H.10.v1	H	$AI(Q) = K_{like}, K_{like} \neq K_{validated}$	current
EQ-015/H.11.v1	H	$K_{like} \rightarrow K_{assumed}$ (dangerous shortcut); repaired: $K_{like} \rightarrow \text{check} \rightarrow K_{checked}$ $\rightarrow \text{independent support} \rightarrow K_{supported}$ $\rightarrow \text{warrant } V^* \rightarrow K_{validated} V^*$	current
EQ-015/H.12.v1	H	$G_s = g(k,d,v,p,r,1-f,a \Theta_s, \Pi_s);$ $T_s = h(c,f,b,o \Theta_s, \Pi_s); \quad \Delta_s = G_s - T_s;$ $\eta_s > 0, \Delta_s > 0 \Rightarrow \text{expansion};$ $\eta_s > 0, \Delta_s < 0 \Rightarrow \text{tunnel}$	current

Code	Dom.	Statement	Status
EQ-015/H.13.v1	H	$H0^* \rightarrow K_like \rightarrow D^{\wedge}eff \rightarrow R^{\wedge}eff \rightarrow H \leftrightarrow AI \rightarrow \chi_recip \rightarrow \mu(C-T \Theta, \Pi) \rightarrow Y^{\wedge}return \rightarrow J^*$	current
EQ-015/H.14.v1	H	$R_s^{\wedge}ep = \rho(I_s, V_s, Q_s);$ $U_s^{\wedge}R = u(C_s^{\wedge}v, T_s^{\wedge}v, A_s^{\wedge}v);$ $R_s^{\wedge}ex = \psi(R_s^{\wedge}ep, U_s^{\wedge}R);$ Resistance quality $\neq$ resistance accessibility	current
EQ-015/H.15.v1	H	$K_s = (\kappa 0, \kappa 1, W0, W1);$ Calibration error $\approx$ $N^{\wedge}\{-1\} \Sigma_i (\kappa_i - y_i)^2$	current
EQ-015/H.16.v1	H	$H_return = \langle G_CTSA, L, M, P, W, \Delta_dir \rangle$	current
EQ-015/H.17.v1	H	$R^{\wedge}return_H = \langle C, T, S, A \rangle$	current
EQ-015/H.18.v1	H	$D > 0, Resist > 0, A_H > 0$	current
EQ-015/H.19.v1	H	$J^*_s = (AUG_s, SYN_s, RET_s);$ $AUG_s = P^{\wedge}joint_s - P^{\wedge}H_s;$ $SYN_s = P^{\wedge}joint_s - \max(P^{\wedge}H_s, P^{\wedge}AI_s)$	current
EQ-015/H.20.v1	H	H1-H6 [Open], §12	current
EQ-015/H.21.v1	H	$\pi^{\wedge}deploy = f(Stakes, LearningNeed, Irreversibility, DependencyRisk, UserSkill);$ $State \rightarrow Challenge \rightarrow Check \rightarrow Own$ (Lite); $Topic$ Entry $\rightarrow$ Triage $\rightarrow$ Minimum Suf- ficient Dialogue $\rightarrow$ Escalate $\rightarrow$ Periodic Return (Critical); $\pi^{\wedge}dialogue = f(TopicMode, Goal, Skill, Stakes, Domain, LearningNeed, Reversibility)$	current
EQ-015/H.22.v1	H	Reset: fresh framing/session/source route; Removal: absence of decisive AI assistance	current
EQ-015/H.23.v1	H	$\Delta H_s = \langle \Delta C_s, \Delta T_s, \Delta S_s, \Delta A_s, \Delta A^{\wedge}corr_H, \Delta A^{\wedge}live_H, s \rangle;$ $F^{\wedge}return = f(Criticality, LearningNeed, FailureCost, DependencyRisk)$	current
EQ-015/H.24.v1	H	$I_s \rightarrow a_s \rightarrow \delta^{\wedge}world_s \rightarrow H_{s+1,0};$ Live Problem $\rightarrow$ Question $\rightarrow$ Dialogue $\rightarrow$ Human Return $\rightarrow$ Action $\rightarrow$ World Feedback $\rightarrow$ Revision or New Problem	current
EQ-015/H.25.v1	H	DCPp, DEpp, AgP [Open]	current
EQ-015/H.26.v1	H	H1-H10, §15.2	current
EQ-015/H.27.v1	H	What unresolved difference deserves my atten- tion? / How much AI/friction does this task require? / What remains with me after AI is removed? / What does the world return when I act?	current
EQ-015/H.28.v1	H	$\mathcal{E} = f(ConstraintPrecision, ContextRecall, SourceTraceability, ErrorRepair)$	current
EQ-015/H.29.v1	H	H1: AI increases access to/reorganization of human interpretive contexts, not event ac- cess; H3: bounded interpretive expansion un- der high $\mathcal{E}$ and preserved S; H4: AI as archi- tectural doorway to interpretive diversity	current
EQ-015/H.30.v1	H	$Z_{HCA,i,n} = \langle P^{\wedge}live_{i,n}, K^{\wedge}life_{i,n}, B^{\wedge}bar_{i,n}, C^{\wedge}cand_{i,n}, C^{\wedge}live_{i,n}, h_{i,n}, R^{\wedge}return_{H,i,n}, \Omega^{\wedge}real_{i,n} \rangle$	current
EQ-015/H.31.v1	H	$K^{\wedge}life_{i,n} = \langle E^{\wedge}econ, F^{\wedge}base, L^{\wedge}lang, D^{\wedge}digital, T^{\wedge}disc, H^{\wedge}health, M^{\wedge}mob, N^{\wedge}mentor, C^{\wedge}cred \rangle$	current
EQ-015/H.32.v1	H	Stable Unaided Return $\uparrow \Rightarrow h^{\wedge}decisive \downarrow$; Attempt $\rightarrow$ Minimal Sufficient Scaffold $\rightarrow$ Feedback $\rightarrow$ Reattempt $\rightarrow$ Fading $\rightarrow$ Unaided Execution $\rightarrow$ Novel Transfer; $R^{\wedge}return_H \uparrow \Rightarrow h^{\wedge}decisive \downarrow$	current

Code	Dom.	Statement	Status
EQ-015/H.33.v1	H	Adult: Purpose+Transparency+Refusal+Revision+Data Minimization; Child: Child Assent+Adult Oversight+Privacy+Safety+No Opaque Persuasion(+Age-Appropriate Design)	current
EQ-015/H.34.v1	H	H1-H6 [Open], §17.2	current
EQ-015/H.35.v1	H	P1: framing residual; P2: iteration without integration is insufficient	current
EQ-015/H.36.v1	H	M0: context-only state change; M1: sparse but not low-rank update; M2: low-rank factorized update (= CAN-054's own $\Delta O_n = B_n A_n$); M3: regularized full-rank update; M4: finite graph rewiring; M5: hybrid fast trace plus slow update	current
EQ-015/H.37.v1	H	$W_n = (F_n, A_n^{\wedge} \text{agency}, M_n^{\wedge} \text{meaning}, R_n^{\wedge} \text{relation}, C_n^{\wedge} \text{competence}, Q_n^{\wedge} \text{repa})$ $C_{\text{mal}}[n] = \text{rig}(a_n) \cdot \text{gen}(a_n) \cdot \text{mis}(a_n) \cdot \text{loss}(a_n)$	current
EQ-015/M.01.v1	M	$S_n = (G_n, \Lambda_n, T_n)$	current
EQ-015/M.02.v1	M	$S_{\{n+1\}} = F(S_n, u_n, c_n, T_n)$	current
EQ-015/M.03.v1	M	retention $\rightarrow$ structure $\rightarrow$ translation $\rightarrow$ readout $\rightarrow$ meaning $\rightarrow$ experience $\rightarrow$ memory $\rightarrow$ belief $\rightarrow$ claim $\rightarrow$ checking $\rightarrow$ status $\rightarrow$ public report (forbidden order: starting from a familiar name and retrospectively presenting the result as knowledge)	current
EQ-015/M.04.v1	M	Retention $\rightarrow$ Structure $\rightarrow$ Translation $\rightarrow$ Readout $\rightarrow$ Meaning $\rightarrow$ Report (compressed); Retention $\rightarrow$ Structure $\rightarrow$ Candidate State $\rightarrow$ Sufficiency $\rightarrow$ Quotient $\rightarrow$ Domain Dynamics $\rightarrow$ Readout $\rightarrow$ Meaning $\rightarrow$ Experience $\rightarrow$ Memory $\rightarrow$ Knowledge $\rightarrow$ Checked Report (MEMK extension)	current
EQ-015/M.05.v1	M	DVP*=ReduceCorrelatedError+ExposeResidualDependence+BottomOutWherePossible Disagreement=>Resolve v Declare; NoHumanAvailable=>DVP $\rightarrow$ K1 $\rightarrow$ HumanAcquisition	current
EQ-015/M.06.v1	M	HorizontalGeneration $\rightarrow$ EpistemicFriction $\rightarrow$ VerticalStrengthening $\rightarrow$ ResourceReturn-Phenomenon	current
EQ-015/M.07.v1	M	Phenomenon $\rightarrow$ AIExploration $\rightarrow$ DVP $\rightarrow$ HumanMaster $\rightarrow$ Integrity $\rightarrow$ K1 $\rightarrow$ {Global,LocalFriction} $\rightarrow$ K2 $\rightarrow$ Revision $\rightarrow$ K3; EpistemicPosition=RecognizableProblem+RepeatHumanNodes+CitableAssets+CorrectionHistory CrediblePath=EarlyTimestamp+ExplicitProvisionality+RapidRevision+ExternalFriction	current
EQ-015/M.08.v1	M	$T_n = \{(x_k, t_k, w_k)\}_{k \leq n}$, Rhythm $_n = \Omega(T_n, B_n)$; $m_{\{t+1\}}(e) = \rho m_t(e) + 1[e_t = e]$, $0 \leq \rho < 1$; $\kappa^{\wedge}\{\text{sem}\}_{\{t+1\}}(e Q) \propto$ $\kappa^{\wedge}\{\text{sem}, (0)\}_t(e Q) \cdot \exp(\beta a_{\{Q, t\}}(e) + \mu m_t(e) - \nu c_{\{A, t, Q\}}(e) + \xi_t(e))$; corrected non-derivation: Rhythm $_t, m_t, a_t, c_t \rightarrow \kappa^{\wedge}\{\text{sem}\}_{\{t+1\}}$	current
EQ-015/M.09.v1	M	$B[t+1] = F(B[t]) + C_H(H[t])$; $H[t+1] = G(H[t]) + C_B(B[t])$; $E[t] = R(B[t], H[t])$	current
EQ-015/M.10.v1	M	$B^{\wedge} \text{use} = \langle \text{Time}, \text{CognitiveLoad}, \text{VerificationCost}, \text{Interruption}, \text{LiteracyDemand} \rangle$; Epistemically optimal $\neq$ Behaviorally adoptable	current

Code	Dom.	Statement	Status
EQ-015/M.11.v1	M	Def-2 Access augmentation ($Y1 \rightarrow (Y1, Y2)$); Def-3 Contrast/relevance operation ($X \rightarrow X' \subseteq X$); Def-4 Inferential commit- ment (a rule changes what can be inferred without new access); Def-5 Decision-policy augmentation (a loss/threshold/rule maps epistemic state to action)	current
EQ-015/M.12.v1	M	$M_A[n] \neq \text{theta}(E)$	current
EQ-015/M.13.v1	M	$\backslash \text{mathbb{L}}^{\wedge n}[X] = (M/2dt) \cdot \Delta X^{\wedge T}(G \otimes I) \Delta X$ $+ (D/2) \cdot \Delta X^{\wedge T}(\backslash \Omega \otimes I) \Delta X$ $- (dt/2) \cdot \Delta X^{\wedge T}(G \otimes (K \cdot L_R$ $+ k_{\{2\}}I))X$	current
EQ-015/M.14.v1	M	$H = M \cdot v \cdot \Phi^{\wedge T} v \cdot \Psi$ $+ K \cdot \Delta \Phi^{\wedge T} L_R \Psi + k_{\{2\}} \cdot \Delta \Phi^{\wedge T} \Psi$ (conserved quan- tity)	current
EQ-015/M.15.v1	M	$H_{nl} = \sum_i M_i v \cdot \Phi_i v \cdot \Psi_i$ $+ K \cdot \Delta \Phi^{\wedge T} L_w \Psi + \Delta \Psi^{\wedge T} \nabla V(\Phi) - J^{\wedge T} \Psi$	current
EQ-015/P.01.v1	P	Planck–Einstein relation $E = \hbar \omega$	current
EQ-015/P.02.v1	P	Schrödinger equation / eigenvalue problem	current
EQ-015/P.03.v1	P	Special relativity / Lorentz invariance	current
EQ-015/P.04.v1	P	General relativity field equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$	current
EQ-015/P.05.v1	P	Schwarzschild metric	current
EQ-015/P.06.v1	P	Regge-Wheeler equation	current
EQ-015/P.07.v1	P	Electroweak mixing: $A = \sin\theta W \cdot W^{\wedge 3} + \cos\theta W \cdot B,$ $Z = \cos\theta W \cdot W^{\wedge 3} - \sin\theta W \cdot B,$ $m_{W^{\wedge 2}} = g^{\wedge 2} v^{\wedge 2} / 4,$ $m_{Z^{\wedge 2}} = (v^{\wedge 2} / 4)(g^{\wedge 2} + g'^{\wedge 2}),$ $\sin^2\theta W = g'^{\wedge 2} / (g^{\wedge 2} + g'^{\wedge 2})$	current
EQ-015/P.08.v1	P	Fermi coupling G_F and $v = (\sqrt{2} G_F)^{-1/2}$ (Higgs vev)	current
EQ-015/P.09.v1	P	Gauge-anomaly cancellation condi- tions $[SU(3)]^{\wedge 2} U(1), [SU(2)]^{\wedge 2} U(1),$ $[grav]^{\wedge 2} U(1), [U(1)]^{\wedge 3}$ fixing the SM hypercharges	current
EQ-015/P.10.v1	P	Peter–Weyl theorem / character convolution (class-function convolution multiplies repre- sentation coefficients)	current
EQ-015/P.100.v1	P	Normal Zeeman effect $dE = m_l \mu_B B$	current
EQ-015/P.101.v1	P	Rutherford differential scattering cross section $d\sigma/d\Omega = (Z_1 Z_2 k_e e^2 / 4E)^2 / \sin^4(\theta/2)$	current
EQ-015/P.102.v1	P	Fermi-Dirac distribution	current
EQ-015/P.103.v1	P	Bose-Einstein distribution	current
EQ-015/P.104.v1	P	Quantum harmonic oscillator canonical par- tition function $Z = 1/(2 \sinh(\hbar \omega/2$ $k_B T))$	current
EQ-015/P.105.v1	P	Planck’s law of blackbody spectral radiance $B(\lambda, T)$	current
EQ-015/P.106.v1	P	Rectangular waveguide TE _{mn} cutoff fre- quency	current

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EQ-015/P.107.v1	P	Oscillating electric dipole radiated power $P = p_0^2 \omega^4 / (12 \pi \epsilon_0 c^3)$	current
EQ-015/P.108.v1	P	Relativistic Doppler shift (radial) $f_{\text{obs}} = f_{\text{source}} \sqrt{(1-\beta)/(1+\beta)}$	current
EQ-015/P.109.v1	P	Vis-viva equation $v^2 = GM(2/r - 1/a)$	current
EQ-015/P.11.v1	P	Wilson-loop area-law criterion for confinement $-\log \langle W(C) \rangle = \sigma \cdot \text{Area}(C)$, and the center role in confinement [K. Wilson (1974); center-symmetry / center-vortex work ('t Hooft 1978; Greensite) 1974; 1978+	current
EQ-015/P.110.v1	P	General-relativistic perihelion/apsidal precession $d\phi = 6 \pi G M / (c^2 a (1-e^2))$	current
EQ-015/P.111.v1	P	Schwarzschild radius $r_s = 2GM/c^2$	current
EQ-015/P.112.v1	P	Schwarzschild static-observer gravitational time dilation $d\tau/dt = \sqrt{1-r_s/r}$	current
EQ-015/P.113.v1	P	Debye model heat capacity, low-temperature T^3 limit $C_V = (12 \pi^4/5) N k_B (T/\theta_D)^3$	current
EQ-015/P.114.v1	P	Resonance quality factor from damping ratio $Q = 1/(2 \zeta)$	current
EQ-015/P.115.v1	P	Ziegler-Nichols closed-loop PID tuning rule ($K_p=0.6 K_u$, $T_i=0.5 P_u$, $T_d=0.125 P_u$)	current
EQ-015/P.116.v1	P	Hagen-Poiseuille equation $Q = \pi r^4 dP / (8 \mu L)$	current
EQ-015/P.117.v1	P	Blasius laminar flat-plate boundary layer similarity solution $(\delta_{99} \approx 5.0x / \sqrt{Re_x})$, $Cf_x = 0.664 / \sqrt{Re_x}$	current
EQ-015/P.12.v1	P	Ginsparg-Wilson relation $\{\Gamma, D\} = a D \Gamma D$ (lattice chirality without naïve anticommutation) and the overlap/sign-operator realization $D = (1/a)(I - V)$, $V = \Gamma \epsilon$	current
EQ-015/P.13.v1	P	Nielsen-Ninomiya fermion-doubling no-go (locality + translation invariance + Hermiticity + exact naïve chiral anticommutation $\implies 2^d$ species)	current
EQ-015/P.14.v1	P	Clifford algebra $\{A_\mu, A_\nu\} = 2\delta_{\mu\nu} I$ / irreducible Dirac-matrix dimension (2×2) Pauli, mutually-orthogonal directions in $\mathbb{R}^3$	current
EQ-015/P.15.v1	P	Unitary $N \times N$ quark-mixing-matrix physical-parameter counting (N^2 real params, $2N-1$ removable by phase redefinition $\implies (N-1)^2$ physical, split into $N(N-1)/2$ mixing angles + $(N-1)(N-2)/2$ CP-violating phases; nonzero phase count requires $N \geq 3$)	current
EQ-015/P.16.v1	P	Gatto-Sartori-Tonin relation $\sin \theta_C \approx \sqrt{m_d/m_s}$ (Cabibbo angle from the down-type quark mass ratio) [R. Gatto, G. Sartori, M. Tonin 1968	current

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EQ-015/P.17.v1	P	Group-averaging / Reynolds ("twirl") projection onto the invariant subspace $\Pi(X)=(1/ G)\Sigma_{R\in G} R^{\{T\}}XR$, here for the signed-permutation (hyperoctahedral) group $B_4=(\mathbb{Z}_2)^4 \rtimes S_4$ $\Rightarrow \Pi_4(X)=(\text{Tr } X/4)I$	current
EQ-015/P.18.v1	P	Osterwalder–Schrader reflection positivity via the Gram / transfer-matrix construction (a half-slab amplitude B gives a positive kernel $K=B^{\dagger}B \geq 0$, i.e. reflection positivity by construction; hidden-midpoint reconstruction)	current
EQ-015/P.19.v1	P	Perron–Frobenius / doubly-stochastic consensus (a primitive doubly-stochastic P has a simple eigenvalue 1 with $P _{1^\perp}$ spectral radius $<1 \Rightarrow$ convergence to the mean)	current
EQ-015/P.20.v1	P	Quasnormal-mode late-time power-law ("Price") tail	current
EQ-015/P.21.v1	P	Late-time tail dependence on initial data	current
EQ-015/P.22.v1	P	Late-time tails, further numerical study	current
EQ-015/P.23.v1	P	Quasnormal mode spectrum (continued-fraction method)	current
EQ-015/P.24.v1	P	Raychaudhuri equation	current
EQ-015/P.25.v1	P	de Broglie's internal clock	current
EQ-015/P.26.v1	P	Dirac equation	current
EQ-015/P.27.v1	P	Zitterbewegung (relativistic)	current
EQ-015/P.28.v1	P	Zitterbewegung interpretation	current
EQ-015/P.29.v1	P	Higgs mechanism	current
EQ-015/P.30.v1	P	Electroweak Higgs mechanism / one-doublet EWSB: residual $Q=T_3+Y$ stabilizer, $m_W^2=g^2v^2/4$, $m_Z^2=(g^2+g'^2)v^2/4$, $m_W=m_Z \cos\theta$, and tree-level custodial $\rho=1$	current
EQ-015/P.31.v1	P	Elitzur's theorem (a local gauge symmetry cannot break spontaneously; only gauge-invariant order parameters like $\langle H^\dagger H \rangle$ are physical)	current
EQ-015/P.32.v1	P	Robb's causal-order axiomatization	current
EQ-015/P.33.v1	P	Zeeman: causality determines Lorentz group	current
EQ-015/P.34.v1	P	Malament: causal order fixes topology	current
EQ-015/P.35.v1	P	One-loop non-abelian gauge β -function with matter, $b_a = (11/3)C_2(G_a) - (2/3)\Sigma T_a(R_{\text{fermion}}) - (1/3)\Sigma T_a(R_{\text{scalar}})$ (the 11/3, 2/3, 1/3 kinematic weights are the gauge-boson+ghost, Weyl-fermion, and complex-scalar one-loop vacuum-polarization contributions)	current
EQ-015/P.36.v1	P	GUT hypercharge normalization $T_1(R) = (3/5)Y^2$ (SU(5) embedding factor needed to compare $U(1)_Y$ on the same Dynkin-index footing as SU(2)/SU(3))	current
EQ-015/P.37.v1	P	Schutz's independent Minkowski axioms	current
EQ-015/P.38.v1	P	Regge calculus	current

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EQ-015/P.39.v1	P	Causal set theory	current
EQ-015/P.40.v1	P	Wolfram hypergraph rewriting project	current
EQ-015/P.41.v1	P	Tree-level Higgs quartic self-coupling relation $\lambda = m_H^2/(2 v^2)$ (from $V(H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2$ minimized at v)	current
EQ-015/P.42.v1	P	QCD running coupling $\alpha_s(M_Z)$ (asymptotic freedom / one-loop-and-beyond running of the SU(3) gauge coupling)	current
EQ-015/P.43.v1	P	QCD vacuum angle θ_{QCD} and the strong CP problem (CP-violating term $\theta (g^2/32 \pi^2) \text{Tr} G \tilde{G}$ in the QCD Lagrangian; neutron EDM as the experimental probe) G. 't Hooft (θ -vacuum structure, 1976); neutron EDM experimental collaborations (bound) 1976 (θ -vacuum); ongoing (EDM bound)	current
EQ-015/P.44.v1	P	Neutrino oscillation mass-squared splittings $\Delta m_{21}^2, \Delta m_{31}^2$ (PMNS-driven flavor oscillation probability depends only on these differences, not absolute masses)	current
EQ-015/P.45.v1	P	Clausius relation $\delta Q = T \delta S$	current
EQ-015/P.46.v1	P	Bekenstein–Hawking area-entropy law	current
EQ-015/P.47.v1	P	Unruh temperature	current
EQ-015/P.48.v1	P	Jacobson’s thermodynamic derivation of Einstein’s equation	current
EQ-015/P.49.v1	P	Verlinde’s entropic gravity	current
EQ-015/P.50.v1	P	Sakharov induced gravity	current
EQ-015/P.51.v1	P	Susskind holographic principle	current
EQ-015/P.52.v1	P	Maldacena AdS/CFT	current
EQ-015/P.53.v1	P	Ryu–Takayanagi entanglement=area	current
EQ-015/P.54.v1	P	Van Raamsdonk: spacetime from entanglement	current
EQ-015/P.55.v1	P	Quasinormal-mode analytic representation	current
EQ-015/P.56.v1	P	QNM review (literature target values)	current
EQ-015/P.57.v1	P	Hyperboloidal-slicing method	current
EQ-015/P.58.v1	P	Perfectly Matched Layer absorbing boundary	current
EQ-015/P.59.v1	P	Kepler’s third law $T^2 = 4 \pi^2 a^3/(G M)$	current
EQ-015/P.60.v1	P	Bernoulli’s equation (steady incompressible streamline energy balance)	current
EQ-015/P.61.v1	P	Continuity equation $A_1 v_1 = A_2 v_2$ (incompressible 1D duct form)	current
EQ-015/P.62.v1	P	2D truss method of joints (joint-equilibrium linear system, $m+r=2j$ determinacy count)	current
EQ-015/P.63.v1	P	Euler–Bernoulli beam theory	current
EQ-015/P.64.v1	P	Cubic-Hermite beam FEM element (4-DOF stiffness matrix, consistent UDL fixed-end forces; exact nodal values for point/uniform loads)	current
EQ-015/P.65.v1	P	1D axial-bar FEM element ($k_e = EA/L$ two-node element)	current
EQ-015/P.66.v1	P	Modal analysis closed forms for uniform spring-mass chains, $\omega_n^2 = (2k/m)(1 - \cos(n \pi/(N+1)))$, and the generalized eigenproblem $K \phi = \omega^2 M \phi$	current

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EQ-015/P.67.v1	P	1D Ising model + transfer-matrix exact solution ($f = -kT \ln \lambda_+$, $\xi = -1/\ln \tanh(\beta J)$) at $h=0$)	current
EQ-015/P.68.v1	P	2D square-lattice Ising critical temperature $T_c = 2J/(k_B \ln(1+\sqrt{2}))$	current
EQ-015/P.69.v1	P	1D Ising real-space decimation RG, $K' = (1/2) \ln \cosh(2K)$ ($b=2$ block-spin)	current
EQ-015/P.70.v1	P	SUVAT kinematics equations ($v=u+at$, $s=ut+\frac{1}{2}at^2$, $v^2=u^2+2as$)	current
EQ-015/P.71.v1	P	Hooke's law $F=kx$ and spring elastic PE $\frac{1}{2}kx^2$	current
EQ-015/P.72.v1	P	Simple-pendulum and mass-spring SHM period ($T=2\pi\sqrt{L/g}$, $T=2\pi\sqrt{m/k}$) and closed-form SHM motion ($x=A \cos(\omega t + \varphi)$)	current
EQ-015/P.73.v1	P	Snell's law of refraction $n_1 \sin \theta_1 = n_2 \sin \theta_2$ and total-internal-reflection critical angle	current
EQ-015/P.74.v1	P	Young's double-slit interference fringe spacing $\Delta y = \lambda L/d$	current
EQ-015/P.75.v1	P	Diffraction grating equation $d \sin \theta = m\lambda$	current
EQ-015/P.76.v1	P	Malus's law $I=I_0 \cos^2 \theta$	current
EQ-015/P.77.v1	P	Radioactive decay law $N=N_0 e^{-\lambda t}$ and half-life $t_{1/2}=\ln 2/\lambda$	current
EQ-015/P.78.v1	P	Bohr model hydrogen energy levels $E_n=-13.6056931/n^2$ eV (full-precision CODATA 2018 Rydberg energy)	current
EQ-015/P.79.v1	P	Planck-Einstein photon energy relation $E=hf=hc/\lambda$	current
EQ-015/P.80.v1	P	Mass-energy equivalence $E=mc^2$	current
EQ-015/P.81.v1	P	Fourier's law of heat conduction $Q/t=kA\Delta T/L$	current
EQ-015/P.82.v1	P	Newton's law of cooling (convective form) $Q/t=hA\Delta T$	current
EQ-015/P.83.v1	P	Stefan-Boltzmann law $P=\epsilon\sigma AT^4$	current
EQ-015/P.84.v1	P	Carnot heat-engine efficiency $\eta=1-T_c/T_h$	current
EQ-015/P.85.v1	P	Linear thermal expansion $\Delta L=\alpha L_0\Delta T$	current
EQ-015/P.86.v1	P	Force on a current-carrying wire $F=BIL$	current
EQ-015/P.87.v1	P	Solenoid magnetic field $B=\mu_0 nI$ and straight-wire field $B=\mu_0 I/(2\pi r)$ (Ampère's law special cases)	current
EQ-015/P.88.v1	P	Faraday's law of induction $\mathcal{E}=-N\Delta\Phi/\Delta t$ and the motional-EMF special case $\mathcal{E}=BLv$	current
EQ-015/P.89.v1	P	Coulomb's law potential of a point charge $V=kq/r$, uniform-field relation $E=V/d$, and capacitor relations ($Q=CV$, $C=\epsilon_0 A/d$, $E=\frac{1}{2}CV^2$, series/parallel capacitance)	current

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EQ-015/P.90.v1	P	Real-source terminal voltage with internal resistance $V = \epsilon - Ir$	current
EQ-015/P.91.v1	P	Rotational dynamics: $\tau = I\alpha$, $L = I\omega$, $KE = \frac{1}{2}I\omega^2$, and standard moment-of-inertia forms (rod, disc, sphere, hoop)	current
EQ-015/P.92.v1	P	Coulomb friction model $f = \mu N$, incline force components, Atwood machine acceleration, Pascal's principle hydraulic press	current
EQ-015/P.93.v1	P	Torsion of circular shafts: shear stress $\tau = Tr/J$, angle of twist $\theta = TL/GJ$, polar moment of a solid/hollow circular section	current
EQ-015/P.94.v1	P	Fast Fourier Transform (Cooley–Tukey algorithm; used here via <code>numpy.fft.rfft</code>)	current
EQ-015/P.95.v1	P	Periodogram / power spectral density estimation (one-sided PSD scaling used here follows the periodogram lineage; Hann window per J. von Hann, popularized by Blackman & Tukey)	current
EQ-015/P.96.v1	P	Total harmonic distortion (THD, ratio of harmonic to fundamental amplitude)	current
EQ-015/P.97.v1	P	Fokker–Planck equation (and its overdamped/Smoluchowski stationary limit $p(x) \propto \exp(-V(x)/D)$)	current
EQ-015/P.98.v1	P	Boltzmann distribution $p \propto \exp(-E/k_B T)$	current
EQ-015/P.99.v1	P	Hydrogen fine-structure energy shift $dE_{fs} = -(Ry \alpha^2/n^4)(n/(j+1/2) - 3/4)$ (relativistic kinetic + spin-orbit, combined first-order result)	current
EQ-015/S.01.v1	S	$R := (T_R, I_R); M(t'+\Delta t') = T_R(M(t'))$; $A(t'+\Delta t') \subseteq I_R(A(t'), M(t'))$; $Etic(A;t') := \exists R_A(t') \in R_{adm}(t'): A \subseteq M(t'), M(t'+\Delta t') = T_{\{R_A\}}(M(t'))$, $A(t'+\Delta t') \subseteq I_{\{R_A\}}(A(t'), M(t'))$	split
EQ-015/S.02.v1	S	$Ethical(A) \Leftrightarrow \exists Choice(A \rightarrow R_A): d/dt' V_{\{A, R_A\}}(M(t')) \leq 0 \wedge \tau_{c'}(R_A) > 0 \wedge \Delta_{spec}(R_A) > 0$ [author-labelled final, locked, will not be weakened or reinterpreted later]	split
EQ-015/S.03.v1	S	$Eth_{col}(G) \Leftrightarrow d/dt' V_G(M) \leq 0 \wedge \min_i \Delta_{spec}(R_i) > 0$; $Conf_{\{ind \rightarrow col\}} \Leftrightarrow d/dt' V_{\{A_i\}} \leq 0 \wedge d/dt' V_G > 0$; $Conf_{\{col \rightarrow ind\}} \Leftrightarrow d/dt' V_G \leq 0 \wedge \exists i: d/dt' V_{\{A_i\}} > 0$; $\Delta_{spec}(R_i \cup R_j) \leq 0$ [regime incompatibility]; structural injustice $\Leftrightarrow \exists i, j: C_{\{A_i, R_i\}} \gg C_{\{A_j, R_j\}}$	split

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EQ-015/S.04.v1	S	objective possibility $\neq$ structural feasibility; feasibility $\neq$ live possibility; live possibility $\neq$ stated preference; stated preference $\neq$ free consent; choice $\neq$ enactment; enactment $\neq$ evaluator-visible agency; option count $\neq$ effective agency; accessibility $\neq$ warrant; resonance $\neq$ consent; repetition $\neq$ legitimacy; retention $\neq$ improvement; norm conformity $\neq$ absence of agency; resistance to norms $\neq$ agency by definition; meaning-shaping $\neq$ domination; live-field narrowing $\neq$ structural violence without causal and normative conditions; AI influence $\neq$ manipulation by definition; coupled performance $\neq$ Human Return	split
EQ-015/S.05.v1	S	P1 objective $\neq$ practical possibility; P2 practical possibility is meaning-shaped; P3 meaning-shaping can alter agency, prior to preference; P4 choice can be real within a narrowed field; P5 expansion of the live field can also be harmful	not_an_equation
EQ-015/S.06.v1	S	$p^*_{\{A,g\}}(h,z;T,B,P) = \max_{\{\pi \in \Pi^{\wedge}\{\text{wit}\}}_{A(g;h,z,T,B)}} \Pr^{\wedge} \pi_{\{P(\text{Read}_g \cap D_g \cap X_g \cap F_g)\}}$, with $p^*_{\{A,g\}} := 0$ when the set is empty; $p^*_A = (p^*_{\{A,g\}})_{\{g \in G\}}$, $C^{\wedge} \alpha_A = \{g : p^*_{\{A,g\}} \geq \alpha\}$; $A^{\wedge}\{\text{corr}\}_A(h,z;T,B,P,w) := \Sigma_g w_g p^*_{\{A,g\}} \in [0,1]$; chain-rule $p_{\{A,g\}}(\pi) = r \cdot d \cdot x \cdot f$	split
EQ-015/S.07.v1	S	$p^*(2)_{\{A,g\}} := \max_{\{z \in J_{\text{feas}}\}} p^*_{\{A,g\}}(h,z;T,B,P)$; $L^{\wedge}\{\text{recoverable}\}_A = \max_{\{z \in J_{\text{feas}}\}} A^{\wedge}\{\text{corr}\}_A(h,z) - A^{\wedge}\{\text{corr}\}_A(h,z_0)$	split
EQ-015/S.08.v1	S	$L^{\wedge}\{\text{live}\}_{\{A,g\}} = \max_{\{z \in J_{\text{feas}}\}} D_L(L^{\wedge} z_A(g), L^{\wedge}\{z_0\}_A(g))$	current
EQ-015/S.09.v1	S	$\Delta \text{spec}(\text{Ri}) > 0 \iff \text{channel}_i = \text{open} \wedge \dot{\text{R}}_i \neq 0$	current
EQ-015/S.10.v1	S	$Y_{\text{obs}} = \text{calm} \wedge p^*_{\text{wit}} = \text{low} \wedge F = \text{blocked}$	split
EQ-015/S.11.v1	S	D1 Potential $\neq$ exercised $\neq$ observed; D2 Declared set $\neq$ witnessed set; D3 Layer1 $\neq$ layer2; D4 Task potential $\neq$ aggregate potential; D5 Feasible $\neq$ permitted; D6 Diagnosis of compression $\neq$ attribution of responsibility; D7 Recoverable gap $\neq$ accumulated loss	split
EQ-015/S.12.v1	S	P-B channel shift (load displaces between coupled tasks while the aggregate stays flat); P-C the excluded path (hidden cost concentration inside a rising mean); P-D layer-2 dominates the structural case	not_an_equation
EQ-015/S.13.v1	S	$R := (T_R, I_R)$	current
EQ-015/S.14.v1	S	$M(t' + \Delta t') = T_R(M(t'))$	current
EQ-015/S.15.v1	S	$A(t' + \Delta t') \subseteq I_R(A(t'), M(t'))$	current
EQ-015/S.16.v1	S	$\text{Etic}(A; t') := \exists R_A(t') \in R_{\text{adm}}(t') : A \subseteq M(t'), M(t' + \Delta t') = T_{\{R_A\}}(M(t')), A(t' + \Delta t') \subseteq I_{\{R_A\}}(A(t'), M(t'))$	current

Code	Dom.	Statement	Status
EQ-015/S.17.v1	S	$V_{\{A,R\}}(M) \geq 0$	current
EQ-015/S.18.v1	S	$\tau_{c'}(R) > 0$	current
EQ-015/S.19.v1	S	$\Delta_{\text{spec}}(R) > 0$	current
EQ-015/S.20.v1	S	$\text{Ethical}(A) \Leftrightarrow \exists \text{Choice}(A \rightarrow R_A) : d/dt' V_{\{A,R_A\}}(M(t')) \leq 0 \wedge \tau_{c'}(R_A) > 0 \wedge \Delta_{\text{spec}}(R_A) > 0$	current
EQ-015/S.21.v1	S	$V^+ := \max(dV/dt', 0)$	current
EQ-015/S.22.v1	S	$\text{Eth_ind}(A) \Leftrightarrow d/dt' V_{\{A,R_A\}} \leq 0 \wedge \Delta_{\text{spec}}(R_A) > 0$	current
EQ-015/S.23.v1	S	$M(t') = M_0 \phi_0 + \sum_k a_k(t') \phi_k$	current
EQ-015/S.24.v1	S	$S_A(R) \subseteq \text{span}\{\phi_k\}$	current
EQ-015/S.25.v1	S	$P_{\{S_A(R)\}} : M \rightarrow S_A(R)$	current
EQ-015/S.26.v1	S	$\limsup_{\{n \rightarrow \infty\}} \ P_{\{S_A(R_A)\}} M(t'+n)\ ^2 > 0$	current
EQ-015/S.27.v1	S	$\lim_{\{n \rightarrow \infty\}} \ P_{\{S_A(R_A)\}} M(t'+n)\ ^2 = 0$	current
EQ-015/S.28.v1	S	$V_G(M) := \sum_{i=1}^N w_i V_{\{A_i,R_i\}}(M), w_i > 0$	current
EQ-015/S.29.v1	S	$\text{Eth_col}(G) \Leftrightarrow d/dt' V_G(M) \leq 0 \wedge \min_i \Delta_{\text{spec}}(R_i) > 0$	current
EQ-015/S.30.v1	S	$\text{Conf}_{\{\text{ind} \rightarrow \text{col}\}} \Leftrightarrow d/dt' V_{\{A_i\}} \leq 0 \wedge d/dt' V_G > 0$	current
EQ-015/S.31.v1	S	$\text{Conf}_{\{\text{col} \rightarrow \text{ind}\}} \Leftrightarrow d/dt' V_G \leq 0 \wedge \exists i: d/dt' V_{\{A_i\}} > 0$	current
EQ-015/S.32.v1	S	$\Delta_{\text{spec}}(R_i \cup R_j) \leq 0$	current
EQ-015/S.33.v1	S	$\exists i,j : C_{\{A_i,R_i\}} \gg C_{\{A_j,R_j\}}$	current
EQ-015/S.34.v1	S	objective possibility $\neq$ structural feasibility	current
EQ-015/S.35.v1	S	feasibility $\neq$ live possibility	current
EQ-015/S.36.v1	S	live possibility $\neq$ stated preference	current
EQ-015/S.37.v1	S	stated preference $\neq$ free consent	current
EQ-015/S.38.v1	S	choice $\neq$ enactment	current
EQ-015/S.39.v1	S	enactment $\neq$ evaluator-visible agency	current
EQ-015/S.40.v1	S	option count $\neq$ effective agency	current
EQ-015/S.41.v1	S	accessibility $\neq$ warrant	current
EQ-015/S.42.v1	S	resonance $\neq$ consent	current
EQ-015/S.43.v1	S	repetition $\neq$ legitimacy	current
EQ-015/S.44.v1	S	retention $\neq$ improvement	current
EQ-015/S.45.v1	S	norm conformity $\neq$ absence of agency	current
EQ-015/S.46.v1	S	resistance to norms $\neq$ agency by definition	current
EQ-015/S.47.v1	S	meaning-shaping $\neq$ domination	current
EQ-015/S.48.v1	S	live-field narrowing $\neq$ structural violence without causal and normative conditions	current
EQ-015/S.49.v1	S	AI influence $\neq$ manipulation by definition	current
EQ-015/S.50.v1	S	coupled performance $\neq$ Human Return	current
EQ-015/S.51.v1	S	$p^*_{\{A,g\}} = \max_{\{\pi \in \Pi_{\text{feas_A}}(g)\}} \Pr(R_g \cap D_g \cap X_g \cap F_g)$	current
EQ-015/S.52.v1	S	$p^*_{\{A,g\}}(h,z;T,B,P) := \max_{\{\pi \in \Pi_{\text{wit_A}}(g;h,z,T,B)\}} \Pr^{\wedge \pi} P(R_g \cap D_g \cap X_g \cap F_g), \text{ with } p^*_{\{A,g\}} := 0 \text{ when the set is empty}$	current
EQ-015/S.53.v1	S	$p^*_A = (p^*_{\{A,g\}})_{g \in G}, C^{\wedge \alpha}_A = \{g : p^*_{\{A,g\}} \geq \alpha\}$	current
EQ-015/S.54.v1	S	$A^{\wedge \text{corr_A}}(h,z;T,B,P,w) := \sum_g w_g p^*_{\{A,g\}} \in [0,1]$	current

Code	Dom.	Statement	Status
EQ-015/S.55.v1	S	$p_{\{A,g\}}(\pi) = r \cdot d \cdot x \cdot f$, with $r = \Pr(Rg)$, $d = \Pr(Dg Rg)$, $x = \Pr(Xg Rg,Dg)$, $f = \Pr(Fg Rg,Dg,Xg)$	current
EQ-015/S.56.v1	S	$p^*(2)_{\{A,g\}} := \max_{\{z \in Jfeas\}} p^*_{\{A,g\}}(h,z;T,B,P)$	current
EQ-015/S.57.v1	S	$L^{\wedge}_{recoverable_A} = \max_{\{z \in Jfeas\}} A^{\wedge}_{corr_A}(h,z) - A^{\wedge}_{corr_A}(h,z0)$	current
EQ-015/S.58.v1	S	$Yobs = calm \wedge p^*wit = low \wedge F = blocked$	current
EQ-015/S.59.v1	S	Potential $\neq$ exercised $\neq$ observed	current
EQ-015/S.60.v1	S	Declared set $\neq$ witnessed set	current
EQ-015/S.61.v1	S	Layer 1 $\neq$ layer 2	current
EQ-015/S.62.v1	S	Task potential $\neq$ aggregate potential	current
EQ-015/S.63.v1	S	Feasible $\neq$ permitted	current
EQ-015/S.64.v1	S	Diagnosis of compression $\neq$ attribution of responsibility	current
EQ-015/S.65.v1	S	Recoverable gap $\neq$ accumulated loss	current
EQ-015/W.01.v1	W	human labour $\rightarrow$ production $\rightarrow$ wage $\rightarrow$ claim on output; labour-based claim $\rightarrow$ human/citizen claim on social production	not_an_equation
EQ-015/W.02.v1	W	$\dot{Z}_t = v_t G_t - \delta_Z Z_t$, $0 \leq v_t \leq 1$	current
EQ-015/W.03.v1	W	$B^{\wedge}RB_t = N^{\wedge}RB_t q^{\wedge}RB_t$; split growth decomposition $B^{\wedge}RB/B^{\wedge}RB = N^{\wedge}RB/N^{\wedge}RB + q^{\wedge}RB/q^{\wedge}RB$; $M_t = (K^{\wedge}M_t)^{\kappa} (A^{\wedge}AI_t)^{\alpha} (B^{\wedge}RB_t)^{\beta}$; $\rho = (\sigma - 1)/\sigma$; $s^{\wedge}L_t = (\omega_H H_t^{\rho})/(\omega_H H_t^{\rho} + \omega_M M_t^{\rho})$	split
EQ-015/W.04.v1	W	$L_t = \langle L^{\wedge}task_t, L^{\wedge}income_t, L^{\wedge}bottleneck_t, L^{\wedge}bargain_t \rangle$	split
EQ-015/W.05.v1	W	$q_t = o_t + \tau_t(1 - o_t)$; split $\Gamma_t = s^{\wedge}L_t + q_t(1 - s^{\wedge}L_t)$; $q^{\wedge}_{min}_t = (\Gamma_t - s^{\wedge}L_t)/(1 - s^{\wedge}L_t)$; $D^{\wedge}rent$, $\Gamma^{\wedge}_{net}$	split
EQ-015/W.06.v1	W	$AD_t = C(\Gamma^{\wedge}eff_t Y_t, m_t) + I_t + G_t + NX_t$; split $\chi^{\wedge}_{dem}_t = \min\{1, AD_t/Y_t\}$; $\Pi^{\wedge}M_t = \chi^{\wedge}_{dem}_t Y_t - Cost^{\wedge}M_t$; $\Gamma^{\wedge}eff Y \rightarrow AD \rightarrow \Pi^{\wedge}M \rightarrow W^{\wedge}M_{\{t+1\}} \rightarrow M_{\{t+1\}}$	split
EQ-015/W.07.v1	W	$W^{\wedge}M_{\{i,t+1\}} = (1 - \delta_W) W^{\wedge}M_{\{i,t\}} + r^{\wedge}M_t W^{\wedge}M_{\{i,t\}} + s^{\wedge}\{M, cap\}_{\{i,t\}} + T^{\wedge}cap_{\{i,t\}}$; $o_t = \Sigma_{\{i \in B\}} W^{\wedge}M_{\{i,t\}} / \Sigma_i W^{\wedge}M_{\{i,t\}}$; current redistribution $\neq$ future ownership reproduction	split
EQ-015/W.08.v1	W	$G^{\wedge}_{conv}_{\{j,t\}}(e) = 1 - [V_t(e j)/V_t(e)]_+$; split $X_{\{i,t\}}(e;j) = \text{clip}(V_{\{i,t\}}(e j)/V_{\{i,t\}}(e), 0, 1)$; $D_{\{i \rightarrow j,t\}}(g) = \Sigma_e w_e(g) G^{\wedge}_{conv}_{\{j,t\}}(e) [1 - X_{\{i,t\}}(e;j)]$; Concentration $\neq$ $G^{\wedge}_{conv} \neq D$; concentration $\neq$ dependency $\neq$ agency loss (conditional)	split
EQ-015/W.09.v1	W	$C_{\{i,t\}} = \langle O_{\{i,t\}}, G_{\{i,t\}}, \Gamma_{\{i,t\}}, A^{\wedge}_{access}_{\{i,t\}}, X_{\{i,t\}}, D_{\{i,t\}}, R^{\wedge}_{rent}_{\{i,t\}} \rangle$	current
EQ-015/W.10.v1	W	$P^{\wedge}H_t = (\Gamma^{\wedge}eff_t)^{\theta} \Gamma(A^{\wedge}_{corr}_{\{H,t\}})^{\theta} A(A^{\wedge}_{live}_{\{H,t\}})^{\theta} \Lambda(r^{\wedge}H_t)^{\theta} R(S^{\wedge}H_t)^{\theta} \dot{Y}_t > 0 \Rightarrow \dot{P}^{\wedge}H_t > 0$	split
EQ-015/W.11.v1	W	$\dot{S}^{\wedge}H_t = s_1 W_t + s_2 N_t - \delta_{SS} S^{\wedge}H_t$; split $1 = l_{wage} + l_{care} + l_{learn} + l_{civic} + l_{leisure}$	split

Code	Dom.	Statement	Status
EQ-015/W.12.v1	W	$\dot{H}^{cap}_t = f(\text{Care}_t, \text{Health}_t, \text{Education}_t, \text{Nutrition}_t, \text{Community}_t) - \delta_{HH} \hat{H}^{cap}_t$ productive necessity of humans $\neq$ so- cial necessity of human reproduction;	split
EQ-015/W.13.v1	W	$\{\text{Care}, \text{Health}, \text{Education}\} \rightarrow H^{cap} \rightarrow \{A^{corr}_{H,r_H,S_H}\} \rightarrow P \rightarrow I_{\{t+1\}}$	
EQ-015/W.14.v1	W	$\Omega_t = w_M(g_M - g_H) + w_{Dg} D + w_{Gg} G - w_{eq} q - w_{\Gamma g} (\Gamma^{eff}) - w_X Xg$ $A^{corr}_{\{H,i,t\}}(g) = \text{current}$ $\max_{\{\pi \in \Pi^{live}_{\{i,t\}}(g)\}}$ $\Pr \pi(R_g \cap D_g \cap X_g \cap F_g)$	
EQ-015/W.15.v1	W	$R^{return}_{\{H,t\}} = \langle C_t, T_t, S^{skill}_t, A^{alt}_t \rangle$	current
EQ-015/W.16.v1	W	Machine expansion $\nRightarrow$ Human expansion; split $C^H_t = \langle \Gamma^{eff}_t, X^H_t, \Lambda^{live}_{\{H,t\}}, A^{corr}_{\{H,t\}}, R^{route}_{\{H,t\}}, W^{world}_{\{H,t\}} \rangle$ $\eta^{HC}_{\{j,t\}} = \partial \ln C^H_{\{j,t\}} / \partial \ln M_t$	
EQ-015/W.17.v1	W	$H \text{ problem} \rightarrow AI \text{ divergence} \rightarrow H \text{ resistance} \rightarrow \text{World test} \rightarrow H \text{ integration} \rightarrow AI \text{ removal} \rightarrow H \text{ return (high-conversion); } H \text{ problem} \rightarrow AI \text{ answer} \rightarrow \text{use} \rightarrow \text{dependence (low-conversion); Maximum AI assistance} \rightarrow \text{Maximum durable human conversion}$	not_an_equation
EQ-015/W.18.v1	W	$B_t = \langle D_t, \dot{G}^{conv}_t, C^{info}_t, 1 - X^H_t, 1 - r_{\{H,t\}}, 1 - \Lambda^{live}_{\{H,t\}}, 1 - A^{corr}_{\{H,t\}}, 1 - \Gamma^{eff}_t, 1 - H^{cap}_t \rangle$	current
EQ-015/W.19.v1	W	$W_j = \{t: C^{rec}_{\{j,t\}} \leq C^-_j \wedge \tau^{rec}_{\{j,t\}} \leq \tau^-_j\};$ $\Delta g_{\{j,t\}} = [g^-_{\{M,t\}} - g^-_{\{Cj,t\}}]_- +;$ $U_{\{j,t\}} = \Delta g_{\{j,t\}} L_{\{j,t\}} S_{\{j,t\}} \tau^{rec}_{\{j,t\}};$ $U_t = \langle U_{\{1,t\}}, \dots, U_{\{J,t\}} \rangle;$ $U^{max}_t = \max_j U_{\{j,t\}}$	split
EQ-015/W.20.v1	W	$B^{RB}_t = N^{RB}_t q^{RB}_t$	current
EQ-015/W.21.v1	W	$B_{dot}^{RB}_t / B^{RB}_t = N_{dot}^{RB}_t / N^{RB}_t + q_{dot}^{RB}_t / q^{RB}_t$	current
EQ-015/W.22.v1	W	$M_t = (K^{M_t})^{\kappa} (A^{AI}_t)^{\alpha} (B^{RB}_t)^{\beta}$	current
EQ-015/W.23.v1	W	$\rho = (\sigma - 1) / \sigma$	current
EQ-015/W.24.v1	W	$s^L_t = (\omega_H H_t^{\rho}) / (\omega_H H_t^{\rho} + \omega_M M_t^{\rho})$	current
EQ-015/W.25.v1	W	$L_t = \langle L^{task}_t, L^{income}_t, L^{bottleneck}_t, L^{bargain}_t \rangle$	current
EQ-015/W.26.v1	W	$q_t = o_t + \tau_t(1 - o_t) = 1 - (1 - o_t)(1 - \tau_t)$	current
EQ-015/W.27.v1	W	$\Gamma_t = s^L_t + q_t(1 - s^L_t)$	current
EQ-015/W.28.v1	W	$q^{min}_t = (\Gamma_{bar} - s^L_t) / (1 - s^L_t)$	current
EQ-015/W.29.v1	W	$D^{rent}_{(i,t)} = (Rent^{(AI,out)}_{(i,t)} - Rent^{(AI,in)}_{(i,t)}) / Y_{(i,t)}$	current
EQ-015/W.30.v1	W	$\Gamma^{net}_{(i,t)} = \Gamma_{(i,t)} - D^{rent}_{(i,t)}$	current
EQ-015/W.31.v1	W	$AD_t = C(\Gamma^{eff}_t Y_t, m_t) + I_t + G_t + NX_t$	current
EQ-015/W.32.v1	W	$\chi^{dem}_t = \min\{1, AD_t / Y_t\}$	current
EQ-015/W.33.v1	W	$\Pi^M_t = \chi^{dem}_t Y_t - Cost^M_t$	current
EQ-015/W.34.v1	W	$W^M_{(i,t+1)} = (1 - \delta_W) W^M_{(i,t)} + r^M_t W^M_{(i,t)} + s^{(M,cap)}_{(i,t)} + T^{cap}_{(i,t)} - Tax^{cap}_{(i,t)}$	current

Code	Dom.	Statement	Status
EQ-015/W.35.v1	W	$o_t = (\sum_{(i \text{ in } B)} W^M_{(i,t)}) / (\sum_i W^M_{(i,t)})$	current
EQ-015/W.36.v1	W	current redistribution != future ownership re- production	current
EQ-015/W.37.v1	W	$G^{conv}_{(j,t)}(e) = 1 - [V_t(e \mid -j) / V_t(e)]_{_+}$	current
EQ-015/W.38.v1	W	$X_{(i,t)}(e; \ j) = \text{clip}(V_{(i,t)}(e \mid -j) / V_{(i,t)}(e), 0, 1)$	current
EQ-015/W.39.v1	W	$D_{(i \rightarrow j,t)}(g) = \sum_{(e \text{ in } E(g))} w_e(g) G^{conv}_{(j,t)}(e) [1 - X_{(i,t)}(e; j)]$	current
EQ-015/W.40.v1	W	Concentration != G^{conv} != D	current
EQ-015/W.41.v1	W	concentration $\Rightarrow$ dependency $\Rightarrow$ agency loss	current
EQ-015/W.42.v1	W	$P^H_t = [(\Gamma^{eff_t})^{\theta_Gamma} (A^{corr_}(H,t))^{\theta_A} (\Lambda^{live_}(H,t))^{\theta_Lambda} (r^H_t)^{\theta_R} (S^H_t)^{\theta_S} (X^H_t)^{\theta_X}] / (1 + D^H_t)^{\theta_D}$	current
EQ-015/W.43.v1	W	$P_dot^H_t / P^H_t = \theta_Gamma (\Gamma^{dot_eff_t} / \Gamma^{eff_t}) + \theta_A (A^{dot_corr_}(H,t) / A^{corr_}(H,t)) + \theta_R (r_dot^H_t / r^H_t) + \theta_Lambda (\Lambda^{dot_live_}(H,t) / \Lambda^{live_}(H,t))$	current
EQ-015/W.44.v1	W	$\dots + \theta_S (S_dot^H_t / S^H_t) + \theta_X (X_dot^H_t / X^H_t) - \theta_D (D_dot^H_t / (1 + D^H_t))$	current
EQ-015/W.45.v1	W	$Y_dot_t > 0 \Rightarrow P_dot^H_t > 0$	current
EQ-015/W.46.v1	W	$AWA_t = < Y(\text{up}), \Gamma^{eff} \text{adequate}, O^H(\text{down}), X^H(\text{down}), D^H(\text{up}), \Lambda^{live_H}(\text{down}), A^{corr_H}(\text{down}), r_H(\text{down}), S_H(\text{down}) >$	current
EQ-015/W.47.v1	W	$S_dot^H_t = s_1 W_t + s_2 N_t - \delta_S S^H_t$	current
EQ-015/W.48.v1	W	$1 = l_wage + l_care + l_learn + l_civic + l_leisure$	current
EQ-015/W.49.v1	W	$H_dot^{cap_t} = f(\text{Care_t}, \text{Health_t}, \text{Education_t}, \text{Nutrition_t}, \text{Community_t}) - \delta_H H^{cap_t}$	current
EQ-015/W.50.v1	W	productive necessity of humans != social necessity of human reproduction	current
EQ-015/W.51.v1	W	Machine expansion $\Rightarrow$ Human expansion	current
EQ-015/W.52.v1	W	$C^H_t = < \Gamma^{eff_t}, X^H_t, \Lambda^{live_}(H,t), A^{corr_}(H,t), R^{route_}(H,t), W^{world_}(H,t), r_H(t), H^{cap_t}, S_H(t) >$	current
EQ-015/W.53.v1	W	$\eta^{HC}_{(j,t)} = d(\ln C^H_{(j,t)}) / d(\ln M_t)$	current
EQ-015/W.54.v1	W	$\eta^{HC_Gamma} > 0, \eta^{HC_rH} < 0, \eta^{HC_X} < 0$	current
EQ-015/W.55.v1	W	$W_j = \{ t : C^{rec}_{(j,t)} \leq C_bar_j \text{ AND } \tau^{rec}_{(j,t)} \leq \tau_bar_j \}$	current
EQ-015/W.56.v1	W	$\Delta g_{(j,t)} = [g_{_}(M,t) - g_{_}(Cj,t)]_{_+}$	current
EQ-015/W.57.v1	W	$U_{(j,t)} = \Delta g_{(j,t)} * L_{(j,t)} * S_{(j,t)} * \tau^{rec}_{(j,t)}$	current
EQ-015/W.58.v1	W	$U_t = < U_{(1,t)}, \dots, U_{(J,t)} >$	current

Code	Dom.	Statement	Status
EQ-015/W.59.v1	W	$\hat{U}_{\max_t} = \max_j U_{_}(j,t)$	current

EQ-016 — Regime-split discriminant

tier: Dr; status: current

$$\lambda_c = D^2/(4MK)$$

EQ-017 — Telegraph/relaxation reduced form

tier: finite_diagnostic; status: current

$$\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R$$

EQ-018 — DRL two-field apparatus state tuple

tier: Dr; status: current

$$\mathfrak{S}_n = (G_n, \Lambda_n, \mathcal{T}_n, \Theta_n)$$

EQ-019 — Reader/record stacked vector

tier: Dr; status: current

$$X_n^A = (\Phi_n^I, \Psi_n^I)^T$$

EQ-020 — Coupled transport operator $\mathbb{G}_n$

tier: Dr; status: current

$$\mathbb{G}_n = L_{\{G_n\}} \otimes I_{\mathcal{F}} + I_{\{G_n\}} \otimes C_{\mathcal{F}} + C_{\text{int},n}$$

EQ-021 — Metric-G skew/symmetric decomposition of $\mathbb{G}_n$

tier: Th_cogc; status: current

$$\mathbb{G}_n^{(+)} = (\mathbb{G}_n + \mathbb{G}_n^{\text{adj}G})/2, \mathbb{G}_n^{(-)} = (\mathbb{G}_n - \mathbb{G}_n^{\text{adj}G})/2 \text{ [retention-metric } G\text{-adjoint, not naive transpose]}; \langle z, \mathbb{G}_n^{(-)} z \rangle_G = 0$$

EQ-022 — Discrete reader/record coupled recurrence

tier: finite_diagnostic; status: current

$$\text{Reader: } M \delta_t^2 \Phi_n + D \delta_t^c \Phi_n + K\mathbb{G}[\Theta_n] \Phi_n + \nabla V(\Phi_n) - J_n = \mathcal{R}_{\Phi,n}; \text{Record: } M \delta_t^2 \Psi_n - D \delta_t^c \Psi_n + K\mathbb{G}[\Theta_n]^T \Psi_n + \nabla^2 V(\Phi_n) \Psi_n = \mathcal{R}_{\Psi,n}$$

EQ-023 — Potential gradient/Hessian (mother potential)

tier: finite_diagnostic; status: current

$$\nabla V_i(\Phi) = a_i \Phi_i + b_i \Phi_i^3; (\nabla^2 V)_{ii} = a_i + 3b_i \Phi_i^2$$

EQ-024 — Reader one-step implicit solve

tier: finite_diagnostic; status: current

$$A_{\Phi} = M/\Delta t^2 + D/2\Delta t; b_{\Phi,n} = \mathcal{R}_{\Phi,n} + J_n - K\mathbb{G}_n \Phi_n - \nabla V(\Phi_n) + (2M/\Delta t^2) \Phi_n + (D/2\Delta t - M/\Delta t^2) \Phi_{\{n-1\}}; A_{\Phi} \Phi_{\{n+1\}} = b_{\Phi,n}$$

EQ-025 — Record one-step implicit solve

tier: finite_diagnostic; status: current

$$A_{\Psi} = M/\Delta t^2 - D/2\Delta t; b_{\Psi,n} = \mathcal{R}_{\Psi,n} - K\mathbb{G}_n^T \Psi_n - \nabla^2 V(\Phi_n) \Psi_n + (2M/\Delta t^2) \Psi_n - (M/\Delta t^2 + D/2\Delta t) \Psi_{\{n-1\}}; A_{\Psi} \Psi_{\{n+1\}} = b_{\Psi,n}$$

EQ-026 — Record/readout/epistemic composite readout

tier: Th_coqc; status: current

$$R_0 = \Gamma_{\{RAR, 0\}}(D_0) = \Omega_A \circ A \circ \Pi \circ T_{\Sigma}(D_0)$$

EQ-027 — N2 — Readout equation

tier: Th_coqc; status: current

$$M_A[n] = K_A \cdot \theta(E[n]) + \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}}$$

EQ-028 — N2 epistemic-error non-collapse law

tier: Th_coqc; status: current

$$M_A[n] \neq \theta(E) \quad \forall n; \quad \epsilon_{\text{tot}} = \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}} > 0 \quad \forall n$$

EQ-029 — bR cross-domain lineage ledger conservation

tier: Dr; status: current

$$I_Q = I_B + O_C + O_P + O_B$$

EQ-030 — Standard Model domain state tuple

tier: Dr; status: current

$$Z_n = (\Phi_n, \Psi_n, \Theta_n, U_n, \Sigma_n, \mathcal{T}_n, \Lambda_n)$$

EQ-031 — Gauge-equivalence readout

tier: Dr; status: current

$$r = 0(X); \quad 0(hX) = 0(X) \quad \forall X \implies h \text{ internal rename}; \quad X \sim X' \iff 0(X) = 0(X'); \quad \text{physical state} = [X]$$

EQ-032 — No-decoder-recovers-state theorem

tier: Th_coqc; status: current

$$\forall X, R, 0, x_1, x_2 : \quad x_1 \neq x_2 \rightarrow 0(x_1) = 0(x_2) \rightarrow \forall D : D(0(x_1)) = x_1 \rightarrow D(0(x_2)) = x_2 \rightarrow \text{False}$$

EQ-033 — Gauge redundancy forces undecodability

tier: Th_coqc; status: current

$$\forall X, R, 0, h, x : \quad h(x) \neq x \rightarrow 0(h(x)) = 0(x) \rightarrow \forall D : D(0(h(x))) = h(x) \rightarrow D(0(x)) = x \rightarrow \text{False}$$

EQ-034 — True state exists but no total decoder

tier: Th_coqc; status: current

$$x_1 \neq x_2 \rightarrow 0(x_1) = 0(x_2) \rightarrow (\exists x : X, x = x_1) \wedge (\exists x : X, x = x_2) \wedge \neg(\exists D : R \rightarrow X, D(0(x_1)) = x_1 \wedge D(0(x_2)) = x_2)$$

EQ-035 — Retained (metric-G) inner product

tier: Dr; status: current

$$\langle x, y \rangle_G = x^\dagger G y, \quad G > 0; \quad \|x\|_G^2 = x^\dagger G x \geq 0$$

EQ-036 — Antisymmetry factor r=-1 (SU(3) closure step 1)

tier: Th_coqc; status: current

$$\Omega(\dots x_i, x_{\{i+1\}} \dots) = r \cdot \Omega(\dots x_{\{i+1\}}, x_i \dots); \quad r^2 = 1 \implies r = \pm 1; \quad r = -1$$

EQ-037 — Minimal odd arity k=3 (SU(3) closure step 2)

tier: Th_coqc; status: current

$$\Omega(x_1 \dots x_k) = \Omega(x_2 \dots x_k, x_1); \quad (-1)^{\wedge\{k-1\}} = 1 \implies k \text{ odd}; \quad k = 3$$

EQ-038 — Minimal dimension 3 (SU(3) closure step 3)*tier: Th_coqc; status: current*

$$\Omega(x,y,z) = 0 \text{ if } \dim V < 3; \dim V \geq 3; \dim V = 3, V \cong \mathbb{C}^3$$

EQ-039 — Unitary-group closure (SU(3) closure step 4)*tier: Th_coqc; status: current*

$$U^\dagger G U = G, G=I \implies U^\dagger U=I \implies U \in U(3)$$

EQ-040 — Determinant-one selection giving SU(3) (SU(3) closure step 5)*tier: Th_coqc; status: current*

$$\Omega(Ux, Uy, Uz) = \det(U) \cdot \Omega(x, y, z); \det U = 1; SU(3) = \{U \in U(3) : \det U = 1\}$$

EQ-041 — su(3) dimension count = 8 (SU(3) closure step 6)*tier: Th_coqc; status: current*

$$9 - 1 = 8 = \dim \mathfrak{su}(3)$$

EQ-042 — Admissible gauge-automorphism group and transition relabeling*tier: Dr; status: current*

$$\mathcal{A} = \{ h : 0h=0, hF=Fh, h^\dagger Gh=G \}; U'_{\{j \leftarrow i\}} = h_j U_{\{j \leftarrow i\}} h_i^{-1}$$

EQ-043 — Loop holonomy and curvature defect*tier: Dr; status: current*

$$H_C = \prod_{e \in C} U_e; H_{C'} = h H_C h^{-1}; K_C = H_C - I$$

EQ-044 — Unified force residual decomposition*tier: Dr; status: current*

$$\mathcal{F}_{\text{all}} = \mathcal{F}_G + \mathcal{F}_{\text{EM}} + \mathcal{F}_W + \mathcal{F}_S + \mathcal{F}_{\text{res}}; \chi^4 = 1 - \|\mathcal{F}_{\text{res}}\|^2 / \|\mathcal{F}_{\text{all}}\|^2$$

EQ-045 — Blind-pipeline gauge-algebra dimension result*tier: finite_diagnostic; status: current*

$$\dim Z(\mathfrak{g}) = 1; \mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3); \dim = 1+3+8 = 12$$

EQ-046 — U(1) phase action*tier: Dr; status: current*

$$x \mapsto e^{i\alpha} x$$

EQ-047 — General representation action and invariant tensor*tier: Dr; status: current*

$$x \mapsto \rho_R(\mathfrak{g})x; C_2(R); \mathcal{I} : R_1 \otimes \dots \otimes R_n \rightarrow 1$$

EQ-048 — Color-singlet tensor decompositions*tier: Dr; status: current*

$$3 \otimes 3^- \rightarrow 1 + \dots; 3 \otimes 3 \otimes 3 \rightarrow 1 + \dots \text{ via } \epsilon_{\{abc\}} x^a y^b z^c$$

EQ-049 — Triality Z_3 labeling and closure*tier: Th_coqc; status: current*

$$P/Q \cong Z_3; \tau(p,q) = p+2q \pmod{3}; \tau(3)=1, \tau(3^-)=2, \tau(8)=0; \tau(R \otimes 8) = \tau(R); 1+2=0, 1+1+1=0$$

EQ-050 — Physical Hilbert subspace at zero constraint

tier: Dr; status: current
 $\mathcal{H}_{\text{physical}} \subseteq \mathcal{H}_{\{\tau=0\}}$

EQ-051 — $Z_3/2D$ confinement area law

tier: finite_diagnostic; status: current
 $z_e \in Z_3; u_p = \prod z_e^{\epsilon}; S = \kappa \Sigma |u_p - 1|^2; |\omega - 1|^2 = 3; r = e^{-3\kappa}; q(\kappa) = (1-r)/(1+2r); \langle W(C) \rangle = q^{\{A(C)\}}; \sigma = -\log q > 0$

EQ-052 — Retained-information vs surface-entropy inequality

tier: finite_diagnostic; status: current
 $N_A \lesssim \mu^A; \Sigma N_A \rho^A \lesssim \Sigma(\mu\rho)^A; \mu\rho < 1 \iff I_{\text{retain}} = -\log \rho > S_{\text{surface}} = \log \mu$

EQ-053 — Character-expansion confinement machinery

tier: Th_coqc; status: current
 $S_p(U) = \kappa \|U - I\|_F^2; \|U - I\|^2 = 6 - 2\text{Re Tr } U; K_\kappa(U) = \exp[\kappa(\chi_3 + \chi_3^-)] = \sum_R c_R(\kappa) \chi_R; c_R(\kappa) = \int_{\{SU(3)\}} K_\kappa \chi_R^*; dK/d\kappa = (\chi_3 + \chi_3^-)K; c_R'(\kappa) = \sum_S (N_{\{3S\}^R + N_{\{3^-S\}^R}) c_S; c_R(0) = \delta_{\{R0\}}; u = c_3/3c_0; v = c_8/8c_0$

EQ-054 — All-order character recursion identity

tier: Th_coqc; status: current
 $c_0'(\kappa) = 2c_3(\kappa)$ [source: InfoAllOrderCharacter.v / all_order_character_v1_0.py -- a different file than EQ-053's SM §11]

EQ-055 — Group-averaging projector idempotence/positivity

tier: untagged; status: current
 $P_e = \int_G \rho_e(h) d\mu(h); P^2 = P; P^\dagger_G = P; \|P_e T\|_G \leq \|T\|_G$

EQ-056 — Reader/record characteristic-root sign split

tier: finite_diagnostic; status: current
reader roots: $M s^2 + D s + K = 0 \implies \text{Re}(s) = -D/(2M)$; record roots: $M s^2 - D s + K = 0 \implies \text{Re}(s) = +D/(2M)$

EQ-057 — Rational growth ratio dissolving the I1 continuum-completeness injection

tier: finite_diagnostic; status: current
 $\rho(n) := r^n, r \in \mathbb{Q}, r > 1$

EQ-058 — Mass-ratio geometric-mean fit values (three distinct sectors)

tier: untagged; status: current
 $r_U = \sqrt{(587.963 \cdot 135.827)} \approx 282.597083; r_D = \sqrt{(20.00 \cdot 44.75)} \approx 29.917803; r_E = \sqrt{(206.85 \cdot 16.81)} \approx 58.970290$ (three distinct numbers, same formula, different inputs -- not an equality)

EQ-059 — Golden-ratio identity (proposed use rejected)

tier: Th_coqc; status: current
 $\phi = (1+\sqrt{5})/2, x^2 - x - 1 = 0$ (the identity itself is Th_coqc; proposed use as item-1's r is Open/REJECTED, not adopted)

EQ-060 — Regime discriminant Δ^2 and root behavior

tier: finite_diagnostic; status: current

$\Delta^2 = D^2 - 4MK$; $\lambda_c < 1$: $\text{Re}(s) = \mp D/2M$ (K-independent); $\lambda_c > 1$: $\text{Re}(s) = [D \pm \sqrt{(D^2 - 4MK)}]/2M$ (K-dependent)

EQ-061 — Step-reader/record implicit coefficients

tier: finite_diagnostic; status: current

step_reader: $A_\Phi = M/dt^2 + D/2dt$; step_record: $A_\Psi = M/dt^2 - D/2dt$

EQ-062 — Reader/record envelope negative finding

tier: finite_diagnostic; status: current

$|\Phi|$ envelope: $1.215839 \rightarrow 0.183275$; $|\Psi|$ envelope: $1.105998 \rightarrow 322.802505$; $\Phi \cdot \Psi$ (n=10) = -1.188008
 $\rightarrow \Phi \cdot \Psi$ (n=1999) = -59.738547 (ratio $\approx 50.28\times$) -- negative finding, refutes its own opening hypothesis

EQ-063 — Mass-as-readout-of- τ_c bridge

tier: finite_diagnostic; status: current

$\tau_c = \hbar/(2mc^2)$ [N4 SCALE BUS]; $m = \hbar/(2c^2\tau_c)$ [readout direction -- mass is what τ_c looks like].
Unit/calibration bridge only, not a derivation of EQ-015's M coefficient.

EQ-064 — M calibration via method-of-moments EIV/replicate-IV

tier: finite_diagnostic; status: current

$M_{\text{hat}} = (a_1^T y_2 + a_2^T y_1) / (2 a_1^T a_2)$ [replicate-IV, preferred]; $M_{\text{joint}} = 1.0004294772248$ (0.042948% error vs known $M_{\text{true}}=1$; 500-seed sweep mean error 0.335552% at $\sigma=1e-5$). Does not close EQ-015's M derivation -- calibration on a declared synthetic tape.

EQ-065 — Declared branch-tape Π_0 construction

tier: untagged; status: current

$\Pi_0 = 3\lambda_U + 3\lambda_D + \lambda_E = 6.328453553357985$ (0.0037% error vs the fixture's known-M reconstruction).
Branch initial conditions explicitly disclosed as arbitrary/uncalibrated; must not be conflated with the independently-derived real-PDG-mass $\Pi_0 \approx 6.9888$ elsewhere in the domain.

EQ-066 — Order-vacuum threshold closure

tier: untagged; status: current

$\Pi_0 > \alpha_{\text{ord}} \Rightarrow \text{ORDERED_READY}$, $r^* = 3.823356105009073$. Caveat: ORDERED_READY on this stepper is structurally guaranteed regardless of branch-tape data ($\lambda_j \in (0,1]$ forces $\Pi_0 \in (0,7]$, $\alpha_{\text{ord}} = -0.5$ sits below that bound) -- not evidence the branch construction is predictive.

EQ-067 — Native vacuum amplitude

tier: untagged; status: current

$v_{\text{native}} = \sqrt{(2 r^*)} = 2.7652689218262565$ (no physical unit attached)

EQ-068 — Three negative findings bridging v_{native} to GeV-scale physics

tier: finite_diagnostic; status: current

(a) fit $\Lambda_{RD \rightarrow \text{GeV}} = 246/v_{\text{native}} = 88.96060634765863$; independent Higgs-mass prediction 218.005 GeV vs real PDG 125.20 GeV -- 74.13% error, fails a 5% band. (b) zero-fitted-parameter ratio test -- identical 74.13% error. (c) zero-external-input internal consistency $m_{\text{from } \tau_c \text{ native}} := 1/(2 \cdot (M_{\text{joint}}/D))$ vs curvature-mass route -- ratio 0.0612, 94% deviation, not consistent. Open question left unresolved: whether this native-vs-GeV comparison is a valid readout-vs-readout comparison or a category error against a non-readout dimensional constant.

EQ-069 — RETRACTED — continuous- Θ graph-accumulation approach

tier: RETRACTED; status: historical

RETRACTED 2026-07-26 (continuum contamination): $\text{cond}\#(G[\Theta])$ is a smooth bijection $\Theta \leftrightarrow R$ -- a coordinate, not an observable -- so a single continuous knob cannot force generations. Number kept reserved so the stream stays stable; see domains/standard_model/item1_exploration/CONTINUUM_ARC_ERROR_NOTE.md.

EQ-070 — RETRACTED — continuous- Θ graph-accumulation approach (duplicate entry)

tier: RETRACTED; status: historical

RETRACTED 2026-07-26, same content/reason as EQ-069, number reserved.

EQ-071 — RETRACTED — continuous- Θ graph-accumulation approach (duplicate entry)

tier: RETRACTED; status: historical

RETRACTED 2026-07-26, same content/reason as EQ-069/EQ-070, number reserved.

ElectroweakDecoder_v0.3 — Unified Force v0.3 — Calibrated Electroweak Decoder (EM $\leftrightarrow$ Weak)

tier: untagged; status: current

Structural core (exact): the neutral obstruction $M^2 = (v^2/4) \cdot \text{outer}((g, -g'), (g, -g'))$ is rank-1, so $\det M^2 = 0$ identically, giving exactly one massless + one massive direction -- the photon's masslessness. EMERGES: photon $\sim (g', g) = \sin(\theta_W) W^3 + \cos(\theta_W) B$ (eigenvalue 0), $Z \sim (g, -g')$ with $m_Z^2 = (v^2/4)(g^2 + g'^2)$, $m_W^2 = g^2 v^2/4$; failing control = a generic rank-2 obstruction has $\det \neq 0$, no massless direction. Calibration (CODATA2022+PDG2025): $v=246.21965$ GeV, $g'/g=0.535802$, $g=0.652824$, $g'=0.349784$; held-out tree-level $M_Z = M_W/\cos(\theta_W) = 91.178$ vs PDG 91.188 (0.011%, radiative layer OPEN).

EtaOmegaMatrices — DRL-Telegraph apparatus: pairing and orientation matrices

tier: Dr; status: current

$\eta = [[0, 1], [1, 0]]$ (symmetric pairing); $\omega = [[0, 1], [-1, 0]]$ (antisymmetric orientation)

F_ep — VI.3 — F_ep (obstruction depth)

tier: untagged; status: current

F_ep = obstruction depth (strength of best support); "the epistemic reading of N3: how far the best supporting readout is from $0 \rightarrow 0$ "

Face.1.CharEq — Face 1: per-mode characteristic equation for mode amplitude a_k

tier: untagged; status: current

$$M s^2 + D s + K \lambda_k = 0$$

Face.1.Decomposition — Face 1: Scalar-Eigenmode Reduction Error correction — symmetric/skew decomposition of L_R

tier: Dr; status: current

$L_R = L_R^+(+) + L_R^(-)$, a symmetric part and a skew part under the retention metric G [Dr, pending test T1, not yet proven]

Face.1.Eigenmode — Face 1: Eigenmode

tier: untagged; status: current

$L_R \phi_k = \lambda_k \phi_k$ -- Meaning: existing structure = retained eigenmode under admissible transport

Face.10.IdentifiabilityGate — Face 10: identifiability gate on the domain-discovery minimal exact law

tier: finite_diagnostic; status: current

$\text{rank}(A) = (\text{number of candidate coefficients}) \rightarrow \text{law is identifiable -- report it; rank}(A) < (\text{number of candidate coefficients}) \rightarrow \text{UNDERDETERMINED -- abstain}$

Code	Dom.	Statement	Status
Face.10.IdentifiabilityGate	De/E.01.Phi	$\text{FI}(q; A1, A2, C, \pi) = 1 - V^*(C \backslash \pi) / V^*(C)$	current

Face.10.StrictGap — Face 10: strict gap (readout-not-truth)

tier: Th_coqc; status: current

$M_A[n] \neq \theta(E) \forall n; \epsilon_{\text{tot}} = \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}} > 0 \forall n$. Every observer's record is lossy; this is not a defect, it is the definition of finite access. [Th_coqc] grounding added 2026-07-25 (formal/Info axiom-free, Print Assumptions $\Rightarrow$ Closed for all three lemmas): the true pre-record state continues to EXIST whenever two distinct true states are collapsed by a non-injective readout -- what fails to exist is a single decoder correct on both (no_decoder_recovers_state, true_state_exists_but_no_total_de

Face.10.bRLedger — Face 10: bR cross-domain lineage ledger conservation

tier: finite_diagnostic; status: current

$r_B = E \cdot A_C \cdot A_P \cdot A_B$ (composite lineage across quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport domain-translations); $I_Q = I_B + O_C + O_P + O_B$ (retained information either survives to the biological end as I_B or is accounted for as an explicit obstruction O at each stage)

Face.11.FailAbleGate — Face 11: FAIL-ABLE gate law

tier: untagged; status: current

A Type-P evidence gate -- a genuine, falsifiable pass/fail test of whether some obstruction has actually vanished -- requires both a passing control and a failing control, both machine-derived, not merely asserted. A gate that can only ever pass is a Type-U convention, not evidence the obstruction target has been hit.

Face.11.SolveTarget — Face 11: Obstruction / Solve-Target Face

tier: untagged; status: current

$O_R(R^{\diamond}) = O_{\{E_R\}}$; $S_R = \|O_R\|^2$. The solve target is the zero-section of an obstruction bundle, not scalar annihilation. Variational lens: minimize $S_R = \|O_R\|^2$.

Face.12 — Face 12: Boundary-Data Face

tier: untagged; status: current

Absolute constants = boundary data. Structure can constrain relations / roles / ratios. Numbers are measured, not derived from zero. Boundary data in this project: $G, \hbar, c, k_B, e, m_e, \alpha, v_{\text{Higgs}} = 246.22 \text{ GeV}; \tau_{\mu} = 2.187 \times 10^{-6} \text{ s}$ (SM-EFF closure, \$XI); $\Sigma_Y = 0$ (hypercharge cancellation, \$XI).

Face.2 — Face 2: Decay / Impermanence

tier: untagged; status: current

$|a_k[n]| \leq |a_k[0]| \exp(-\gamma_k n \Delta\theta), \gamma_k > 0$ -- Retained structures are finite and decaying unless driven or repaired.

Face.3.CharEq — Face 3: Dispersion Split — per-mode characteristic

tier: Th_coqc; status: current

$$M s^2 + D s + K \lambda = 0$$

Face.3.CriticalSplit — Face 3: Dispersion Split — critical split (λ_c, k_c)

tier: Th_coqc; status: current

$$\lambda_c = D^2 / (4 M K); k_c = 1 / (2 \sqrt{(\tau_c D)})$$

Face.3.Dispersion — Face 3: Dispersion Split — discrete/canon dispersion relation and regime split

tier: Th_coqc; status: current

$\lambda_{\pm}(k) = (-1 \pm \sqrt{(1 - 4\tau_c D k^2)}) / (2\tau_c)$; $k < k_c \rightarrow$ overdamped $\rightarrow$ classical-like; $k > k_c \rightarrow$ oscillatory $\rightarrow$ quantum-like. Machine-checked axiom-free: RDL_SpineStability.split_classical/quantum ✓

Face.4 — Face 4: Stability / Energy

tier: Th_coqc; status: current

$dE/dt = -D \|v\|^2 \leq 0$ -- Energy is monotone non-increasing; strict decay when velocity is nonzero. Machine-checked: RDL_SpineStability.energy_nonincreasing / strict_decay ✓

Face.5.GraphSpeed — Face 5: graph-native wave speed

tier: Open; status: current

$c^2 = K/M$ -- the graph-native wave speed, read off the $M \partial_t^2 \Phi$ and $K \cdot L_R \Phi$ terms. Identifying this graph-native c with the measured physical speed of light is [Open].

Face.5.LightCone — Face 5: Finite-Speed / Relativity — light-cone / finite-front readout

tier: untagged; status: current

$\|x\| \leq v t$; $v = \sqrt{D / \tau_c}$. Light-cone / finite-front readout. $\tau_c > 0 \rightarrow$ finite propagation guaranteed.

Face.6.MassMemory — Face 6: Mass / Memory — mass-memory formula

tier: Dr; status: current

$m = \hbar / (2 \tau_c c^2)$ [mass-memory formula; Dr/readout tier]. Status: structural / design-analogy. Do not treat as a new measured mass prediction.

Face.6.ModeMass — Face 6: Mass / Memory — mode-mass formula

tier: untagged; status: current

$$m_{k^2} = (K + D_s \lambda_k) / \tau_c$$
 [mode-mass formula; structural/readout]

Face.7.Landauer — Face 7: Landauer energy benchmark

tier: untagged; status: current

$$E_{\min}(\text{erase 1 bit}) = k_B T \ln 2$$
 (declared benchmark, not derived from zero)

Face.7.Residual — Face 7: residual discipline

tier: untagged; status: current

$$\text{spine_residual} = (\text{inertia} + \text{damping} + \text{geometry} + \text{nonlinear}) - (\text{source} - \text{noise})$$

Face.7.RootForce — Face 7: Force / Energy / RDU Readback Gate — root-force equation

tier: untagged; status: current

$$F_{\text{root}} = -K L_R \Phi - \nabla V(\Phi)$$

Face.7.SINewtonGate — Face 7: SI Newton readback gate

tier: untagged; status: current

force_to_SI_newton(F_{root}) iff all of: $\dim(F_{\text{root}}) = \text{force}$; source certificates present; dt nonzero certificate; residual discipline explicit; semantic unit compatibility; claim-tier boundary stated. Verified by: formal/Quantum_Relativity_Formal_DAG_*.v + scripts/stress_master_equations.py

Face.8.CurvatureForce — Face 8: Curvature and force restated on this face (companion/response field Ψ)

tier: Dr; status: current

Curvature is the record of an order that could not be swapped (τ_c 's non-commutative ordering, read out geometrically as the failure of parallel transport to commute). Force is that same record acting back on the channel that produced it (the $J - \eta$ term read at this face's geometric resolution), proposed to be carried by a companion/response field Ψ (Bateman-doubled / CTP-Keldysh construction). Does not extend RDL_MetricReadout.metric_form_readout's [Th_coqc] certificate.

Face.8.Guardrail — Face 8: operator-to-metric guardrail

tier: Th_coqc; status: current

operator $\rightarrow$ metric $\checkmark$; spectrum alone $\rightarrow$ metric $\times$ (ill-posed)

Face.8.MetricReadout — Face 8: Operator-to-Metric Geometry — canonical reading and operational shorthand

tier: Th_coqc; status: current

metric = principal symbol of the second-difference / Laplacian operator; $g^{ij} \approx \frac{1}{2} \text{Hessian}(\text{principal symbol})$; $D2\text{dir}(Q) = 2h^2 \cdot v^T H v$ (exact, numpy-computable). Machine-checked: RDL_MetricReadout.metric_form_re $\checkmark$ axiom-free

Face.9 — Face 9: CPTP / Quantum-Channel Face — Kraus completeness

tier: Th_coqc; status: current

$\sum_j K_j^\dagger K_j = I$ -- Kraus completeness + trace preservation + complete positivity. Machine-checked in RDL_StarRig* C34-C39 $\checkmark$

FalsifierFhuman — VIII.8 — Falsifier F-human

tier: Th_coqc; status: current

Claim under test: the τ_c^H loop (VIII.1-VIII.5) with the r0/r1 resonance gain and the first-order relaxation memory kernel $K_{\tau H}(s)$ correctly predicts which transduced frequencies ω_A produce strong agency-level response via $\chi_H = \omega_A \cdot \tau_c^H$. Passing control: the five worked transduction rows of VIII.5's table each fall in the range giving high resonance gain under $\tau_c^H \approx 0.1\text{--}3$ s. Failing control (any one refutes the section): (1) a genuinely ringing, second-order, M-bearing signature in human working-memory decay data; (2) a domain where the naive $\chi_H = \omega_{\text{raw}} \cdot \tau_{c_H}$ correctly predicts agency-level response; (3) empirically measured coupling between the five Step-5 policy terms that a scalar argmin genuinely cannot represent even approximately.

FourForceCirculation_v0.2 — Four-Force Circulation v0.2 — directed-response identity and Omega recovery

tier: untagged; status: current

On a four-sector ring fixture $A = H + \Omega$ (H symmetric reciprocal load, Ω antisymmetric circulation load), $\chi = A^{-1}$: $\text{spec}(H) = \{6, 4, 4, 2\} > 0$, $\text{spec}(A) = \{6, 2, 4 + i\}$, $\text{Re } \lambda(A) > 0$; exact directed-response identity $\chi - \chi^T = -2 \chi^T \Omega \chi$; $\Omega_{\text{recovered}} = \Omega_{\text{planted}}$ read as the unique antisymmetric part of χ^{-1} (exact by construction), with reciprocal ($\Omega=0 \Rightarrow \chi=\chi^T \Rightarrow \Omega_{\text{rec}}=0$) and missing-edge failing controls. Baseline $\Omega=0$ recovers Onsager reciprocity.

FrameMixingFromAction_v1.11 — Frame-mixing weights from the action (v1.11) — reflection-positive Gram derivation

tier: Th_cogc; status: current

A primitive frame-rewrite grammar (identity, swap, cycle, cycle^{-1} , sign-flip; order-384 frame group F_4) carries retained-rewrite costs $\text{eps}(m)$, giving half-step amplitudes $b_m = e^{-\{\text{eps}(m)/2\}}$; a hidden-midp slab makes the frame transfer a Gram operator $K_{\text{fr}} = B^{\dagger} B$ (reflection-positive by construction), whose relative-frame expansion yields weights $p(h) = \sum_{m^{-1} n = h} b_m b_n / (\sum b)^2$ with $p(h) \geq 0$, $\sum p(h) = 1$, $p(h) = p(h^{-1})$. The induced metric channel $M_b(G) = \sum_h p(h) h^T G h$ has only fixed metric the scalar line $R.I_4$ -- isotropy is derived, not assumed. Equal-cost fixture: anisotropy contracts at exact rate $\rho_{\text{aniso}} = 0.7361824549886247... < 1$ (largest root of $244140625 \lambda^6 - 371093750 \lambda^5 + 190234375 \lambda^4 - 41812500 \lambda^3 + 4299375 \lambda^2 - 201750 \lambda + 3481$), a stronger contraction than v1.10's 0.8580.

Gate1 — Gate 1 — No-Free-Domain-Law

tier: untagged; status: current

$\delta_R + \text{Retention} \Rightarrow (F, C, V, \theta)_{\text{domain}} \mathfrak{M}_{\text{compatible}} = \{ \mathcal{M} : \mathcal{M} \models \delta_R, \text{Retention}, \mathcal{K} \}$ If $|\mathfrak{M}_{\text{compatible}}| > 1$, no domain-specific law may be declared from retention and the tape alone -- an additional interaction tape, observation, or postulate is required first.

Gate2 — Gate 2 — Three-Valued Admissibility

tier: untagged; status: current

$C(\xi \mid c, \mathcal{T}) \in \{1, 0, \perp\}$ 1 admitted · 0 obstructed · $\perp$ unresolved $\text{Adm}(\xi) = 1 \text{ (T} \wedge \text{I} \wedge \text{B} \wedge [C=1]) / 0 \text{ (}[C=0]\text{)} / \perp \text{ (}[C=\perp]\text{)}$ $\text{Adm} = 1 \Rightarrow \text{expand}$; $\text{Adm} = 0 \Rightarrow \text{record the obstruction}$; $\text{Adm} = \perp \Rightarrow \text{record it as unresolved and do not guess}$.

Gate3 — Gate 3 — Context-Indexed Law

tier: untagged; status: current

$F_n = F_n(\mathfrak{G}_n, c_n, \mathcal{T}_n)$ $\mathbb{G}_n = \mathbb{G}_n(\mathbb{G}_n, c_n, \mathcal{T}_n)$ $\text{Adm}_n = \text{Adm}_n(\xi; c_n, \mathcal{T}_n)$. Valid in context c does not imply valid in context c' .

Gate4 — Gate 4 — State-Sufficiency Gate (three-valued)

tier: untagged; status: current

$\text{Suff}_{\{\alpha, L\}}(\ddagger_{\alpha}^{\text{cand}}; c, \mathcal{T}) \in \{1, 0, \perp\}$ Retention $\rightarrow$ State Proposal $\rightarrow$ Sufficiency $\rightarrow$ Discovery $\rightarrow$ Quotient $\ddagger^{(k+1)} = \ddagger^{(k)} \boxplus \Delta \ddagger_{\text{missing}}$ (if insufficient). A threshold or decoder may never be used to compensate for a distinction the state has already lost -- the correction is always to enlarge $\ddagger$, never to patch the readout downstream.

Gate5 — Gate 5 — Invariant-Completion Gate

tier: untagged; status: current

$D^2(g) = D^2(g')$ but $\chi(g) \neq \chi(g')$ (one invariant set can be insufficient) $\text{Inv}_{\alpha}(z) = (I_1(z), \dots, I_m(z))$ exact quotient requires: $\text{Inv}_{\alpha} = \text{Inv}_{\alpha} \circ q_{\alpha}$ $q(z) = q(z')$ but $\text{Inv}_{\alpha}(z) \neq \text{Inv}_{\alpha}(z') \Rightarrow \text{refine } C_a \rightarrow C_{\{a,1\}} \sqcup C_{\{a,2\}}$. Apparent symmetry is not valid quotient symmetry.

Gate6 — Gate 6 — Query-Relative Symmetry Group

tier: untagged; status: current

$\mathcal{H}_\alpha = \{ h : 0_\alpha(hz) = 0_\alpha(z), hF = Fh \}$ $\mathcal{D}_\alpha = \mathbb{Z} / \mathcal{H}_\alpha$. Only transformations that fix this question's readout may be quotiented away.

Gate7 — Gate 7 — Calibration Firewall

tier: untagged; status: current

$y_\alpha^{\text{known}} = U_\alpha(r_{RD} ; \theta_\alpha, c, C_\alpha^{\text{cal}})$ $H_{\text{cal}} = \text{Hash}(U_\alpha, \theta_\alpha, \text{training}, \text{units}, \text{protocol})$ (frozen before checking) $\text{Checker}(H_{\text{cal}}, Y_{\text{heldout}}) \rightarrow \epsilon_{\text{cal}}$. A native informational functional is not a physical observable until an independently checked calibration has been applied.

Gateway.GatewayPDE — IV.6 PGFT-RDU Native-Unit Gateway — the gateway PDE (the calculator)

tier: finite_diagnostic; status: current

$\partial_\Theta \phi = -D_R L_R \phi - \mu_R \phi + S_R$ -- a first-order simplification of the native master PDE, VERDICT: PASS on its own acceptance tests; unit ledger requires every term to share $\delta_R^{\sim}(e)/\Theta_R$.

Gateway.InputAdapter — IV.6 PGFT-RDU Native-Unit Gateway — 14-unit-type input adapter

tier: finite_diagnostic; status: current

$I_{\text{nat}} = E / (k_B T)$ where E is energy-equivalent. Supported units: eV, keV, MeV, GeV, TeV, PeV, EeV, J, kg, g, kcal, cal, nat, bit.

Gateway.NativeMasterPDE — IV.6 PGFT-RDU Native-Unit Gateway — native master PDE

tier: Dr; status: current

$\mathcal{M}_R \partial_\Theta^2 \Phi + \mathbb{D}_R \partial_\Theta \Phi + \mathbb{K}_R L_R \Phi + \nabla_\Phi V_R = J_R - \eta_R$ -- the same trunk as II.3's spine PDE, relabeled in native RD units so SI never has to be assumed foundational.

Gateway.OpenBridgeGuard — IV.6 PGFT-RDU Native-Unit Gateway — Schwarzschild/Unruh non-identity guard

tier: Open; status: current

$\tau_c^{\sim}(E) = ? \quad \tau_c^{\sim}(U)$ -- [OpenBridge], NOT an identity. Open until an observer/redshift translation map is supplied and passes the commuting-square test (X.4). Symbol equality is not semantic equality; verdict on any premature unification = INVALID_COMPARISON (equivalently OPEN_MISSING_TRANSLATION), status OPEN, not REFUTED, NOT = FAIL and NOT = PASS.

Gateway.OutputAdapters — IV.6 PGFT-RDU Native-Unit Gateway — output adapters

tier: finite_diagnostic; status: current

$E_{\text{out}} = k_B T \times I_{\text{out}}$ (native readout decoded back to energy); $\ell = c\tau$ (readout length); $F_\lambda = E/\lambda$ (readout force-at-wavelength). ($\tau = \hbar/2E$, $m = E/c^2$, and $\alpha_{\text{QG}} = G \cdot E^2/(\hbar c^5) = (E/E_{\text{Planck}})^2$ are the same output table's members already carried in IV.1/IV.2.)

Gateway.PrimitiveTuple — IV.6 PGFT-RDU Native-Unit Gateway — native-unit primitive tuple

tier: Dr; status: current

$R\Diamond = (\Gamma, \mathcal{D}, \mathcal{T}, \mathcal{L}, \Pi, \mathcal{A}, \mathbb{G}, \eta)$ -- Γ history, $\mathcal{D}$ retained distinctions, $\mathcal{T}$ tape, $\mathcal{L}$ transport/Laplacian structure, Π projection, $\mathcal{A}$ admissibility, $\mathbb{G}$ the operator (II.8a), η residual

Gateway.SchwarzschildIdentity — IV.6 PGFT-RDU Native-Unit Gateway — Schwarzschild energy identity

tier: untagged; status: current

F_Genesis_SchwarzschildEnergy: $r_s = 2GE/c^4$ [DeclaredFormula] (declared, not derived) -- restated in Coq syntax in Genesis_Canon_Bridge_v2_4_0, not a proved theorem, therefore not Th_coqc.

Gateway.SemanticLanes — IV.6 PGFT-RDU Native-Unit Gateway — semantic-lane definitions $\tau_c(E)$, $\tau_c(U)$

tier: untagged; status: current

$\tau_c(E) = \hbar / 2E$ energy / mass-memory lane [DeclaredFormula] (IV.1); $\tau_c(U) = \pi c / a_{\text{local Unruh}}$ / acceleration lane [DeclaredFormula] (this gateway)

Gateway.UnruhIdentity — IV.6 PGFT-RDU Native-Unit Gateway — Unruh τ_c identity

tier: untagged; status: current

F_Genesis_UnruhTauC: $\tau_c(T_{\text{Unruh}}) = \pi c/a$ [DeclaredFormula] (declared, not derived)

GaugeFaces — Four forces as gauge faces

tier: Dr; status: current

EM $\rightarrow$ U(1), A_μ = photon field; Weak $\rightarrow$ SU(2), $W^\pm$, Z_0 ; Strong $\rightarrow$ SU(3), gluons; Gravity $\rightarrow$ diffeomorphism invariance (readout-facing structure)

GaussJordanStepper — DRL-Telegraph apparatus: explicit Gauss–Jordan stepper with FAIL rule

tier: finite_diagnostic; status: current

$A_\Phi = M/\Delta t^2 + D/2\Delta t$; $b_{\Phi,n} = \mathcal{R}_{\Phi,n} + J_n - K \mathbb{G}_n \Phi_n - \nabla V(\Phi_n) + (2M/\Delta t^2)\Phi_n + (D/2\Delta t - M/\Delta t^2)\Phi_{\{n}$
 $A_\Phi \Phi_{\{n+1\}} = b_{\Phi,n}$. $A_\Psi = M/\Delta t^2 - D/2\Delta t$; $b_{\Psi,n} = \mathcal{R}_{\Psi,n} - K \mathbb{G}_n^T \Psi_n - \nabla^2 V(\Phi_n) \Psi_n + (2M/\Delta t^2)\Psi_n - (M/\Delta t^2 + D/2\Delta t)\Psi_{\{n-1\}}$; $A_\Psi \Psi_{\{n+1\}} = b_{\Psi,n}$ -- solved by partial-pivot Gauss-Jordan; the stepper is required to FAIL when a pivot falls below tolerance, no silent regularization.

Guard1 — Guard 1 — Package health smoke check

tier: untagged; status: current

python -c "from anse_spine import NuclearCore, Domain, Spine; print('anse_spine OK')"

Guard10 — Guard 10 — Skew-L_R / metric-G well-posedness probe (pre-T1)

tier: Dr; status: current

Sanity-checks that the PROPOSED $L_R = L_R^{(+)} + L_R^{(-)}$ decomposition under retention metric G is at minimum well-defined (symmetric/antisymmetric split exists and is unique) on the linearized two-field fixtures used so far. This is NOT test T1 itself -- passing this probe does not promote the tier above PROPOSED/Dr. python scripts/skew_lr_wellposed_probe.py -fixtures chemostat_linearized,lv_linearized

Guard11 — Guard 11 — FAIL-ABLE gate audit

tier: untagged; status: current

Scans the guard suite itself for Type-U gates lacking a constructed failing control, and blocks any PR that adds a new gate without one. python scripts/failable_gate_audit.py -suite scripts/

Guard12 — Guard 12 — CFL sufficiency-not-necessity control

tier: untagged; status: current

Proves the II.1 / step-13 CFL bound is a CONSERVATIVE SUFFICIENT gate, not a tight instability threshold. On the 4-node fixture $W = [[0,1,.5,0],[1,0,1,.5],[.5,1,0,1],[0,.5,1,0]]$ ($L_R=D_W-W$, $\lambda_{\max}=3.618034$, $\gamma=0.4$, $D=3$): PASSING control $\Delta\theta=0.55$ -- VIOLATES the file bound ($\Delta\theta_{\text{CFL}}=0.4244865$) yet stays stable: per-mode update spectral radius $\rho=0.883176 < 1$, energy non-increasing, $\|X\|$ decays $3.162 \rightarrow 1.1e-3$ over 50 ticks ($0.55 < \text{the true per-operator boundary } h_{\text{exact}}=0.6827093$). FAILING control $\Delta\theta=0.75$ -- past the true spectral threshold: $\rho=2.610975$, $\|X_{50}\| \approx 2.6e21$, blows up. Together they falsify "exceeding the bound $\Rightarrow$ unconditionally unstable" while leaving "respecting the bound $\Rightarrow$ stable" intact. h_{exact} is a spectral/asymptotic boundary (transient growth as $\Delta\theta \rightarrow h_{\text{exact}}$ via the non-normal update matrix), NOT a safe operating point -- keep a margin. `python scripts/cfl_sufficiency_guard.py -passing 0.55 -failing 0.75 -check energy-non-increasing -check spectral-radius`

Guard13 — Guard 13 — Schwarzschild/Unruh semantic-lane type-checker

tier: untagged; status: current

Refuses any `compare(LocalAcceleration, SurfaceGravity)` not mediated by a declared, tested `ObserverMap`. Acceleration readouts must carry mandatory metadata; a bare scalar `a` is rejected at the gateway: `AccelerationReadout = {value, observer_class, normalization_point, local_or_asymptotic, redshift_map, semantic_lane}`. PASSING control: a comparison mediated by `SurfaceGravity = ObserverMap(LocalAcceleration` with a supplied, checked redshift map -- allowed. FAILING control: the $\tau_c^*(E)=\tau_c^*(U) \Rightarrow a=\kappa$ identification with NO `ObserverMap` -- must return `INVALID_COMPARISON/OPEN`, and must NOT return `FAIL` or `PASS`. `python scripts/bh_gateway_lane_check.py -reject-unmapped-compare -require-metadata observer_class`

Guard14 — Guard 14 — Domain-discovery identifiability gate

tier: finite_diagnostic; status: current

The discovery engine's concrete realization of the three-valued admissibility gate ($\perp$ = do not guess): before reporting any "minimal exact law", require `rank(design_matrix) == n_candidate_coefficients` over $\mathbb{Q}$, else abstain. NOT a new root rule -- the `finite_diagnostic` instance of the state-sufficiency / $\perp$ -unresolved principle already in A.13/VI, applied to the tool. PASSING control: the N-S/8 tape, declared basis $\{1, N, S, N^2, N \cdot S, S^2\}$ (6 rows, rank 6) -- identifiable, law reported. FAILING control: the same basis with any row removed (rank $5 < 6$) -- MUST return `UNDERDETERMINED/abstain`, NOT a guessed law. `python scripts/discovery_identifiability_gate.py -require-full-rank -abstain-on-deficient`

Guard15 — Guard 15 — Geometry-dominance diagnostic

tier: untagged; status: current

The computable defect bound the living-geometry mixing current (II.8a) already demands: for a moving-basis step $\Phi_n = V_n \phi_n$, $R_{\text{geo}} = |(V_{n+1} - V_n) \phi_n| / |\Delta \Phi_n|$; reject a fixed-basis (drop-the-basis-motion-term) approximation when $R_{\text{geo}} \gtrsim 1$. This is a diagnostic ON an existing master-equation term, not an imported rule. PASSING control: $V_{n+1} = V_n$ (no basis motion) $\rightarrow R_{\text{geo}} = 0$, fixed-basis allowed. FAILING control: $V_n = I$, $V_{n+1} = 3\text{-}4\text{-}5$ rotation, $\phi_n = (2,1) \rightarrow R_{\text{geo}} \approx 1.099 + \text{sign-flip}$; fixed-basis approximation MUST be rejected. `python scripts/geometry_dominance_check.py -reject-above 1.0`

Guard2 — Guard 2 — PGFT v0.6 reproducible PDE lane smoke

tier: untagged; status: current

`python scripts/pgft_v0_6_smoke.py`

Guard3 — Guard 3 — RDU readout suite

tier: untagged; status: current

`python scripts/run_rdu_readout_suite.py -stress-samples 100 -trial-count 100 -grid-sizes 2`

Guard4 — Guard 4 — Docs root anchor integrity

tier: untagged; status: current

python scripts/verify_docs_root.py

Guard5 — Guard 5 — Omega_H horizon transduction smoke (27 checks)

tier: untagged; status: current

python scripts/smoke_omega_h.py

Guard6 — Guard 6 — Jupyter notebook smoke (7 cells)

tier: untagged; status: current

jupyter nbconvert -to notebook -execute notebooks/anse_spine_smoke.ipynb

Guard7 — Guard 7 — Label-inflation grep

tier: untagged; status: current

Flags any `Th_coqc` claim paired with a continuum physics proper noun where the `.v` object is a scalar/finite reduction, not the named continuum structure itself. Run before any interpretation card ships. Type-P: passing control = the two surviving cards from the 2026-07-21 audit; failing control = the six retracted cards. `python scripts/label_inflation_guard.py -cards docs/claims/*.md`

Guard8 — Guard 8 — Domain-discovery adversarial battery

tier: untagged; status: current

Must recover $N\cdot S/8$, N^2 , affine, and the 3-variable case exactly (passing controls); must REFUSE $N\cdot S/16$ and any non-polynomial candidate law (failing controls). A battery run that accepts a wrong law is a hard CI failure, not a warning. `python scripts/domain_discovery_battery.py -holdout-exact -refuse-no-`

Guard9 — Guard 9 — bR lineage ledger conservation check

tier: untagged; status: current

Verifies $I_Q = I_B + O_C + O_P + O_B$ holds along the quantum→chemical→protein→biological-transport chain for every synthetic fixture; flags any fixture where q_Q alone (skipping the intermediate quotients) spuriously commutes, since that would itself be a reduction-error false positive. `python scripts/br_lineage -chain quantum,chemical,protein,biological`

HumanAgencyTuple — VIII.1 — The Human Agency Tuple A_H

tier: Dr; status: current

$A_H = (B_H, \tau_c^H, L_H, \Omega_H, K_H, \Phi_H, \Pi_H, U_H)$ -- B_H : body-ground (embodied substrate); τ_c^H : human causal-memory timescale (~ 0.1 -3 s species-level); L_H : language/symbol layer (compression + framing); Ω_H : world-model (retained internal map); K_H : framing operator (perception → record); Φ_H : phenomenalization function; Π_H : policy function (action selection); U_H : utility / value function.

Hypercharge_Z6_v1.5 — Hypercharge + Z6 global structure — anomaly closure (v1.5)

tier: Th_coqc; status: current

Under the minimal one-generation skeleton (Q, u^c, d^c, L, e^c, H) , interaction closure fixes charges up to two free scales ($u=-q-h$, $d=-q+h$, $l=-3q$, $e=h+3q$); anomaly cancellation forces the ratios: gravitation anomaly $A_{\text{grav}} = 6q+3u+3d+2l+e = h-3q$ and cubic $A_{111} = (h-3q)^3$ factor into the SAME condition $h=3q$ ($A_{111} = (A_{\text{grav}})^3$). At $h=3q$, primitive $q=1$: $y = 6Y = (1, -4, 2, -3, 6, 3) \Rightarrow Y = (1/6, -2/3, 1/3, -1/2, 1, 1/2)$ and $Q_{\text{em}} = T3+Y = (2/3, -1/3, 0, -1)$. Center-lock $2t+3s+y \equiv 0 \pmod{6}$ holds for every record $\Rightarrow$ the invisible common center is exactly Z_6 , so $G_{\text{phys}} = [\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)]/Z_6$. Global $\text{SU}(2)$ (Witten): $\# \text{doublets} = 3Q+L = 4$ is even (control: colored- Q -only = 3 odd would FAIL). Negative control: adding

a right-handed ν^c makes $A_{\text{grav}} = A_{111} = 0$ for all $(q, h) \Rightarrow Y$ and B-L degenerate, lifted only by the extra self-pair ν^c tensor $\nu^c \rightarrow 1 \Rightarrow 2n=0$.

I — Forced Set I — retained distinction forces the graph Laplacian

tier: Th_cogc; status: current

$L_R = D_W - W$ -- retained distinction forces the graph-Laplacian form; among all candidate operators, only L_R passes every required structural test (InfoRetainedDistinctionForcesLaplacian)

II — Forced Set II — action stationarity under node removal

tier: untagged; status: current

$\delta S / \delta Z_n |_{\text{removed}} = \delta S / \delta Z_n |_{\text{kept}} + \text{boundary term}$, exactly -- removing a retained node changes the stationary residual by exactly the boundary term -- no free growth (InfoActionStationarity)

III — Forced Set III — nonnegative strain / back-reaction

tier: untagged; status: current

$\text{strain}(x, e) \geq 0$; $\text{strain}(x, e) = 0 \iff x_u = x_v$ -- strain across any edge is nonnegative, and zero iff there is no retained distinction across it (InfoBackReaction)

IV — Forced Set IV — cone inheritance under stepping

tier: untagged; status: current

$\text{step} \Rightarrow$ same local stencil -- one operator step inherits its own local stencil $\Rightarrow$ the finite causal cone survives composition (InfoConeInheritance)

IX — Forced Set IX — frame-covariant causal quadratic form

tier: untagged; status: current

$Q_v = n_+ \cdot n_-$ is frame-covariant -- the causal quadratic form built from the cone's edges is covariant under frame change; signature comes from the causal order $\prec$ (InfoLorentz)

ImaginationTransport — VIII.6a — Imagination as Reversible Transport Across Living Modes ($\lambda \neq 0$)

tier: Dr; status: current

Imagination is the ability to mix, swap, replace, splice, extend, and transform multiple finite meaning-se into new forms. The mixing/swapping/splicing operation imagination performs is the same object, algebraically, as the skew part of the multimode operator, $L_R^{\prec}(-)$, introduced at V.13a and XIV.4 as the proposed $L_R = L_R^{\succ}(+) + L_R^{\prec}(-)$ decomposition under the retention metric G . $L_R^{\prec}(-)$ is exactly reversible and norm-preserving ($z^T L^{\prec}(-) z = 0$ for the retention metric's inner product), operating only in the living-mode regime $\lambda \neq 0$ (eigenmodes of Face 1 that have not collapsed to the kernel of L_R).

IntertwinerOrderVacuum_v1.13 — Intertwiner and order-vacuum closure (v1.13) — corrected $\Pi_0 > \alpha$ criterion

tier: Th_cogc; status: current

Invariant multiplicities $\nu_U = \nu_D = \nu_E = 1$ (each $\text{Hom}_G(\dots, 1)$ one-dimensional); fixing the order carrier H , the linear maps $M_j(H)$ satisfy $M_j M_j^{\dagger} = r \cdot I_{\{d_j\}}$ with rank $(d_U, d_D, d_E) = (3, 3, 1)$ giving 7 closure singular modes and one primitive ordered-tape history per branch. Correction: an earlier exponential ansatz $Z_j(r) = 1 + \zeta_j e^{\{\kappa_j r\}}$ wrongly kept weight $\zeta_j > 0$ at $r=0$ and is superseded by the representation-derived fermionic Fock determinant $Z_j(r) = \det(I + K_j) = (1 + \lambda_j r)^{d_j}$. With $V_{\text{eff}}(r) = \alpha r + \beta r^2 - 3 \log(1 + \lambda_U r) - 3 \log(1 + \lambda_D r) - \log(1 + \lambda_E r)$, $V_{\text{eff}}'(0) = \alpha - \Pi_0$, $\Pi_0 = 3 \lambda_U + 3 \lambda_D + \lambda_E$; $\Pi_0 > \alpha \Rightarrow$ order. Convexity $V_{\text{eff}}''(r) > 0$ holds automatically for $\beta > 0$. No-go: $0 < \lambda_j \leq$

1 => Pi_0 <= 7 => alpha >= 7 forces r*=0. The gauge-invariant statement is $\langle H^\dagger H \rangle > 0$ (Elitzur); NOT a prediction of the W/Z/Higgs masses.

InverseArrow — Inverse arrow: domain-discovery engine converges on the same MQ.08/L_R node

tier: finite_diagnostic; status: current

raw transition tape $(N, S) \rightarrow (N', S')$, no domain name, no law handed in $\rightarrow$ domain-discovery engine [law-DSL candidate-basis grammar $\{1, N, S, N^2, N \cdot S, S^2, \dots\}$; identifiability gate $\text{rank}(A) = \# \text{coeff}$ else $\perp = \text{abstain}$; interaction test (mixed 2nd difference $\neq 0$); exact closure over $\mathbb{Q}$ + conservation] $\rightarrow$ minimal exact update law, e.g. $N' = N + N \cdot S/8$, $C = N + S$ conserved -- the SAME MQ.08 / L_R object, recovered from data, never fabricated

IsotropicFixedPoint_v1.10 — Isotropic fixed point — Lorentz shadow from coarse-graining (v1.10)

tier: Th_cogc; status: current

The finite signed-permutation twirl $\text{Pi}_4(X) = (\text{Tr}(X)/4)$. I (over the order-384 group $B_4 = (Z_2)^4$ semidirect S_4) projects onto isotropy; a partial twirl gives the exact contraction $\Delta_{\{n+1\}} = (1-\alpha) \Delta_n$. A five-move local mixer has exact spectral radius $\rho_{\text{frame}} = 0.858010754587974 \dots < 1$ on the 9-dim traceless space (largest root of $15625 \lambda^6 - 25000 \lambda^5 + 8125 \lambda^4 + 3500 \lambda^3 - 1625 \lambda^2 - 100 \lambda + 59$, rational Bolzano bracket $p(858/1000) < 0 < p(8581/10000)$, $p(1) = 584 > 0$). Coupled sectors mix by a doubly-stochastic map, so $c_{\text{gauge}} = c_{\text{matter}} = c_{\text{order}} = c_{\text{tape}} \rightarrow c_*$, giving a 3+1 Lorentz shadow $E^2 = c_*^2 |p|^2 + m_s^2 c_*^4$ (Lorentz universality without mass universality).

L0 — L0 — NUCLEUS

tier: untagged; status: current

the 5 equations + the Coq floor (RDL*.v); minimal, domain-independent, machine-checked where stated

L1 — L1 — SCALE BUS

tier: untagged; status: current

the τ_c atlas (37 disciplines) -- connect any domain by memory time [tau_c/tau_c_master.py]

L2 — L2 — READOUT

tier: untagged; status: current

retrieval = readout · confidence = obstruction · ABSTAIN below ϵ_{tot} [rag_spine]

L3 — L3 — ADAPTERS

tier: untagged; status: current

per-domain (M, D, K, K_A, L_R); physics / bio / econ / IR / AI plug in identically

L4 — L4 — AGENCY

tier: untagged; status: current

argmin 0 preserving repairability · bend information by choice [ai_agency, info_bending]

L5 — L5 — GOVERNANCE

tier: untagged; status: current

readout-not-truth · bounded-judge · machinic_core (only Coq+lib-verified is core; everything else is diagnostic or open)

LR_skew_split — V.13a Scalar-Eigenmode Reduction Error — L_R split under the retention metric G

tier: untagged; status: current

$L_R = L_R^{(+)} + L_R^{(-)}$ (split under the retention metric G, not naive transpose) -- $L_R^{(+)}$ carries the symmetric (diagonal-dominant) part, $L_R^{(-)}$ carries the skew, rotational, cross-channel coupling a scalar reduction throws away.

Layer.3 — Stack correction: Layer 3 — LP-NS audit (nonlinear paraproduct diagnostic, a checker not a generator)

tier: untagged; status: current

a CHECKER on candidate solutions, not a native generator

Layer3 — Three stacked layers — Layer 3: LP-NS audit (nonlinear paraproduct diagnostic)

tier: finite_diagnostic; status: current

Examines the nonlinear paraproduct structure $(u \cdot \nabla)u$ inside ∇V (Littlewood-Paley / Navier-Stokes-flavored audit machinery) to check whether a given solution or regime stays inside admissible bounds. Explicitly a checker, not a native generator of dynamics.

LayerStack — VIII.0 — Three-layer stack placement (DRL-Telegraph / RTPE / LP-NS)

tier: finite_diagnostic; status: current

Layer 1 -- DRL-Telegraph root (2nd order, the generator): $M \partial^2 \Phi + D \partial \Phi + K \cdot L_R \cdot \Phi + \nabla V = J - \eta$.
 Layer 2 -- RTPE turbulence relaxation (1st order, the $M \rightarrow 0$, $V \rightarrow 0$ limit): $\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R$ [PASS_WITH_LIMITS, finite_diagnostic].
 Layer 3 -- LP-NS audit (nonlinear paraproduct CHECKER, not a generator): diagnoses $(u \cdot \nabla)u / \nabla V$ for blow-up risk; does not itself produce dynamics. The human memory kernel $K_{\tau H}(s) = (1/\tau_c^H) e^{-s/\tau_c^H}$ introduced in VIII.2 is a first-order exponential relaxation kernel -- structurally the same shape as the RTPE relaxation law, not the second-order Telegraph generator.

LivingGeometry — DRL-Telegraph apparatus: living-geometry recurrence

tier: finite_diagnostic; status: current

$\Theta_{\{n+1\}} = A_{\Theta} \Theta_n + B_{\Theta \Phi} \Phi_n + B_{\Theta \Psi} \Psi_n + u_{\Theta, n}$; $\mathbb{G}[\Theta_n] = \mathbb{G}_0 + \Sigma_a \Theta_n^a \mathbb{G}_a$ (operator read from the geometry state)

MQ08-CFL — MQ.08 CFL / stability condition

tier: finite_diagnostic; status: current

$\Delta\theta \leq 2 / (\gamma + \sqrt{\gamma^2 + 4 \cdot \lambda_{\max} \cdot D_s})$

MQ08-stepper — MQ.08 discrete stepper

tier: finite_diagnostic; status: current

$\gamma = 1/\tau_c$; $D_s = D \cdot \Delta\theta$; $V[n+1] = V[n] + \Delta\theta \cdot (-\gamma \cdot V[n] - D_s \cdot (L_R @ X[n]))$; $X[n+1] = X[n] + \Delta\theta \cdot V[n+1]$

Code	Dom.	Statement	Status
MQ08-stepper/P.01.v1	P	Retention discretizes into ticks $t=n \cdot \Delta\theta$; the finite causal graph and retention operator L_R are built on those ticks, stepped by MQ.08.	current

MassGap_v1.3_v1.4 — Mass gap — finite-transfer gap theorem (v1.3) and universal reflection-positive slab (v1.4)

tier: *Th_cogc*; status: *current*

v1.3: given a positive self-adjoint physical transfer operator T_{phys} with a unique vacuum that strictly contracts the nonvacuum sector ($q = \|T_{\text{phys}} P_{\text{perp}}\| < 1$), the gap $\Delta = -(1/a) \log(q) > 0$ and connected gauge-invariant correlators decay exponentially -- with negative controls (massless-diff gap closes as $L \rightarrow \infty$; vacuum degeneracy $\Rightarrow \Delta = 0$). v1.4: gauge $(U(1)/SU(2)/SU(3))$ + scalar + generic fermionic Fock kernels are Gram-positive $\Rightarrow T_{\text{UF}} \geq 0$, and the same reader $m = -(1/a) \log(\lambda)$ classifies every mass TYPE from the spectral-measure shape; a PDG-2026 unit test shows the extractor reads the input masses back (tests the READER, not a prediction); held-out test: mass ratios $m_i/m_j = \log(\lambda_i)/\log(\lambda_j)$ are lattice-scale independent ($\mu/e = 206.768, \dots$).

MasterEqCanon — MQ.08 master equation, Canon §3 state-space form

tier: *finite_diagnostic*; status: *current*

$d/dt [X; V] = [[0, I], [-K L_R, -\gamma]] [X; V] + [0; J - \eta]$; expanded: $V[n+1] = V[n] + \Delta\theta(-\gamma \cdot V[n] - D_s \cdot L_R \cdot X[n])$; $X[n+1] = X[n] + \Delta\theta \cdot V[n+1]$

MeasuringAgency — VIII.4 — Measuring Agency (epistemic bandwidth, repair rate, agency gradient)

tier: *Dr*; status: *current*

Epistemic bandwidth: $I(A_H; E) / \tau_c^H$ (bits per unit time). Repair rate: $dR_H/dt = -\mu \cdot \|\text{Res}_H\|^2$. Agency gradient: $\partial\pi^*/\partial\eta_H$ (sensitivity to readout noise). Claim tier: *Dr* (open theoretical; falsifiable -- §11 of docs/claims/HUMAN_AGENCY_SINGLE_TAU_C.md, and see F-human, VIII.8).

MemoryKernel — Memory-kernel reading of the telegraph equation

tier: *finite_diagnostic*; status: *current*

$K(\tau) = (1/\tau_c) e^{(-\tau/\tau_c)}$; $\int_0^\infty \tau K(\tau) d\tau = \tau_c$

MixingCurrent — DRL-Telegraph apparatus: geometric mixing current

tier: *finite_diagnostic*; status: *current*

$\Delta\Phi_n = V_{\{n+1\}} \Delta\phi_n + (V_{\{n+1\}} - V_n) \phi_n$ -- dropping the second term (basis-motion contribution) requires a proof or a stated defect bound; it is not a free simplification.

ModeEquation — Regime split: mode-by-mode damped oscillator

tier: *untagged*; status: *current*

$M \ddot{\phi}_k + D \dot{\phi}_k + K \lambda_k \phi_k = 0$

N1-Layer3 — N1 Layer 3 — LP-NS audit

tier: *finite_diagnostic*; status: *current*

nonlinear paraproduct diagnostic over the $\nabla V / (u \cdot \nabla)u$ term

N2-ScalarEigenmode — N2 repair — Scalar-Eigenmode Reduction Error split of L_R

tier: *Open*; status: *current*

$L_R = L_R^+(+) + L_R^-(-)$, split under a retention metric G into symmetric (ordinary damped/relaxation) and antisymmetric (skew, rotational) parts, so a naive scalar/naive-transpose reduction does not silently discard off-diagonal/skew coupling before N2's readout

N3 — N3 — OBSTRUCTION

tier: Th_coqc; status: current

$dE/dt = \langle \partial\Phi, J \rangle - D \|\partial\Phi\|^2 \leq 0, 0 \rightarrow 0$; consistency-seeking; the monotone (Coq-checked)

N5 — N5 — INVARIANTS

tier: untagged; status: current

anomaly ratios, $2/\alpha^2$, π , ϕ ; the dimensionless readouts that DO close (readout-invariants)

NewtonGate — VII.2 — Newton gate

tier: untagged; status: current

`force_to_SI_newton(F, dim) → value` iff `dim == [1,1,-2,0,0,0,0]`; else `→ None` (gate closes; readback refused)

ObstructionCertificate — VI.6 — bR ledger obstruction certificate

tier: finite_diagnostic; status: current

$I_Q = I_B + O_C + O_P + O_B$; the information present at the quantum end equals what survives at the biological end plus what was obstructed at each intermediate translation

OmegaH — VIII.5 — Ω_H , the Horizon Transduction Operator

tier: Dr; status: current

The resonance equation $\chi_H = \omega * \tau_{c_H}$ must use the agency-side frequency, not the raw physical carrier. WRONG: $\chi_H = \omega_{\text{raw}} * \tau_{c_H}$ (light at 4×10^{14} Hz $\rightarrow$ gain $\approx 0 \rightarrow$ but humans see!). CORRECT: $\chi_H = \omega_A * \tau_{c_H}$, $\omega_A = \text{Freq}(\Omega_H(R_{\text{raw}}))$, $\Omega_H = C_H \circ E_H \circ T_H \circ S_H$. Ω_H pipeline: $R_{\text{raw}} \rightarrow S_H$ (select) $\rightarrow T_H$ (transduce) $\rightarrow E_H$ (extract envelope) $\rightarrow C_H$ (compress) $\rightarrow \omega_A$. Gain functions: r_0 default $G_{r_0} = 1.0 / (1.0 + (\chi - 1.0)**2)$ [at $\chi \rightarrow 0$ yields $G=0.5$, a known asymmetry]; r_1 log-normal $G_{r_1} = \exp(-\log(\max(\chi, 1e-12))**2/2.0)$ [symmetric suppression both sides].

OrderHiggsClosure_v1.12 — Minimal Order/Higgs (v1.12) — Higgs rep forced, electroweak mass matrix

tier: Th_coqc; status: current

From the blind matter skeleton, the minimal order carrier is forced: a color-singlet, an $SU(2)$ doublet (2 tensor R_2 contains 1 iff $R_2=2$), and hypercharge $y_H=3$ fixed by every matter closure ($1+h-4=0$, $1-h+2=0$, $-3-h+6=0$) -- i.e. $H=(1,2)_{\{1/2\}}$, the Higgs rep derived, not fed. A nonzero order H_* has stabilizer $Q_{\text{res}} = T_3+Y$ (dim 1), so 4-1=3 of the four $SU(2) \times U(1)$ generators are broken; the neutral (W^3, B) mass matrix $(v^2/4)[[g^2, -gg'], [-gg', g'^2]]$ has det 0, rank 1 $\Rightarrow$ one massless photon $A \sim (g', g)$ and $m_Z^2 = (g^2 + g'^2)v^2/4$, charged pair gives $m_W^2 = g^2 v^2/4 \Rightarrow m_W = m_Z \cos(\theta)$ and tree-level $\rho=1$; degree-of-freedom count balances $8+4 = 2+9+1$.

PythonAPI — VII.4 — Python API (anse_spine.units)

tier: untagged; status: current

`dimension_of`, `is_dimension_compatible`, `readout_rdu`, `to_newton`, `validate_unitmap` -- five functions; `readout_rdu(tau) → dict` of $\tau/\ell/f/E/m/\epsilon$ values; `to_newton(value, dim) → float | None`; example `readout_rdu(→ {'tau':..., 'length':..., 'energy':...})`

QuantumDomain_Born_target — V.18 Registered Domain: Quantum — Born rule as a target, not an axiom

tier: untagged; status: current

The verifier shows refinement-consistency selects the quadratic $w(P_i|\psi) = \langle \psi, P_i \psi \rangle / \langle \psi, \psi \rangle$ (a linear $p=1$ rule fails: $\sqrt{25} \neq \sqrt{9} + \sqrt{16}$) -- but uniqueness across all admissible refinements is [TARGET -- NOT YET DERIVED], so measurement/Born stays RED.

QuantumDomain_complexification — V.18 Registered Domain: Quantum — complexification gate (i as a readout, with failing control)

tier: untagged; status: current

complexification gate: i is a READOUT of a closed real oriented mode-pair $J^2 = -I$, $J^{\dagger} J = -J$ (FAILING control: $J_{\text{bad}}^2 = +I \Rightarrow \text{NO_COMPLEX_QUANTUM}$)

QuantumDomain_norm_reversibility — V.18 Registered Domain: Quantum — positive norm and reversible evolution

tier: untagged; status: current

positive norm $N_Q = \langle \psi, \psi \rangle_G$; reversible $U^{\dagger} G U = G$

QuantumDomain_osc_split — V.18 Registered Domain: Quantum — oscillatory split (green node)

tier: untagged; status: current

oscillatory split $\lambda_c = D^2/4MK$ (osc iff $\lambda > \lambda_c$)

RDU-E_c — VII.2 — RDU E_c readout

tier: untagged; status: current

E_c readout $\rightarrow$ energy $[2,1,-2,0,0,0,0]$, unit: J ; $E = \hbar/\tau_c$

RDU-ell_c — VII.2 — RDU ℓ_c readout

tier: untagged; status: current

ℓ_c readout $\rightarrow$ length $[1,0,0,0,0,0,0]$, unit: m ; $\ell = c \cdot \tau_c$

RDU-eps_L — VII.2 — RDU ϵ_L (thermal) readout

tier: untagged; status: current

ϵ_L readout $\rightarrow$ thermal $[2,1,-2,0,0,0,0]$, unit: J ; $\epsilon = k_B T \ln 2$ -- the SI-adaptor reading of the Landauer floor (VI.4 bound B1)

RDU-f_c — VII.2 — RDU f_c readout

tier: untagged; status: current

f_c readout $\rightarrow$ frequency $[0,0,-1,0,0,0,0]$, unit: Hz ; $f = 1/\tau_c$

RDU-m_c — VII.2 — RDU m_c readout

tier: untagged; status: current

m_c readout $\rightarrow$ mass $[0,1,0,0,0,0,0]$, unit: kg ; $m = \hbar/(2\tau_c c^2)$ -- mass sits in the middle of the table, computed FROM τ_c , not the other way round

RDU-tau_c — VII.2 — RDU τ_c readout

tier: untagged; status: current

τ_c readout $\rightarrow$ time scale $[0,0,1,0,0,0,0]$, unit: s (the input row; every row beneath it in the RDU table is downstream of τ_c)

RTPEOperatorSplit — DRL-Telegraph apparatus: RTPE turbulence operator's own symmetric/skew/nonlinear decomposition

tier: finite_diagnostic; status: current

$\mathcal{L}_R[I, \Theta] I = \mathcal{L}_R^{(+)}[\Theta] I + \mathcal{L}_R^{(-)}[\Theta] I + \mathcal{N}_R[I]$; $(\mathcal{L}_R^{(+)})^T = \mathcal{L}_R^{(+)} \succeq 0$ (symmetric part positive-semidefinite); $(\mathcal{L}_R^{(-)})^T = -\mathcal{L}_R^{(-)}$ (skew part antisymmetric); $(\mathcal{N}_R[I])_i = \sum_{jk} B_{ijk} I_j I_k$ (quadratic nonlinear transfer). Conservative-transfer check: $\langle I, \mathcal{N}_R[I] \rangle = 0$; $\langle I, \mathcal{L}_R^{(-)} I \rangle = 0$ (both to be verified, not assumed).

Re_ep — VI.3 — Re_ep (epistemic Reynolds)

tier: untagged; status: current

Re_ep = epistemic Reynolds (spread = contestedness / turbulence)

ReaderRecordEqs — DRL-Telegraph apparatus: reader and record equations

tier: finite_diagnostic; status: current

Reader equation: $M \delta_t^2 \Phi_n + D \delta_t^c \Phi_n + K \mathbb{G}[\Theta_n] \Phi_n + \nabla V(\Phi_n) - J_n = \mathcal{R}_{\Phi, n}$. Record equation: $M \delta_t^2 \Psi_n - D \delta_t^c \Psi_n + K \mathbb{G}[\Theta_n]^T \Psi_n + \nabla^2 V(\Phi_n) \Psi_n = \mathcal{R}_{\Psi, n}$. Note the sign flip on the damping term (+D reader, -D record) and the transpose on the operator's coupling to Ψ (Bateman-doubled structure).

RelativityDomain_Qv — V.17 Registered Domain: Relativity — null-coordinate invariant Q_v

tier: untagged; status: current

null coords $n+/- = vt +/- x$; $Q_v = n+ . \quad n- = v^2 t^2 - x^2$ (from the cone's edges); gates 01-04, 04 = RD-neutrality ($n+' \quad n-' = n+ \quad n-$)

RelativityDomain_calculator_identities — V.17 Registered Domain: Relativity — Schwarzschild/Unruh calculator identities (NOT derived)

tier: untagged; status: current

Schwarzschild $r_s = 2GE/c^4$ and Unruh $\tau_c = \pi c / a$ -- derived (they stay [DeclaredFormula] calculator identities, not established as a derivation of real physics).

RelativityDomain_dynamic_metric_source — V.17 Registered Domain: Relativity — dynamic metric-source law

tier: untagged; status: current

dynamic metric-source step law: $M_{\Theta} * \delta_t^2 \Theta + \text{grad}(U_{\Theta}) + K . \Phi^T (d G / d \Theta) \Psi = 0$ -- with minimal closed state $Z_n = (\Phi_n, \Phi_{\{n-1\}}, \Psi_n, \Psi_{\{n-1\}}, \Theta_n, \Theta_{\{n-1\}})$, exact fixture $\Theta_{\{n+1\}} = 3/8$, and a failing control (dropping the record source Ψ gives residual $1 \neq 0$ -- Ψ is load-bearing).

RelativityDomain_lapse_Lorentz — V.17 Registered Domain: Relativity — lapse, Lorentz-form transforms, redshift, horizon

tier: untagged; status: current

$B = N . \text{diag}(\kappa^{-1}, \kappa)$, $N = \sqrt{\det B} = \text{lapse}$; $\Gamma_R = 1/\sqrt{1-\beta^2}$, $x' = \Gamma_R(x - u t)$, $t' = \dots$ (clock/length/simultaneity/composition); redshift $\nu_o = N . \nu_i$; horizon $\Leftarrow N = 0$ (rank-loss, finite)

ResidualsAndPotential — DRL-Telegraph apparatus: residual decomposition and cubic potential

tier: finite_diagnostic; status: current

$\mathcal{R}_\Phi = \mathcal{J}_{\text{cut},\Phi} + \mathcal{J}_{\text{syn}} + \mathcal{J}_{\text{geo}}$; $\mathcal{R}_\Psi = \mathcal{J}_{\text{cut},\Psi} + \mathcal{J}_\Lambda$; $\nabla V_i(\Phi) = a_i \Phi_i + b_i \Phi_i^3$ (executable cubic potential); $(\nabla^2 V)_{ii} = a_i + 3 b_i \Phi_i^2$

Rgeo — DRL-Telegraph apparatus: geometry-dominance defect bound R_{geo}

tier: finite_diagnostic; status: current

$R_{\text{geo}} = |(V_{\{n+1\}} - V_n) \phi_n| / |\Delta \Phi_n|$ -- measures how much of the true step is basis-motion a fixed-basis approximation would discard. When $R_{\text{geo}} \gtrsim 1$ the drop-the-second-term approximation must be auto-rejected, not silently taken (can flip the sign of the change, not merely its magnitude). 2026-07-21 control: $R_{\text{geo}} = 8/\sqrt{53} \approx 1.099$, a 109.89% relative error.

RootNativeChirality_v1.7 — Root-native chirality (v1.7) — chirality grading and exact no-go

tier: Th_coqc; status: current

The chirality grading Γ_T GROWS from the ordered triple $(\Gamma^2=I, \Gamma^\dagger=\Gamma, R \Gamma = -\Gamma)$, no Γ^5 imported; an exact no-go shows an unbroken tape-reversal makes the two orientation sectors carry equivalent weak reps, so grading alone gives no chiral gauge asymmetry -- it takes an orientation-odd order χ , and $P_w = (I - \chi \Gamma)/2$ makes $SU(2)$ act on one sector.

RootToTrunk — Root-to-trunk progression diagram

tier: untagged; status: current

$P0: \exists a, b : a \neq b \rightarrow \text{distinguishability} \rightarrow \text{asymmetry} \rightarrow \text{temporal ordering} \rightarrow \text{retention/causal memory}$
 $\tau_c > 0 \rightarrow \text{finite causal graph } L_R \rightarrow \text{ROOT-3: discrete stepper MQ.08} \rightarrow \text{telegraph coarse-grain} \rightarrow \text{UNIVERSAL SPINE PDE}$

SI7Vectors — VII.1 — SI 7-Base Dimension Vectors

tier: untagged; status: current

Dimension vector: $[L, M, T, I, \Theta, N, J]$ with $L=\text{length(m)}$, $M=\text{mass(kg)}$, $T=\text{time(s)}$, $I=\text{current(A)}$, $\Theta=\text{temperature(K)}$, $N=\text{amount(mol)}$, $J=\text{luminosity(cd)}$. Derived examples: force $[1, 1, -2, 0, 0, 0, 0] \rightarrow \text{Newton}$; energy $[2, 1, -2, 0, 0, 0, 0] \rightarrow \text{Joule}$; power $[2, 1, -3, 0, 0, 0, 0] \rightarrow \text{Watt}$; pressure $[-1, 1, -2, 0, 0, 0, 0] \rightarrow \text{Pascal}$

SM_DAG — V.21 Standard Model — root-native DAG and gauge-as-automorphism ($\text{Aut}(F, O)$)

tier: Th_coqc; status: current

$\text{Aut}(F, O) = \{ h : 0 \circ h = 0, h \circ F = F \circ h \}$ -- the internal symmetry, grown from the exact-quotient discipline, never posited. Spine of the DAG: $\delta_R \rightarrow \text{typed state} \rightarrow \text{ordered paths} \rightarrow [\text{R3 readout-presence}, \text{automorphism}, \text{FIRST GATE}] \rightarrow \text{localized transport} \rightarrow \text{connection} \rightarrow \text{holonomy} \rightarrow \text{curvature} \rightarrow \text{gauge algebra} / \text{quantum completion} / \text{scale flow} \rightarrow \text{representations} \rightarrow \text{chirality/anomaly} \rightarrow \text{order-parameter breaking } G \rightarrow H \rightarrow \text{mass/coupling structure} \rightarrow \text{generations/mixing} \rightarrow \text{empirical group-ID} \rightarrow \text{bounded SM claim}$. Verdict: Root-Native Standard-Model Architecture: Substantial Conditional and Node-Level Closure, not a complete result; end-to-end physical SM from the UNRESTRICTED root is still 0% OPEN.

SM_Discovery_Pipeline_v0.4 — SM Discovery Pipeline v0.4 — finite blind discovery of hypercharges and cubic anomaly cancellation

tier: Th_coqc; status: current

BLIND discriminators (group name, reps, charges, Left/Right never inputs) recover on finite fixtures: the algebra (center 1 + ideals $8+3 \Rightarrow u(1)+su(3)+su(2)$; negative control $1+3+3 \Rightarrow \text{FAIL}$); chirality ($\eta_\chi = 1$ chiral vs 0 vectorlike, conditional on a given orientation op); the reps $(3,2)+(1,2)+(3,1) \times 2 +$

and the hypercharges $q=1/6$, $l=-1/2$, $u=2/3$, $d=-1/3$, $e=-1$ FORCED by anomaly + coupling closure with the cubic $[U(1)]^3$ anomaly = 0 exact (Th_coqc). A finite radiative log-det engine gives raw curvatures $(r_1, r_2, r_3) = (-153/20, -243/20, -216/5)$ and $\text{Tr } K_0^{-1} = 27/10$ -- finite diagnostic, NOT physical beta-function

SM_G0_order_defect — SM-G0/G0.6 — order-defect from ordered composition (borrow #2 discharged)

tier: Th_coqc; status: current

InfoOrderDefectFromComposition.v (Print Assumptions Closed over Q): ordered composition is plain 2x2 rational matrix product (no non-commuting term written in); associativity is a theorem; $K(X, Y) = XY - YX$ is bilinear + antisymmetric; Jacobi is DERIVED from associativity; non-commutativity is emergent (a witness pair fails to commute); commuting transports imply zero defect (AP20's own control, now derived). $c_{\text{self}}/c_{\text{geo}} = 1$ is not root-derived; the non-abelian input is relocated, not eliminated.

SelfClosure — VIII.6 — Self as a Closure Property, Not a Substance

tier: Dr; status: current

Self, as used here, is not a substance, a location, or a homunculus. It is the closure property of the $\tau_c \hat{H}$ loop: the fact that Ω_H both (a) results from the loop's own prior passes and (b) conditions the loop's next pass, at a stable, characteristic causal-memory timescale $\tau_c \hat{H}$.

SemanticLanes — VII.3 — Semantic Lanes

tier: untagged; status: current

Lanes: MECHANICAL (forces, momenta, kinematics), THERMAL (temperature, entropy, heat, Landauer), ELECTROMAGNETIC (field, current, voltage, flux), INFORMATION (bits, entropy, τ_c obstruction), BIOLOGICAL (body-scale τ_c , biochemical rates), FINANCIAL (τ_R , price, volatility)

SignBridge — Continuum limit and provenance: PGFT v0.5 $\rightarrow$ current form, sign bridge

tier: untagged; status: current

PGFT v0.5 (old): $M \Phi_{tt} + D \Phi_t - K \Delta \Phi + \nabla V(\Phi) = J$; Current form: $M \Phi_{tt} + D \Phi_t + K L_R \Phi + \nabla V(\Phi) = J - \eta$; Sign bridge: $-K \Delta \Phi \leftrightarrow K L_R \Phi$ (graph-to-continuum gate declared)

SpinePDE — The universal spine PDE (Eq.49)

tier: untagged; status: current

$M \partial_t^2 \Phi + D \partial_t \Phi + K \cdot L_R \Phi + \nabla V(\Phi) = J - \eta$ (memory $\cdot$ smoother $\cdot$ graph/operator $\cdot$ nonlinear = source $\cdot$ residual). Gauge-covariant reading: $\partial_\mu \rightarrow \mathcal{D}_\mu = \partial_\mu + A_\mu$

Step2Record — VIII.3 Step 2 — RECORD (N2 record law + three-way noise decomposition)

tier: untagged; status: current

$M_H[n] = K_H \cdot \theta(E[n]) + \eta_H[n]$ (always lossy; $\epsilon_{\text{tot}} = \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}} > 0$). η_{sel} -- selection loss ($K_H \theta$ admits only part of $E[n]$ past threshold); η_{map} -- mapping loss (K_H 's finite-resolution map folds distinct raw states together); η_{self} -- self-noise (B_H 's own retained history injects noise not originating in $E[n]$).

Step3Phenomenalize — VIII.3 Step 3 — PHENOMENALIZE

tier: Dr; status: current

$P_H[n] = \Phi_H(M_H[n])$ (qualia / subjective-report layer). The record is turned into whatever a subjective report is made of.

Step4Resonance — VIII.3 Step 4 — RESONANCE

tier: untagged; status: current

$G_i^H = 1 / (1 + (\omega_i \tau_c^H - 1)^2)$ (information at matching frequency is amplified; others decay). This is the r0 default gain function (see VIII.5 for the r1 log-normal alternative).

Step5Policy — VIII.3 Step 5 — POLICY (optimal action)

tier: Dr; status: current

$\pi^*[n] = \operatorname{argmin} [E\|\operatorname{Res}\|^2 + \lambda C + \rho \operatorname{Risk} - \nu \operatorname{Val} + \mu \operatorname{RepairLoss}]$, where $E\|\operatorname{Res}\|^2$: epistemic residual; λC : cost of action; $\rho \operatorname{Risk}$: risk of irreversible damage; $\nu \operatorname{Val}$: value of outcome; $\mu \operatorname{RepairLoss}$: loss of repairability. New reading, $[Dr]$, flagged PROPOSED and untested (T1): if the terms are genuinely coupled, Π_H is better modeled as a multimode L_R operator over a policy-state vector, with off-diagonal skew coupling captured by $L_R = L_R^+(+) + L_R^*(-)$ under the retention metric G (not a naive transpose); the scalar form is kept as the working, currently-used approximation.

Step6ActRepair — VIII.3 Step 6 — ACT + REPAIR

tier: untagged; status: current

Action updates world-model Ω_H ; residuals drive repair. Whatever residual Res remains unresolved after the action becomes the input the next repair cycle works against; $dR_H/dt = -\mu \cdot \|\operatorname{Res}_H\|^2$ (VIII.4) is the rate law for this repair.

SurfaceAutomaton_v1.1 — Surface automaton for $\mu_4^{\text{admissible}}$ (v1.1) — lower-bound spectral radii

tier: Th_coqc; status: current

$\mu_4^{\text{admissible}}$ = the Perron growth of a frontier automaton whose state b_s is exactly the minimal sufficient quotient. First real 4D spectral radii (single-sheet sector): canonical connector $z^3 + z^2 + 3z - 1 = 0 \Rightarrow \mu_{\text{can}} = 3.38298$; shortest-paths $2z^3 - z^2 + 4z - 1 = 0 \Rightarrow \mu_{\text{short}} = 3.87513$ (finite strips $H=0..4$ give 1, 2.846, 3.348, 3.555, 3.660, converging). This is a LOWER bound (omits bubbles/handles/branching), giving bracket $3.87513 \leq \mu_4^{\text{admissible}} \leq 54.366$; does not yet replace 20e in the certificate.

T1 — T1 — metric-G skew-decomposition abstract closure (retention_skew_quadratic_form_v1)

tier: Th_coqc; status: current

$z^T (L_R^*(-)) z = 0$ under the retention inner product, for the G -adjoint (not naive-transpose) skew part -- proven as a general fact about any G -adjoint decomposition, given G invertible. InfoRetentionMetric theorem `retention_skew_quadratic_form_vanishes`, axiom-free (Print Assumptions: Closed under the global context), fully abstract (no Section/Variable/Hypothesis, explicit forall-premises). Companion numeric fixture `t1_metric_g_skew_verify.py` confirms it on a concrete $G \neq I$ case AND confirms the negative control: the naive-transpose split does NOT have this property when $G \neq I$.

T1b — T1b — concrete instantiation gap, and T1b milestone #1 (relativity domain, $G=I$ derived)

tier: finite_diagnostic; status: current

What T1 does NOT yet close: it does not construct the concrete retention metric G for the real G_n operator of II.8a ($L_{\{G_n\}} \text{ tensor } I_F + I_{\{G_n\}} \text{ tensor } C_F + C_{\text{int},n}$), confirm G is invertible there, or verify that G_n 's actual G -adjoint matches the assumed form. T1b milestone #1 (2026-07-24, finite_diagnostic relativity domain, proof of concept): `T1b_relativity_instantiation_verify.py` closes the first concrete instance of the full chain -- a scalar retained-load functional $F \rightarrow \text{Face 8's Hessian-readout (RDL_MetricR metric_form_readout, Th_coqc)} \rightarrow \text{the retention metric } G \rightarrow \text{T1's skew-decomposition} \rightarrow \text{using only pre-existing, independently verified objects. Result: Face 8 reads } G=I \text{ exactly off obstruction's Hessian at every sampled base point and step size -- } G=I \text{ is derived, not assumed, matching the same } G=I \text{ basis convention SM_INFORMATION_PHILOSOPHY_MASTER.md section 3.2 already uses to derive SU(3). } U_C \text{'s skew part then vanishes under this derived } G \text{ exactly as T1 predicts.}$

T2 — T2 — endogenous state-dependent $L_R[I_R]$ (remaining open two-field-wall gap)

tier: Open; status: current

What remains open is the harder, nonlinear case: endogenous state-dependent $L_R[I_R]$, where the operator itself depends on the state it is acting on. Multimode L_R with the G-metric skew split handles the linearized cases of two-field systems without any change to the master PDE; the general nonlinear case is not handled. (Elsewhere, V.19's biology registration reports T2 endogenous $L_R[\Theta]$ CLOSED via operator-memory augmentation: I-only fails, augmented (I, Θ) closes -- a domain-specific instance, while the general nonlinear case named here remains open.)

TapeKineticOperator_v1.8 — Tape kinetic operator (v1.8) — Ginsparg-Wilson relation and no species doubling

tier: Th_cogc; status: current

A root-native D_T obeys an exact Ginsparg-Wilson relation $\Gamma(I-V) + (I-V)\Gamma = (I-V)\Gamma(I-V)$, so chirality survives at finite spacing without naive anticommutation; the free fixture has no species doubling ($0 < m_0 < 2r \Rightarrow$ one physical zero; $16 \rightarrow 1+15$ at $d=4$) -- the Clifford relation $\{A_\mu, A_\nu\} = 2\delta$ is forced by isotropy, not imported.

Telegraph — Coarse-grain trunk: telegraph equation

tier: finite_diagnostic; status: current

$$\tau_c \partial_t^2 u + \partial_t u = D \nabla^2 u$$

TermTier-D — Term-by-term tier: $D \partial_t \Phi$ (damping/smoothing/drain)

tier: finite_diagnostic; status: current

$D \partial_t \Phi$ -- damping/smoothing/drain. Well-behaved as a linear damping term at the telegraph/spine level; deeper identity is $1/\tau_c$ -flavored memory decay. D is NOT the same object as the first-order relaxation constant τ_R that governs RTPE turbulence.

TermTier-J — Term-by-term tier: J (source/drive)

tier: Dr; status: current

J -- source/drive. External forcing/injection, always supplied from outside the trunk -- never derived from M , D , K , L_R , V themselves.

TermTier-KLR — Term-by-term tier: $K \cdot L_R \Phi$ (graph/operator transport)

tier: Th_cogc; status: current

$K \cdot L_R \Phi$ -- graph/operator transport; principal symbol $\rightarrow$ metric readout. L_R is a full operator on a multimode state, not a scalar $\lambda\phi$ (Scalar-Eigenmode Reduction Error if narrowed). Proposed fix: $L_R = L_R^{(+)} + L_R^{(-)}$ under retention metric G .

TermTier-M — Term-by-term tier: $M \partial_t^2 \Phi$ (memory/inertia)

tier: finite_diagnostic; status: current

$M \partial_t^2 \Phi$ -- memory/inertia. M is POSITED, not derived (8 failed forcing attempts). Mass is a readout of τ_c via $m = \hbar/(2c^2\tau_c)$; τ_c is discrete and logically prior to mass. Only the quantum regime meaningfully exercises M (D/M vs QuTiP residual 7.6×10^{-4}). Turbulence does NOT live in this term.

TermTier-eta — Term-by-term tier: η (residual/turbulence/irreducible mismatch)

tier: finite_diagnostic; status: current

η -- residual/turbulence/irreducible mismatch. The slot that keeps the equation honest about what it does not close. A FAIL-ABLE gate law applies: a Type-P evidence gate needs both a passing and

a failing machine-derived control; a claim resting on η alone with no demonstrated failure mode is Type-U convention, not evidence.

TermTier-gradV — Term-by-term tier: $\nabla V(\Phi)$ (nonlinear potential)

tier: finite_diagnostic; status: current

$\nabla V(\Phi)$ -- nonlinear potential/cascade/restoring structure. Turbulence correction (2026-07-21): turbulence is NOT carried by the linear $M \partial^2_t \Phi$ term; it lives here, in the paraproduct term $(u \cdot \nabla)u$. The inertial quantity governing turbulence is the first-order relaxation constant τ_R , not M .

TrialitySpectralFlow_v0.6 — Triality spectral flow (v0.6) — Block-Scale Existence Theorem

tier: Th_cogc; status: current

Serial blocking is convolution $\Rightarrow$ Peter-Weyl $\Rightarrow a_R^{\{m\}} = a_R^m \Rightarrow \rho_t(b) = \rho_t^{b^2}$, so information cost grows as area $I_t(b) = b^2 * I_t(1)$. Block-Scale Existence Theorem: for any $0 < \rho_t < 1$ and finite μ_4 there is a coarse scale $b_* > \sqrt{\log(\mu_4) / -\log(\rho_t)}$ with $c_t(b_*) < 1$ -- spectral contraction beats surface entropy (analytic PASS: $\kappa=0.01$, $b=2 \Rightarrow c \leq 0.004276$).

UnifiedForceClosure_v0.1 — Unified Force Closure v0.1 — internally closed at finite/typed level, chi4 completeness score

tier: untagged; status: current

One action, one block master equation, four forces as orthogonal projection readouts $F_{\text{all}} = F_G + F_{\text{EM}} + F_W + F_S + F_{\text{res}}$, with a falsifiable completeness score χ_4 ($\chi_4 = 1$ on a strong+weak fixture; $\chi_4 < 1$ when a hidden fifth interaction leaks). Witnessed: automorphism group, connection-law-holonomy invariants, order-defect+Jacobi, covariant Laplacian, the selected-state h/m split -- with six failing controls. Physical four-force unification is OPEN.

UnifiedForce_action — V.22 Unified Force — one action, four projections (master statement)

tier: Th_cogc; status: current

$S_{\text{UF}} = S_{\text{DRL}} + S_{\text{Theta}} + S_U + S_{\text{Sigma}} + S_{\text{cut/tape}}$, $\Delta S_{\text{UF}} / \Delta Z_n = J_{\{C,n\}}[Z_n]$; $F_D = P_D(-\text{grad}_Z U_{\text{UF}} + J_C)$: $P_{\text{geo}} \rightarrow \text{gravity}$. $P_{\text{center}} \rightarrow \text{EM}$. $P_{\text{chiral,broken}} \rightarrow \text{weak}$. $P_{\text{simple,unbroken}} \rightarrow \text{st}$.

UpperAutomaton_v1.2 — Upper automaton (v1.2) — entropy ceiling 54 to 7.084

tier: Th_cogc; status: current

A 4D edge touches 6 plaquettes (1 in, 5 free x_i in Z_3); Z_3 link closure $\sum x_i = 2 \pmod 3$ has $3^4=81$ solutions; splitting by the number k of nonzero new plaquettes gives the branching polynomial $B(z) = 5z + 10z^2 + 30z^3 + 25z^4 + 11z^5$ ($\text{Sum}=81$; the $k=2$ coefficient 10 is the Z_3 triple junction $1 \rightarrow 1+1$). A scalar majorant makes $B(z) < 1$ a contraction; critical root $B(z_+) = 1 \Rightarrow z_+ = 0.141161 \Rightarrow \mu^+ \leq 7.084097$ (upper bound, over-counts non-closing surfaces). Squeezes the bracket to 3.87513 $\leq \mu_4^{\text{admissible}} \leq 7.08410$ and widens the certificate window to $\kappa < 0.321687$ (about 6x the old 20e window).

V — Forced Set V — cross-term dominance in a local cell

tier: untagged; status: current

$\text{cell_diag} \perp \text{cell_cross_polarization}$ -- the cross term in a 2×2 local cell separates cleanly from the diagonal term (`InfoCrossTermDominance`)

V.14.FailAbleGateLaw — V.14 The FAIL-ABLE Gate Law (Type-P vs Type-U evidence)

tier: untagged; status: current

A genuine, Type-P evidence gate for any bridge or reduction claimed in this Part requires both a passing control and a failing control, and both must be machine-derived -- not asserted, not merely imagined as a contrast case. A gate that can only be shown passing, with no demonstrated way for the same machinery to fail on a genuinely different input, is Type-U, a convention dressed as a finding. Where a leaf claims a bridge with only a passing case shown and no demonstrated failing case, that claim is Type-U convention and should be read at Dr tier at best until a failing control is produced.

V0_DomainDiscoveryEngine — V.0 Domain-discovery engine (finds q_α from a raw transition tape)

tier: finite_diagnostic; status: current

Given a raw tape of transitions $(N, S) \rightarrow (N', S')$, the engine discovers: (1) how many variables the domain minimally needs via an exact-gate closure test; (2) whether channels interact via a mixed second-difference test; (3) the minimal exact update law over Q that reproduces every observed transition with no residual; (4) conservation laws as the kernel of the discovered update. Adversarial battery: correctly recovered true laws $N*S/8$, $N^2/8$, an affine law, and a three-variable case; never fabricated a plausible-looking wrong law such as $N*S/16$; held out exact on unseen transitions; refused to fit non-polynomial laws.

VI — Forced Set VI — cubic linearization residual

tier: untagged; status: current

$qshift(cubic\ residual) \geq 0$ -- the residual linearized from the cubic potential is nonnegative, strictly positive off the fixed point (InfoCubicLinearization)

VI.7.FailAbleGateLaw — VI.7 — The Fail-Able Gate Law

tier: untagged; status: current

a gate only counts as Type-P (a genuine evidence-bearing predicate) if it has both a machine-derived passing control AND a machine-derived failing control; a gate with only passing cases must be labeled Type-U (a convention or definition dressed up as evidence) until a genuine failing control is produced and correctly rejected

VII — Forced Set VII — cut monotone under edge growth

tier: untagged; status: current

cut monotone under edge growth -- screened strain stays nonnegative as the graph grows (InfoCutGrowth)

VIII — Forced Set VIII — graph Noether theorem

tier: untagged; status: current

graph permutation symmetry $\Rightarrow$ conserved quantity -- a symmetry under graph permutation forces a conserved quantity, Noether, on a graph (InfoGraphNoether)

VerdictGate — VI.3 — Verdict gate (DECIDE / ABSTAIN / ESCALATE)

tier: untagged; status: current

DECIDE when F_{ep} high AND Re_{ep} low $\rightarrow$ strong + uncontested; ABSTAIN when F_{ep} low AND k_{ep} low $\rightarrow$ no signal (for free); ESCALATE when Re_{ep} high $\rightarrow$ contested $\rightarrow$ pay the LLM. Qualitative result: only the contested/under-damped steps pay the large model.

WP.S1.ForbiddenOrder — Forbidden order

tier: untagged; status: current

Textbook Name -> Import Final Law -> Fit Parameters -> Declare Derived

WP.S1.InformationPhilosophyConstitution — Information philosophy constitution (P1-P10)

tier: untagged; status: current

InformationPhilosophyConstitution: P1 no_built_in_domain_labels ('The root contains no built-in labels such as physics, chemistry, biology, mind, AI, or society.');

P2 difference_before_entity ('Difference precedes the name of what differs.');

P3 history_before_container_time ('Ordered change precedes time treated as a container.');

P4 retention_before_persistence_semantics ('A state becomes reportable only when some distinction is retained.');

P5 reader_creates_domain_access_not_root_truth ('A reader makes structure accessible; it does not retroactively create the root.');

P6 domain_is_translation ('A domain is a minimal sufficient dynamically closed translation.');

P7 meaning_after_readout ('Meaning arises after readout under context; it is not stored in the root.');

P8 translation_must_account_for_loss ('Every non-exact translation must identify where distinctions are lost.');

P9 claim_cannot_borrow_certainty ('A claim at one layer may not borrow certainty from another layer.');

P10 report_is_last ('A report is emitted only after retention, structure, translation, calibration, and checking.');

each with a stated machine_consequence.

WP.S1.MandatoryDerivationOrder — Mandatory derivation order

tier: untagged; status: current

Retention -> Structure -> Translation -> Meaning -> Report

WP.S10.ContractionBound — Contraction certificate — operator bound

tier: untagged; status: current

$0 \leq T_RG \leq I \text{ (or) } ||T_RG||_op \leq 1$

WP.S10.NormalizedTransfer — Normalized transfer operator

tier: untagged; status: current

$T_RG = T_+ / \text{zeta}$, $\text{zeta} > 0$, $\text{zeta} \geq \text{lambda_max}(T_+)$. 'The normalization zeta MUST be declared and recorded; it may not be hidden.'

WP.S10.PhysicalProjector — Physical projector conditions

tier: untagged; status: current

$P_phys^\dagger = P_phys$, $P_phys^2 = P_phys$. 'Projector construction MUST be traceable to a declared redundancy or constraint sector.'

WP.S10.PhysicalProjectorGate — Physical projector gate

tier: untagged; status: current

PhysicalProjectorGate checks: hermitian, idempotent, removes_only_declared_redundancy, preserves_declared_fail_code INVALID_PHYSICAL_PROJECTOR.

WP.S10.PositiveKernelDefinition — Positive kernel definition

tier: untagged; status: current

$K(X_+, X_+) = \sum_Y A(Y|X_+) * \text{conj}(A(Y|X_+))$

WP.S10.PositiveKernelPositivity — Positive kernel positivity

tier: untagged; status: current

$K = A^{\dagger} A \geq 0$

WP.S10.PositiveTransferOperator — Positive transfer operator

tier: untagged; status: current

$T_+ = P_{\text{phys}} e^{-V_{\text{hat}}/2} K e^{-V_{\text{hat}}/2} P_{\text{phys}} \geq 0$, $V_{\text{hat}} = V_{\text{hat}}^{\dagger}$

WP.S10.ReflectionPositiveTerminology — Reflection-positive terminology requirement

tier: untagged; status: current

'The term reflection-positive is permitted only when the following are supplied: reflection map, positive half-algebra, gluing rule, compatible measure/action. If only $A^{\dagger} A$ is established, use the term positive midpoint kernel.'

WP.S10.ResonanceExtension — Resonance extension requirement

tier: untagged; status: current

'A resonance as a complex pole does not follow directly from a positive self-adjoint spectrum; it requires a declared extension.' ResonanceExtension requires: resolvent_or_scattering_object, analytic_continuation_sheet_and_boundary_conditions; status_if_absent NOT_AVAILABLE.

WP.S10.SpectralClassification — Spectral classification

tier: untagged; status: current

SpectralClassification: isolated_eigenvalue -> stable_retained_mode; isolated_below_continuum -> bound_or_composite_mode; zero_edge -> scale_free_or_gapless_mode; positive_continuum_edge -> threshold_perfect; no_projected_support -> no_standalone_readout.

WP.S10.SpectralEquation — Spectral reader eigenequation

tier: untagged; status: current

$T_{\text{RG}} |\phi_s\rangle = \lambda_s |\phi_s\rangle$

WP.S10.SpectralGapFormula — Spectral gap readout

tier: untagged; status: current

For $a > 0$ and $0 < \lambda_s \leq 1$: $\epsilonpsilon_s = -(1/a) \log(\lambda_s)$

WP.S11.BarePotential — Bare potential

tier: untagged; status: current

$V_0(r) = \alpha r + \beta r^2$, $\beta > 0$

WP.S11.EffectivePotential — Effective potential

tier: untagged; status: current

$V_{\text{eff}}(r) = V_0(r) - \sum_{\mu} \log Z_{\mu}(r)$

WP.S11.EffectivePotentialDiagonal — Effective potential, diagonalized eigenvalue form

tier: untagged; status: current

For eigenvalues $\lambda_{\mu} r$ with multiplicity d_{μ} : $V_{\text{eff}}(r) = \alpha r + \beta r^2 - \sum_{\mu} d_{\mu} \log(1 + \lambda_{\mu} r)$

WP.S11.EffectivePotentialSlope — Effective potential slope at origin

tier: untagged; status: current

$$V_{\text{eff}}'(0) = \alpha - \sum_{\mu} d_{\mu} \lambda_{\mu}$$

WP.S11.FluctuationDeterminant — Fluctuation determinant

tier: untagged; status: current

$$Z_{\mu}(r) = \det(I + K_{\mu}(r))$$

WP.S11.FluctuationKernel — Fluctuation kernel

tier: untagged; status: current

$$K_{\mu}(r) = B_{\mu}(r) B_{\mu}(r)^{\dagger}$$

WP.S11.OrderOnsetCriterion — Local order onset criterion

tier: untagged; status: current

$$\sum_{\mu} d_{\mu} \lambda_{\mu} > \alpha$$

WP.S11.OrderParameter — Order parameter

tier: untagged; status: current

$$r = \text{Sigma}^{\dagger} G \text{Sigma} \geq 0$$

WP.S11.OrderSelectionClaimScope — Order selection claim scope

tier: untagged; status: current

OrderSelectionClaim: exact_scope [declared_effective_potential, local_slope_at_origin]; not_implied [global_uniqueness, empirical_phase_identity, physical_energy_interpretation].

WP.S12.CoarseGrainingDefect — Coarse-graining defect

tier: untagged; status: current

$$\epsilon_{\text{cg}} = || O_D^{\wedge}(b) C_b - O_D ||$$

WP.S12.CoarseGrainingMap — Coarse-graining map

tier: untagged; status: current

$$Z^{\wedge}(b) = C_b[Z]$$

WP.S12.EffectiveAction — Optional formal effective action

tier: untagged; status: current

$S_{\text{RG}}^{\wedge}(b) = -\log \int \{C_b[Z] = Z^{\wedge}(b)\} e^{\{-S_{\text{RG}}[Z]\}} d\mu[Z]$ ('OPTIONAL_FORMAL unless the Domain Card supplies the measure, domain of integration, finiteness/integrability conditions, and a computable approximation rule.')

WP.S12.FixedPoint — Renormalization fixed point

tier: untagged; status: current

$$R_b(\theta_*) = \theta_*$$

WP.S12.IrrelevantDistinctionCertificate — Irrelevant distinction certificate

tier: untagged; status: current

$$|| \delta \theta_n || \leq \rho^n || \delta \theta_0 ||, \rho < 1$$

WP.S12.ModalReductionDefect — Modal reduction defect

tier: untagged; status: current

$\epsilon_{\text{mode}} = ||(I-P)GP|| + ||PG(I-P)||$

WP.S12.ModalReductionGate — Modal reduction admissibility gate

tier: untagged; status: current

A scalar or single-mode reduction is admissible only when: (1) the quotient is valid; (2) one mode is sufficient for the declared readout; (3) off-diagonal, skew, and moving-basis currents are below tolerance; (4) geometry is fixed or externally prescribed; (5) no required retained nonlinear transfer exists between modes. ModalReductionGate: pass_when $\epsilon_{\text{mode}} \leq \text{preregistered_tolerance}$ and all five conditions hold; fail_code SCALAR_EIGENMODE_REDUCTION_ERROR.

WP.S12.NoEarlyCollapseLaw — No-early-collapse law

tier: untagged; status: current

If an encoding E satisfies $E(z) = E(z')$ but $z \neq z'$, no downstream solver can recover the distinction from that input alone. NoEarlyCollapse rule: 'compress only after equivalence is audited'; correction_if [enlarge_candidate_state, restore_lineage_or_tape, increase_resolution]; forbidden_correction [redefine_output_target_law].

WP.S12.ReadoutPreservingCoarseGraining — Readout-preserving coarse-graining condition

tier: untagged; status: current

$0_D(b) \circ C_b = 0_D$

WP.S13.DomainCard — Executable Domain Card schema

tier: untagged; status: current

DomainCard schema_version 1.0 with required blocks: identity (domain_id, version, parent_root_version, status); question (target, horizon, resolution, interventions, tolerance, context); candidate_state (encoder, slot_activation, fields, memory_slots, boundary_slots, lineage_slots); carriers; readers; invariants; symmetries; constraints; dynamics (root_kernel, graph_operator, field_operator, coefficients, currents, allowed/forbidden reductions, physical_projector, midpoint_amplitude, potential_operator); calibration (native_units, decoder, freeze_hash, held_out_outputs); semantics (labels ATTACH_AFTER_TRANSLATION, defect_metrics; evidence (positive/negative controls); provenance (root_derived, architecture_declared, empirical_input, calibration_only, external_comparison, open_hypothesis); forbidden_imports [domain_final_target_answer_as_parameter, semantic_name_before_gate, fit_each_output_separately, physical_unit_without_checker_outcome_visible_to_maker].

WP.S14.Gate1 — Gate 1 — No-Free-Domain-Law

tier: untagged; status: current

$\Delta_R + \text{Retention} \not\Rightarrow (F, C, V, \theta)_{\text{domain}}$. 'If more than one compatible model remains: status UNRESOLVED, required_next_input [interaction_tape, intervention, observation, explicit_postulate].'

WP.S14.Gate2 — Gate 2 — Three-Valued Admissibility

tier: untagged; status: current

'Return only ADMITTED, OBSTRUCTED, or UNRESOLVED; do not guess.'

WP.S14.Gate3 — Gate 3 — Context-Indexed Law

tier: untagged; status: current

$F_n = F_n(Z_n, c_n, T_n)$. 'Validity in c does not imply validity in c_prime.'

WP.S14.Gate4 — Gate 4 — State Sufficiency

tier: untagged; status: current

Suff_{D,L}(Z_D[^]cand; c, T) in {1, 0, perp}. Audit at least: readout sufficiency, dynamic sufficiency, intervention sufficiency, delayed-return/memory sufficiency, coupling sufficiency, geometry/basis sufficiency, boundary sufficiency.

WP.S14.Gate5 — Gate 5 — Invariant Completion

tier: untagged; status: current

'If a quotient merges states separated by a required invariant, refine the partition.'

WP.S14.Gate6 — Gate 6 — Query-Relative Symmetry

tier: untagged; status: current

H_D = { h : O_D(h z) = O_D(z) , h F = F h }

WP.S14.Gate7 — Gate 7 — Calibration Firewall

tier: untagged; status: current

y_D[^]known = U_D(r_{native}; theta_D, c, C_D[^]cal) ; H_{cal} = Hash(U_D, theta_D, training, units, protocol).

'A native informational value is not a physical, chemical, or biological observable before independent calibration.'

WP.S15.CompileDomainAlgorithm — COMPILE_DOMAIN algorithm

tier: untagged; status: current

FUNCTION COMPILE_DOMAIN(root, raw_tape, domain_card_draft): validate root/philosophy; capture phenomenon without mechanism labels; strip domain vocabulary into retained roles; freeze question/horizon/intervention; build candidate state encoder; run state sufficiency audit (enlarge+restart on FAIL, abstain on UNRESOLVED); discover minimal field count and coupling structure; discover candidate partition via future readout signatures; discover query-relative automorphisms; build quotient q_D; build coarse dynamics F_D#; compute epsilon_dyn/read/inv; refine partition until exact/tolerated/impossible; enumerate invariant closure channels; build full operator without premature diagonalization; audit modal and moving-basis defects; run native dynamics; build positive transfer if requested; calibrate only after native outputs are frozen; run independent held-out checker; return DomainRunReport + ClaimLedger.

WP.S15.FieldCountRule — Minimal field count rule

tier: untagged; status: current

m_D = min { m : O_D and F_D close on X^(1:m) }. 'The number of fields is not the number of domains.'

WP.S16.BridgeError — Bridge error

tier: untagged; status: current

epsilon_{A<-B} = d_{Delta}(D_A , hat(D)_A)

WP.S16.BridgeLevels — Bridge evidentiary levels L0-L5

tier: untagged; status: current

BridgeLevels: L0 surface_analogy; L1 shared_variables_or_relations; L2 shared_transition_signature; L3 readout_preserving_map; L4 dynamically_commuting_quotient; L5 recoverable_bidirectional_equivalence. 'Equations that merely look alike may be reported at no more than L0/L1 until stronger gates pass.'

WP.S16.BridgeMap — Cross-domain bridge map

tier: untagged; status: current

K_{A<-B} : D_B x T -> hat(D)_A

WP.S16.BridgeReadout — Exact bridge — readout condition

tier: untagged; status: current

$O_{A\#} K_{\{A<-B\}} = D_{\{A<-B\}} O_{B\#}$

WP.S16.ExactBridgeDynamics — Exact bridge — dynamics commutation

tier: untagged; status: current

$K_{\{A<-B\}} F_{B\#} = F_{A\#} K_{\{A<-B\}}$

WP.S17.DefectRules — Defect classification rules

tier: untagged; status: current

DefectRules: exact = 'every required component equals zero in exact arithmetic'; numerical_pass = 'every required component <= preregistered tolerance'; fail = 'any required component > tolerance'; unresolved = 'a required component cannot be identified'; report rule = 'never collapse named defects into one scalar without retaining the full vector'.

WP.S17.DefectVector — Canonical defect vector

tier: untagged; status: current

epsilon_D = (epsilon_suff, epsilon_dyn, epsilon_read, epsilon_inv, epsilon_bridge, epsilon_cal, epsilon_mode, epsilon_geo, epsilon_solve)^T

WP.S18.FailAbleGateLaw — Fail-able gate law

tier: untagged; status: current

FailAbleGateLaw: Type_P_evidence_requires: [machine_derived_passing_control, machine_derived_failing_control]; Type_U_convention_when: [only_passing_control_exists, failure_case_cannot_change_gate_output]; tier_consequence = "may support finite or machine-checked status within scope", Type_U_pass = "supports at most architecture/conditional wording". "This law tests whether a gate can discriminate; Maker-Checker tests leakage and process bias. Both are required."

WP.S18.MakerCheckerInfoZero — Maker-Checker mutual information condition

tier: untagged; status: current

Info(Maker; Y_checker) = 0. "This is a protocol condition, not a proof that statistical mutual information is zero merely because a hash exists."

WP.S18.MakerCheckerRoles — Maker-Checker firewall roles

tier: untagged; status: current

Registrar freezes: question, allowed_inputs, forbidden_inputs, metrics, thresholds, failure_rules, dataset_split, software_versions. Maker may: propose_state, discover_partition, build_operator, calibrate_on_training_only, produce_prediction; must_not_see: held_out_outputs, checker_only_labels, answer_derived_threshold, post_freeze_report. Freeze hash_contents: model, code, configuration, parameters, prediction, score_orientation, normal_form, dependencies. Checker may: verify_hash, open_held_out, evaluate_controls (and further checker duties beyond the excerpted range).

WP.S18.ProtocolDeviation — Protocol deviation record update

tier: untagged; status: current

$T_{\{n+1\}} = T_n \boxplus \text{DeviationRecord}$. "Deviation records are append-only and MUST NOT be overwritten."

WP.S19.ClaimLedger — Claim ledger schema

tier: untagged; status: current

Claim: id REQUIRED; statement REQUIRED; status one_of [ROOT_AXIOM, DEFINITION, DERIVED_EXACT, MACHINE_CHECKED, EXACT_WITHIN_ARCHITECTURE, CONDITIONAL, FINITE_FIXTURE, NUMERICAL, CALIBRATION_ONLY, EMPIRICALLY_HELD_OUT, OPEN, FAILED_CONTROL, SUPERSEDED]; layer one_of [ONTOLOGY, STRUCTURE, DYNAMICS, TRANSLATION, EPISTEMIC, REPORT]; assumptions.explicit REQUIRED; provenance {root_parents, imported_inputs, calibration_inputs, external_comparisons} REQUIRED; derivation {parents, equations, algorithm_version} REQUIRED; verification {proof_or_derivation_artifact REQUIRED_IF_DERIVED_OR_MACHINE_CHECKED, positive_control REQUIRED_IF_EVIDENT, negative_control REQUIRED_IF_EVIDENTIARY, checker_independent REQUIRED_IF_HELD_OUT_OR_EMPIRICAL, defects REQUIRED_IF_COMPUTED_OR_TRANSLATED}; scope {finite_or_continuum, exact_or_toleranced, local_or_global, domain_specific_or_universal, context} REQUIRED; forbidden_upgrade.text REQUIRED.

WP.S19.LegacyTierMapping — Legacy tier mapping

tier: untagged; status: current

LegacyTierMapping: Ax -> ROOT_AXIOM; Df -> DEFINITION; Th_coqc -> MACHINE_CHECKED_SCOPE; finite_diagnostic -> FINITE_FIXTURE; Dr -> CONDITIONAL; Open -> OPEN. "A legacy Th without a proof artifact is not promoted automatically. Map it to CONDITIONAL or DERIVED_EXACT according to the derivation actually verified."

WP.S2.ArchitectureLayers — Architecture layers L0-L5

tier: untagged; status: current

ArchitectureLayers: L0_ontology {retained_distinction, ordered_history, reader_capability, admissibility, positive_retained_pairing} - 'What is retained, and what remains distinguishable?'; L1_structure {typed_state, equivalence_class, automorphism, transport, holonomy, closure} - 'What structure follows from retention and readout?'; L2_dynamics {action_interface, finite_reference_kernel, constraints, transfer_operator} - 'How does the structure evolve?'; L3_translation {domain_card, sufficiency, quotient, bridge, calibration} - 'How does a domain read the dynamics?'; L4_epistemic {registrar, maker, freeze, checker, controls} - 'Was the claim checked independently?'; L5_report {claim_ledger, scope, defect_vector, reproducibility_record} - 'What may actually be claimed?'

WP.S2.LayerNonBorrowingRules — Layer non-borrowing rules

tier: untagged; status: current

LayerNonBorrowingRules: 'L0 axiom does not prove L3 domain semantics'; 'L1 mathematical closure does not prove L2 is nature's unique dynamics'; 'L2 finite execution does not prove continuum existence'; 'L3 calibration fit does not equal held-out prediction'; 'L4 clean protocol does not prove the model true'; 'L5 wording MUST use the weakest load-bearing status'.

WP.S20.DomainRunReport — Mandatory Human-AI Run Report schema

tier: untagged; status: current

DomainRunReport: document_version, root_hash, domain_card_hash, input_hash, code_or_algorithm_hash REQUIRED; philosophy_guards {forbidden_import_check, semantic_order_check, lineage_preservation_check} PASS_FAIL; types {state_shapes, units, semantic_lanes} REQUIRED; solver {algorithm, time_step, pivot_min, condition_estimate, convergence, iterations}; quotient {partition, epsilon_suff, epsilon_dyn, epsilon_read, epsilon_inv} REQUIRED; operator {full_operator_used, epsilon_mode, epsilon_geo, holonomy_defects}; transfer {positivity_certificate, contraction_certificate, eigenvalues, persistence_costs} OPTIONAL; controls {positive, negative, expected_vs_observed} REQUIRED; calibration {used, freeze_hash, held_out_scope, epsilon_cal}; final.status one_of [PASS_EXACT, PASS_TOLERANCED, FAIL, UNRESOLVED, PROTOCOL_FAIL, DRIFT]; failure_codes, claim_ids, reproducibility_command.

WP.S21.GlobalRuntimeDAG — Global runtime DAG (text form)

tier: untagged; status: current

PHILOSOPHY CONSTITUTION -> ROOT PRIMITIVES (delta_R, Tape, Reader, Admissibility, G) -> TYPED RETAINED STATE (Phi, Psi, Theta, Sigma, L_cl) -> STATE SUFFICIENCY AUDIT [FAIL -> enlarge state; UNRESOLVED -> abstain; PASS ->] -> READER EQUIVALENCE -> QUERY-RELATIVE AUTOMORPHISM -> TRANSPORT U -> HOLONOMY H_C + DEFECT K_C -> CLOSURE INTERTWINERS -> FULL MULTIMODE OPERATOR [modal defect audit; moving-basis defect audit] -> DRL READER-RECORD STEP -> PHYSICAL PROJECTOR -> POSITIVE TRANSFER [positivity certificate contraction certificate] -> SPECTRUM / PERSISTENCE -> COARSE-GRAINING -> EXACT QUOTIENT AUDIT -> DOMAIN SEMANTICS + CALIBRATION -> MAKER FREEZE -> INDEPENDENT CHECKER -> DEFECT VECTOR [PASS; FAIL; UNRESOLVED] -> BOUNDED CLAIM LEDGER.

WP.S22.RG_EXECUTE — Top-level execution pseudocode formal steps

tier: untagged; status: current

From FUNCTION RG_EXECUTE(root_config, domain_card, observations, checker_bundle): key formal audit/build calls embedded in the control flow -- AUDIT_SUFFICIENCY(candidate_state, domain_card.question) returning status in {OBSTRUCTED, UNRESOLVED, ...}; BUILD_FULL_OPERATOR(graph_operator, field_operator, closures, interaction_operator, geometry=Theta, metric=G); AUDIT_EXACT_QUOTIENT(native_run, partition, readers, invariants, metrics) returning fail/unresolved; native_hash = HASH(native_run, quotient_report, transfer_r, freeze_hash = FREEZE_ALL(root_config, domain_card, native_hash, predictions); status = CLASSIFY(defects, domain_card.defect_metrics); claims = BUILD_CLAIM_LEDGER(status, ALL_PROVENANCE()). Pseudocode control-flow itself is excluded per task scope; only these formal calls/equational content are recorded.

WP.S23.FixtureA — Fixture A — Exact quotient (passing)

tier: untagged; status: current

fine_states [0,1,2,3]; quotient {0:A,1:A,2:B,3:B}; fine_dynamics {0:2,1:3,2:2,3:3}; reader quotient_label; expected coarse_dynamics {A:B,B:B}, epsilon_dyn=0, epsilon_read=0, status=PASS. Reason: fine states 0 and 1 are both in cell A and both transition to B; states 2 and 3 remain in B.

WP.S23.FixtureB — Fixture B — Quotient failing control

tier: untagged; status: current

fine_states [0,1,2,3]; quotient {0:A,1:A,2:B,3:B}; fine_dynamics {0:2,1:0,2:2,3:3}; expected reason "states in A have inconsistent coarse successors", failure_code LOST_INFORMATION, status FAIL. "This is the failing control for the exact quotient gate."

WP.S23.FixtureC — Fixture C — Holonomy

tier: untagged; status: current

representation 2x2_real_rotation; cycle [U_2_from_1: rotation_degrees_30, U_3_from_2: rotation_degrees_m, U_1_from_3: rotation_degrees_minus_20]; expected_pass: total_rotation_degrees=0, defect_frobenius_approx; status=PASS. failing_control: replace U_1_from_3 with rotation_degrees_minus_15; expected: total_rotation_degrees=15, trace_approx=1.9923893962, determinant_approx=1, defect_frobenius_approx=0.1233742584, status=FAIL_IF_FLAT

WP.S23.FixtureD — Fixture D — Positive transfer

tier: untagged; status: current

A=[[0.8,0.0],[0.0,0.5]]; V=[[0,0],[0,0]]; P_phys=identity_2; a=1.0; expected K=[[0.64,0.0],[0.0,0.25]], eigenvalues [0.64,0.25], persistence_costs_approx [0.4462871026, 1.3862943611], positivity PASS, contraction PASS. Failing control: supplied_K=[[1.0,2.0],[2.0,1.0]], expected_eigenvalues [3.0,-1.0], expected positivity FAIL, failure_code NON_POSITIVE_KERNEL.

WP.S23.FixtureE — Fixture E — Two-node DRL reader step

tier: untagged; status: current

L = [[1,-1],[-1,1]]. drl_two_node_linear: parameters M=1.0, D=0.5, K=1.0, dt=0.1; operator=[[1,-1],[-1,1], potential gradient/hessian zero; sources J=[0,0], R_Phi=[0,0]; initial Phi_{-1}=[1,-1], Phi_0=[1,-1]; expected Phi_1_approx=[0.9804878049,-0.9804878049], solver_status PASS. Reference continuation: Phi_0=[1,

Phi_1=[0.9804878049,-0.9804878049], Phi_2=[0.9427959548,-0.9427959548], Phi_3=[0.8885467107,-0.8885467107]
 Phi_4=[0.8196062829,-0.8196062829]. "For the declared discrete energy definition, the implementation
 SHOULD show a non-increasing trend after a preregistered transient tolerance. This fixture does
 not support a continuum theorem." Record-channel passing control drl_record_zero_invariant: M=1.0,D=0.5,K
 Psi_{-1}=[0,0], Psi_0=[0,0], R_Psi=[0,0], potential_hessian zero; expected Psi_1=[0,0], status PASS.
 Fail-closed record solve control drl_record_singular_step: M=1.0, D=20.0, dt=0.1, derived_A_Psi_scalar=0.
 expected status FAIL, failure_code SINGULAR_RECORD_SOLVE, silent_regularization_allowed false.

WP.S23.FixtureF — Fixture F — Modal reduction

tier: untagged; status: current

modal_reduction_pass: operator=[[2.0,0.0],[0.0,5.0]], projector=[[1,0],[0,0]], expected epsilon_mode=0,
 status PASS. Failing control modal_reduction_fail: operator=[[2.0,1.0],[-1.0,5.0]], projector=[[1,0],[0,0]],
 expected epsilon_mode_positive true, failure_code SCALAR_EIGENMODE_REDUCTION_ERROR, status FAIL_IF_SCALAR_

WP.S23.FixtureG — Fixture G — Geometry dominance

tier: untagged; status: current

moving_basis_no_motion: V_n=identity_2, V_next=identity_2; expected R_geo=0, fixed_basis ALLOWED.
 moving_basis_sign_flip: V_n=identity_2, V_next=rotation_matrix{cos:0.6, sin:0.8}, phi_n=[2.0,1.0],
 delta_phi_n=[0.5,-0.25]; expected true_delta_Phi=[-1.1,1.45], dropped_basis_term_first_component=0.5,
 R_geo_approx=1.0988845116, fixed_basis REJECTED.

WP.S24.AffineDecoder — Frozen affine decoder

tier: untagged; status: current

Phi_n = (r_n - 500*1) / 100.

WP.S24.ArchivalDisclosureRule — Archival disclosure rule for published held-out data

tier: untagged; status: current

ArchivalDisclosureRule: original_run held_out_status UNSEEN_BEFORE_FREEZE, evidence_status VALID_FOR_RECO
 after_publication held_out_status PUBLIC, evidence_status REPRODUCTION_ONLY; model_revision required
 [new_domain_card_version, new_freeze_hash, new_hidden_held_out_bundle, repeated_positive_and_negative_cont
 "Because the held-out values are printed in this archival whitepaper, they now support reproduction
 but no longer constitute untouched evidence for future revisions. Any parameter, operator, encoder,
 or threshold change MUST use a new unrevealed checker bundle and a new freeze hash."

WP.S24.CalibrationDesign — Calibration design equations (y_n , X_n , θ -hat OLS)

tier: untagged; status: current

For each training equation index $n=1, \dots, 9$: $y_n = -M \delta t^2 \Phi_n$, $X_n = [\delta t^c \Phi_n, L_{\{P_4\}} \Phi_n]$.
 After stacking all four components and all training indices: $\theta\text{-hat} = [D\text{-hat}; K\text{-hat}]$
 $= (X^T X)^{-1} X^T y$.

WP.S24.CalibrationProtocol — Calibration protocol results (D, K fit)

tier: untagged; status: current

CalibrationProtocol: split training_observations [0,10], training_equations [1,9], held_out_observations
 [11,17]; fixed_before_fit list (dataset_hash, calibration_certificate_hash, split_indices, M, dt,
 L_graph, zero_source, zero_potential, no_intercept, evaluation_thresholds, positive_control, negative_cont
 fitted_parameters D estimate 0.348925 (D>=0), K estimate 0.221249 (K>=0); solver ordinary_least_squares_no
 float64, design_rank 2, expected_rank 2, singular_values [1.681640, 0.455975], condition_number 3.688012,
 residual_variance_estimate 0.0000208442, parameter_standard_error D=0.009894 K=0.003119, approximate_95_pe

D=[0.329533,0.368318] K=[0.215135,0.227363]; training_fit equation_rmse 0.004437, equation_max_abs_error 0.010863, status PASS_TOLERANCED; freeze sha256 88b431490789ad33508efa6f09f7b73fce6762423909477fd0ef61abc6

WP.S24.CanonicalDataHash — canonical_data_sha256 anchor for frozen raw dataset

tier: untagged; status: current

canonical_data_sha256: 43b1aa7081ce1fbf71fa05f45dd35e24b09720479b1afa1f128b93852fbd5231; hash_scope: UTF-8 bytes from the header row through the final data row, including the final LF and excluding these comment lines. Dataset: 18 rows (t=0..17) of columns t,r0,r1,r2,r3 (four-channel counts), e.g. row t=0: 599.581,500.136,499.999,500.098; row t=17: 524.490,526.098,524.749,524.455.

WP.S24.ConcreteDomainRunReport — Concrete domain run report (RG-RP4C-001-RUN-001)

tier: finite_diagnostic; status: current

ConcreteDomainRunReport: run_id RG-RP4C-001-RUN-001, root_version RG-UTW-1.2.0, domain_card_version RG-RP4C-001-1.0.0, dataset_sha256 43b1aa7081ce1fbf71fa05f45dd35e24b09720479b1afa1f128b93852fbd5231, calibration_certificate_sha256 0ab2b3b56d33cd14eefa90e5a79289c3197fbc389706ae5dbbcd992af9b76de2, freeze_hash 88b431490789ad33508efa6f09f7b73fce6762423909477fd0ef61abc63f89ec; state_dimensions Phi 4, memory_depth 2, active_dynamic_fields 1; operator path_laplacian_P4 shape [4,4], symmetry_defect 0.0, row_sum_defect 0.0; calibration D_hat 0.348925, K_hat 0.221249, design_rank 2, condition_number 3.688012, training_equation_rmse 0.004437; held_out horizon 7, rmse 0.004945, mae 0.004030, max_abs_error 0.009580; controls positive_control PASS, negative_control FAIL_AS_REQUIRED; defects epsilon_suff 0.0, epsilon_dyn 0.0, epsilon_read 0.0, epsilon_inv 0.0, epsilon_bridge NOT_REQUESTED, epsilon_cal 0.004437, epsilon_held_out 0.004945; final_status PASS_TOLERANCED; claim tier finite_diagnostic, statement "The frozen finite benchmark pipeline predicts its seven-step held-out trace within preregistered tolerances.", exclusions [empirical_real_world_validation, continuum_claim, universal_domain_claim, unique_operator_derivation].

WP.S24.ConcreteExecutionDAG — 24.6 Concrete execution DAG

tier: untagged; status: current

INFORMATION PHILOSOPHY CONSTITUTION -> (forbids semantic and target leakage) RAW FOUR-CHANNEL COUNTS + DATASET HASH -> (verify order, completeness, lineage) INDEPENDENT AFFINE CALIBRATION CERTIFICATE -> (decode only by frozen baseline and gain) ENCODED RETAINED TAPE Phi[0:17] -> (fixed split before fitting) TRAIN Phi[0:10] and DECLARED P4 LAPLACIAN FIXED M=1, dt=1 -> BUILD LINEAR CALIBRATION SYSTEM X theta = y -> (rank and conditioning gate) FIT D, K -> (freeze payload and SHA-256) RECURSIVE PREDICTION Phi_hat[11:17] -> (checker reveals held-out values) RMSE + MAE + MAX-ERROR -> POSITIVE CONTROL (must PASS) / NEGATIVE CONTROL (must FAIL) -> PASS_TOLERANCED / FAIL / UNRESOLVED -> FINITE, BOUNDED CLAIM ONLY.

WP.S24.ConcreteOperatorChecks — Concrete operator invariant checks

tier: untagged; status: current

ConcreteOperatorChecks: symmetry expression "norm(L - transpose(L), Frobenius)" expected 0.0; zero_row_sum expression "max_abs(L @ ones_4)" expected 0.0; positive_semidefinite expression "min_eigenvalue(L)" expected_lower_bound -1.0e-12; retained_metric expression "min_eigenvalue(G)" expected_lower_bound 1.0; quotient_map_identity, epsilon_dyn_expected 0.0, epsilon_read_expected 0.0, epsilon_inv_expected 0.0.

WP.S24.ConcreteReaderRecurrence — Concrete reader recurrence (second-order DRL kernel, RP4C)

tier: untagged; status: current

$M * \delta t^2 \Phi_n + D * \delta t^c \Phi_n + K * L_{\{P_4\}} \Phi_n = 0$, with $M=1$, $\Delta t=1$. Equivalently: $(M + D/2) \Phi_{\{n+1\}} = -K L_{\{P_4\}} \Phi_n + 2M \Phi_n + (D/2 - M) \Phi_{\{n-1\}}$.

WP.S24.ConcreteReferenceStatus — Concrete reference benchmark scope card

tier: untagged; status: current

ConcreteReferenceStatus: benchmark_id RG-RP4C-001, name "Four-Node Retained Pulse Chain"; purpose [exercise a complete Domain Card; expose a concrete operator and data encoder; calibrate unknown native coefficients without target leakage; evaluate recursively on a frozen held-out segment; require a passing positive control and a failing negative control]; epistemic_scope {data_type: deterministic_synthetic_independent_fixture, evidence_type: held_out_synthetic_evidence, external_instrument_validation: false}; admissible_claim ["the declared finite pipeline is reproducible", "the declared operator coefficients are identifiable on this training split", "the frozen model predicts the declared held-out segment within preregistered tolerances"]; forbidden_claim ["this benchmark proves a universal law of nature", "synthetic held-out evidence is external empirical validation", "the four-node interpretation is present in the root"].

WP.S24.DomainCard — 24.1 Concrete Domain Card (RG-RP4C-001)

tier: untagged; status: current

DomainCard schema_version 1.0; identity {domain_id: RG-RP4C-001, version: 1.0.0, parent_root_version: RG-UTW-1.2.0, status: FROZEN_REFERENCE}; question {target: "predict the next seven four-component retained readouts after tape index 10", horizon 7_tape_steps, resolution 1_tape_step, tolerance held_out_max 0.015, held_out_mae_max 0.012, held_out_max_abs_error_max 0.030, operator_symmetry_max 1.0e-12, operator_residual_max 1.0e-12, solve_residual_max 1.0e-12; context topology fixed_four_node_path, boundary closed_endpoints, source zero, potential zero}; candidate_state slot_activation Phi ACTIVE, Psi INACTIVE_WITH_REASON, Theta FIXED_BACKGROUND=four_node_path, Sigma INACTIVE_WITH_REASON, Tape ACTIVE_APPEND_ONLY, G ACTIVE_identity_4, L_c1 ACTIVE_EMPTY_CLOSURE_LEDGER, U/Gamma/Xi INACTIVE_WITH_REASON; fields Phi real_vector_length_4; carriers dimensions Phi[4], graph_operator[4,4], retained_metric[4,4], primitive_relations [[0,1],[1,2],[2,3],[3,4]], readers node_vector_reader 0_vector(Phi_n)=Phi_n, total_reader 0_total(Phi_n)=sum_i Phi_n[i], equivalence_tolerance 1.0e-12; invariants set [L_graph = L_graph^T; L_graph * ones = 0; L_graph positive semidefinite; G = identity_4 positive definite]; dynamics root_kernel finite_DRL_reader_equation_with_Psi_inactive, graph_operator path_laplacian_P4 matrix [[1,-1,0,0],[-1,2,-1,0],[0,-1,2,-1],[0,0,-1,1]], field_operator zero_4x4, coefficients dt=1.0 FROZEN, M=1.0 FROZEN_GAUGE_CHOICE, D=0.348925 TRAINING_CALIBRATED, K=0.221249 TRAINING_CALIBRATED; calibration decoder equation "Phi_n = (r_n - b) / g", baseline_counts [500,500,500,500], gain_counts_per_native_unit [100,100,100,100], calibration_certificate_sha256 0ab2b3b56d33cd14eefa90e5a79289c3197fbc389706ae5dbbcd992af9b76de2, parameters fixed [M, dt, L_graph, baseline_counts, gain_counts_per_native_unit], fitted [D, K], training_data tape_indices [0,10], equation_indices [1,9], freeze_hash 88b431490789ad33508efa6f09f7b73fce6762423909477fd0ef61abc63f89ec, held_out_outputs tape_indices [11,17] visibility FORBIDDEN.

WP.S24.ExternalEvidenceBundle — External evidence adapter contract

tier: untagged; status: current

ExternalEvidenceBundle: bundle_id REQUIRED; source_manifest {producer, acquisition_protocol, acquisition_raw_file_sha256, schema, access_or_license_status} REQUIRED; calibration_certificate {certificate_id, calibration_method, valid_range, uncertainty, certificate_sha256} REQUIRED; split_manifest {rule_frozen_benchmark_validation: REQUIRED_TRUE, training_ids, held_out_ids_or_commitment_hash, maker_visibility TRAINING_ONLY, checker_visibility HELD_OUT_ONLY}; model_freeze {root_version, domain_card_hash, code_hash, parameter_hash, prediction_hash} REQUIRED; checker {independent_identity_or_process, preregistered_metrics, preregistered_thresholds, positive_control, negative_control} REQUIRED; empirical_claim_gate PASS_when [provenance_complete, no_train_test_leakage, calibration_valid_for_observed_range, held_out_thresholds_pass, positive_control_pass, negative_control_fails_as_required], otherwise FAIL_OR_UNRESOLVED.

WP.S24.HeldOutCheckerReport — Held-out and control evidence report

tier: finite_diagnostic; status: current

HeldOutCheckerReport: freeze_hash_verified true; recursive_forecast held_out_indices [11,17], observed_validation false; metrics rmse 0.004945, mae 0.004030, max_abs_error 0.009580; thresholds rmse_max 0.015, mae_max

0.012, max_abs_error_max 0.030; positive_control recovered_D 0.350000, recovered_K 0.220000, status PASS; negative_control (K=0, refit D) refitted_D 0.016637, held_out_rmse 0.123705, required_rmse_min 0.050, status FAIL_AS_REQUIRED; final_status PASS_TOLERANCED; claim_status finite_diagnostic; cannot_claim [external_empirical_validation, universal_domain_law, unique_root_derivation_of_the_operator]. Held-out predicted-vs-observed table for tape indices 11-17 across Phi_0..Phi_3 given in text.

WP.S24.MetricAndLaplacian — Retained metric and graph operator (path Laplacian P4)

tier: untagged; status: current

$G = I_4$; $L_{\{P_4\}} = [[1, -1, 0, 0], [-1, 2, -1, 0], [0, -1, 2, -1], [0, 0, -1, 1]]$.

WP.S24.RawObservation — Raw observation vector at tape index n

tier: untagged; status: current

$r_n = (r_{\{n,0\}}, r_{\{n,1\}}, r_{\{n,2\}}, r_{\{n,3\}})^T$.

WP.S25.PassFailUnresolvedTable — PASS/FAIL/UNRESOLVED decision table and precedence

tier: untagged; status: current

Status precedence is fail-closed: DRIFT or PROTOCOL_FAIL > FAIL > UNRESOLVED > PASS_TOLERANCED > PASS_EXACT. "A higher-precedence status overrides every lower-precedence status." Table rows: All required exact defects are zero -> PASS_EXACT (emit exact-within-scope claim); All required numerical defects within frozen tolerance -> PASS_TOLERANCED (emit numerical/finite claim); A required defect exceeds tolerance -> FAIL (name defect and failure code); Candidate state insufficient -> FAIL_STATE (enlarge state; do not patch output); Identifiability rank deficient -> UNRESOLVED (abstain; acquire more distinctions); Positive control passes but no failing control exists -> TYPE_U_ONLY (at most conditional wording); Maker saw held-out answer -> PROTOCOL_FAIL (invalidate independent evidence claim); Root contains domain-final law -> DRIFT (reject run before calculation); Calibration absent for named physical quantity -> UNCALIBRATED (return native value only); Scalar reduction drops coupling -> REDUCTION_FAIL (use full operator); Projector not Hermitian/idempotent -> FAIL_PROJECTOR (rebuild physical sector); Transfer positive but not contraction-certified -> POSITIVE_ONLY (do not claim nonnegative persistence costs); Resonance requested without analytic extension -> NOT_AVAILABLE (report missing extension).

WP.S26.AntiDriftRules — Anti-drift rule set (DRIFT_01 through DRIFT_16)

tier: untagged; status: current

DriftRules (16 rules, each a prose statement, no formal equations): DRIFT_01_ROOT_IMMUTABILITY "Do not alter root to match one domain answer."; DRIFT_02_NO_SEMANTIC_PREMISE "Attach domain names only after a passed translation."; DRIFT_03_NO_IMPORTED_FINAL_LAW "Imported law must be EXTERNAL_COMPARISON or CALIBRATION_ONLY."; DRIFT_04_NO_FIT_BACK "Held-out target cannot become a parameter, eigenvalue or closure weight."; DRIFT_05_EXACT_SCOPE "Exact must name axioms, architecture, arithmetic and domain scope."; DRIFT_06_FINITE_NOT_CONTINUUM "Finite verifier is not a continuum proof."; DRIFT_07_HESSIAN_NOT_P "Local curvature is not mass or resonance without spectral calibration."; DRIFT_08_MULTIPLICITY_SEPARATION "Keep invariant, rank, generation and ordering multiplicities distinct."; DRIFT_09_POSITIVE_KERNEL "Cross-midpoint interaction needs Gram or positive-type certificate."; DRIFT_10_NEGATIVE_CONTROL "Every evidentiary gate needs a genuine failing control."; DRIFT_11_SUPERSEDED_FENCED "Old formulas remain only with SUPERSEDED status and replacement pointer."; DRIFT_12_ONE_ROOT_MANY_READOUTS "Domain-spec action requires a declared parent map to S_RG."; DRIFT_13_NO_SILENT_REGULARIZATION "A failing solve must fail visibly."; DRIFT_14_NO_PREMATURE_DIAGONALIZATION "Audit discarded coupling before scalar mode use."; DRIFT_15_CALIBRATION_SEPARATION "Native output, unit conversion and semantic report are distinct stages."; DRIFT_16_PHILOSOPHY_TRACE "Every major implementation gate must cite the constitution principle it enforces." Merged into one row: these are individually enumerated but none contains a distinct formal/mathematical statement beyond prose rules, so recorded as a single combined item per task guidance.

WP.S27.LegacyMigration — Legacy migration ledger

tier: untagged; status: current

LegacyMigration: carried_forward_as_normative [commuting_square, failure_taxonomy, state_sufficiency_before, no_early_collapse, maker_checker_firewall, prediction_freeze, fail_able_gate_law, defect_vector, full_multimode_operator, reader_record_pair, living_geometry, moving_basis_current, explicit_fail_closed_loop, modal_reduction_audit, unit_and_semantic_lane_separation, append_only_tape_and_deviation_record]; carried_forward_as_reference_realization [discrete_DRL_stepper, telegraph_like_reduction, two_field_action_finite_fixture_guards]; not_promoted_to_universal_root [tau_c_equals_hbar_over_2mc2, mass_as_universal_memory, four_forces_as_gauge_faces, classical_quantum_identity_from_damping_regime, turbulence_domain_interpretation, Standard_Model_specific_claims, black_hole_specific_claims, biological_or_chemical_semantics, SI_constants, named_physical_equations]; treatment rule "Keep as DomainCard hypothesis, external comparison, calibration or open frontier."

WP.S28.ReferenceAPI — Minimal implementation interfaces (ReferenceAPI)

tier: untagged; status: current

ReferenceAPI function signatures (input -> output): validate_root(root_config) -> guard_report; encode_state(ordered_tape, encoder) -> candidate_state, provenance; audit_sufficiency(candidate_state, question) -> ADMITTED_OBSTRUCTED_UNRESOLVED, epsilon_suff; discover_partition(candidate_state, dynamics, readers, interventions) -> partition, refinement_trace; audit_quotient(partition, fine_dynamics, coarse_dynamics, readers, invariants) -> epsilon_dyn, epsilon_read, epsilon_inv; discover_automorphisms(representations, dynamics, G) -> group_generators, certificates; build_transport(local_frames, automorphisms) -> U; compute_holonomy(U, ordered_cycles) -> H, K, invariants; enumerate_closure(representations, automorphism_group) -> intertwiners, multiplicities, obstructions; build_operator(graph_operator, field_operator, interaction, Theta, G) -> full_operator, symmetric_part, skew_part; run_drl(state_history, operator, coefficients, currents) -> next_state, residuals, solve_report; build_transfer(midpoint_amplitude, potential, projector) -> T_positive, positivity, contraction; spectral_read(T, scale_a) -> spectrum, persistence, support_report; coarse_grain(state, coarse_map, readers) -> coarse_state, epsilon_cg; calibrate(native_output, frozen_decoder, training_calibration) -> named_output, epsilon_cal; check(freeze_hash, held_out, controls) -> checker_report; report(all_reports) -> DomainRunReport, ClaimLedger.

WP.S29.AIContract — AI execution contract

tier: untagged; status: current

AIContract: before_run [freeze_root_primitives, state_question_and_tolerance, declare_available_and_forbidden, strip_early_domain_semantics]; during_run [preserve_full_operator_until_reduction_is_audited, preserve_linearity, return_named_defects, distinguish_zero_obstructed_unresolved, fail_closed_on_singular_or_untyped_operation, separate_native_output_from_calibration]; after_run [include_positive_and_negative_controls, use_weakest_linearity, state_finite_continuum_scope, state_simulated_empirical_status, output_reproducibility_record]; forbidden [invent_missing_constants, call_analogy_derivation, call_synthetic_data_empirical_validation, call_machine_learning, hide_nonzero_residual, silently_change_domain_card].

WP.S29.AIResult — AI output template (AIResult)

tier: untagged; status: current

AIResult: interpretation_order {retention, structure, translation, meaning, report} REQUIRED_SUMMARY each; calculation {inputs REQUIRED_TYPED_LIST, equations_used REQUIRED_LIST, steps REQUIRED_ORDERED_LIST, outputs REQUIRED_TYPED_LIST}; verification {invariants REQUIRED_MAP, defects REQUIRED_VECTOR, positive_controls REQUIRED_IF_EVIDENTIARY, negative_control REQUIRED_IF_EVIDENTIARY, checker_status REQUIRED}; claim {status REQUIRED_ENUM, scope REQUIRED_MAP, cannot_claim REQUIRED_LIST, next_bottleneck REQUIRED_OR_NONE}.

WP.S3.CanonicalState — Canonical state vector Z_n

tier: untagged; status: current

$Z_n = (\Phi_n, \Psi_n, \Theta_n, \Sigma_n, T_n, G_n, L^{cl}_n ; U_n, \Gamma_n, \Xi_n)$. Semicolon separates left (primary retained-state slots: Φ proposed_distinction_state, Ψ companion_record_response, Θ relational_geometry_translation_state, Σ retained_order_reference, Tape append_only_history,

G positive_retained_metric, L_cl closure_and_lineage_ledger) from right (derived/reified operational slots retained for reproducibility: U transport, Gamma orientation_grading with Gamma^{dagger}=Gamma, Gamma²=I, Xi orientation_selecting_order, odd under declared tape reversal; lambda_n^a constraint multiplier is auxiliary, not part of L_cl).

WP.S3.NotRootByDefault — Items not root primitives by default

tier: untagged; status: current

NotRootByDefault: spacetime, particle, molecule, gene, neuron, institution, physical_energy, mass, force, probability, continuum, SI_units, domain_final_equation.

WP.S3.PositiveMetricNorm — Retained inner product and norm

tier: untagged; status: current

$\langle x, y \rangle_{G_n} = x^{\dagger} G_n y$; $\|x\|^2_{G_n} = x^{\dagger} G_n x \geq 0$

WP.S3.PositiveMetricPositivity — Positive retained metric positivity

tier: untagged; status: current

$G_n = G_n^{\dagger}$, $G_n > 0$ (Hermitian, positive definite). PositiveMetricGate checks: hermitian_within_tolerance, smallest_eigenvalue_above_tolerance; pass = both true; fail = any false -> INVALID_STATE_METRIC.
'retained load MUST NOT be called energy until a domain-specific semantic calibration has passed.'

WP.S3.RootPrimitives — Root primitives

tier: untagged; status: current

RootPrimitives: distinction_record (symbol delta_R, 'record that at least one distinction is retained', primitive:true); ordered_tape (symbol T, 'append-only order of events, rewrites, residues and deviations', primitive:true); reader_family (symbol O, 'declared maps that can distinguish, store or act on retained records', primitive:true); admissibility_language (symbol C, 'rules stating admitted, obstructed or unresolved constructions', primitive:true); positive_retained_pairing (symbol G, 'positive pairing used to measure retained load', primitive:true).

WP.S30.HumanChecklist — Human execution checklist

tier: untagged; status: current

HumanChecklist: philosophy ["Did I begin from retained distinctions rather than a domain name?", "Did I separate readout from reality and meaning from phenomenon?"]; model ["Are all variables typed?", "Does every state slot have an update, constraint or background declaration?", "Did I preserve lineage and order?"]; translation ["Was sufficiency checked before quotient?", "Does the square commute?", "Which defect is nonzero?", "Is the symmetry query-relative?"]; computation ["Was the full operator used?", "Was scalar reduction audited?", "Did the solver fail visibly when invalid?", "Is positivity distinct from contraction?"]; evidence ["Is there a failing control?", "Was the prediction frozen?", "Was the checker independent?", "Are calibration and held-out evaluation separate?"]; report ["Is the claim no stronger than its weakest support?", "Are finite, continuum, simulated and empirical scopes explicit?"].

WP.S31.OfficialStatus — Official status declaration

tier: untagged; status: current

OfficialStatus: name Readout_Genesis_Universal_Technical_Whitepaper, version 1.2.0, language English_only peer_review_status INTERNAL_TECHNICAL_REVIEW_RESOLVED; closed [philosophy_led_translation_order, anti_drift_constitution, typed_root_and_state_contract, reader_equivalence_contract, exact_quotient_and_deautomorphism_transport_holonomy_contract, closure_grammar, modular_action_interface, executable_finite_readout, positive_transfer_requirements, coarse_graining_and_reduction_gates, domain_compilation_protocol, maker_checker_firewall, claim_and_run_report_schemas, standalone_reference_fixtures, concrete_reference_domain, concrete_operator_and_data_encoding, frozen_parameter_calibration, held_out_checker_with_controls];

conditional_or_open [unique_concrete_universal_action, unique_domain_emergence_from_root, generalized_mach, full_endogenous_geometry_backreaction, universal_continuum_limit, universal_empirical_calibration, measurement_and_probability_foundations, spin_statistics, complete_cross_domain_predictions, primitive_con, final_boundary valid_description ["Universal Research and Derivation Architecture", "Philosophy-Governed Human-AI Executable Specification", "Finite Reference Kernel with Fail-Closed Verification"], invalid_desc: ["Complete Final Law of Nature", "Empirically Proven Theory of Every Domain"]].

WP.S32.OneBoxSummary — One-box summary diagram (boxed equation)

tier: untagged; status: current

boxed{ Information Philosophy $\Downarrow$ $\Delta_R, T, O, C, G \Downarrow Z_n = (\Phi, \Psi, \Theta, \Sigma, T, G, L^{cl}; U, \Gamma, \Xi)_n \Downarrow$ Sufficiency $\rightarrow$ Equivalence $\rightarrow$ Automorphism $\rightarrow$ Transport $\rightarrow$ Closure $\Downarrow$ $P_{phys}[\Delta S_{RG} + \lambda \Delta C] = 0 \Downarrow T_+ = P_{phys} e^{-V/2} A^\dagger A e^{-V/2} P_{phys} \succeq 0 \Downarrow$ Quotient + ϵ_D + Calibration Freeze + Independent Checker $\Downarrow$ PASS / FAIL / UNRESOLVED $\Downarrow$ Bounded Claim }.
Summary text: "Differences are retained before they are named. Structure is checked before it is translated. Meaning follows readout. Claims are emitted only after computation, defect control, freezing, and independent checking."

WP.S4.DomainDefinition — Domain as sufficient quotient

tier: untagged; status: current

$D_D = Z_D / \sim_D$, where Z_D MUST be sufficient before quotienting.

WP.S4.DomainGateDefects — Domain gate defects

tier: untagged; status: current

$\epsilon_{dyn} = ||q_{n+1} F_n - F_n q_n||$; $\epsilon_{read} = ||O_n - O_n q_n||$; $\epsilon_{inv} = ||Inv_n - Inv_n q_n||$. ExactDomainGate classification: exact_when sufficiency_status==ADMITTED and all three ϵ 's ==0; approximate_when sufficiency ADMITTED and all required defects within frozen tolerances with at least one nonzero; fail_when sufficiency OBSTRUCTED or any defect exceeds frozen tolerance; unresolved_when sufficiency UNRESOLVED or norm/tolerance missing or identifiability not decidable.

WP.S4.DomainReadableState — Domain-readable state

tier: untagged; status: current

$[Z]_{\sim_D}$ (the equivalence class of Z under $\sim_D$)

WP.S4.DomainRequiredProperties — Domain definition — required vs not-sufficient properties

tier: untagged; status: current

DomainDefinition phrase: 'minimal sufficient dynamically closed translation'. required_properties: state_sufficiency, readout_factorization, dynamic_commutation, invariant_preservation, declared_context, lineage_preservation. not_sufficient: subject_name, equation_shape_similarity, dimensional_similarity, one_successful_fit, one_passing_control.

WP.S4.ExactDomainGate — Exact domain gate — dynamic commutation

tier: untagged; status: current

$q_{D,n+1} \circ F_n = F_n q_{D,n}$

WP.S4.InvariantFactorization — Exact domain gate — invariant factorization

tier: untagged; status: current

$Inv_{D,n} = Inv_n q_{D,n}$

WP.S4.LocalReaderEquivalence — Local reader equivalence

tier: untagged; status: current

$x \sim_{\{0_D\}} y \iff 0_\alpha(x) = 0_\alpha(y) \text{ for all } \alpha$

WP.S4.ReaderFamily — Declared reader family

tier: untagged; status: current

$0_D = \{0_\alpha\}_{\alpha \in R_D}$

WP.S4.ReadoutFactorization — Exact domain gate — readout factorization

tier: untagged; status: current

$0_{\{D,n\}} = 0^{\#}_{\{D,n\}} \circ q_{\{D,n\}}$

WP.S4.StrongReaderEquivalence — Strong (horizon) reader equivalence

tier: untagged; status: current

$q_D(z) = q_D(z') \Rightarrow 0_{\{D,n:n+L\}}(z,u) = 0_{\{D,n:n+L\}}(z',u) \text{ for all } u \text{ in } U_D$

WP.S4.TranslationFailureTaxonomy — Translation failure taxonomy

tier: untagged; status: current

TranslationFailure allowed_codes: MISTRANSLATION ('map itself is wrong or ill-posed'); LOST_INFORMATION ('source quotient erased a required distinction'); INSUFFICIENT_RESOLUTION ('retained resolution is too coarse'); TARGET_LACKS_VARIABLES ('target state has no slot for required structure'); NO_CLOSURE ('target has no well-posed update'); CONTEXT_MISMATCH ('law valid in c but applied in c_prime'); CALIBRATION_FAILURE ('native output cannot be independently decoded'); IDENTIFIABILITY_UNRESOLVED ('more than one compatible law remains'). 'A report saying only the reduction fails without a failure code is incomplete.'

WP.S5.AutomorphismGroup — Query-relative automorphism group

tier: untagged; status: current

$A_{\{D,n\}} = \{ h_n : 0_{\{D,n\}} h_n = 0_{\{D,n\}} , h_{\{n+1\}} F_n = F_n h_n , h_n^\dagger G_n h_n = G_n \}.$ 'This group is relative to the declared question and readers; it is not an absolute symmetry group.'

WP.S5.CurvatureDefect — Curvature defect

tier: untagged; status: current

$K_C = H_C - I$

WP.S5.Holonomy — Holonomy around a cycle

tier: untagged; status: current

$H_C = \text{product}_{\{e \in C, \text{ordered}\}} U_e$

WP.S5.HolonomyInvariants — Readable holonomy invariants

tier: untagged; status: current

$\text{Tr}(H_C), \det(H_C), \text{spec}(H_C).$ 'Semantic labels such as curvature, reaction-path defect, reasoning drift, or institutional contradiction may be attached only by a validated Domain Card.'

WP.S5.Transport — Transport map

tier: untagged; status: current

$U_{\{j \leftarrow i\}} : X_i \rightarrow X_j$

WP.S5.TransportGaugeTransform — Transport gauge transform

tier: untagged; status: current

$U'_{\{j<-i\}} = h_j U_{\{j<-i\}} h_i^{-1}$

WP.S5.TransportMetricCompatibility — Transport metric compatibility

tier: untagged; status: current

$U_{\{j<-i\}}^\dagger G_j U_{\{j<-i\}} = G_i$ (equality for isometric transport)

WP.S5.TransportMetricDefect — Transport metric-compatibility defect

tier: untagged; status: current

$\epsilon_{\{G,U\}} = || U^\dagger G_j U - G_i ||$

WP.S6.ClosureMultiplicity — Closure invariant multiplicity

tier: untagged; status: current

$\text{nu_alpha} = \dim \text{Hom}_A(R_1(x) \dots (x) R_k, 1)$. If the Hom-space is zero: `closure_status=OBSTRUCTED`, `interaction_allowed=false`.

WP.S6.InvariantClosureMap — Invariant closure map

tier: untagged; status: current

$I_\alpha : R_1(x) \dots (x) R_k \rightarrow 1$

WP.S6.MultiplicityLedger — Multiplicity ledger

tier: untagged; status: current

MultiplicityLedger: `invariant_multiplicity = number_of_independent_invariant_closure_maps`; `rank_multiplicity = operator_or_matrix_rank_degeneracy`; `generation_multiplicity = repeated_retained_response_classes`; `raw_ordering_count = number_of_unquotiented_sequences`. 'Do not infer physical channel count from raw ordering count before quotienting by: cyclic start-point equivalence, reversal equivalence when the reader is orientation-insensitive, gauge equivalence, reader equivalence, structural sharing without lineage erasure.'

WP.S6.ThreeValuedAdmissibility — Three-valued admissibility

tier: untagged; status: current

$C(\xi | c, T)$ in $\{1, 0, \text{perp}\}$. Admissibility: '1' $\rightarrow$ status ADMITTED, action construct; '0' $\rightarrow$ status OBSTRUCTED, action append_obstruction_certificate; 'bottom' $\rightarrow$ status UNRESOLVED, action abstain_and_request; 'UNRESOLVED is not equivalent to OBSTRUCTED.'

WP.S7.ActionContracts — Action term contracts

tier: untagged; status: current

ActionContracts: `S_DRL` consumes $[\text{Phi}, \text{Psi}, \text{G}, \text{Tape}, \text{declared_coefficients}]$, must be typed/variational-or-expl; `S_geo` consumes $[\text{Theta}, \text{Phi}, \text{Psi}, \text{Sigma}, \text{Tape}]$ for relation_update/basis_motion/backreaction; `S_tr` consumes $[\text{U}, \text{Theta}, \text{G}]$ built from relative_transport/holonomy/retained_mismatch; `S_int` consumes $[\text{closure_maps}, \text{Phi}, \text{Psi}]$ restricted to admitted invariant closure channels; `S_ord` consumes $[\text{Sigma}, \text{Xi}, \text{U}, \text{G}]$ for retained_reference/or; `S_cut` consumes $[\text{boundary}, \text{cut_currents}, \text{closure_ledger}]$ for unresolved_boundary_obligation/information_cross; `S_tape` consumes $[\text{Tape}, \text{Gamma}, \text{Xi}]$ for order_sensitivity/reversal_status/sequence_memory; `S_read` consumes $[\text{reader_family}, \text{state}, \text{coarse_readout}]$ for readout_consistency/microscopic_to_retained_factorization. ActionStatus: `universal_modular_interface EXACT_WITHIN_ARCHITECTURE`, `unique_concrete_action_for_all_domains`, `OPEN`, `finite_reference_realization_below EXECUTABLE_REFERENCE`.

WP.S7.ConstrainedEquation — Constrained variational equation

tier: untagged; status: current

$P_{\text{phys}} [\delta S_{\text{RG}} / \delta Z_n + \sum_a \lambda_n^a \delta C_a[Z] / \delta Z_n] = 0$, subject to equality constraints $C_a^{\text{~}}[Z]=0$ and inequality constraints $C_b^{\text{~}}[Z] \leq 0$.

WP.S7.KKTConditions — KKT conditions for inequality constraints

tier: untagged; status: current

$\lambda_b \geq 0$, $C_b^{\text{~}}[Z] \leq 0$, $\lambda_b C_b^{\text{~}}[Z] = 0$

WP.S7.ModularAction — Universal modular action

tier: untagged; status: current

$S_{\text{RG}} = S_{\text{DRL}} + S_{\text{geo}} + S_{\text{tr}} + S_{\text{int}} + S_{\text{ord}} + S_{\text{cut}} + S_{\text{tape}} + S_{\text{read}}$

WP.S7.VariableEquationCoverageGate — Variable equation coverage gate

tier: untagged; status: current

'If G, Theta, U, Gamma, or Xi is dynamical, it MUST have an equation or update rule. Otherwise it MUST be marked background, externally_prescribed, or derived_cache.' VariableEquationCoverageGate: for_each_state_slot require_one_of [variational_equation, recurrence, algebraic_constraint, declared_background, derived_cache_with_recompute_rule]; fail_code UNCOVERED_STATE_VARIABLE.

WP.S8.CentralDifference — Discrete central time-difference operator

tier: untagged; status: current

$\delta_t^c X_n = (X_{n+1} - X_{n-1}) / (2 \Delta t)$

WP.S8.DRLStepAlgorithm — DRL_STEP reference runtime algorithm

tier: untagged; status: current

FUNCTION DRL_STEP(run, state_n_minus_1, state_n): validate $dt > 0$, G positive definite, provenance/type on every slot; update geometry candidate; build operator; split into metric-symmetric/skew sectors and check residuals; compute moving-basis current and R_{geo} , fail GEOMETRY_REDUCTION_INVALID if fixed-basis requested and $R_{\text{geo}} > \text{tolerance}$; compute closure channels and append obstruction certificates; assemble R_{Phi} , R_{Psi} by named current channels; build and solve $A_{\text{Phi}} \Phi_{\text{next}} = b_{\text{Phi}}$ and $A_{\text{Psi}} \Psi_{\text{next}} = b_{\text{Psi}}$ via fail-closed solver; recompute Θ_{next} /transport U if needed; compute residuals, constraints, invariants; append to Tape; return FAIL if any hard invariant fails, UNRESOLVED if any required fact unidentified, else PASS_WITH_OUTPUT.

WP.S8.DiscreteAction — Discrete action / Lagrangian sum

tier: untagged; status: current

$S_{\text{DRL}} = \sum_n L^n$, with reference (real-coordinate) Lagrangian: $L^n = (1/\Delta t) \Delta \Phi_n^T M_n \Delta \Psi_n + (1/2)(\Phi_n^T D_n \Delta \Psi_n - \Psi_n^T D_n \Delta \Phi_n) - \Delta t [K_n \Phi_n^T G_n \Psi_n + \Psi_n^T \text{grad } V_n(\Phi_n) - J_n^T \Psi_n]$. (Replace transpose by declared Hermitian/metric adjoint in complex coordinates.)

WP.S8.FullOperator — Full operator on graph-times-field space

tier: untagged; status: current

$G_n = L_{\{\text{graph}, n\}}(x) I_F + I_{\{\text{graph}, n\}}(x) C_F + C_{\{\text{int}, n\}}$

WP.S8.GeometryDominanceRatio — Geometry-dominance ratio

tier: untagged; status: current

$R_{\text{geo}} = || (V_{\{n+1\}} - V_n) \text{ phi}_n || / || \Delta \text{ Phi}_n ||$. MovingBasisGate: fixed_basis_allowed_when $R_{\text{geo}} \leq \text{preregistered_tolerance}$; fixed_basis_rejected_when $R_{\text{geo}} > \text{tolerance}$; unresolved_when denominator zero and numerator nonzero or basis provenance missing.

WP.S8.LinearSolveGate — Linear solve gate

tier: untagged; status: current

LinearSolveGate: method reference = partial_pivot_gauss_jordan_or_stable_equivalent; must_report [pivot_min, rank, condition_estimate, solver_residual]; fail_when [pivot_min < pivot_tolerance, rank < expected_rank, solver_residual > solve_tolerance]; must_not [silently regularize, replace singular solve with guessed output].

WP.S8.LivingGeometry — Living geometry update

tier: untagged; status: current

$\Theta_{\{n+1\}} = A_{\Theta} \Theta_n + B_{\{\Theta, \Phi\}} \Phi_n + B_{\{\Theta, \Psi\}} \Psi_n + u_{\{\Theta, n\}}$

WP.S8.MetricAdjoint — Metric-adjoint

tier: untagged; status: current

$A^{\{\dagger\}_G} = G^{\{-1\}} A^{\dagger}_G$

WP.S8.MovingBasis — Moving basis representation

tier: untagged; status: current

$\Phi_n = V_n \text{ phi}_n$

WP.S8.MovingBasisDelta — Moving-basis increment decomposition

tier: untagged; status: current

$\Delta \text{ Phi}_n = V_{\{n+1\}} \Delta \text{ phi}_n + (V_{\{n+1\}} - V_n) \text{ phi}_n$

WP.S8.OperatorGeometryDependence — Operator dependence on geometry state

tier: untagged; status: current

$G[\Theta_n] = G_0 + \sum_a \Theta_n^a G_a$

WP.S8.OperatorSplit — Metric symmetric/skew split

tier: untagged; status: current

$G^{(+)} = (1/2)(G + G^{\{\dagger\}_G})$; $G^{(-)} = (1/2)(G - G^{\{\dagger\}_G})$

WP.S8.OperatorSplitChecks — Operator split required checks

tier: untagged; status: current

$(G^{(+)})^{\{\dagger\}_G} = G^{(+)}$, $(G^{(-)})^{\{\dagger\}_G} = -G^{(-)}$. OperatorSplitStatus: algebraic_split_for_given DERIVED_EXACT; universal_physical_interpretation CONDITIONAL; state_dependent_endogenous_operator_closure OPEN_UNLESS_INSTANCE_PROVED.

WP.S8.PhiOneStepSolve — Phi one-step linear solve

tier: untagged; status: current

$A_{\Phi} = (M/\Delta t^2 + D/(2 \Delta t)) I$; $b_{\{\Phi, n\}} = R_{\{\Phi, n\}} + J_n - K G_n \Phi_n - \text{grad } V(\Phi_n) + (2M/\Delta t^2) \Phi_n + (D/(2 \Delta t) - M/\Delta t^2) \Phi_{\{n-1\}}$; $A_{\Phi} \Phi_{\{n+1\}} = b_{\{\Phi, n\}}$

WP.S8.PsiOneStepSolve — Psi one-step linear solve

tier: untagged; status: current

$A_{\Psi} = (M/\Delta t^2 - D/(2 \Delta t)) I$; $b_{\{\Psi,n\}} = R_{\{\Psi,n\}} - K G_n^{\{\dagger_G\}} \Psi_n - \text{grad}^2 V(\Phi_n) \Psi_n + (2M/\Delta t^2) \Psi_n - (M/\Delta t^2 + D/(2 \Delta t)) \Psi_{n-1}$; $A_{\Psi} \Psi_{n+1} = b_{\{\Psi,n\}}$

WP.S8.ReaderEquation — Reader field equation

tier: untagged; status: current

$M \Delta t^2 \Phi_n + D \Delta t^c \Phi_n + K G[\Theta_n] \Phi_n + \text{grad} V(\Phi_n) - J_n = R_{\{\Phi,n\}}$

WP.S8.RecordEquation — Record field equation

tier: untagged; status: current

$M \Delta t^2 \Psi_n - D \Delta t^c \Psi_n + K G[\Theta_n]^{\{\dagger_G\}} \Psi_n + \text{grad}^2 V(\Phi_n) \Psi_n = R_{\{\Psi,n\}}$

WP.S8.ResidualChannels — Residual channel decomposition

tier: untagged; status: current

$R_{\Phi} = J_{\{\text{cut},\Phi\}} + J_{\text{syn}} + J_{\text{geo}}$; $R_{\Psi} = J_{\{\text{cut},\Psi\}} + J_L$; cut current: $J_C = J_{\text{transport}} + J_{\text{return}} + J_{\text{readout}} + J_{\text{tape}}$

WP.S8.SecondDifference — Discrete second time-difference operator

tier: untagged; status: current

$\Delta t^2 X_n = (X_{n+1} - 2 X_n + X_{n-1}) / \Delta t^2$

WP.S8.SkewNearZero — Skew-sector near-zero readout condition

tier: untagged; status: current

$\langle z, G^{(-)} z \rangle_G \approx 0$

WP.S9.LegacyStepper — Legacy single-field stepper (reduced reference)

tier: untagged; status: current

$\gamma = 1/\tau_c$; $D_s = D * \Delta \theta$; $V[n+1] = V[n] + \Delta \theta * (-\gamma V[n] - D_s * (L_R @ X[n]) + \text{source}[n] - \text{residual}[n])$; $X[n+1] = X[n] + \Delta \theta * V[n+1]$

WP.S9.LegacyStepperRules — Legacy stepper status and forbidden claims

tier: untagged; status: current

LegacyStepperRules: status REDUCED_REFERENCE; cfl_bound meaning sufficient_not_necessary; must_not_claim [‘exceeding the bound implies unconditional instability’, ‘single-field stepper is the complete two-field ontology’, ‘scalar eigenmode represents the full operator without discarded-coupling audit’].

WP.S9.StabilityBound — Legacy stepper stability bound

tier: untagged; status: current

$\Delta \theta \leq 2 / (\gamma + \sqrt{\gamma^2 + 4 \lambda_{\max} D_s})$ (sufficient, not necessary)

WeldDSL — V.20 THE WELD — master equation and domain-DSL are one object (commuting square)

tier: finite_diagnostic; status: current

$q_D \circ F = F \#_D \circ q_D$ -- a domain q_D is an admissible SENTENCE of the DSL iff it commutes with the master stepper F . Witnessed by scripts/unification_weld.py (exact-rational, PASS): a quotient q_D that closes the square is admitted as a real sentence; a one-entry perturbation of F breaks the square and the quotient is correctly refused.

WhyD4_v1.9 — Why d=4 (v1.9) — derived spacetime dimension

tier: Th_coqc; status: current

The minimal discrete carrier (orientation bit tensor incidence bit, dim 4) + first-order isotropy + kinetic-closure gate + no-spectator derive d=4: $d \leq 4$ (≤ 3 orthogonal directions in R^3), d even (parity $(-1)^{d-1} = -1$), $d \neq 2$ (incidence spectator), with explicit realization $A_0 = \text{tau}_1$ tensor I , $A_i = \text{tau}_2$ tensor sigma_i , and $\text{Gamma}_T = -A_0 A_1 A_2 A_3$.

X — Forced Set X — boost-invariant box quadratic form

tier: untagged; status: current

box_quad is boost-invariant ($g^2(1-v^2) = 1$) -- the box quadratic form is unchanged under a boost -- narrow claim only, not the wider "unification" this book refuses (InfoLorentzInvariance)

X.1 — X.1 — What Is Machine-Checked (seven readouts of L_R)

tier: Th_coqc; status: current

Machine-checked theorems (axiom-free unless disclosed): growth ladder: $\forall n, \exists a_{n+1} > a_n$; record law: $\forall r, \epsilon(r) > 0$; CPTP completeness: completeness list $\rightarrow \Sigma K \uparrow K = I$; operator $\rightarrow$ metric: principal_symbol $L \rightarrow \text{metric_form}$; energy stability: $dE/dt = -D\|v\|^2$ (energy_nonincreasing / strict_decay); dispersion split: split_classical / split_quantum; unit-node grammar: force readback structure; domain_rescale_covariant; constant-coefficient spine (M, D, K, λ held constant) rescales covariantly -- [Th_coqc]. Files: formal/Quantum_Relativity_Formal_DAG_Deep_ConnectedClosure_MassFreeVisualBH_CognitiveRhythm_NoImpact.v; formal/RDL_SpineStability.v, RDL_MetricReadout.v, RDL_StarRig*.v, RDL_DtN.v.

X.2 — X.2 — The Coq-Physics Boundary

tier: untagged; status: current

Machine-checked structure $\neq$ empirical physics truth. Structure can be Coq-verified with disclosed assumptions. Numbers require laboratory measurement. The Coq floor closes STRUCTURE. Measurement closes NUMBERS.

X.3 — X.3 — Assumption Ledger (Th_coqc naming corollary)

tier: Th_coqc; status: current

The full axiom disclosure lives in formal/AXIOM_STATUS.md. Key principle: a claim is Th_coqc only when the Coq artifact builds AND assumptions are publicly disclosed. Corollary (sharpened 2026-07-21): a claim is Th_coqc only if the object being named in English matches the object actually quantified over in the .v file -- not a same-named neighbor, not an intended future generalization, not a scalar stand-in for a full operator.

X.4a — X.4 — The admissibility square (InfoQuotientCompressionExactness / Kemeny-Snell lumpability)

tier: Th_coqc; status: current

A domain-bridge translation $T_{\{a \rightarrow b\}}$ must commute with the two domains' generators: $T_{\{a \rightarrow b\}} \cdot F\#_a = F\#_b \cdot T_{\{a \rightarrow b\}}$, and must preserve readout. This square -- the admissibility square -- $q_{\{n+1\}} \cdot F\#_n = F\#_n \cdot q_n$ -- is the machine-checked theorem InfoQuotientCompressionExactness, and is the formal restatement of Kemeny-Snell lumpability (a quotient of a Markov chain is itself Markov exactly when this square commutes for the chain's generator), applied to L_R instead of a generic Markov generator, as the test for whether a claimed cross-domain reduction is honest or mistranslation.

X.4b — X.4 — Failure modes of the admissibility square + the Buckingham trap / correct order

tier: untagged; status: current

FAILURE MODES when $T_{\{a \rightarrow b\}} \cdot F\#_a \neq F\#_b \cdot T_{\{a \rightarrow b\}}$: mistranslation -- the map $T_{\{a \rightarrow b\}}$ itself is wrong/ill-posed; lost information -- q_a discarded exactly the structure b needs; insufficient resolution -- a's readout

grid is too coarse for b's dynamics; target lacks variables -- b's state space has no slot for what a is sending; no closure -- b cannot form a well-posed update law from the image. FORBIDDEN (the Buckingham trap): textbook-name $\rightarrow$ stuff-parameters-in $\rightarrow$ declare "derived". CORRECT: Retention $\rightarrow$ Structure $\rightarrow$ Domain-translation $\rightarrow$ Meaning $\rightarrow$ Report.

XI — Forced Set XI — memory precedes mass (and its converse)

tier: untagged; status: current

rate $\cdot \tau_c = 1$; τ_c equal $\Leftrightarrow$ dynamics identical -- equal memory (τ_c) forces identical dynamics regardless of how (M,D) split (M,D,dt nonzero) -- memory precedes mass (InfoMemoryBeforeMass); the converse -- one agreeing readout forces shared τ_c , and retains nothing finer than τ_c -- closed 2026-08-29 (InfoSharedReadoutForcesSharedMemory, axiom-free)

XI.1a — XI.1 — Claim-Tier Table

tier: untagged; status: current

Ax = Root axiom / primitive stance (P_0 , $\tau_c > 0$, $\Delta\theta > 0$). Df = Definition ($\gamma = 1/\tau_c$, $D_s = D \cdot \Delta\theta$). Th = Theorem-level claim in project canon (Asymmetry, temporal ordering). Th_coqc = Machine-checked structural theorem (Energy stability, CPTP, metric). finite_diagnostic = Computed diagnostic in code units (RDU readout, PGFT v0.6 smoke). Dr = Design-analogy / derived reading needing bridge assumptions (Mass-memory, human agency). Open = Explicitly open or not closed by current proofs (Quantum gravity, beyond-SM).

XI.1b — XI.1 — NEW rule: Th_coqc may never be paired with an unmatched continuum-physics noun (label inflation)

tier: Th_coqc; status: current

NEW rule, binding after the 2026-07-21 hollow-label audit: Th_coqc may never be paired with a continuum-physics proper noun ("quantum," "relativistic," "the Navier-Stokes equation," "turbulence") unless the .v file's quantified object is itself that continuum object, not a finite/discrete analogue that merely shares its name or its limiting behavior. When a finite analogue is what was actually verified, the correct tag is finite_diagnostic, and the write-up must say so in the same sentence that mentions the physics name. Six of the project's eight physics-interpretation cards were found in violation on 2026-07-21 and downgraded pending re-audit; two survived because their Coq object was the literal finite structure their name claimed to be (energy monotonicity, CPTP completeness).

XI.2a — XI.2 — Readout Invariants (π , ϕ , α)

tier: untagged; status: current

$\pi = 3.14159\dots$ (readout-invariant: ratio of geometry, not derived from dynamics). $\phi = 1.61803\dots$ (golden ratio: readout-invariant growth ratio). $\alpha = 1/137.036$ (fine-structure: measured boundary datum). These are readout-invariants -- they appear in multiple contexts but are not manufactured by the spine; they are boundary inputs or structural ratios.

XI.2b — XI.2 — Quantity is not a pre-existing $\mathbb{R}$ (Dr capstone)

tier: Dr; status: current

There is no $\mathbb{R}$ hiding behind the odometer. What is real is the readout that an agency operator actually returns -- bounded, discrete, at finite resolution h -- never a Platonic magnitude the readout merely approximates from outside. A quantity is a coordinate/projection onto whatever readout space the system can actually give, not a pre-existing continuum value measurement politely uncovers. Two projections can carry identical arithmetic and land on the same numeric value and still be different quantities: a quantity's identity is the role that number plays in the bounded, causal, inferential readout graph it came from, not its momentary number. This is exactly what InfoQuotientCompressionExactness commuting-square condition is built to make precise: stated over quotient roles, not raw numeric coincidence.

XI.3a — XI.3 — What This Project Claims (original list)

tier: Th_coqc; status: current

CLAIMS (explicitly): primordial distinguishability as root (Ax); $\tau_c > 0$ as retention axiom (Ax); MQ.08 as fundamental computational equation (Th); universal spine PDE unifies domains structurally (Th); energy monotone non-increasing (Th_coqc); dispersion split exists at λ_c (Th_coqc); metric from operator principal symbol (Th_coqc); CPTP completeness (Th_coqc); every observer's record is lossy (Th, N2); Newton readback admissible through the gate (finite_diagnostic); τ_c atlas spans 37 disciplines, ~85 orders (finite_diagnostic); human agency = τ_c^H instantiation (Dr, open).

XI.3b — XI.3 — Additions carried forward from the 2026-07-21 working session

tier: finite_diagnostic; status: current

Three governing layers are structurally distinct and must not be merged (Th): DRL-Telegraph root (2nd order) / RTPE turbulence relaxation (1st order, the $M \rightarrow 0$, $V \rightarrow 0$ limit) / LP-NS paraprodut audit (a CHECKER, not a generator). Turbulence enters through the nonlinear $\nabla V / (u \cdot \nabla)u$ paraprodut term, not the linear $M \cdot a$ inertial term (Th, 2026-07-21 correction). τ_R (first-order relaxation-memory) is the inertia that governs turbulence behavior; M is a posited parameter that quantum alone exercises meaningfully (finite_diagnostic: D/M vs QuTiP agree to $7.6e-4$). Mass is a readout of discrete τ_c via $m = \hbar / (2c^2 \tau_c)$; τ_c is discrete and prior to mass (Ax, founder-locked). Multimode L_R with skew decomposition $L_R = L_R^+ + L_R^-$ under the retention metric G handles linearized two-field cases without altering the master equation (Dr, PROPOSED, gated on pending test T1). A domain-discovery engine recovers the minimal closed update law, variable count, channel-mixing, and conservation laws from a raw (N,S) transition tape with zero imported domain formulas, and provably refuses to fabricate a wrong law under an adversarial battery (finite_diagnostic). A cross-domain lineage ledger (bR) tracks retained information across a quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport chain as a multiplicative obstruction account $I_Q = I_B + O_C + O_P + O_B$ (finite_diagnostic architecture; not yet validated against event-resolved real data). A Type-P evidence gate requires BOTH a machine-derived passing control and a machine-derived failing control; a gate with only one side is Type-U convention, not evidence (Th, the FAIL-ABLE gate law).

XI.4a — XI.4 — What This Project Does NOT Claim (original list)

tier: untagged; status: current

NON-CLAIMS (explicitly): derives G , $\hbar$, c , k_B , α from first principles; derives particle masses from zero; proves a new physics theory; solves quantum gravity; provides clinical guidance; provides nuclear authorization; provides orbital control authorization; provides hardware-control authorization; replaces standard cosmological measurement; replaces CODATA, laboratory measurement, or domain-specific safety review.

XI.4b — XI.4 — Non-claims added from the 2026-07-21 session

tier: untagged; status: current

Does NOT claim the skew- L_R / metric- G antisymmetry algebra is proven -- it is PROPOSED, gated on test T1, and must not be cited as settled. Does NOT claim the two-field wall (chemostat / Lotka-Volterra / MHD) is fully closed -- the remaining gap (endogenous state-dependent $L_R[I_R]$) is real and gated on test T2. Does NOT claim the domain-discovery engine has been validated on real chemostat or any real biological data -- pilot runs are synthetic/adversarial only; real-data validation is pre-registered future work. Does NOT claim the bR lineage ledger is a first-principles derivation of biological transport from quantum mechanics -- it is a finite_diagnostic accounting architecture over an obstruction ledger, awaiting event-resolved empirical test. Does NOT claim any "machine-checked" physics-interpretation card is checked in the continuum sense unless its Coq object literally is that continuum object.

XI.5-F1 — XI.5 — Falsifier F1 (retention axiom)

tier: untagged; status: current

F1: Show that $\tau_c = 0$ is consistent with retained structure (falsifies the retention axiom).

XI.5-F2 — XI.5 — Falsifier F2 (graph-operator reading)

tier: untagged; status: current

F2: Show that transport over a finite causal graph is impossible without L_R (falsifies the graph-operator reading).

XI.5-F3 — XI.5 — Falsifier F3 (Face 9, CPTP completeness)

tier: untagged; status: current

F3: Show that CPTP completeness fails for admissible channels (falsifies Face 9; Coq refutation would close this).

XI.5-F4 — XI.5 — Falsifier F4 (Face 4, energy stability)

tier: untagged; status: current

F4: Show that energy can increase monotonically under pure damping (falsifies Face 4; Coq would catch this).

XI.5-F5 — XI.5 — Falsifier F5 (Face 8, operator-to-metric)

tier: untagged; status: current

F5: Show that operator principal symbol gives no metric readout (falsifies Face 8; Coq refutation would close this).

XI.5-FT1 — XI.5 — Falsifier F-T1 (skew-L_R / metric-G algebra, test T1)

tier: Dr; status: current

F-T1: Test T1 -- construct a two-field system where the claimed skew-L_R / metric-G decomposition is well-defined, and show it FAILS to reproduce known linearized coupling behavior (falsifies the PROPOSED skew-L_R algebra of XIV.3; until T1 runs and passes, the algebra stays PROPOSED).

XI.5-FT2 — XI.5 — Falsifier F-T2 (two-field-wall gap, test T2)

tier: Open; status: current

F-T2: Test T2 -- construct an endogenous, state-dependent L_R[I_R] system (the remaining two-field-wall gap) and show no admissible closure of the master equation reproduces its qualitative behavior (would confirm the wall is real and not merely under-explored; a PASS on T2 instead would shrink the wall further).

XI.5-Fhuman — XI.5 — Falsifier F-human (τ_{c^H} / Dr claim)

tier: Dr; status: current

F-human: Measure τ_{c^H} directly and find it is not in the predicted range for multiple channels; or show the residual model breaks down systematically across populations (falsifies §VIII / Dr claim).

XI.6 — XI.6 — The FAIL-ABLE Gate Law

tier: untagged; status: current

A gate G is Type-P (evidence) only if BOTH exist and are machine-derived: (a) a PASSING control -- an input on which G is known/constructed to pass; (b) a FAILING control -- an input on which G is known/constructed to fail. If only (a) exists (or only a "should pass" story with no constructed failing case), G is Type-U (convention) -- it documents an agreement, not a discriminating test, and MUST NOT be cited as evidence for a claim. Tier consequence: Type-P gate passing $\rightarrow$ supports Th_coqc / finite_diagnostic tagging; Type-U gate passing $\rightarrow$ supports at most Dr, and must say "convention, not evidence" in the same sentence.

XII — Forced Set XII — metric is the energy readout

tier: untagged; status: current

`qform(L(edges), x)  $\equiv$  energy(edges, x)` -- the metric's Hessian and the energy functional are the same object (`InfoMetricIsEnergyReadout`)

XIII — Forced Set XIII — quantum-relativity dispersion identity survives boost

tier: untagged; status: current

`M $\cdot\omega^2$  = K $\cdot\lambda \iff$  box_quad = 0` (survives boost) -- the dispersion identity, and this null-condition survives a boost -- narrow claim only (`InfoQuantumRelativityUnification`)

XIII.1 — XIII.1 — Discrete Calderón / EIT Recovery

tier: Th_coqc; status: current

This project's discrete version of the Calderón problem asks: given a finite boundary readout (a Dirichlet-to-Neumann map, `RDL_DtN.v` in the formal floor), does it determine a hidden interior "IFF-tree" with no appeal to completeness at infinity? Answer: boundary readout does pin down the interior tree exactly, for the finite graphs tested -- the spine PDE's operator-to-metric face (F8, Appendix B) read through the inverse-problem quotient. Tier: the DtN machine-checked structure is `Th_coqc`; the "recovers the interior tree" instance results are `finite_diagnostic` until a broader battery is run.

XIII.2 — XIII.2 — Neural-Network-From-Root + Training Convergence

tier: finite_diagnostic; status: current

A network whose update rule is derived from the discrete stepper `MQ.08` rather than borrowed from backpropagation folklore trains to convergence over exact rational arithmetic ($\mathbb{Q}$ -closed -- no floating point rounding hides behind the convergence claim). Not a claim that this is a better network; it is a claim that the learning-domain quotient `q_learning` admits a translation from the spine's discrete stepper that commutes with gradient descent's own update structure on the tested fixtures. Tier: `finite_diagnostic`, $\mathbb{Q}$ -closed convergence proven on tested architectures and fixture sizes; general architecture-independent convergence remains Open.

XIII.3 — XIII.3 — DEC Toolkit (Discrete Exterior Calculus)

tier: Th_coqc; status: current

The Discrete Exterior Calculus toolkit -- `grad`, `div`, `curl`, Hodge star, harmonic decomposition, Sobolev bounds, and a discrete maximum principle, all built directly around `L_R` rather than around a continuum manifold. Each operator is a readout of `L_R` under a different classical-calculus quotient: `grad` and `div` through the boundary-operator quotient, `curl` and Hodge through the exterior-algebra quotient, the maximum principle through the monotone-operator quotient that also produces the energy-stability theorem in X.1. Tier: the toolkit's structural identities are `Th_coqc`; any specific PDE solved with it inherits `finite_diagnostic` at the instance level.

XIII.4 — XIII.4 — Black-Hole / τ_c -of-Agency Knife-Edge

tier: Dr; status: current

The horizon-transduction smoke suite (`smoke_omega_h.py`, 27 checks) exercises a reading of black-hole-like behavior as an extreme regime of the same spine PDE: a region where the finite cone condition $\|x\| \leq vt$ (Face 5) becomes the effective causal horizon, and where the record law ($\epsilon(r) > 0$) forces every crossing observer's record to be lossy in a way that reads, under the physics quotient, like horizon information loss. The same finite-cone/record-loss machinery, read under the agency quotient instead, produces the τ_c^H human-agency model (`HUMAN_AGENCY_SINGLE_TAU_C.md`). Both readings sit at the same structural boundary of the spine (the regime where the finite cone shrinks toward degeneracy) and diverge only in which quotient is applied. Neither reading derives the other; they are siblings under one trunk. Tier: firmly `Dr` for both readings; the underlying finite-cone and record-loss theorems that both depend on are `Th_coqc`.

XIII.5 — XIII.5 — Discrete Navier–Stokes Theorems (Stokes limit / advection cubic-form)

tier: Th_coqc; status: current

Stokes-limit theorem: in the regime where the nonlinear advection term is dropped, the discrete spine PDE's solution structure reduces to a discrete Stokes system, machine-checked (Th_coqc). Advection cubic-form result: the nonlinear $(u \cdot \nabla)u$ term, expressed in the DEC toolkit's discrete calculus, takes a cubic form whose sign properties were checked numerically across tested grid sizes and found consistent with the energy-stability requirement of X.1 -- finite_diagnostic, not yet a general machine-checked theorem, because the check was run on tested instances, not universally quantified and proven in Coq. This pair is the direct formal ancestor of the 2026-07-21 turbulence correction in PART XIV.

XIV — Forced Set XIV — torsion seeds the gauge algebra

tier: untagged; status: current

torsion $\neq 0 \Rightarrow$ non-abelian group of rank N -- a nonzero torsion witness generates a non-abelian group of rank N -- the gauge-algebra seed (InfoSeedTorsionGroupAndRankN)

XIV.1 — XIV.1 — Three Stacked Layers — Do Not Merge

tier: untagged; status: current

1. DRL-Telegraph root (2nd order): $M \partial^2 \Phi + D \partial \Phi + K \cdot L_R \cdot \Phi + \nabla V = J - \eta$ -- the trunk equation.
2. RTPE turbulence (1st-order relaxation): $\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R$ -- the $M \rightarrow 0, V \rightarrow 0$ LIMIT of layer 1, a genuine reduction, not an independent equation. Status: PASS_WITH_LIMITS.
3. LP-NS audit (nonlinear paraprodut diagnostic): A CHECKER, not a native generator -- audits whether a candidate solution to layers 1/2 is consistent with the nonlinear paraprodut structure; does not itself generate dynamics. Discipline: nothing may cite "the turbulence equation" as if on equal ontological footing with the trunk PDE.

XIV.2 — XIV.2 — The Turbulence Correction (supersedes prior reading)

tier: untagged; status: current

Turbulence does not live in the linear M-a inertial term. The corrected reading, following directly from the discrete Navier–Stokes advection-cubic-form result (XIII.5): turbulence lives in the nonlinear $\nabla V / (u \cdot \nabla)u$ paraprodut term. The inertial quantity that actually governs turbulent behavior is τ_R , the first-order relaxation-memory constant from the RTPE limit (XIV.1, layer 2) -- not M.

XIV.3 — XIV.3 — M Posited; Mass Is a Readout of Discrete τ_c

tier: finite_diagnostic; status: current

M, the inertial mass parameter in the trunk equation, is posited, not derived -- eight separate attempts to force M out of more primitive structure this session all failed. Corrected reading, founder-locked: mass is a readout of τ_c , via $m = \hbar / (2 c^2 \tau_c)$, with τ_c itself discrete and prior to mass (Ax, P0). The D/M ratio was checked against QuTiP and agreed to 7.6×10^{-4} (finite_diagnostic), giving some confidence the posited-M reading is numerically consistent in the one regime where M matters, without claiming M has been derived. QSL discriminator: this book's τ_c is NOT the textbook quantum speed limit (QSL) time τ (Mandelstam-Tamm / Margolus-Levitin) -- the two share only an algebraic shadow (both traffic in $\hbar/2E$ -flavor expressions), while τ_c is a discrete, prior, structural retention time and the textbook QSL τ is a derived bound on how fast a quantum state can evolve.

XIV.4 — XIV.4 — Scalar-Eigenmode Reduction Error and Skew- L_R

tier: Dr; status: current

The error: L_R is a full operator acting on a multimode state. A recurring shortcut treated it via its scalar eigenmode reduction, $\lambda \phi$, as if that told the whole story -- it does not, discarding the off-diagonal and skew coupling between modes. The proposed fix: decompose the operator as $L_R = L_R^+(+) + L_R^-(-)$ under the retention metric G -- not a naive transpose; G makes "symmetric"/"antisymmetric"

mean something physically meaningful in this setting. This metric-G / L_R-antisymmetry algebra is PROPOSED, not proven, and gated on test T1 (XI.5, F-T1). Nothing downstream may cite this decomposition as settled.

XIV.5 — XIV.5 — The Two-Field Wall Has Shrunk, Not Vanished

tier: Open; status: current

The "two-field wall": systems like chemostats, Lotka-Volterra predator-prey dynamics, and MHD couple two interacting state fields, which earlier analysis could not fit into the single-field spine PDE without changing the master equation itself. The multimode-plus-skew-L_R reading (XIV.4) handles the linearized cases of all three without touching the master equation. What remains is a real, named gap: endogenous, state-dependent L_R[I_R] -- cases where the operator itself depends on the state it acts on. This residual gap is gated on test T2 (XI.5, F-T2) and must be reported as an open gap. Cross-reference: A.10 ("number of fields $\neq$ number of domains") is the reason this is not a contradiction of any prior single-field claim.

XIV.6 — XIV.6 — Domain-Discovery Engine

tier: finite_diagnostic; status: current

Given nothing but a raw tape of (N, S) state transitions -- no imported biology, no imported chemistry, no hand-written domain formula -- the engine discovers, purely from the data: (a) how many variables the domain minimally needs (via an exact-gate closure test); (b) whether channels interact (via a mixed second-difference test); (c) the minimal exact update law over $\mathbb{Q}$ (rational arithmetic -- no floating-point rounding to hide behind); (d) conservation laws. The adversarial battery (FAIL-ABLE gate law, XI.6) includes passing controls (recovers N·S/8, N^2 , an affine law, a genuine 3-variable case) and failing controls it must reject (refuses to fabricate N·S/16, refuses non-polynomial candidate laws). Holdout data is matched exactly, not approximately. Not yet run on real chemostat or any other real biological dataset -- every result so far is synthetic and adversarial-battery-only.

XIV.7 — XIV.7 — The bR Cross-Domain Lineage Ledger

tier: finite_diagnostic; status: current

Tracks retained information along a claimed reduction chain quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport expressed as $r_B = E \cdot A_C \cdot A_P \cdot A_B$ (a product of an SI-decode factor E, external/measured, and per-stage admissibility factors A_C , A_P , A_B for the chemical, protein, and biological-transport translations). Key diagnostic result: the single quantum-domain quotient q_Q does not commute for the biological question -- trying to answer a biological-transport question using only the raw quantum quotient, skipping the intermediate chemical and protein translations, fails the admissibility square of X.4. The ledger's obstruction certificates conserve retained lineage as $I_Q = I_B + O_C + O_P + O_B$ -- total quantum-origin information equals what survives to the biological readout (I_B) plus what was lost at each intermediate obstruction (O_C , O_P , O_B). E is treated as external throughout. A real test needs event-resolved empirical data, pre-registered but not yet run.

XIV.8 — XIV.8 — Consolidated Status Table

tier: untagged; status: current

Three-layer separation (Telegraph/RTPE/LP-NS): Th, settled this session. Turbulence in ∇V paraproduct, not M: Th, supersedes prior reading. M posited; mass = readout of discrete τ_c : Ax+finite_diagnostic, gate D/M vs QuTiP 7.6e-4, status consistent, not derived. Scalar-eigenmode reduction error identified: Th, settled finding. Skew-L_R / metric-G decomposition: Dr, PROPOSED, gate T1, pending. Two-field wall (linearized cases): Th, shrunk, closed for linearized. Two-field wall (endogenous L_R[I_R]): Open, gate T2, remaining gap, real. Domain-discovery engine: finite_diagnostic, gate adversarial battery (Type-P), passed; real-data validation pending. bR lineage ledger: finite_diagnostic architecture gate event-resolved test, not yet run. FAIL-ABLE gate law: Th, self-applying, binding, PR #29.

XIX — Forced Set XIX — selected state minimizes the action

tier: untagged; status: current

selected state = argmin(action) -- the selected state minimizes the action -- the general variational principle (InfoSeedArgminActionCost)

XV — Forced Set XV — coercive potential, bounded closure

tier: untagged; status: current

∇V coercive, bounded closure -- the potential's coercivity is bounded -- wshare/wdeg closure (InfoCoercivi

XVI — Forced Set XVI — Clairaut symmetry of the metric lift

tier: untagged; status: current

$\partial^2(\text{metric})/\partial x \partial y = \partial^2(\text{metric})/\partial y \partial x$ -- mixed-partial (Clairaut) symmetry of the metric lift (InfoAnalysisLif
-- solver arc (private) copy; NOT the causal-quantum-gravity same-named file, which imports Schwarzschild
and is refused)

XVII — Forced Set XVII — seed readout vanishes iff homogeneous

tier: untagged; status: current

SeedReadout = 0 $\iff$ Φ homogeneous -- velocity+coupling+damping combine into one seed readout that vanishes iff the field is homogeneous (InfoSeedUnifiedMasterEquation)

XVIII — Forced Set XVIII — dispersion relation is scale-gauge invariant

tier: untagged; status: current

dispersion relation is scale-gauge invariant -- the dispersion relation's sign/structure is invariant under a scale-gauge transform (InfoScaleGaugeNonReadout)

XX — Forced Set XX — closed loop pure gauge iff flat

tier: untagged; status: current

closed loop pure gauge $\iff$ flat -- a connection from any frame difference; genuine curvature is exactly the non-coboundary part (InfoConnectionFromFrame)

XXI — Forced Set XXI — discrete Riemann curvature from second differences

tier: untagged; status: current

flat $\iff$ second difference = 0 -- discrete curvature from second differences (InfoDiscreteRiemannCurvature

XXII — Forced Set XXII — curvature recovered from transport commutator

tier: untagged; status: current

curvature recovered from the transport commutator -- a nonvacuous witness recovers curvature from the transport commutator (InfoDiscreteRiemannCommutator)

XXIII — Forced Set XXIII — λ_c as the spine's own crossover

tier: untagged; status: current

λ_c is the spine's own crossover -- the classical/quantum crossover λ_c -- a distinct object from GR redshift, not a claim of unifying the two (InfoTelegraphHorizonUnification)

XXIV — Forced Set XXIV — Jacobi from associativity of composition

tier: untagged; status: current

$K(X,Y) = XY - YX$ bilinear+antisym.; Jacobi holds -- Jacobi follows from associativity of ordered composition alone -- no Lie algebra imported; the algebraic identity is general, the specific noncommuting witness pair is hand-exhibited (InfoOrderDefectFromComposition)

bRLedger — VI.6 — bR cross-domain lineage ledger

tier: finite_diagnostic; status: current

$r_B = E \cdot A_C \cdot A_P \cdot A_B$; tracing quantum $\rightarrow$ chemical $\rightarrow$ protein $\rightarrow$ biological-transport as a chained readout

bR_Ledger_qQ_negative_result — V.13 bR Ledger — negative result: a single quantum quotient q_Q does not commute to biology

tier: finite_diagnostic; status: current

A single quantum-domain quotient q_Q does not commute, on its own, all the way to a biological question -- attempting to answer a biological-transport question using only q_Q reproduces exactly the kind of commuting-square failure IV.5 predicts for an under-resolved bridge, recorded as a second, independent confirmation of the Scalar-Eigenmode Reduction Error.

bR_Ledger_r_B — V.13 bR Cross-Domain Lineage Ledger — lineage ratio

tier: finite_diagnostic; status: current

$r_B = E \cdot A_C \cdot A_P \cdot A_B$ -- where E is the external SI-decode factor (explicitly external), and A_C , A_P , A_B are the retention-preserving amplitudes attached to the chemical, protein, and biological-translation steps respectively.

k_ep — VI.3 — k_ep (consistency coupling)

tier: untagged; status: current

k_{ep} = consistency coupling (coherence of the supporting set); "the epistemic reading of N2 through a multimode L_R : how coherently the supporting set of readouts agrees"

lambda0_collapse — V.5 Complex Systems — the $\lambda=0$ collapse (kernel of L_R as irreversible-collapse boundary)

tier: Dr; status: current

As λ_2 is driven toward zero, the graph approaches disconnection: the kernel of L_R (the connected-component null space that Fiedler's $\lambda_2 \rightarrow 0$ limit detects) is the point past which the collapse cannot be reversed from inside the graph. At that point the readout genuinely is zero, not merely small -- the zero-side of the same zero-infinity distinction the book insists on for every other non-readout limit; a boundary a system can approach and even name, but never a magnitude the readout itself ever actually reports as a finite number produced by dynamics.

pi_phi_page_curve — V.15 Open bridges — π/ϕ retained-history fusion identity (external contribution)

tier: Open; status: current

$\tau(x) \tau \approx 1$ (+) $\tau \Rightarrow \text{FPdim}(\tau) = \phi$ -- closed only under 7 declared gates H1-H7; root-forcing of H1-H7 in any natural domain is untested; a candidate golden-ratio measurement schedule shows opposite results under two different protocols (positive AND negative controls both kept); physical Page-curve/black-claims are explicitly [Open]/not demonstrated. Nothing here enters THE FORCED SET.

q1_hollow_card_downgrade — V.1 Quantum — 6-of-8 hollow physics-interpretation-card downgrade

tier: Open; status: current

A self-audit of the physics interpretation cards attached to the quantum and GR leaves found that 6 of 8 cards were hollow -- labeled with continuum physics names while the actual content was an arithmetic tautology dressed up to look machine-checked. Those 6 cards are downgraded to [Open] as of this edition; the remaining 2 of 8 stand as previously tiered. The CHSH/Tsirelson guard and RDL_Heisenberg are unaffected.

q_biology_eigenmode — V.6 Biology — organism as driven self-sustained eigenmode

tier: untagged; status: current

organism = driven self-sustained eigenmode (τ_c = body scale); diffusion, osmotic, Stokes, Nernst, Michaelis-Menten readouts; homeostasis = obstruction reduction; mortality = irreversible decay

q_biology_engine_home_leaf — V.6 Biology — domain-discovery engine's home leaf (adversarial battery run as a biological N,S tape)

tier: finite_diagnostic; status: current

The V.0 finite_diagnostic engine's adversarial battery (recovering $N^2S/8$, $N^2/8$, affine, and 3-variable laws; never fabricating $N^2S/16$; exact holdout; refusing non-polynomial fits) was run in the shape of a biological population/nutrient tape (N, S). This is a success at the level of the engine, not yet a biology result -- it has not been run against real chemostat data.

q_chemistry_RD — V.7 Chemistry — reaction-diffusion smoother

tier: untagged; status: current

reaction-diffusion = $\text{grad}(V) + K \cdot \text{grad}^2$ smoother; kinetics and RD patterns; quantum chem binding (τ_c = molecular)

q_cognitive_recordlaw — V.8 Cognitive — mind's record law

tier: untagged; status: current

record law $M_A = K_A \theta(E) + \eta$; mutual information / residual; finite-access readout ($\epsilon_{\text{tot}} > 0$ always)

q_complex_fiedler — V.5 Complex Systems — graph-spectral translation (Fiedler value, criticality)

tier: untagged; status: current

graph spectrum, Fiedler λ_2 ; criticality at λ_c ; synchronization / diffusion readout

q_cosmology — V.11 Cosmology — expansion as growth in accessible causal relations

tier: untagged; status: current

cosmic τ_c ; ordering / scale unfolding; expansion = growth in accessible causal relations

q_finance_RTPE — V.10 Finance — RTPE relaxation equation

tier: untagged; status: current

RTPE $\tau_R dI/dt + L_R I = S + \eta$; OU / mean-reversion face; η = market turbulence (residual)

q_formal — V.12 Formal Math — the leaf that checks the honesty of the other leaves' translations

tier: Th_coqc; status: current

metric readout (Coq: RDL_MetricReadout); CPTP completeness (Coq: RDL_StarRig*); spine stability (Coq: RDL_SpineStability); telescope / no-creation; disclosed Coq axiom tiers (formal/AXIOM_STATUS.md)

Code	Dom.	Statement	Status
q_formal/M.01.v1	M	Forman(-Ricci) combinatorial curvature	current
q_formal/M.02.v1	M	Ollivier-Ricci curvature	current
q_formal/M.03.v1	M	Ricci curvature of graphs (Forman/Ollivier relation)	current
q_formal/M.04.v1	M	Rayleigh quotient / theory of sound	current

Code	Dom.	Statement	Status
q_formal/M.05.v1	M	Anderson–Morley eigenvalue bound $\lambda_{\max}(\deg u + \deg v)$	current
q_formal/M.06.v1	M	Algebraic connectivity / Fiedler value	current
q_formal/M.07.v1	M	Diameter-based algebraic-connectivity floor $\lambda_{\{2\}} \geq 4/(nD)$	current
q_formal/M.08.v1	M	Spectral graph theory (survey)	current
q_formal/M.09.v1	M	Can one hear the shape of a drum?	current
q_formal/M.10.v1	M	Graph Laplacian $\rightarrow$ Laplace–Beltrami convergence	current
q_formal/M.11.v1	M	Crystallographic restriction theorem (only n -fold rotational symmetries with n in $\{1,2,3,4,6\}$ are compatible with a lattice; equivalently $\text{trace} = 2\cos(2\pi/n) \in \mathbb{Q} \implies n \in \{1,2,3,4,6\}$)	current
q_formal/M.12.v1	M	Euler–Lagrange first variation	current
q_formal/M.13.v1	M	Noether’s theorem	current
q_formal/M.14.v1	M	Courant–Friedrichs–Lewy stability condition	current
q_formal/M.15.v1	M	Green’s identity	current
q_formal/M.16.v1	M	Gauss–Ostrogradsky divergence theorem	current
q_formal/M.17.v1	M	Störmer–Verlet leapfrog integrator	current
q_formal/M.18.v1	M	Marsden–West discrete mechanics	current
q_formal/M.19.v1	M	Lax–Richtmyer stability equivalence	current
q_formal/M.20.v1	M	Polyak’s heavy-ball momentum method	current
q_formal/M.21.v1	M	Nesterov’s accelerated gradient method	current
q_formal/M.22.v1	M	ODE model of Nesterov’s method	current
q_formal/M.23.v1	M	Integral quadratic constraints for optimizer analysis	current
q_formal/M.24.v1	M	Onsager reciprocity relations (symmetric linear-response/susceptibility when circulation vanishes)	current
q_formal/M.25.v1	M	Discrete cosine transform (DCT-II) diagonalization of the free/Neumann path-graph Laplacian	current
q_formal/M.26.v1	M	Discrete sine transform (DST) diagonalization of the pinned/Dirichlet path-graph Laplacian	current
q_formal/M.27.v1	M	Chebyshev polynomial expansion of a matrix exponential (modified-Bessel coefficients)	current
q_formal/M.28.v1	M	Lanczos/Krylov-subspace approximation of matrix functions	current
q_formal/M.29.v1	M	expm_multiply scaling-and-squaring / Krylov baseline algorithm	current
q_formal/M.30.v1	M	Regular-polygon planar central configuration (equal masses, rigid rotation)	current
q_formal/M.31.v1	M	Errors-in-variables (EIV) attenuation: $\hat{M}_{\text{OLS}}/\hat{M}_{\text{true}} \approx \text{Var}(a_{\text{true}})/(\text{Var}(a_{\text{true}}) + \text{Var}(\text{noise})),$ method-of-moments correction by subtracting known noise variance from the observed regressor second moment	current
q_formal/M.32.v1	M	Instrumental-variable (IV) estimation via two independent noisy replicate measurements of the same latent regressor, $\hat{M} = (a_1^2 + a_2^2)/(2 + a_1^2 + a_2^2)$	current

q_formal_commuting_square — V.12 Formal Math — commuting-square identity (InfoQuotientCompressionExactness)

tier: Th_coqc; status: current

$q_{\{n+1\}}$. $F_n = F\#_n$. q_n (InfoQuotientCompressionExactness) -- one of q_formal 's Th_coqc entries; the commuting-square criterion of IV.5 gets its teeth here

q_geometry_LR_Laplacian — V.3 Geometry/Manifold — L_R as (minus) the Laplace-Beltrami operator

tier: Th_coqc; status: current

$L_R \rightarrow -\Delta_g$ (operator-first metric); principal symbol $\rightarrow$ metric

q_quantum_CHSH — V.1 Quantum — CHSH / Tsirelson bound (quantum-correlation guard)

tier: untagged; status: current

CHSH / Tsirelson bound (quantum-correlation guard)

q_quantum_CPTP — V.1 Quantum — CPTP channel readout

tier: untagged; status: current

CPTP channel readout $\sum K\text{-dagger } K = I$

q_quantum_Heisenberg — V.1 Quantum — uncertainty guard

tier: Th_coqc; status: current

uncertainty guard $\Delta x * \Delta p \geq \hbar/2$ (Coq: $RDL_Heisenberg$)

q_quantum_lambda_c — V.1 Quantum — dispersion split (critical wavenumber-associated eigenvalue)

tier: untagged; status: current

dispersion split $\lambda_c = D^2 / 4MK$

q_quantum_osc — V.1 Quantum — oscillatory modes threshold

tier: untagged; status: current

oscillatory modes $k > k_c$

q_relativity_cone — V.2 Relativity/GR — finite causal cone

tier: untagged; status: current

finite cone $\|x\| \leq vt$, $v = \sqrt{D/\tau_c}$

q_relativity_metric — V.2 Relativity/GR — metric read from the operator symbol

tier: untagged; status: current

metric from operator symbol $g^{\{ij\}} = 1/2 \text{ Hess } \sigma_{\{L_R\}}$

q_social — V.9 Social — civilization as the spine at social τ_c

tier: untagged; status: current

civilization = spine at social τ_c ; testimony / institutions / repair; diffusion / agent-network readout

q_turbulence_fullPDE — V.4 PDE/Turbulence — full PDE with nonlinear cascade term

tier: untagged; status: current

full PDE $M \Phi_{tt} + D \Phi_t + K L_R \Phi + \text{grad}(V) = J - \eta$

step1 — Step 1 — Bare distinction (P0)

tier: untagged; status: current

$\exists a, b : a \neq b$ -- Something is distinguishable from something else.

step10 — Step 10 — Graph diffusion coefficient

tier: untagged; status: current

$D_s = D \cdot \Delta\theta$ -- Graph diffusion coefficient.

step11 — Step 11 — Velocity update (MQ.08)

tier: untagged; status: current

$V[n+1] = V[n] + \Delta\theta(-\gamma V[n] - D_s L_R X[n])$ -- Velocity update. The engine of change.

step12 — Step 12 — State update (MQ.08)

tier: untagged; status: current

$X[n+1] = X[n] + \Delta\theta V[n+1]$ -- State update. History accretes.

step13 — Step 13 — CFL / stability condition

tier: Th_coqc; status: current

$\Delta\theta \leq 2 / (\gamma + \sqrt{\gamma^2 + 4 \lambda_{\max} D_s})$ -- Sufficient stability gate (conservative). Step too large removes the guarantee -- not proof of breakdown.

step14 — Step 14 — Telegraph equation (DRL-Telegraph root)

tier: untagged; status: current

$\tau_c \partial_t^2 u + \partial_t u = D \nabla^2 u$ -- Coarse-grain shorthand of the stepper. Keeps memory + finite speed. [2026-07-21] identified as the DRL-Telegraph root; the sibling RTPE turbulence relaxation $\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R$ is the $M \rightarrow 0, V \rightarrow 0$ limit of this equation, and was found today to PASS_WITH_LIMITS as a controlled reduction; the two, plus the LP-NS nonlinear-paraproduct checker, are stacked layers, not to be merged.

step15 — Step 15 — Memory kernel

tier: untagged; status: current

$K(\tau) = (1/\tau_c) e^{-(\tau/\tau_c)}, \int_0^\infty \tau K(\tau) d\tau = \tau_c$ -- Memory kernel. The past fades exponentially.

step16 — Step 16 — Universal spine PDE

tier: untagged; status: current

$M \partial_t^2 \Phi + D \partial_t \Phi + K L_R \Phi + \nabla V(\Phi) = J - \eta$ -- THE universal spine. One equation for everything. [2026-07-21, central correction of the day] turbulence lives in the nonlinear $\nabla V(\Phi) / (u \cdot \nabla)u$ paraproduct term, not the linear inertial term $M \partial_t^2 \Phi$ (eight forcing attempts on M failed); the inertia relevant to turbulence is τ_R (RTPE), not M; the LP-NS audit is a checker on the $\nabla V(\Phi)$ term, not a generator.

step17 — Step 17 — Gauge-covariant extension

tier: untagged; status: current

$\partial_\mu \rightarrow \mathfrak{D}_\mu = \partial_\mu + A_\mu$ -- Gauge-covariant extension. Forces arise from this.

step18 — Step 18 — Face 1: eigenmode decomposition

tier: untagged; status: current

$L_R \phi_k = \lambda_k \phi_k$ -- Eigenmodes of the causal graph. Stable structures. [2026-07-21, second major correction] this scalar reading is named the Scalar-Eigenmode Reduction Error: L_R is a full multimode operator; proposed fix $L_R = L_R^{(+)} + L_R^{(-)}$, skew part defined against the retention metric G , [PROPOSED pending test T1]; endogenous state-dependent $L_R[I_R]$ tracked as test T2.

step19 — Step 19 — Face 2: mode decay

tier: untagged; status: current

$|a_k[n]| \leq |a_k[0]| e^{(-\gamma_k n \Delta\theta)}$ -- Modes decay. Impermanence is structural.

step2 — Step 2 — Discriminator exists

tier: untagged; status: current

$\exists A : A \text{ discriminates } E_1 \neq E_2$ -- A discriminator exists. The observer relation is introduced.

step20 — Step 20 — Face 3: dispersion split

tier: Th_cogc; status: current

$\lambda_c = D^2/(4MK)$, $k_c = 1/(2\sqrt{(\tau_c D)})$ -- Critical split: below = classical, above = quantum-like.

step21 — Step 21 — Face 4: energy monotonicity

tier: Th_cogc; status: current

$dE/dt = -D \|v\|^2 \leq 0$ -- Energy monotone non-increasing. Stability theorem.

step22 — Step 22 — Face 5: light-cone

tier: untagged; status: current

$\|x\| \leq vt$, $v = \sqrt{(D/\tau_c)}$ -- Light-cone. Causal horizon.

step23 — Step 23 — Face 6: mass-memory

tier: Dr; status: current

$m = \hbar/(2 \tau_c c^2)$ -- Mass = inverse memory. Every particle has a characteristic timescale. [2026-07-21] mass is a readout of τ_c ; τ_c is discrete and prior to mass; M (the inertial coefficient in step 16's spine) is [Dr]/posited, not derived, and is exercised only by the quantum face (D/M vs QuTiP discrepancy 7.6×10^{-4} , finite_diagnostic).

step24 — Step 24 — Face 7: root force

tier: untagged; status: current

$F_{\text{root}} = -K L_R \Phi - \nabla V(\Phi)$ -- Root force = geometric + nonlinear imbalance.

step25 — Step 25 — Face 7: Newton readback gate

tier: untagged; status: current

$\text{force_to_SI_newton}(F_{\text{root}})$ iff gate passes -- Newton readback is admissible only through the unit/source/read gate.

step26 — Step 26 — Face 8: operator→metric

tier: Th_cogc; status: current

$g^{ij} \approx \frac{1}{2} \text{Hess}(\text{principal symbol of } L_R)$ -- Metric = symbol of the operator. Geometry from dynamics.

step27 — Step 27 — Face 9: CPTP completeness

tier: Th_cogc; status: current

$\Sigma_j K_{j\dagger} K_j = I$ -- CPTP completeness. Quantum channel structure.

step28 — Step 28 — Face 10: record law (readout-not-truth)

tier: untagged; status: current

$M_A[n] = K_A \cdot \theta(E[n]) + \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}}, \epsilon_{\text{tot}} > 0$ -- Record law. Every observation is lossy.

step29 — Step 29 — Face 11: obstruction

tier: untagged; status: current

$O_R(R^{\diamond}) = O_{\{E_R\}}, S_R = \|O_R\|^2$ -- Obstruction = residual inconsistency. Solving = zeroing the bundle.

step3 — Step 3 — Asymmetry

tier: untagged; status: current

$A \rightarrow B \neq B \rightarrow A$ -- Asymmetry. The direction of relation matters.

step30 — Step 30 — Face 12: boundary discipline

tier: untagged; status: current

absolute constants = boundary data (measured, not derived) -- $G, \hbar, c, k_B, e, m_e, \alpha$ are inputs, not outputs.

step31 — Step 31 — Scale bus: τ_c per domain

tier: untagged; status: current

$\tau_c = \hbar/(2mc^2)$ for each domain's mass/scale -- The bridge connecting all domains. 37 disciplines, 85 orders. [2026-07-21] audited by the bR cross-domain lineage ledger: $r_B = E \cdot A_C \cdot A_P \cdot A_B$; $I_Q = I_B + O_C + O_P + O_B$; a single quantum-domain quotient q_Q does NOT commute directly to a biological-level question -- intermediate chemical/protein layers each contribute their own obstruction term.

step32 — Step 32 — One trunk, many leaves (canon scale principle)

tier: untagged; status: current

$\forall \text{ domain } d: \text{PDE}(d) = \text{SPINE_PDE at } \tau_c(d), L_R(d)$ -- Every domain is the same equation read differently. [2026-07-21] domain-discovery engine result (finite_diagnostic): given a raw (N,S) transition tape, without importing a domain formula in advance, discovers minimal variable count (exact-gate closure test), channel interaction (mixed second differences), minimal exact update law over $\mathbb{Q}$, and conservation laws; recovered $N \cdot S/8, N^2/8$, an affine law, and a genuine 3-variable case; never fabricated a plausible wrong law (e.g. $N \cdot S/16$); holdout data matched exactly; non-polynomial laws correctly refused.

step33 — Step 33 — Record genesis (RAR V1.4)

tier: untagged; status: current

$R_O = \Omega_A \circ A \circ \Pi \circ T_{\Sigma}(D_O)$ -- How a record is formed from raw domain data D_O .

step34 — Step 34 — Human readout (species-level instantiation)

tier: Dr; status: current

$M_H[n] = K_H \theta(E[n]) + \eta_H[n]$ -- Human readout. Species-level τ_c^H instantiation.

step35 — Step 35 — Optimal policy preserves repairability

tier: Dr; status: current

$\pi^* = \operatorname{argmin} [E\|\operatorname{Res}\|^2 + \lambda C + \rho \operatorname{Risk} - \nu \operatorname{Val} + \mu \operatorname{RepairLoss}]$ -- Optimal policy preserves repairability.

step36 — Step 36 — Three CPU epistemic scalars

tier: finite_diagnostic; status: current

$\operatorname{Re_ep}, \operatorname{F_ep}, \operatorname{k_ep}$ (three CPU scalars) -- Any reasoning step reduces to these three numbers.

step37 — Step 37 — Verdict gate (DECIDE/ABSTAIN/ESCALATE)

tier: Dr; status: current

DECIDE | ABSTAIN | ESCALATE -- Verdict gate over the three scalars. [2026-07-21] FAIL-ABLE gate

law: a genuine Type-P evidence gate needs both a machine-derived passing control and a machine-derived failing control; a gate with only passing or hand-picked-failing examples is Type-U (a convention dressed as evidence, not a validated threshold).

step38 — Step 38 — Coq growth ladder

tier: Th_coqc; status: current

Coq growth ladder: $\forall n, \exists a_{n+1} : a_{n+1} > a_n$

step39 — Step 39 — Coq record law

tier: Th_coqc; status: current

Coq record law: $\forall r \in \operatorname{Records}, \epsilon(r) > 0$

step4 — Step 4 — Time as transition count

tier: untagged; status: current

$t(s) := \min \# \operatorname{steps}(s_0 \rightarrow s) \in \mathbb{N}$ -- Time = counting admissible transitions. Not a container.

step40 — Step 40 — Coq CPTP completeness

tier: Th_coqc; status: current

Coq CPTP completeness: $\forall K_list, \operatorname{completeness} K_list \rightarrow \sum K^\dagger K = I$

step41 — Step 41 — Coq operator $\rightarrow$ metric

tier: Th_coqc; status: current

Coq operator $\rightarrow$ metric: $\forall L, \exists g = \operatorname{readout_metric}(L)$

step42 — Step 42 — SM-EFF closure (boundary data)

tier: finite_diagnostic; status: current

$v_{\operatorname{Higgs}} = 246.22 \operatorname{GeV}$ (CODATA/measured input); $\sum Y = 0$ (hypercharge cancellation -- structural discipline); $\tau_{\mu} = 2.187 \times 10^{-6} \operatorname{s}$ (muon lifetime -- measured boundary datum). SM-EFF: standard-model effective readout using declared measured inputs. Not a derivation from zero. Boundary data closes the number, structure closes the roles/relations. There is no step 43 -- the stream is exactly and only 42 steps.

step5 — Step 5 — Retention axiom

tier: untagged; status: current

$\tau_c > 0$ -- Some structure persists between transitions. Memory exists.

step6 — Step 6 — Discreteness of time

tier: untagged; status: current

$t = n \Delta\theta$, $\Delta\theta > 0$ -- World is discrete before it is continuous.

step7 — Step 7 — L_R defined as retained transport graph

tier: Definition; status: current

L_R = retained graph of admissible transport -- The causal graph emerges. What can reach what?

step8 — Step 8 — Finite causal propagation speed

tier: untagged; status: current

$v = \sqrt{(D/\tau_c)} < \infty$ -- Causal propagation is finite-speed.

step9 — Step 9 — Damping definition

tier: untagged; status: current

$\gamma = 1/\tau_c$ -- Damping = inverse memory time.

v011.1 — APP-v011.1 Hidden-memory kernel + observed/hidden state split

tier: untagged; status: current

$x_{\{0,n+1\}} = F_{\{00,n\}} x_{\{0,n\}} + R_n x_{\{H,n\}} + B_{\{0,n\}} u_n x_{\{H,n+1\}} = W_n x_{\{0,n\}} + F_{\{H,n\}} x_{\{H,n\}} + B_{\{H,n\}} u_n K_{\text{mem}}[m] = R_{\{n+m\}} (\prod_{j=1}^{m-1} F_{\{H,n+j\}}) W_n$ (x_0 = observed sub-state, x_H = hidden sub-state, $K_{\text{mem}}[m]$ = memory kernel at lag m)

v011.10 — APP-v011.10 Continuum readout map + validity gate

tier: Open; status: current

$\Phi_\epsilon(t) = C_{\{\alpha,\epsilon\}}(\Phi_0, \Phi_1, \dots)$ [Df] $\delta_{-t^2} \Phi_n \rightsquigarrow \partial_{-t^2} \Phi$, $\delta_{-t^c} \Phi_n \rightsquigarrow \partial_{-t} \Phi$ (only when bridge error is controlled) [Open validity] (The native order remains discrete; a continuum field is a coarse-grain readout, valid only when the bridge error is controlled -- otherwise the arrow substitution is unlicensed. Correct order: retained discrete synthesis $\rightarrow$ DRL recurrence $\rightarrow$ continuum readout.)

v011.11 — APP-v011.11 Five explicit conditions for a valid scalar reduction

tier: untagged; status: current

The frozen scalar equation $R = M a + D v + K \lambda \phi + V' - j + \eta$ is obtained only when all of the following hold: 1. a valid quotient has already been found; 2. one invariant mode v_a is sufficient for the declared readout; 3. off-diagonal, skew and moving-basis currents vanish in that sector; 4. geometry is fixed or externally prescribed; 5. no retained nonlinear transfer between modes is required. (The scalar equation is a legitimate reduced theorem domain, not the complete ontology of the generalized architecture.)

v011.12 — APP-v011.12 15-step runtime protocol

tier: Dr; status: current

1. Register the domain semantic card without importing its governing equation. 2. Encode observations into RD and declare decoder/calibration separately. 3. Discover the partition and multimode state space. 4. Audit the exact quotient and readout factorization. 5. Build the full operator; do not diagonalize prematurely. 6. Measure symmetric, antisymmetric and discarded modal sectors. 7. Generate only admissible syntheses and append every obstruction certificate. 8. Evolve the DRL-Telegraph reader-record pair. 9. Evolve retained geometry when it is endogenous. 10. Construct the turbulence retained-state readout when requested. 11. Audit the DRL-to-RTPE bridge defect. 12. Apply the LP/NS layer only as a separately labelled checker. 13. Audit conservation, lineage, tape, recovery and identifiability. 14. Decode the final RD report into domain units only after calibration. 15. Compare with preregistered external benchmarks using a checker independent of the maker.

v011.13 — APP-v011.13 Named open items

tier: Open; status: current

axiom-free Coq lift from scalar rationals to generalized operator modules; exact derivation of RTPE from the full DRL-Telegraph architecture; endogenous living-geometry action with complete backreaction; derivation of nonlinear transfer without importing benchmark laws; continuum-uniform turbulence residual bound; external DNS/experimental validation and component ablation; full chemostat, Lotka-Volterra and MHD derivations from native RD dynamics; a unified hybrid action deriving discovery, admissibility, tape and turbulence together.

v011.2 — APP-v011.2 Injectivity laws (output-residue transition, tape-encoder)

tier: untagged; status: current

$(z_n, u_n) \neq (z_n', u_n') \Rightarrow (z_{\{n+1\}}, r_n) \neq (z_{\{n+1\}}', r_n')$ [Ax] no distinction lost at the step
 $E_{\{T,n\}}(r) = E_{\{T,n\}}(r') \Rightarrow r = r'$ [Ax] tape encoder is injective

v011.3 — APP-v011.3 DRL-Telegraph $\rightarrow$ RTPE conditional bridge

tier: Open; status: current

$C_{\text{turb},n+1} \circ F_n = F_{\{R,n\}} \circ C_{\text{turb},n}$ [Df] exact-bridge condition $\epsilon_{\text{bridge},n} = C_{\text{turb},n+1}(F_n \mathbin{\&\circ} \mathcal{G}_n)$
– $F_{\{R,n\}}(C_{\text{turb},n} \mathbin{\&\circ} \mathcal{G}_n)$ [Df] bridge defect (if only approximate) $\eta_{\{R,n\}} = \eta_{\text{unresolved},n} + E_{\text{bridge}}(\epsilon_{\text{br}}$
[Df] defect folded into the RTPE residual (That RTPE is an exact reduction of the full DRL-Telegraph architecture is [Open].)

v011.4 — APP-v011.4 LP/NS finite-diagnostic residual (checker only, never native dynamics)

tier: finite_diagnostic; status: current

$H_{\text{q}}^{\text{rem}} = [H_{\text{q}}^{\text{hid}} - C_{\text{q}}^{\text{tr}} - C_{\text{q}}^{\text{p}} - C_{\text{q}}^{\text{inc}} - C_{\text{q}}^{\text{nonalign}} - C_{\text{q}}^{\text{hf}} - C_{\text{q}}^{\text{para}}]_+$ (six correction terms: transport C_{q}^{tr} , pressure C_{q}^{p} , incompressibility $C_{\text{q}}^{\text{inc}}$, non-alignment $C_{\text{q}}^{\text{nonalign}}$, high-frequency C_{q}^{hf} , paraproduct $C_{\text{q}}^{\text{para}}$)

v011.5 — APP-v011.5 Readability / identifiability formalism + strict report hierarchy

tier: untagged; status: current

$Y_L = \mathcal{G}_L x_0 + N_L$, $d_{\text{struct}}(L) = \text{rank } \mathcal{G}_L$ [Df] $I_{\text{read}} = \mathfrak{J}_{\text{dom}}(Y_L, \mathcal{G}_L, \Sigma_x, \Sigma_N, \alpha, \rho)$ [Df]
 $\Sigma_\alpha(s) = \Sigma_\alpha(s') \Rightarrow s \sim_{\{\Delta, \alpha\}} s'$ [Df] $P_{\{\alpha, n\}} \neq Y_{\{\alpha, n\}} \neq 0_\alpha(\mathcal{G}_n) \neq \mathcal{G}_n$ [Df] strict report hierarchy

v011.6 — APP-v011.6 Modal-reduction discarded-coupling audit + admissibility rule

tier: untagged; status: current

$\epsilon_{\text{mode}} = \|(I-P) \mathbin{\&\circ} P\| + \|P \mathbin{\&\circ} (I-P)\|$ [Df] Rule [Ax] (source §13, verbatim): "A scalar eigenmode is admissible only when the discarded coupling is zero or bounded below the preregistered tolerance for the declared readout."

v011.7 — APP-v011.7 Total-cost decomposition + replication depth/branching

tier: Dr; status: current

$c_{\text{URS}} = C_{\text{discover}} + C_{\text{audit}} + C_{\text{normalize}} + C_{\text{frontier}} + C_{\text{lineage}} + C_{\text{tape}} + C_{\text{geometry}} + C_{\text{turb}} + C_{\text{recovery}}$ $r_!(\pi) = \max \# \{ \text{nested licensed replications} \}$, $\mathcal{B}_!(\pi) = \prod_{\{j=1\}}^{\{r_!(\pi)\}} \kappa_j$ Rule [Dr]: intrinsic exponential phase is declared only when BOTH licensed branching and the count of distinguishable admissible normal forms grow exponentially.

v011.8 — APP-v011.8 Theorem-compatibility regimes

tier: Dr; status: current

Regime 1, fixed linear operator [Dr]: if $\mathbb{G}$, $\mathcal{L}_R$, M , D , K are fixed linear operators, sources are declared and nonlinear currents vanish -- superposition holds for solutions using the same operator trajectory under homogeneous boundary conditions; the zero state is invariant when all sources and currents vanish; rescale covariance holds under the declared weights; an antisymmetric operator sector preserves linearity and contributes no quadratic self-energy. Regime 2, time-dependent exogenous operator [Dr]: if the operator trajectory is prescribed independently of the state, common-trajectory superposition survives, but the autonomous theorem must be restated. Regime 3, endogenous geometry or nonlinear transfer [Open]: if $\mathbb{G}_n = \mathbb{G}[\Theta_n]$ with $\Theta_{n+1} = F_\Theta(\Theta_n, X_n)$, or $\mathcal{N}_R \neq 0$, global-solution superposition generally fails -- expected, not a refutation of the linear skeleton (the existing T2 gap / two-field wall). Honesty line: Regimes 1-2 are elementary linear-algebra facts stated but NOT machine-checked in our Coq floor; they are [Dr] here, not [Th_coqc].

v011.9 — APP-v011.9 Consolidated master box

tier: untagged; status: current

$\delta_R: \Gamma \neq \Gamma' \Rightarrow \mathfrak{R}(\Gamma) \neq \mathfrak{R}(\Gamma') \quad \Gamma \vdash_{\{\alpha, \rho, \kappa\}} \phi, A \not\Rightarrow A \otimes A, !_\kappa A \vdash A^{\wedge \{\otimes m\}} \quad \mathcal{P}_n = \text{FixRefine}(\mathcal{M}_n, Q_\alpha) \quad q_{n+1} F_n = F_n \# q_n, O_{\{\alpha, n\}} = O_{\{\alpha, n\}} \# q_n \quad \text{Adm} = T \wedge I \wedge B \wedge [O_\rho = 0] \quad \mathfrak{S}_n = (G_n, \Lambda_n, \mathcal{T}_n, \Theta_n) \quad X_n A = (\Phi_n I, \Psi_n I) T, A = (\sigma, I) \quad G_n = L_{\{G_n\}} \otimes I + I \otimes C_{\mathcal{F}} + C_{\{\text{int}, n\}} \quad G_n = G_n^{\wedge \{+\}} + G_n^{\wedge \{-\}} \quad \Theta_{n+1} = F_\Theta(\Theta_n, X_n, \Lambda_n, \mathcal{T}_n, u_n) \quad M \delta_t^2 \Phi + D \delta_t^c \Phi + K G[\Theta] \Phi + \nabla V - J = P_\Phi \mathcal{J}_C + \mathcal{J}_{\text{syn}} + \mathcal{J}_{\text{geo}} \quad M \delta_t^2 \Psi - D \delta_t^c \Psi + K G[\Theta] T \Psi + \nabla^2 V \Psi = P_\Psi \mathcal{J}_C + \mathcal{J}_\Lambda \quad I_R = C_{\text{turb}}(\mathfrak{S}, X) \quad \tau_R \delta_t I_R + \mathcal{L}_R[I_R, \Theta] \quad I_R = S_R + \eta_R \mathcal{L}_R I_R = \mathcal{L}_R^{\wedge \{+\}} I_R + \mathcal{L}_R^{\wedge \{-\}} I_R + \mathcal{N}_R[I_R; \Theta] \quad \mathcal{J}_C = \mathcal{J}_{\text{transport}} + \mathcal{J}_{\text{return}} + \mathcal{J}_{\text{readout}} + \mathcal{J}_{\text{tape}} \quad \mathcal{T}_{n+1} = \mathcal{T}_n \boxplus E_T(r_n) \quad \mathfrak{S}^{\wedge}_{\{\alpha, n\}} = K_{\{\alpha \leftarrow \beta\}}^{\wedge \{m\}}(Y_{\{\beta, n-m: n\}}, \mathcal{T}_n) \quad P_{\{\alpha, n\}} = \Pi_\alpha(Y_{\{\alpha, n\}}, \delta_\alpha, \nu_\alpha, D_\alpha, \theta_\alpha) \quad (\text{This is a consolidation, not new math -- the whole architecture boxed together for one-point reference, reproduced verbatim from the v0.11 source box.})$

weld — The one-line master equation (the weld)

tier: Th_coqc; status: current

$\delta_R = (a \# b) \vdash [\text{Th_coqc}] \quad L_R = D_W - W \vdash [\text{Dr}] \quad F \text{ (MQ.08 stepper)} \equiv \{ q_D : q_D \circ F = F \#_D \circ q_D \}$

Code	Dom.	Statement	Status
weld/E.01.v1	E	$\text{Know}_A(W) = 1$ iff $\text{Dist}(R_A[n], R_A^{\wedge \nu}[n]) \leq \epsilon_K$ for all admissible variations ν on window W . Independently echoed as: Knowing = Coupled System + Selection + Retention + Model + Error + Correction.	current
weld/E.02.v1	E	SC: $K(S, p) \rightarrow \text{Subject}(S)$; the collapse the paper rejects, HSC: $\text{Epi}(X, p) \rightarrow \text{Knower}(X, p)$; Possession-Constitution Collapse named; Knower fetishism: treating a knower's recognized identity/credentials as possessing epistemic value in itself once the evidential/relational work behind it is abstracted away	current
weld/E.03.v1	E	$\sigma_K(p) = \text{Admit}_E(p \text{Agent}, D, C, O, \text{Access}, \text{Language}, \text{Tools}, \text{Rights}, \text{Prov}, \text{Ev}, \text{Method}, I)$ $K_p = \langle p, C_p, P_p, E_p, M_p, I_p, U_p, \Omega_p, O_p, L_p, \sigma_K(p) \rangle$; status space $\sigma_K(p) \in \{\text{ADMITTED}, \text{LOCAL}, \text{TRANSPORTABLE}, \text{UNRESOLVED}, \text{O}$ three levels $K_{\text{prop}}(p) = \sigma_K(p) \subset$ $K_{\text{op}}(p) = \langle K_{\text{prop}}, \text{Task}, \text{Completion} \rangle \subset$ $K_{\text{trans}}(p) = \langle K_{\text{op}}, \Delta \Theta, \Delta I, \Delta U, \Delta \text{Action}, \text{Persistence}, \text{Transport}, \text{Viability} \rangle$	current
weld/E.04.v1	E	$\Pi_{\text{prac}}(p) = \text{TestPerformance}(Y, \hat{Y}, \text{intervention}, C, O)$ $\sigma_K(p) \neq \Pi_{\text{prac}}(p)$	current

Code	Dom.	Statement	Status
weld/E.05.v1	E	$\chi_G \in \{1,0,\perp\}$, 1=ADMITTED, 0=OB- STRUCTURED, $\perp$ =UNRESOLVED (thirteen admission gates G0–G13); independently, $\chi_t(d) \in \{\text{FORCED, DERIVED, POSITED,}$ $\text{BORROWED, OPEN}\}$ as a per-distinction provenance ledger $L_Q = \{(d, \chi_t(d))\}$	current
weld/E.06.v1	E	$\text{Suff_E, L}\}(Z_E^{\wedge \text{cand}}; Q, O, c, T) \in \{1,0,\perp\}$; $\text{Inv_E}(z) \neq \text{Inv_E}(z') \implies q_E(z) \neq q_E(z')$	current
weld/E.07.v1	E	$\tau_public(p) \leq \inf\{g \in G_p\} \tau(g)$ (Weakest- link claim ceiling, proved)	current
weld/E.08.v1	E	$K_local = K _{\Omega_local}$; $T^{\wedge} Y_{ij} \circ K_i \cong$ $K_j \circ T^{\wedge} C_{ij}$, $\epsilon_bridge = d(T^{\wedge} Y_{ij} \circ K_i,$ $K_j \circ T^{\wedge} C_{ij})$; $\Omega(K) = \{(g, d, l, P, O, c): \text{all re-}$ $\text{quired gates pass}\}$	current
weld/E.09.v1	E	$S_n : X_n \rightarrow Z_n$; $z_n = \Pi_n S_n(X_n) +$ $\eta_n, z_n \in \mathbb{Q}^{\wedge m_n}$; $W_n = \{n-h_n+1,$ $\dots, n\}$; $z^{\sim}_n = \Sigma_{j=0}^{\wedge \{h_n-1\}} a_{n,j}$ z_{n-j}	current
weld/E.10.v1	E	Bridge Burden: any inference from source metadata to a change in epistemic standing must identify the mediating relation that does the work; where none is supplied, the inference is pedigree substitution rather than source- sensitive epistemology.	current
weld/H.01.v1	H	$S_A[n] = q_self(F^{\wedge n}[\delta_R, T_A, c_{0:n}]) = (A_A[n], \Delta_A[n], H_A[n], \text{Phen_A}^{\wedge str}[n],$ identity update	current
weld/H.02.v1	H	$I_A[n+1] = q_id(I_A[n] \oplus A_A[n] \oplus P_A^{\wedge lived}[n] \oplus \Pi_A^{\wedge star}[n] \oplus \text{Res_A}[n] \oplus \Delta T_A[n])$ $H_dyn: \Delta_A(\lambda) = D_A^2 - 4M_AK_A\lambda = 0$; $H_info(A) = \{z \in Z_A:$ $\text{Rec}(z) \implies (E_b < \infty, R_p > 0)\}$; $H_phen(A) = \{r = A_A^{\wedge acc} \Pi_A T_A(\delta_R):$ $\text{PhenGate}(r) = 1\}$; $H_dyn \dashv DI_K \dashv H_info$ $\dashv IP_K \dashv H_phen$	current
weld/H.03.v1	H	$R_A^{\wedge \{1\}}[n] = O_A(q_A(Z_A[n]); c_n, T_n)$; current $R_A^{\wedge \{2\}}[n] = O_A(R_A^{\wedge \{1\}}[n]; c_n, T_n)$; $G_A^{\wedge MR}[n] = R_{\tau_g}(O_A(O_A(q_A(Z_A[n])))$, $C_A^{\wedge ret}, L_A^{\wedge acc}, F_A, \Pi_A$; no-free-governance law: CurrentRead- $\text{out} + \text{SelectionPriority} + \text{SelectionStability} + \text{SelfReport}$ $\not\equiv G_A^{\wedge MR}$; boundedness $B_G = I(R_A^{\wedge \{2\}}; Z_A) / \tau_c A < \infty$, $\ker$ $O_A^{\wedge \{2\}} \neq \emptyset$	current
weld/H.04.v1	H	$G_H[n] = (V_H[n], E_H[n], w_H[n])$, $w_H[n](e) \in \mathbb{Q}_{>0}$, $\phi_n \in \mathbb{Q}^{\wedge V_H[n] }$; $\mu_H \delta^2 \phi_n + d_H \delta \phi_n + \kappa_H L_H[n]$ $\phi_n + \partial_Q V_H(\phi_n) = J_H[n] - \eta_H[n]$	current
weld/H.05.v1	H	$T_{H \leftarrow AI} \circ K_AI \cong K_H \circ T_C$, with ex- plicit defects for semantic loss, source omis- sion, authority laundering, uncertainty com- pression, and context mismatch	current
weld/H.06.v1	H	$\epsilon_H = \text{Def}(q^{\sim}_H \circ F, F \#_H \circ q^{\sim}_H, O_H,$ $\text{Inv_H})$	current
weld/H.07.v1	H	$\Delta \Omega^{\sim} H_s = B_s A_s$, $\text{rank} \ll d_H$; $\Omega^{\sim} H_{s+1,0} = \Omega^{\sim} H_{s,0} + \eta_s \Delta \Omega^{\sim} H_s$; $Y^{\wedge return}_{s+\Delta} = (R_rec, R_disc, T_new, Q_next)$; $\text{RET} = (P^{\wedge post}_{H, AI} - P^{\wedge pre}_{H, AI}) - (P^{\wedge post}_{H, C} - P^{\wedge pre}_{H, C})$	current

Code	Dom.	Statement	Status
weld/H.08.v1	H	$D_s^{\wedge} \text{eff} = C_s / \sim_R $, $d_s = D_s^{\wedge} \text{eff} / C_s $;	current
weld/H.09.v1	H	$N_distinct = \{C1, \dots, Cn\} / \sim_Q $ retained experiential reorganization → possible later CTSA crystallization; retained sensitivity → C_return , retained reweighting → A_return , retained reorganization → $Q_next^{\wedge} \text{better}$	current
weld/H.10.v1	H	$A := \text{capacity to decide under context while retaining accountability}$; $S := (\text{goals, constraints, stakes, role, local evidence, lived situation})$; $F := \text{externally induced context-frame}$; $AI\text{-Empowerment}(H) \Leftrightarrow S$ preserved $\wedge$ high $\mathcal{E}$; S replaced by language $\Rightarrow F$ dominates $\Rightarrow$ [agency] degrades	current
weld/H.11.v1	H	$M0 = \text{formal option count} + \text{stated preference}$; $M1 = \text{affordance/capability} + \text{cost}$; $M2 = \text{constructed-preference/salience}$; $M3 = \text{predictive/active-inference}$; $M4 = \text{readout-native live-field model}$	current
weld/H.12.v1	H	$C \rightarrow T \rightarrow \text{Workflow} \rightarrow S \rightarrow A \rightarrow C'$; Frozen Human baseline $\rightarrow K_like \rightarrow \text{Difference/Resistance} \rightarrow H \leftrightarrow AI \rightarrow \text{Retention gate} \rightarrow \text{CTSA Return Profile} \rightarrow \text{Unaided Return Test} \rightarrow \text{Gain/Loss} \rightarrow \text{Regulation} \rightarrow \text{Provenance/Warrant} \rightarrow \text{Epistemic direction}$	current
weld/M.01.v1	M	$\delta_R = (a \# b) \vdash [\text{Th_coqc}] \quad L_R = D_W - W \vdash [\text{Dr}] \quad F$ (MQ.08 stepper); concretely $S_{n+1} = F(S_n, u_n, c_n, T_n)$; independently corroborated by the discrete telegraph form $V[n+1] = V[n] + \Delta\theta \cdot (-\gamma \cdot V[n] - D_s \cdot L_R \cdot X[n])$ (MQ.08) forced from three postulates including finite propagation $v = \sqrt{(D/\tau_c)} < \infty$, with universal eigenmode decay $ a_k[n] \leq a_k[0] e^{\{-\gamma_k n \Delta\theta\}}$ (Impermanence Theorem).	current
weld/M.02.v1	M	$q_{\{D, n+1\}} \circ F_n = F_{\#} \{D, n\} \circ q_{\{D, n\}}$; $O_{\{D, n\}} = O_{\#} \{D, n\} \circ q_{\{D, n\}}$; equivalently $q_D(F(z, u, c, T)) = F_D(q_D(z), u, c, T)$ and $O_D(z; Q, c) = O_{\#} \{D, n\} \circ q_{\{D, n\}}$ (general); $q_{\{HCA, n+1\}} \circ F_n = F_{\#} \{HCA, n\} \circ q_{\{HCA, n\}}$ (HCA domain instance, READOUT_GENESIS_CORE.md one-line weld)	current
weld/M.03.v1	M	$z \sim_{\{Q, O, c, L\}} z' \iff O(F^{\wedge k} z) = O(F^{\wedge k} z')$ for all $k \leq L$	current
weld/M.04.v1	M	$M0 = \text{arousal/intensity}$ only; $M1 = \text{familiarity/exposure}$; $M2 = \text{prediction/surprise}$; $M3 = \text{language/category construction}$; $M4 = \text{memory/schema retrieval}$; $M5 = \text{the present readout-retention model}$	current
weld/M.05.v1	M	$r_t = A_t \epsilon_t - \delta_t$, $V_t = (1/2) r_t^{\wedge} T$ $W_t \quad r_t$; $P_t := \text{Retain}_{\{\Pi_P\}}(r_t)$; $Q_{\{\Pi_Q\}}(P_t) = d_{\{Q, t\}}$	current

Code	Dom.	Statement	Status
weld/M.06.v1	M	$0 \leq \kappa_{\{A,t,Q\}}(s' s) \leq 1;$ $\mu_{\{A,t\}}(\pi Q) = \prod \kappa_{\{A,t,Q\}}(s_{\{k+1\}} s_k);$ $\text{Acc}_{\{A,t\}}(H Q) = \sup_{\pi} \mu_{\{A,t\}}(\pi Q);$ $\text{reachable}(H) = 1 \not\Rightarrow \text{Acc}(H Q) \text{ high};$ re- structuring operator $B_t: D^{\wedge}_{\text{sem}_{\{t\}}}$ $\rightarrow D^{\wedge}_{\text{sem}_{\{t+\}}};$ insight signature $\text{Acc}_{\{t+\}}(H^* Q) \gg \text{Acc}_{\{t\}}(H^* Q)$	current
weld/M.07.v1	M	$G^{\wedge}K_{\{A,t\}}(Q) = \langle V_{\{A,t\}}, E_{\{A,t\}}, \omega_{\{A,t,Q\}} \rangle;$ $U_{\{A,t\}}(Q) = \{[H] \in H^{\wedge}_{\text{disc}_{\{A,t\}}}(Q):$ $W_t(H) \geq w_0, \exists u \in U_{\{D,t\}} \text{ with}$ $\Delta_u(H, H') > 0\};$ $\tau_U = \inf\{n \geq 0:$ $\Gamma_Q(S_n) \in U_{\{A,n\}}(Q)\};$ $T_U(G^{\wedge}K, Q) = E[\tau_U G^{\wedge}K, Q];$ $\pi^{\wedge}_{\text{first}_{\{G^{\wedge}K, Q\}}}(C_j) = \Pr[H_{\{\tau_U\}} = C_j G^{\wedge}K, Q];$ $\text{usable} \neq \text{true};$ central proposal $G^{\wedge}K \Rightarrow$ $(L(\tau_U), \pi^{\wedge}_{\text{first}})$	current
weld/M.08.v1	M	discriminating action u^* sat- isfies $\delta^{\wedge}_i(u^*) \neq \delta^{\wedge}_j(u^*);$ $\delta^*_{\{t+1\}} = O_D(Z_{\{t+1\}}; u^*);$ $r^*_i = \delta^{\wedge}_i(u^*) - \delta^*_{\{t+1\}};$ $(Z_{\{t+1\}}, A_{\{t+1\}}, D^{\wedge}_{\text{sem}_{\{t+1\}}}) = \text{Update}(Z_t, A_t, \delta^*_{\{t+1\}}, \Pi_R)$	current
weld/M.09.v1	M	ClaimStrength $\leq$ EvidenceStrength	current
weld/M.10.v1	M	Phenomenon $\rightarrow$ Existing Explanations $\rightarrow$ Precise Inadequacy $\rightarrow$ Mechanism $\rightarrow$ Boundary $\rightarrow$ Engine A: Phenomenon $\rightarrow$ Theoretical Inadequacy $\rightarrow$ Mechanism $\rightarrow$ Conceptual Contribution Engine B: Practice $\rightarrow$ Observation $\rightarrow$ Intervention $\rightarrow$ Evidence $\rightarrow$ Implementation; Bridge = $1 \Leftrightarrow \text{Practice} \Delta \text{Theory};$ Conceptual Paper $\rightarrow$ Critical Empirical Test $\rightarrow$ Theory Revision	
weld/M.11.v1	M	K0: private candidate; K1: K0+public timestamp/provenance/provisionality; K2: K1+substantive external friction; K3: K2+review/certification/empirical con- straint; $K0 \rightarrow K1 \rightarrow K2 \rightarrow K3$	current
weld/M.12.v1	M	$\chi_t = \lambda_{\text{mint}} / (\lambda_{\text{conv}} + \epsilon);$ current $B_{\{t+1\}} = B_t + M_t - \omega X_t; \quad M_t \leq \lambda X_t;$ $C_{\{t+1\}} = (1 - \delta)C_t + G_t + \Phi_t(\text{Stock})I_{\text{TP}} \hat{\tau}_{\{r\}} S_t;$ $V_C = dE[C]/dt \approx (\text{Credit Creation} \times \text{Retention} \times \text{Conversion}) / H_{\text{critical}};$ $V \propto \prod x_j; \quad dV/dx_j = V/x_j;$ $\text{Priority}_j = (dV/dx_j) / (\text{Marginal Human Cost}_j + \epsilon);$ $E[C_{\{t+k\}}] > C_t;$ $V_C^* = (\dots) / (H_{\text{critical}} \times (\text{Debt} + \text{Frag} + \text{Bias} + \text{COI} + \epsilon))$	current
weld/M.13.v1	M	$A_0 \subseteq A_1 \subseteq \dots \subseteq A_m; l(d_0\{x'\}) = \min\{j : x' \notin \Gamma_j\}$	current
weld/M.14.v1	M	$C_D = G_D \circ K^*$ (worked identity); Retained-record route (proposition): if a supposedly fixed background corpus contains a current-case-specific retained record $D(x)$ varying with the target alternative, available through a route not contained in the stated source K^* , the effective architecture contains additional access not admitting $C = G \circ K^*$ with G independent of that record.	current

Code	Dom.	Statement	Status
weld/S.01.v1	S	$L_R = D_W - W$ (forced Laplacian); $A := L_R + \Gamma$; $s[n+1] = s[n] + dt(-A s[n] + J)$. L1 (invariance/recurrence): a bounded amplitude-only intervention cannot move the fixed point $s^* = A^{-1}J$, and the state returns above threshold after a finite computable number of ticks once the intervention ends. L2 (the bill): holding the state at threshold requires a non-vanishing per-tick force, cumulative cost growing linearly in time held. L3 (operator change): raising dissipation γ_i (an operator-level change, not amplitude suppression) can push the fixed point permanently below threshold at zero ongoing force. L4 (mutation): suppressing one node under permanent resets produces a strictly positive long-run floor in a coupled neighbour node.	split
weld/S.02.v1	S	$M(t') \in M$ [Axiom I: Reality-as-Record]; $A_i(t') \subseteq M(t')$ [Axiom II: Agency-as-Choice]; $G(t') := \{A_1(t'), \dots, A_N(t')\} \subseteq M(t')$ [Axiom IV: Collective as Coupled Agencies]; $R \in R_adm(t')$ [Admissible Regime Set]	split
weld/S.03.v1	S	$C_{\{A,R\}}[t1,t2] := \int_{t1}^{t2} [\alpha V_{\{A,R\}}^+(M(t')) + \beta \chi_{causal}(R) + \gamma \chi_{spec}(R)] dt'$, $\alpha, \beta, \gamma > 0$; $Resp(A) \equiv C_{\{A,R_A\}}$; Cost-Survival Link: $\lim_{T \rightarrow \infty} C_{\{A,R\}}[0,T] = \infty \Rightarrow \lim_{n \rightarrow \infty} \ P_{\{S_A\}} M(t'+n)\ ^2 = 0$ [Theorem]; choice-gate: $\neg \exists Choice(A \rightarrow R_A) \Rightarrow$ no ethics/responsibility/cost attribution; Tragic regime class: every available regime is either inadmissible or incurs strictly positive unavoidable cost, so ethics reduces to $R \uparrow_A \in \argmin C_{\{A,R\}}$ s.t. admissibility (cost-minimization with residue, not pure pass/fail)	split
weld/S.04.v1	S	$Bel_{\{i,p,n\}} = Rel_B(a_{i,p} x_{i,\mu}, E_{i,M}, \Theta_{split}, S_i^{active}, c_i)$; canonical belief vector $b_{\{i,p,n\}} = (e_{\{i,p\}}, c_{\{i,p\}}, s_{\{i,p\}}, a_{\{i,p\}}, \eta_{\{i,p\}}, g_{\{i,p\}}, r_{\{i,p\}})$; update $b_{\{i,p,n+1\}} = U_B(\dots)$; group stabilization $Bel_{\{g,d,n\}}^{\wedge(l)}(p) = Stabilize_B(\{b_{\{i,p,n\}}\}_{i \in G}, C_B)$; Belief-scale nonpromotion (proved): increasing distribution scale of $Bel(p)$ does not by itself raise $\sigma_K(p)$	
weld/S.05.v1	S	$Z_{\{G,t\}} = \langle Z_{\{A_i,t\}} \rangle_{i=1}^n$, $T_{\{G,t\}}, A_{\{G,t\}}, C_{\{G,t\}}$; $S_{\{G,t\}} = q_{\{sem,G,\Omega_{\{G,t\}}\}}(Z_{\{G,t\}})$; $m_{\{G,t\}}^{\wedge}(e) = \rho_G$ $m_{\{G,t\}}^{\wedge}(e) + \sigma_{\{G,t\}}(e)$; $W_{\{G,t\}}(H) = W_G(R_{\{G,\leq t\}}, independence, defects, calibration, objection channels)$	split
weld/S.06.v1	S	Power may alter agency by altering meaning/accessibility before overt choice	not_an_equation

Code	Dom.	Statement	Status
weld/S.07.v1	S	Experience→Interpretations→Absent Perspective→Observable Evidence→Direct Human Conversation	not_an_equation
weld/S.08.v1	S	P1-P8 [Open], §7	not_an_equation
weld/S.09.v1	S	H1 live possibility beyond option count; H2 meaning access beyond resources; H3 history- shaped re-entry; H4 resonance without con- sent; H5 structural closure before overt prohi- bition; H6 evaluator visibility; H7 human-AI live-field carryover; H8 corrigibility predicts durable agency	not_an_equation
weld/S.10.v1	S	$s(x,t) \in \mathbb{R}^n$	current
weld/S.11.v1	S	$V(x,t) = \Phi(s(x,t))$	current
weld/S.12.v1	S	$\tau \partial_t j + j = -D \nabla s$	current
weld/S.13.v1	S	$\partial_t s = -\nabla \cdot j - \Gamma(s) + S_{env}$	current
weld/S.14.v1	S	$\partial_t s = L \tau s$	current
weld/S.15.v1	S	$s(t) = \sum_k c_k e^{-\lambda_k t} r_k$	current
weld/S.16.v1	S	$L = [[-\alpha, \epsilon], [\epsilon, -\beta]], \alpha, \beta > 0$	current
weld/S.17.v1	S	$\lambda_{\pm} = -(\alpha+\beta)/2 \pm \sqrt{((\alpha-\beta)/2)^2 + \epsilon^2}$)	current
weld/S.18.v1	S	$\Gamma \uparrow$ (increase dissipation)	current
weld/S.19.v1	S	$S_{env} \downarrow$ (reduce load)	current
weld/S.20.v1	S	$L_{ij} \downarrow$ (limit propagation)	current
weld/S.21.v1	S	$\tau \downarrow$ (shorten memory)	current
weld/S.22.v1	S	$LR = DW - W$ (forced Laplacian); $A := LR$ $+ \Gamma$; $s[n+1] = s[n] + dt(-A s[n] + J)$	current
weld/S.23.v1	S	$s^* = A^{-1} J$ is the unique fixed point independent of u ; $\ s[n] - s^*\ \leq$ $\rho^{n-nB} \ s[nB] - s^*\ $ after the last nonzero tick of a bounded intervention; if $s^*_i >$ θ there is an explicit finite N^* after which $s_i[n] > \theta$ for all $n \geq nB + N^*$	current
weld/S.24.v1	S	$ u_i[n] = dt[(\gamma_i + D_i)(s^*_i - \theta) - \sum_{k \neq i}$ $W_{ik}(s^*_k - s_k[n])]$, converging to a rate $F_i \hat{\infty} > 0$ whenever $s^*_i \geq \theta$, so cumula- tive force is $F_i \hat{\infty} \cdot n + o(n)$	current
weld/S.25.v1	S	$s^{*'}_i = s^*_i / (1 + \Delta B_{ii})$ with $B = A^{-1}$ (Sherman-Morrison); any $\theta < s^*_i$ is reached below with $\Delta > \Delta^* = (s^*_i/\theta - 1)/B_{ii}$, and zero ongoing force	current
weld/S.26.v1	S	$s_j \gtrsim (J_j + W_{ij} s^-_i)/(\gamma_j + D_j)$, with s^-_i an explicit increasing function of node i 's own inflow, under permanent periodic resets of node i	current
weld/S.27.v1	S	$M(t') \in M$	current
weld/S.28.v1	S	$A_i(t') \subseteq M(t')$	current
weld/S.29.v1	S	$G(t') := \{A_1(t'), \dots, A_N(t')\} \subseteq M(t')$	current
weld/S.30.v1	S	$R \in R_{adm}(t')$	current
weld/S.31.v1	S	$\chi_{causal}(R) := 1[\tau_{c'}(R) \approx 0]$	current
weld/S.32.v1	S	$\chi_{spec}(R) := 1[\Delta_{spec}(R) \leq 0]$	current
weld/S.33.v1	S	$C_{\{A,R\}}[t_1,t_2] := \int_{t_1}^{t_2} [\alpha$ $V^+_{\{A,R\}}(M(t')) + \beta \chi_{causal}(R) +$ $\gamma \chi_{spec}(R)] dt', \alpha, \beta, \gamma > 0$	current
weld/S.34.v1	S	$Resp(A) \equiv C_{\{A,R_A\}}$	current

Code	Dom.	Statement	Status
weld/S.35.v1	S	$C_G[t1,t2] := \Sigma_{\{i=1\}^N} w_i$	current
weld/S.36.v1	S	$\lim_{\{T \rightarrow \infty\}} C_{\{A,R\}}[0,T] = \infty$	current
weld/S.37.v1	S	$\lim_{\{T \rightarrow \infty\}} C_{\{A,R\}}[0,T] = \infty \Rightarrow$ $\lim_{\{n \rightarrow \infty\}} \ P_{\{S_A\}} M(t'+n)\ ^2 = 0$	current
weld/S.38.v1	S	$Eth(A) \Leftrightarrow C_{\{A,R_A\}}[t,\infty) = 0$	current
weld/S.39.v1	S	$R^* \in \operatorname{argmin}_{\{R \in R_adm\}} C_{\{system,R\}}$ s.t. $\tau_c'(R) > 0, \Delta_spec(R) > 0$	current
weld/S.40.v1	S	$\neg \exists \text{ Choice}(A \rightarrow R_A) \Rightarrow$ no ethics attri- bution, no responsibility attribution, and no moral cost attribution	current
weld/S.41.v1	S	$\forall R \in R_available(t') : \neg[\tau_c'(R) > 0 \wedge$ $\Delta_spec(R) > 0] \vee C_{\{A,R\}}[t',t'+T] > 0$, for all $T > 0$	current
weld/S.42.v1	S	$R^\dagger_A \in \operatorname{argmin}_{\{R \in R_available(t')\}}$ $C_{\{A,R\}}[t',t'+T]$ s.t. $\tau_c'(R) > 0$, $\Delta_spec(R) > 0$	current
weld/S.43.v1	S	$Bel_{\{i,p,n\}} = Rel_B(a_i, p \mid x_i, \mu_i,$ $E_i, M_i, Theta_i, I_i, S_i^{\wedge}active, c_i)$	current
weld/S.44.v1	S	$b_{\{i,p,n\}} = (e_{\{i,p\}}, c_{\{i,p\}}, s_{\{i,p\}},$ $a_{\{i,p\}}, \eta_{\{i,p\}}, g_{\{i,p\}}, r_{\{i,p\}})$	current
weld/S.45.v1	S	$b_{\{i,p,n+1\}} = U_B(b_{\{i,p,n\}}, Ev_{\{i,p\}},$ $M_i, \mu_i, E_i, Theta_i, I_i, Trust_i,$ $Affect_i, Utility_i, Repetition_i, Auth_i,$ $Pow_i, Access_i, Language_i, Algorithm_i,$ $Obj_i, c_i)$	current
weld/S.46.v1	S	$Bel_{\{g,d,n\}}^{\wedge}\{(l)\}(p) =$ Stabi- $lize_B(\{b_{\{i,p,n\}}\}_{i \in G}, C_B)$	current
weld/S.47.v1	S	$Z_{\{G,t\}} := < \{Z_{\{A_i,t\}}\}_{i=1}^n,$ $T_{\{G,t\}}, A_{\{G,t\}}, C_{\{G,t\}} >$	current
weld/S.48.v1	S	$S_{\{G,t\}} := q_{\{sem,G,\Omega_{\{G,t\}}\}}(Z_{\{G,t\}})$	current
weld/S.49.v1	S	$m^{\wedge}G_{\{t+1\}}(e) = \rho_G * m^{\wedge}G_t(e) +$ $\sigma_{\{G,t\}}(e)$	current
weld/S.50.v1	S	$W_{\{G,t\}}(H) = W_G(R_{\{G,<=t\}}, indepen-$ dence, defects, calibration, objection chan- nels)	current
weld/W.01.v1	W	$c^{\wedge}(PE \rightarrow H)_{\{i,t\}} = B^{\wedge}(PE \rightarrow H)(Z_t; i, g)$	current
weld/W.02.v1	W	candidate generation $\rightarrow$ validated	not_an_equation
		$K/A^{\wedge}AI \rightarrow$ robotic embodiment $\rightarrow M_t \rightarrow Y_t \rightarrow L_t \rightarrow s^{\wedge}L_t \rightarrow \Gamma_t \rightarrow W^{\wedge}M_t \rightarrow o_{\{t+1\}} \rightarrow \Gamma^{\wedge}eff_t \rightarrow AD$ bridge $\rightarrow \{\Pi^{\wedge}live, A^{\wedge}corr_H, r_H, S_H\} \rightarrow H^{\wedge}cap_t \rightarrow P^{\wedge}H_t \rightarrow P_t \rightarrow$ collective rules $\rightarrow$ next system state	

Part 2 — Rootless items (target: none)

None at this build.